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2022 K-Innovation Partnership Program with Lao PDR

National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022



2022 K-Innovation Partnership Program with Lao PDR

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Program Investigator	Ji Hyun KIM, Research Fellow, STEP I Wangdong KIM, Senior Research Fellow, STEP I
Project Manager	Inkyoung Sun, Research Fellow, STEP I
Researchers	Inkyoung SUN, Research Fellow, STEP I Sung Chul CHUNG, CEO, Wonjung Institute Su Gwan DOH, Professor, University of Ulsan Jinwon KANG, Research Fellow, Korea Institute of S&T Evaluation and Planning Soojung SOHN, Senior Research Fellow, STEP I Yoojung SHIN, Assistant Professor, Jeonbuk National University Soumana CHOULAMANY, Director General, Dept. of Technology and Innovation (DTI), Ministry of Technology and Communications (MTC), Lao PDR Khamhong SICHANTHAVONG, Deputy Director General, MTC, Lao PDR Vilavsone BOUPHALATH, Deputy Director General, MTC, Lao PDR Sengpasith HOUNGALOUNE, Lecturer, National University of Lao PDR Phanpakit OUPHANHDALA, Deputy Director General, Lao People's Revolutionary Party, Lao PDR Vimontha KHIEOVONGPHACHANH, Professor, National University of Lao PDR Anousone INTHAVONG, Deputy Director, MTC, Lao PDR Thanva SISOU LATH, Director, MTC, Lao PDR
Project Coordinator	Jiyong AN, Senior Researcher, STEP I

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5th~7th Flrs., Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si, 30147.

Republic of Korea

Tel: +82-44-287-2000 Fax: +82-44-287-2068

www.stepi.re.kr

For more information, visit

stepi.re.kr/global

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Preface

The title of the 2022 K-Innovation Partnership Program with Lao PDR is the “National Master Plan for STI R&D and Application in Lao PDR.” As a three-year project (2020-2022), the project aims to support the development of the National STI Master Plan of Lao PDR. For the last two years, our STEPI-MTC (Ministry of Technology and ICT in Lao PDR) team has focused on analyzing the national innovation system in Lao PDR and sharing STI development experiences and current issues of both countries. In 2021, our team particularly explored public STI research institutes, higher education and university systems, and the STI roles in the agriculture and biotech sectors of both Korea and Lao PDR, and published the report, which was presented to the national government of Lao PDR.

In order to develop a framework and drafts of the National STI Master Plan of Lao PDR by the end of the final year of the project, STEPI and MTC collaboration have placed the top two priorities in 2022 in 1) policy consultation in three strategic areas—technology commercialization, SME innovation, and international STI collaboration—, 2) capacity building of Laotian government officials in STI policy planning through a series of workshops and seminars held throughout the year of 2022 in Korea and Lao PDR.

This report is the output of a year-long collaboration between STEPI and MTC on STI policy consultation and a series of capacity-building workshops. Starting with an extensive introduction of the National S&T Basic Plan in Korea in Chapter 1, the report provides policy analyses in three strategic areas of technology commercialization, SME innovation, and international STI collaboration of both Lao PDR and Korea from Chapter 2 to Chapter 7. Each chapter explores the international and domestic environments of the designated strategic area and identifies challenges and opportunities in the area by examining statistical data, governance, stakeholders and required resources and current

policy issues of both countries. In particular, those chapters about each policy area in Lao PDR provide drafts of each policy area of the National Master Plan for STI R&D and Application. The Draft of the National Master Plan is summarized in Chapter 9. Given the significance of highly skilled STI human resources in the context of national development, in addition, it is addressed how the Korean government dealt with the problem of STI brain drain through the advanced S&T institution building in Chapter 8. All these chapters on Korea and Lao PDR deliver valuable information for mutual learning and insightful messages for the implementation of the National STI Master Plan presented in this report.

As highly recognized several times already by local media for its work on identifying strategic areas of the STI master plan and developing a comprehensive framework of STI development, the K-Innovation Partnership Program with Lao PDR has contributed to defining the STI roles aligned with the national socio-economic development framework. Despite the end of our joint project in December 2022, the STEPI is going to support further developments and implementation of the STI master plan of Lao PDR through continuous communications with its partner MTC. Finally, I would like to express my deepest gratitude to all the experts who participated in our bilateral project for hard work and enthusiasm.

Dr. Inkyoung Sun, Research Fellow, STEPI

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CHAPTER 1

National S&T Basic Plan in Korea

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022



Chapter 1. National S&T Basic Plan in Korea

Dr. Sungchul Chung, CEO, Wonjung Institute

1. Introduction

1.1 Overview

National development plan is a formal document that presents the visions and goals of development of a nation as well as a systematic and comprehensive set of strategies to attain the goals, along with the plans for the mobilization of resources required for the implementation of the plan. A national development plan, in general, comprises three components: (1) At the aggregate level, the national planning authority draws up a macroeconomic plan; (2) The macroeconomic plan provides a kind of direction or guidance for sectoral ministries to draft sectoral plans; and (3) To put the plans into action, sectoral ministries prepare project-level investment plans, in consultation with the planning authority. Science, technology and innovation (STI) policy plan is a sectoral plan that focuses on the issues related to the development and promotion of science, technology and innovation.

Basically, governments formulate and implement development plans as a means for government intervention in social and economic development. The need for national development plans is particularly emphasized in the case of developing economies, where government intervention is required to make up for the deficiency of private sectors' capability. James Wolfensohn, former President of the World Bank stressed that "All developing countries should have a comprehensive development framework as they need more integrated approach to development based on a framework articulated and owned by the country itself" (Mallaby, 2004, pp. 232-260). In the same vein,

governments draw up and carry out STI plans as a tool for systematic intervention in the processes of STI development and related policy areas.

Korea's development planning exercise started with the Five-year Economic Development Plan which was first launched in 1962 and concluded with the completion of the Seventh Plan in 1996. The plans provided not only clear directions for national development but also inspired the Korean people to actively engage in the processes of economic development for more than three decades. Each of the seven plans, from the first to the seventh, was geared to attaining a self-sustainable economy, even though the priorities of the plans varied in response to the changes in internal needs and external environments. STI has been in the core of the Korean development strategy from the very beginning. The First Five-year Development Plan was supplemented by a separate plan on STI development (the Five-year Technology Promotion Plan, 1962-1966), and science and technology has since been on the top of the development agenda throughout the entire development period.

This report is to review and draw policy lessons from Korea's policy experience in planning STI development, with a focus on the Science and Technology Framework Plan which was launched first in 2002. The Science and Technology Framework Plan is unique in several ways: (1) It is a plan that has its own legal base – the Framework Act on Science and Technology (2001); (2) Even though it is a sectoral plan, it is not subject to the directions or guidelines of any higher-level plans, and thus sets its own visions and goals to pursue; (3) It is a government-wide plan designed and formulated through inter-ministerial cooperation and consultation; (4) It is a plan to prepare Korea for the new stage of development or 'post-catchup' stage, where, unlike at the catchup stage, creative research becomes more important; and (5) It is a five-year plan that provides a mid-term visions for STI development and strategies to attain the visions, and therefore, it is more about providing an overall vision and directions of STI policy rather than setting specific scientific and/or technological targets.

The Science and Technology Framework Plan (2002-2006) is Korea's attempt to design an S&T policy plan free from the frameworks and/or directions of higher-level plans or

policy, and therefore offers unique lessons for those seeking to prepare an S&T policy plan not stringed to other national development plans. This report begins with a discussion on the rationale for national development planning, including STI planning, structure of STI plan, and what it requires to formulate and implement S&T plans successfully. Then we review the Korean experience in formulating the S&T Framework Plan (2001-2006), covering discussions on how STI planning began in Korea and how it had evolved over time, as well as the planning processes and structure of the plan. Then finally, we assess the achievements of the first S&T Framework Plan, evaluate the overall performance of the plan and conclude the report by deriving policy lessons and issues.

2. Why STI Plan?

2.1 Development Planning: Pros and Cons

Many newly independent countries in the 1950s and 1960s resorted to development planning as a major tool for national development. Then, economic theory was interpreted more in favor of market failure, supporting an interventionist economic policy. In ideological terms, the post-World War period saw a broad consensus in favor of state intervention in the economy on both sides of the Cold War, largely because of the weakness or lack of the capacity of the private sectors in developing countries.¹ They also found superiority of interventionist economic policy in the success of the Marshall Plan in rebuilding the Western European economy.

The effectiveness of the interventionist policies of the newly independent countries based on development planning, however, was far short of sustaining the supports for the interventionist arguments. Beginning in the 1970s, disappointment with the interventionist approach toward development began to spread. Some argue that plan-based development efforts are bound to fail because plans distort resource allocation by replacing more efficient private players with failing state agencies (Agarwala, 1983; Bauer, 1982). The argument was confirmed by the collapse of the Soviet bloc around the end of the 1980s and the remarkable economic success of China's adoption of a market-oriented economic system. Most of the former Soviet republics abandoned their planning systems in the 1990s and early 2000s (Yazlyyev, 2017). Low-income, aid-dependent countries also took similar actions in the 1980s and 1990s: Many national planning agencies and economic development ministries were abolished or merged into other ministries, and their status was marginalized in terms of power, staffing, budget and relevance to policy. After two or so decades of the plan-based policy exercise, in the 1970s and 1980s, they began to lean more toward the concepts of government failure

¹ The success of Soviet industrialization after five-year plans started in 1928 was proclaimed as evidence of the superiority of planned over free market economies (Baran, 1962).

and rent-seeking activities caused by government intervention in the economy than the market failure arguments.

Despite the increased disillusionment with development planning, many countries still rely on planning for the promotion of development. Some of the fast-growing economies, such as China, Malaysia, India and others maintain their national planning systems, while seeking free trade and the globalization of the economic systems. It appears that they do so because unlike the Western countries, they find the sources of the failure of development planning in the lack of policy capacity of the government – lack of political will, inadequate information, inability to grasp the actual needs and capabilities of the economy, weak planning capacity, etc. rather than in government failure (Faber & Seers, 1972). The proponents of planning argue that government intervention is inevitable, if and when private sectors are not capable of dealing with the problems they face in the market (because of the lack of information, improper institutions, immature systems, and others) and/or market fails to work because of externality, imperfect information, uncertainty, spillovers, transaction costs, etc.

Entering into the 1990s, economic theory became more in favor of state intervention in the economic affairs. Many economic theorists point out new flaws in the idea of market mechanism that makes optimal resource allocation, and argue that government responsibility goes beyond the “night-watchman state” (Barr, 1992; Kahneman, 2011; Thaler & Sunstein, 2009; North, 1990). Interestingly, Bates (1995) said that “an older generation (of scholars and advocates) who had emphasized the importance of market failures in development economics finds in the new institutionalism new justification for their interventionist beliefs” (p. 27). However, this is by no means in line with the old generation theories supporting the old-style central economic planning. They argue that new forms of state intervention are in order, which is a new trend in defiance of the neo-liberalism of the 1980s.

The World Bank which had led the drive toward neo-liberalism for more than 20 years, came up with a survey result in 2007 that 62 countries had development plans (Chimhowu et al., 2019). In a 2016–17 survey that replicated the World Bank survey, they found that the number of countries that had national development plans more

than doubled to 134 (Chimhowu et al., 2019). The spread of national development plans over the recent decade is a notable policy trend that requires explanation and exploration. Chimhowu et al. (2019) found three reasons for the re-emergence of national plans²: (1) to meet the conditions for development aids; (2) to respond to the SDGs of 2000; and (3) to meet the challenges and reap the opportunities of globalization. In essence, the re-emergence of national plans suggests that many countries have no better options than relying on state-initiated planning to initiate and implement national efforts to enhance the standard of living of the people under the ever-changing global environments. This is particularly so for the countries, where the private sectors lack capacity to navigate the fast-changing markets.

2.2 Why and How governments intervene in STI

STI policy refers to government measures designed for the creation, funding, support, and mobilization of resource for science, technology and innovation. STI policy covers a very broad range of activities including the production, diffusion, and utilization of knowledge, in both public and private sectors. The scope, aims, subjects, and mechanisms of STI policies have varied widely over time.

Why should governments intervene in the affairs related to STI? Various schools of STI policy offer diverse rationales for government intervention in STI processes, but the most widely accepted one derives from the neo-classical economic theory. The neo-classical economics shows that market is bound to fail in providing optimal solutions if market transactions involve the problems of externality, uncertainty, indivisibility, and/or appropriability. STI policy or government intervention in STI is justified by the theoretical prediction that market is bound to fail in making efficient resource allocation because:

- (1) social returns to STI investment exceed private returns, and therefore private players tend to invest less than social optimality requires, and
- (2) STI involves uncertainty, and therefore, private players tend to invest in R&D and innovation only when expected returns exceed opportunity costs.

² Chimhowu et al. (2019) suggest four reasons, but in fact two of them are about dealing with the SDGs of 2000

For optimal solution, therefore, the theory requires government to step in and correct the market failure by providing subsidies or other forms of assistance.

According to the national innovation system theory, innovation results from interactions between and among innovation players, say, universities, business enterprises, and research organizations, and therefore, the theory emphasizes the importance of systems linkage, such as industry-science interface or industry-university cooperation. The issue is that in the absence of government intervention, the system may fail, keeping new knowledge generated by universities from being utilized by industry, due to the asymmetry of information among the innovation players. To promote the interactions between and among innovation players, government may provide facilities and/or programs, such as ‘techno-polis’ that provides physical and institutional conditions for collaboration between the creators and users of knowledge/technology.

Government interventions in STI, in general, aim at providing the following:

(1) Infrastructure for STI:

- Creation of and support for STI institutions and building STI capacities
- Provision of rules and/or institutions for the promotion of STI: IPR system, information system, standards, etc.
- Promotion of industry-science interface: promotion of clusters, techno-polis, etc.

(2) International STI linkages:

- Provision of institutional and/or political base for international STI linkages with foreign countries
- Diplomatic supports for international STI cooperation
- Promoting and facilitating participation in global STI activities

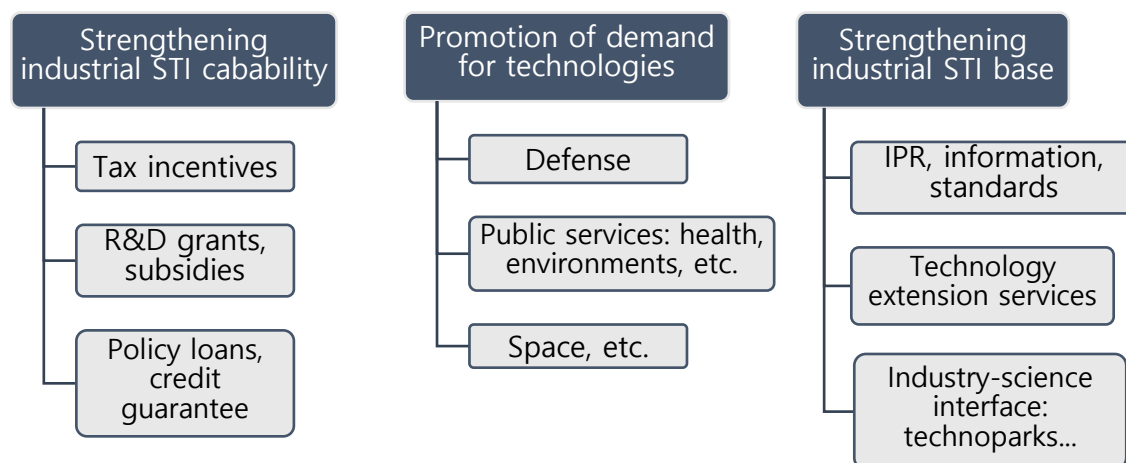
(3) Provision of environments conducive to innovation:

- Provision of stable social and economic policy framework conducive to long-term R&D and innovation: economic and social stability, STI awareness of the society, etc.
- Promotion of market competition and exposure of domestic industries to international competition.

- Provision of financial and non-financial assistance to encourage and facilitate private investments in R&D and innovation

Policy measures for the promotion of industrial STI can also be categorized into: (1) policy measures to strengthen STI supply capacity; (2) creation of the demand for technologies; and (3) actions to strengthen the foundation for STI development. Government can promote industrial STI activities by providing direct (R&D grants, loans, etc.) and indirect subsidies (say, tax incentives) for industrial R&D and innovation. Government can stimulate industrial STI as a consumer of by implementing programs related to the basic function of the state, such as defense, public health, the environments, space, and others. In addition, government can facilitate industries' STI activities by providing an appropriate IPR services, technology extension programs, technology and market information system, and building and operating technology parks for the promotion and facilitation of industry-science interface.

[Figure 1-1] The structure of government supports for industrial STI



Source: Author

2.3 STI Policy Plan

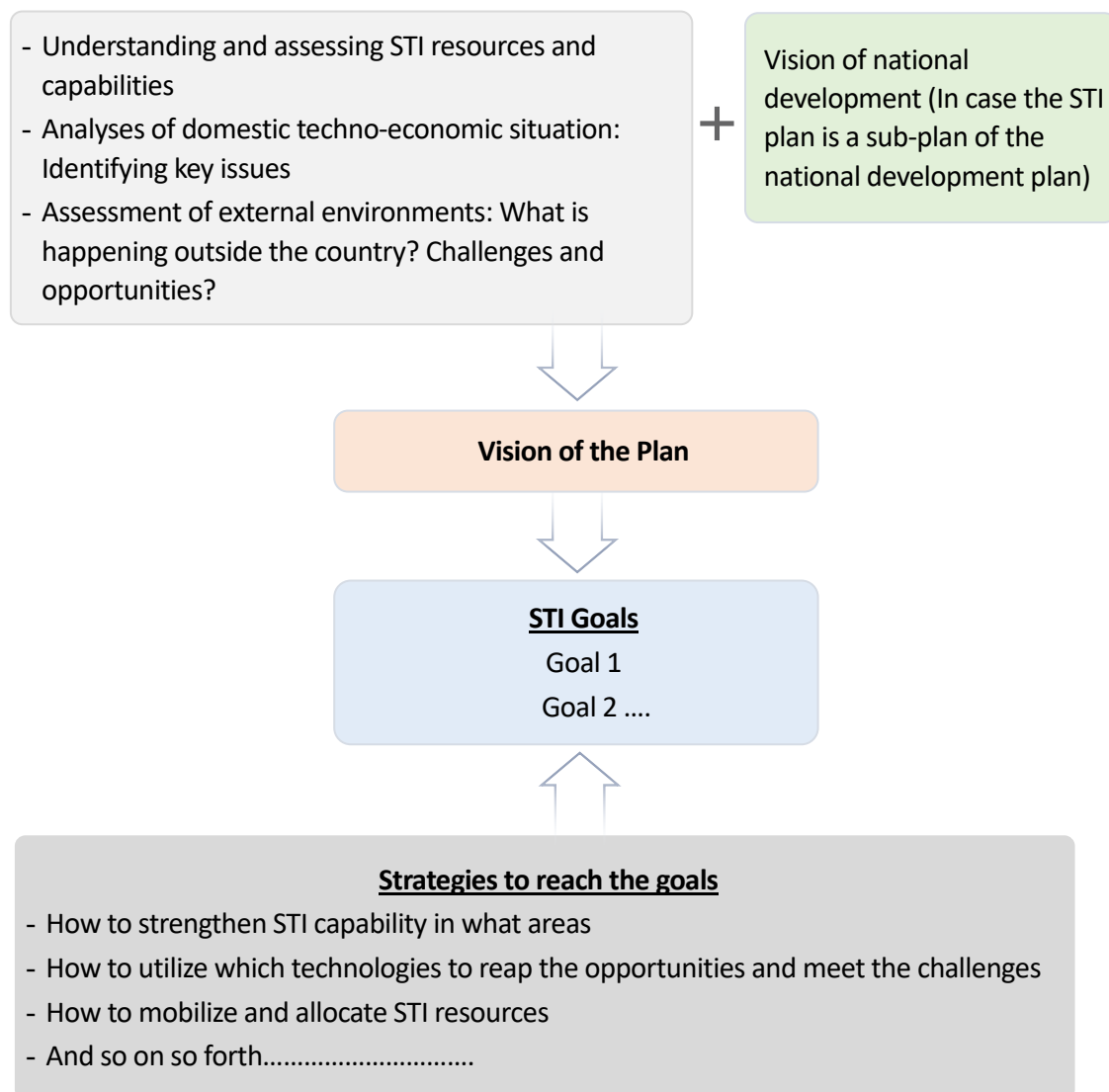
National development plan contains goals of and strategies for national development, on the basis of which government intervene in the development process. National development strategy is about how the government, in cooperation with the private

sectors, best leverages the country's endowments for national development, under the given (1) resource endowments, the capacity of the society, and (3) social, political and economic attitude of the members of the society. This suggests that countries choose different strategies for development because they differ in (1) resource endowments that they can exploit for economic development, (2) the attitude towards development and well-being, that is, motivation and/or desire for change and development, and (3) in terms of capacity to initiate or deal with changes. The same logic applies also to STI plan, which means that STI planning process should begin with the following:

- (1) Understanding and assessing the STI resources and capabilities of the nation: the strength and weaknesses
- (2) Analyzing the domestic situation – identifying issues and diagnosing the nature of the problems
- (3) Assessing the situations surrounding STI, including S&T trends and developments and the ways countries respond to the changes and trends
- (4) Identifying challenges and opportunities based on the above analyses
- (5) Assessment of the nation's capability to deal with the current issues and future challenges

The analyses are to assess the situation where the country stands in terms of STI, evaluate the STI potential of the country, and explore what should be done in terms of STI development to realize the vision for national development. The key player in this process (the ministry or agency responsible for STI policy) should consult with the national planning ministry or authority in order to make sure that the direction of the process is well aligned with the national development vision, and should collaborate with stakeholders and policy experts, and S&T community to make the plan tuned to the needs and actionable. The logical process of planning can be described as [Figure 1-2].

[Figure 1-2] Logical structure of STI plan



Source: Author

The vision for STI development is set in support of the vision for national development set forth in the higher-level plan, if any, taking into consideration the STI resources, capability, other conditions. The goals of the STI plan refer to what the plan pursues to achieve in terms of STI development that are instrumental in realizing the vision. Strategies are actions or measures to take in order to reach the goals that may include:

- Selection of the key technologies that are required to attain the goals

- How to acquire the key technologies: Who do what?
- How to mobilize the resources required – human, financial and organizational
- How to allocate the resources
- How to utilize the technologies: Who do what?
- How to motivate industries for STI
- How to make the environments right for STI
- And others

For the plan to be actionable, strategies should be based on reality -- the resource availability, STI capability, STI trends, and the techno-economic environments where the country stands, for the understanding and assessment of which it requires strong policy capacity. Strong policy capacity is essential not only to the formulation of effective plans but also to the consistent implementation of the plans.

2.4 Policy Capacity: What it requires to make STI Plans Effective?

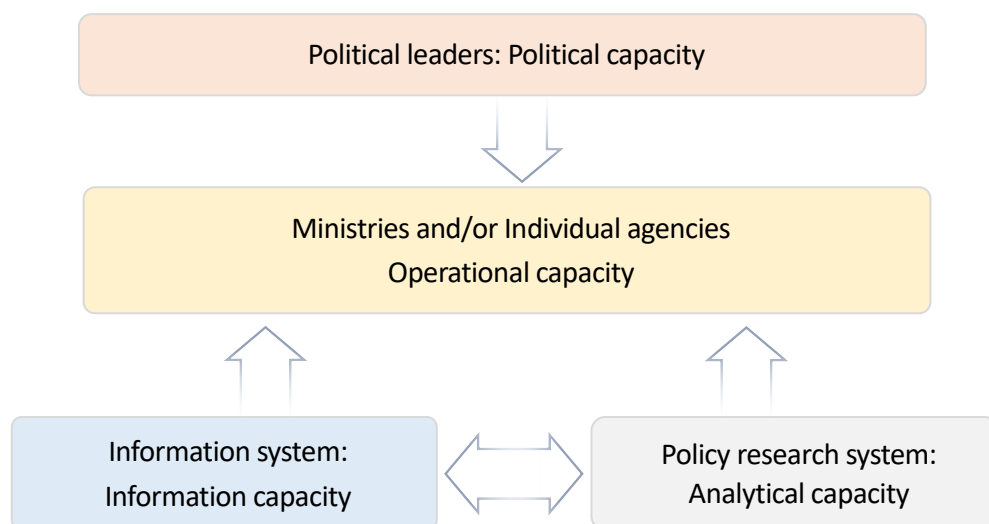
Government intervention in STI development is justified because the unique nature of STI makes it impossible for market to function as a mechanism for optimal resource allocation, and also because in the case of developing economies, not only they lack STI capacity, but also private industries are simply incapable to deal with STI issues they face in the market. As no sectors in developing countries are technologically prepared in the early stage of development, government has to intervene in almost every sector for STI capacity building, which requires solid policy capacity.

Policy capacity includes the capacities to systematically diagnose policy problems, to formulate set of measures to respond to the problems, and to put the policy measures into action in a consistent and effective manner. To understand the nature of policy issues, they need information on STI situation inside and outside the country, for which they need a system that collect and disseminate STI data and information. Then, they need to analyze the information to diagnose the real nature of the issues, for which they need to have policy research capability. Then, based on the understanding, we should

be able to formulate policy measures to deal with the issues identified, which requires organizational apparatus to formulate policy measures, plus technical competence of bureaucrats. Finally, the policy measures should be put into action, for which it requires not only resources to implement the policy measures – financial, human, and organizational resources, but also political leadership to ensure long-term policy commitment to STI, which depends very much on the political competence of STI leadership of the country.

So, policy capacity is composed of a set of resources and skills that are required to perform policy functions -- such as analytical competence, operational competence, and political competence. Each of the competences involves resources and/or capabilities at individual, organizational and systemic levels. In short, policy capacity is defined here as an integration of capability to systematically diagnose policy problems; capability to formulate sets of measures to respond to the problems; and capability to put policy measures into action in a consistent and effective manner.

[Figure 1-3] Policy capacity



Source: Author

3. Korean Experience: The Science and Technology Framework Plan

3.1 Emergence of STI as a Policy Issue and STI Planning in Korea

Science and technology have been in the core of Korea's development strategy from the very beginning. In 1961, a draft of the First Five-year Economic Development Plan was reported to the political leadership. The political leaders were very satisfied with the draft plan that was very ambitious and tuned to the needs of the nation. Expressing satisfaction, however, they raised the following question:

I wonder if there isn't any problem in acquiring technologies required to attain the development goals. Would it be possible for us to build and operate the new factories with our own technologies and technical manpower? If not, are there any additional plans to back up the plan in terms of technology and technical manpower?

(The Innopolis Foundation, 2013, p.28)

No one from the planning ministry expected that such a practical but critically important question to be raised by the political leaders. They could only evade the moment by reporting that a separate back-up plan on technology and manpower was being prepared (Chung, 2016). From that moment on, technology and research have been placed in the center of Korea's development policy. It did not take long for the planning ministry to prepare a supplement to the First Five-year Economic Development Plan (1962-1966) that provided policy measures to acquire technologies required for the industrialization and to build STI capacity. As a central agency responsible for formulating and implementing STI policy, they created the Ministry of Science and Technology (MOST) in 1967.

Since MOST was created in 1967, there have been three strands of STI policy planning activities:

- (1) S&T sector plans of the Five-year Economic Development Plans (Four plans covering the period 1962-1981) and the Five-year Social and Economic Development Plans (Three plans covering the period 1981-1996);

- (2) As the Korean economy grew, the government began to reduce direct intervention in economic affairs and put an end to 'plan-based development' policy in 1997 with the completion of 7th Five-year Social Economic Development Plan (1992-1996). But the government deemed that state intervention is still required in the areas of STI, and thus maintained the plan-based development strategy, such as the Five-year Science, Technology and Innovation Plan (1997-2001), and the Science and Technology Framework Plans (five-year plan that revolves with the five-year term government, 2002-); and
- (3) Along with the five-year plans, MOST has also been formulating longer-term plans that set forth visions for STI development. The first one was the Comprehensive Long-term Plan for Science and Technology Development (1967-1986), and since then MOST has been issuing long-term plans revising mid-long-term development visions for STI development, taking into consideration the scientific and technological advances inside and outside the country. For example, in 1980, seven years before the end of the first long-term plan, MOST issued the Long-term Science and Technology Development Plan (1980-1991). So, the longer-term plans have been prepared and published when the government deemed it necessary to reflect the changes in STI and the new trends and revise visions for STI development.

The above shows that the Korean development has been highly reliant on STI, for the development of which the government played a key role from capacity building to technology development, and to the promotion of industry-science linkages. The strategy Korea opted for was a "STI-based, government-led" development. Such a choice was in a sense inevitable, considering the conditions Korea faced in the early stage of the development, such as:

- (1) Korea was (and still is) very poor in natural resources, but was blessed with well-educated, highly-motivated human resources;
- (2) In terms of development capacity, the private sector was extremely weak, while the government sector was relatively strong in terms of policy capacity; and
- (3) The Korean people then had a highly positive attitude toward national development and had a strong desire for change and development.

So, Korea opted for a strategy based on STI, which is a very brain-intensive approach. The government led the development as the leading actor or orchestrator to make up for the weakness of the private sectors' capacity. And as a means to mobilize national resources for development, to communicate with the tax payers on the need for development, and to engage the society in the development efforts, the government relied very much on planning which required strong policy capacity. Indeed, the plans provided the people with a clear picture of the national development and inspired them to actively participate in the process of economic development for more than three decades. How much the Korean government stressed the role of STI as a strategic tool for attaining the national development goals is well reflected in the Five-year Economic Development Plans. The plans placed STI on the top of the development agenda throughout the entire development period (1962-1996). The S&T policies contained in the plans deserve special attention for the following reasons:

- (1) The plans were very specific about the goals and strategies and clearly tuned to the national development vision, and
- (2) The plans were based on clear understanding and assessment of the capacity and needs of the industries in setting the goals: Starting with promoting technological learning in the early stage of development and then moving to promoting domestic R&D for technology acquisition...
- (3) In identifying target technologies for development, they monitored and assessed both domestic and international trends in technology development, through technology foresight exercise and other analyses.
- (4) The plans were fully backed up by specific investment plans.

The above means that the plans were practical and action-oriented, and thus the STI plans could fulfill the role of facilitating industrial development. The following table lists the STI plans implemented over the past five decades:

[Table 1-1] List of the STI plans

Category	Title of the plan	Plan period
S&T sector plan of the 5-year Economic Development Plan	The 1 st 5-year Technology Promotion Plan	1962-1966
	The 2 nd 5-year Technology Promotion Plan	1967-1971
	The 3 rd S&T Development Plan	1972-1976
	The 4 th S&T Sector Plan	1977-1981
	The 5 th S&T Sector Plan	1982-1986
	The 6 th S&T Sector Plan	1987-1991
	The 7 th S&T Sector Plan*	1992-1996
Legal mandate**	The 5-year S&T Innovation Plan	1997-2001
Legal mandate***	The S&T Framework Plan+	2002-2006
	The S&T Framework Plan of the Participatory Govt	2003-2007
	The S&T Framework Plan of the Lee MB Govt	2008-2012
	The S&T Framework Plan of the Park KH Govt	2013-2017
	The S&T Framework Plan of the Moon JI government	2018-2022
S&T Long-term vision	The Long-term S&T Development Plan	1967-1986
	The Long-term Vision for S&T Development toward the 2000s	1987-2001
	The Long-term Vision for S&T Development toward the Year 2025	2002-2025
	Korea's Dream and Challenges: S&T Vision	2010-2040

*The plan was revised in 1993 into the S&T sector plan of the Five-year New Economy Plan (1993-1997)

**The Special Law for Science, Technology and Innovation (1997)

***The Framework Act on Science and Technology (2002)

+Revised and extended into the S&T Framework Plan of the Participatory Government (2003-2007)

Source: Taken and compiled by the author from Moon et al (2010) and others

3.2 Economic Development and the Evolution of STI Plans: Transition from the ‘S&T Development Plan’ to the ‘S&T Framework Plan’

Korea, which was one of the poorest economies in the world as late as in the 1960s, emerged in the early 2000s as a dynamic, high-technology-based industrial economy. In the course of the growth, Korea has gone through two important structural transformations: one from a traditional agrarian economy to an industrial one based low- and mid-technologies during the of 1960s-1980s, and the other one from a low- and mid-technology-based industrial economy into a high-technology-based economy during the period of 1980s -2000s. By the early 2000s, Korea had successfully made a structural transformation into a high-technology-based economy, entering into a post-catchup stage of development.³ Various factors have worked together to make the transformation possible, of which STI deserves a large share of the credit.

[Table 1-2] Stages of development and STI plans 1962-2001: Korea

Stages of development	STI plans
Industrialization based on technological catch-up (1962-1986)	- The Five-year Technology Promotion Plan, Supplement to the Five-year Economic Development Plan (1st-3rd plans) - The S&T Sector Plan, The Five-year Economic Development Plan (4th-5th plans)
Structural transformation: Promotion of high-tech industries (1987-2001)	- The S&T Sector Plan, The Five-year Social Economic Development Plan (6th-7th plans) - The S&T Innovation Plan (1997-2001) based on the S&T Innovation Law

Source: MOST (various years)

While STI plans deserves credits for the development Korea has made, it does not mean that the validity of the plans remains unchanged in the new stage of development. In order to move ahead with another transition toward a creative innovation economy, it

³ The transformations raised Korea's GDP per capita from a mere \$94 in 1961 to \$35,000 in 2020. Indeed, Korea has accomplished within half a century what it took almost two centuries for the Western industrial economies to achieve.

was imperative for Korea to attain world top-level competitiveness in the strategic technology areas. In addition, the government needed a new frame of STI development because the Five-year national development plan that had provided the visions and frames for STI planning was completed in the end of the 1990s. STI needed a new frame for development. It was, therefore, critically important for the government to put forward a new vision for STI development and design a systematic plan to realize the vision. It was in response to these needs that the government enacted the Framework Act on Science and Technology (2001)⁴ that pursues the following value:

The fundamental value that this Act pursues is to promote science, technology and innovation as a source of social and economic development, based on the principle of human dignity and in harmony with the natural environments and the social ethical value; to promote autonomy and creativity of scientists and engineers; and to realize balanced development of natural science, humanity and social sciences.

(Article 2 of the Act)

The Framework Act emphasizes science- and information-based policy and requires the government to take policy actions to promote the formation of an efficient national innovation system that can usher the nation into a knowledge-based economy. In other words, the Framework Act mandates the government to make policy efforts to transform the nation into an innovation-friendly society that provides a fertile land where creativity blooms and comes to rich fruition – an ecosystem prone to innovation. In addition, the Act mandates the government to formulate an inter-ministerial mid-term STI plan every five years (the Science and Technology framework Plan), the finalization of which is subject to the deliberation and approval of the National Science and Technology Committee.

The Framework Act finds the need for a new plan (the Framework Plan) in the emerging new policy issues that could not be dealt effectively with the past frame of policy. The

⁴ The Framework Act replaced the Science and Technology Promotion Act (1967) and the Science, Technology and Innovation Special Law (1997)

new policy issues and demands, and the changes in policy situation are summarized in [Table 1-3].

[Table 1-3] Economic development and new STI policy issues

Policy situation around the end of the 1990s and the early 2000s	New policy issues and demands
Successful transformation of the country into a hi-tech based economy	Need to make another transition to a higher level of development – move toward a knowledge-based economy
Limit to further growth based on learning and borrowing foreign technologies	Need to invest more in creative research to advance to the technological frontier in the strategic areas
Rapid technological advances and emergence of new industries	Need to reap the opportunities of the emerging technologies
The growth of innovation capabilities of the private sectors, and decreasing effectiveness of the industry-oriented R&D strategy	Need to shift the emphases of R&D policy from meeting short-term industrial demand more toward future-oriented research, including basic scientific research
Rising demand for technologies to deal with social issues, such as health, safety, environment, etc.	Need to broaden the scope to S&T and R&D policy and build STI capability to deal with social issues.
Growth of R&D investment to the level of advanced countries (around 2.5%) in the early 2000s	Need to adopt new R&D system and strategy to ensure the long-term efficiency
STI policy devoted more to the utilization of S&T for social and economic development, and far less to the generation of knowledge.	Need to find a balance between ‘policy for S&T’ and ‘S&T for policy’
With the growth of the private sector, the government can no longer play the lead role in social and economic development	Need to engage the private sectors more in S&T policy process and S&T development to link the creativity of private sectors to a higher level of development.

Source: Author

Therefore, the S&T Framework Plan differs from the past S&T plans in several aspects:

- (1) It is a government-wide plan mandated by the Framework Act on Science and Technology (2001) and formulated through inter-ministerial consultation and public-private cooperation;
- (2) It is a five-year plan that provides a mid-term visions for S&T development and strategies to attain the visions, which suggests that the Framework Plan is more about providing an overall vision and directions of policy. Rather than setting specific targets, the Plan is more indicative and perspective-oriented, suggesting a mid-term direction for the government and private sectors' consideration in making strategic decisions;
- (3) The plan is finalized with the deliberation and approval of the National Science and Technology Committee chaired by the president.

3.3 The Planning Process and Structure of the S&T Framework Plan

Article 7 of the Framework Act on Science and Technology mandates the government to formulate and implement the S&T framework Plan every five years. It is a government-wide plan that synthesizes and systemize S&T policies of diverse ministries and agencies, setting forth the goals and direction of S&T development along with strategies to attain the goals, such as S&T investment, focus areas for investment, resource allocation and directions for sub-sector development plans. The first Framework Plan was put into action in 2002. This section describes the processes of the planning and structure of the first S&T Framework Plan.⁵

3.3.1 Process and Organization of the Planning

The key player in the planning process is the National Science and Technology Committee. The Committee provides the direction of the planning and organizes the

⁵ The plan was revised with the change of the government in 2003, but the essence of the plan remained the same, while the plan period was changed to 2003-2008 from the original plan period 2002-2006. So, the discussion here stick to the original one.

planning apparatus that is composed of (1) Steering Committee for the S&T Framework Plan; (2) Sectoral Planning Sub-Committee; and (3) Secretariat.

The Steering Committee is responsible for analyzing and assessing the S&T trends, techno-economic changes, and other social, political and cultural issues, inside and outside the country that might influence the directions or patterns of development. They derive from the analysis implications and suggestions for Korean development, on the basis of which the visions and goals of the plan are set.

The Sectoral Sub-Committees are organized based on the findings of the above analysis – How many committees for what areas, for example. The Sub-Committees work on issues related to the specific technology or policy areas assigned to them (1) to support the overall planning activities, and (2) to formulate plans for the areas assigned. The Sub-Committees are composed of experts from private and public sectors. For the First Framework Plan, they organized ten Sub-Committees – three responsible for planning national strategic technology development, and the remaining seven assigned to work on how to strengthen S&T capacity. What follows are the Sub-Committees and the scope of the work assigned to them:

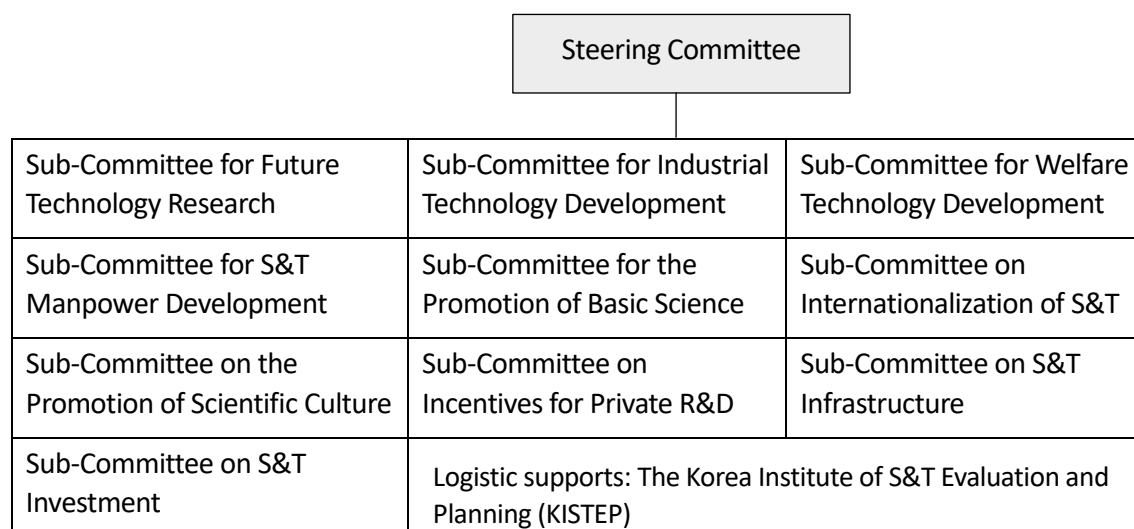
(1) National strategic technology development

- Sub-Committee for Future Technology Research: 6 Ts that include Information technology, Bio-technology, Nanotechnology, Space technology, Environmental technology, Cultural technology.
- Sub-Committee for Industrial Technology Development: Ways and means to promote value creation by linking new technologies to traditional industries; Improving and strengthening the competitiveness of the key industries by commercializing new technologies.
- Sub-Committee for Welfare Technology Development: How to strengthen technological capability to deal with unforeseen disasters – epidemics, typhoon, flood, etc. Development of health, medical, marine, meteorological, disaster prevention technologies. Development of food and energy technologies, along with defense technologies.

(2) Strengthening S&T capacity

- Sub-Committee for Manpower Development: Devising incentives to boost the morale of researchers, improving S&T manpower development system, promotion of inter-sectoral mobility of researchers.
- Sub-Committee for the Promotion of Basic Science: Designing stable and steady supports for basic scientific research, Development of a basic science support system, Inter-regional and inter-disciplinary balance in supports for basic science, Diffusion of the results of scientific research to promote inter-linkages in basic scientific research.
- Sub-Committee on the Internationalization of S&T: Active utilization of foreign S&T resources, International cooperation to solve the issues of global challenge, Promotion of strategic international cooperation at the regional and national levels, Improvement of institutional and regulatory systems to facilitate international S&T cooperation.
- Sub-Committee on the Promotion of Scientific Culture: Enhancing the sense of responsibility of science, Promoting citizens' engagement in S&T, Raising public awareness of S&T, Developing a institutional base for the promotion of scientific culture.
- Sub-Committee on Incentives for Private R&D: Promotion of industrial R&D by strengthening incentives, Promotion of the diffusion of technologies of high commercial potential, Development a system to strengthen the R&D capacity of the private sectors.
- Sub-Committee on S&T Infrastructure: Upgrading the system for testing, analysis, and assessment, Development of an S&T information system, Strengthening overall S&T infrastructure including those in the regions, Strengthening the role of the National Science and Technology Committee in coordinating S&T activities.
- Sub-Committee on S&T Investment: Increasing government investment in R&D, Strategic prioritization of government R&D resource allocation, Strengthening the analyses and evaluation of the government R&D programs, Systematic planning of the government R&D based on technology foresight.

[Figure 1-4] Organization of the S&T Framework Planning



Source: MOST (2000)

Committee members included experts from industries, academia, research community and the government. About 140 experts participated in the committee activities.

The works of the Sub-Committees are put together into a draft plan by the Steering Committee and reported to the National S&T Committee. The draft plan is then reviewed by the ministries and agencies as a process of an inter-ministerial consultation. Then, the next step is holding a public hearing, an opportunity for stakeholders from the civil society, media, S&T community and the government to exchange views on the draft plan. The draft that has taken into account the public opinion is handed over to the National S&T Committee for final deliberation. If the Committee approves, the plan becomes official.

In formulating the first Framework Plan, the Prime Minister's Office (Office of Government Policy Coordination) and 16 ministries collaborated and they are the official co-authors and co-publishers of the plan. But *de facto* key player was the Ministry of Science and Technology. The ministries that participated in the planning are: Ministry of Finance and Economy, Ministry of Education and Human Resources, Ministry of National Unification, Ministry of Foreign Affairs and Trade, Ministry of Defense, Ministry of Science and Technology, Ministry of Culture and Tourism, Ministry of Agriculture and

Forestry, Ministry of Planning and Budget, Ministry of Industry and Resources, Ministry of Information and Communications, Ministry of Health and Welfare, Ministry of Environments, Ministry of Labor Affairs, Ministry of Construction and Transportation, Ministry of Fisheries and Marine Affairs, Office of Government Policy Coordination (Prime Minister's Office).

3.3.2. Structure of the Plan: How the Content was structured

The Framework Plan followed a logical sequence starting from the analyses of current techno-economic situation and predictions on future developments, both internal and external, from which the basic direction of the plan was derived. They combined the analyses on the global trends of techno-economic changes, the responses of the major countries to the changes, and domestic situation – not just science and technology but also economic and social, on the basis of which they derived the visions and goals for S&T development and new direction of S&T policy. Then, the plan set forth strategies to attain the goals. What follows are a very brief skeletal description of the first S&T Framework Plan (2002-2006).

■ The global trends of techno-economic changes:

- Rapid transition toward knowledge society
- Globalization and intensifying competition
- Emergence of new technologies and acceleration of technological progress through inter-disciplinary fusion
- Demand for increased role of technology to meet the increasing desire for higher living standard
- Increased role of technology in national security – defense, food, energy, etc.
- Increased need to raise public awareness of science and technology and social responsibility of science and technology

■ Policy responses of major countries

- Overall responses:
 - Science and technology to meet social needs
 - Strategic investments to attain technological leadership in such areas as ICT, biotechnology, nanotechnology, environmental technology, etc.
 - Adoption of comprehensive coordination system to ensure efficiency in public R&D
 - Public-private cooperation in R&D and technology
- U.S.A.: Focus on improving the framework condition for innovation – Infrastructure, S&T education, etc.
- Japan: Focused R&D investment in such areas as ICT, Biotechnology, nanotechnology, and materials
- EU: Promotion of the mobility of S&T resources, open research system; and increased investment in such technology areas as ICT, biotechnology, nanotechnology and space technology
- China: Selection 12 strategic areas for focused R&D

■ Situation in Korea

- Established a system of R&D within a short period of time (app. 40 years) – Emerged as major player in global R&D landscape (GERD/GDP, from 0.38% in 1967 to 2.68% in 2000)
- Despite the growth of R&D expenditures, analysis shows that R&D explains only 18% of Korea's economic growth (KDI, 1991-2000)⁶, in contrast to 26% in the USA and 22% in Japan
- Increasing demand for the role of science and technology in meeting the challenges that Korea faced, such as quality of life, national defense, and other social issues
- Increasing demand for closer science-society interface

⁶ Re-cited from MMOST (2001)

■ Visions and the direction of the plan

- Visions:
 - “Realization of GDP per capita of \$15,000 by the year 2006”
- Goal
 - “Elevation of Korea’s S&T competitiveness to the level of world top 10 countries”
- Strategies:
 - Focus on the development of strategic technologies
 - Development of a system for the development and utilization of creative S&T manpower
 - Promotion of the opening and international networking of the S&T system
 - Enhancing public awareness of S&T
 - Improving the efficiency of R&D investments
 - Invest a total of \$35 billion in government R&D during the plan period (2002-2006)
- Basic direction of S&T policy
 - Continuous expansion of S&T investments
 - Efficient utilization of the resources available
 - Promotion of creative research
 - Promotion of a culture that places a higher value on S&T
 - Promotion of the globalization and localization of S&T

■ Key contents of the plan

- Development strategic technologies
 - Focus on strategic technologies
 - Application of new modern technologies to traditional industries
 - Development technologies for social welfare
- Strengthening S&T capability
 - Nurturing creative manpower who can lead the information, knowledge society
 - Promotion of basic science
 - Internationalization of S&T activities

- Promotion of scientific culture
- Strengthening policy supports for industrial R&D and innovation
- Strengthening S&T infrastructure
- Continuous increase of S&T investments

■ Investment plan

The S&T Framework Plan also included a plan for government S&T budgeting. The plan estimated that a total of KRW 35,331.6 billion would be required to implement the policy programs, of which 68.9% or KRW 24,369.4 was for the development of strategic technologies. Even though the lion share of the financial resources was planned to be allocated for the development of strategic technologies, the share was actually smaller than the actual share of 2001 (71.4%), which means that emphasis on strategic technology development was slightly reduced in the plan. Still, it was planned that the average annual government S&T budget would increase 1.32 times the actual spending of 2001 during the plan period.

[Table 1-4] Investment plan for the Framework S&T Plan: KRW billion

Areas or sectors	2001 (actual)	Plan period (2002-2006)*	
		Total	Per annum
Development of strategic technologies	3,676.6	24,369.4	4,873.9
- Future promising techs	1,571.8	12,838.0	2,567.6
- Techs for tradition industries	498.6	3,272.4	654.5
- Public welfare technologies	1,606.2	8,259.0	1,651.4
S&T manpower	297.1	2,204.7	440.9
Promotion of basic sciences	455.2	3,046.7	609.3
Internationalization of S&T	43.3	280.0	56.0
S&T culture	11.1	442.0	88.4
Supports for private R&D	236.2	1,096.6	219.3
S&T infrastructure	426.6	3,892.2	778.4
Total	5,146.2	35,331.6	7,066.3

*Actual investment is determined by the annual plan with the approval of the National S&T Committee

Source: Lee (2001)

3.3.3. Implementation and Ex Post Evaluation

What distinguishes the S&T Framework Plan from the past S&T sector plans is that the Framework Plan is more about providing a frame for actions and overall directions of S&T development and resource allocation rather than specific programs for action. For the implementation of the plan, it requires each of the ministries to draw up specific programs for action based on the frame set in the plan. Each of the ministries is financially and policy-wise responsible for the drafting and implementing the action programs under its purview. At the end of the plan period, the performance of the S&T Framework Plan is assessed based on the criteria set by the planning authority. And the results are fed in the process of designing the next plan.

The performance of the S&T Framework Plan was to be evaluated at the end of the plan period based on the following indicators.⁷

- Inputs: *Has the plan been effective in mobilizing financial resources and developing S&T manpower?*
 - R&D expenditures/GDP
 - R&D manpower: Number of researchers per 10,000 population
 - Government R&D expenditures/Government budget
 - Government expenditures on basic research/Government R&D expenditures
- Outputs: *Scientific and technological outputs*
 - Scientific publications: number of SCI publications, world rank, citation
 - Patents: Domestic patents, foreign patents (USPTO, PCT, etc.)
- Outcome: *impacts of the outputs on S&T competitiveness, social and economic development*
 - S&T competitiveness: IMD and other international reports on competitiveness
 - National competitiveness: IMD and other international reports on social and economic competitiveness
 - GDP per capita

⁷ Subject to change depending on the decision of the planning authority

4. Achievements and Policy Issues

4.1 Acheivements

The achievements of the S&T Framework Plan can be assessed using several criteria – resource mobilization, resource allocation, R&D output, scientific, technological and economic impacts. In terms of resource mobilization, it is fair to say that the plan was successful as [Table 1-5], [Table 1-6] and [Table 1-7] show. Government S&T expenditures reached KRW 36,450 billion, which exceed the target (KRW 35,331 billion) set by the plan. The gross expenditures on R&D also increased from KRW 16,110.5 billion (2.28% of GDP) in 2001 to KRW 27,345.7 billion (2.72% of GDP). The increases in R&D expenditures were accompanied by remarkable increases in research manpower – from 178,937 in 2001 to 256,598 (1.43 times increase) in 2006.

[Table 1-5] Government S&T budgets (KRW Billion)

Category	2001	2002	2003	2004	2005	2006
Budgets	5,733.9	6,141.7	6,515.4	7,082.7	7,799.6	8,909.6
Growth	-	7.1	6.1	8.7	10.1	14.2

Source: NTIS

[Table 1-6] Growth of the Gross Expenditures on R&D during the plan period

Category	2001	2002	2003	2004	2005	2006
GERD (KRW billion)	16,110.5	17,325.1	19,168.7	22,185.3	24,155.4	27,345.7
Growth (%)	-	7.5	10.6	15.7	8.9	13.2
GERD/GDP (%)	2.28	2.21	2.28	2.44	2.52	2.72

Source: NTIS

[Table 1-7] Growth of the number of researchers during the plan period

Category	2001	2002	2003	2004	2005	2006
Number	178,937	189,888	198,171	209,979	234,702	256,598
Growth	-	5.8	4.4	6.0	11.8	9.3

Source: NTIS

Regarding resource allocation, the S&T Framework Plan emphasizes (1) focus on the development of strategic technologies, (2) promotion of basic research, and (3) strengthening science-society linkages. [Table 1-8] shows that expenditures on strategic technologies (what the Korean government dubbed 6 T's, including information technology, biotechnology, nanotechnology, space technology, environmental technology, and cultural technology) account for 35-40% of the total government R&D expenditures during the plan period, which is consistent with the government's commitment to the strategic areas of technology. The promise to increase expenditures on basic scientific research has also been implemented – As shown in [Table 1-9], the share of basic research in the government R&D expenditures rose from 18.7% in 2002 to 23.4% in 2006.

Yet, it does not seem that all the promises of the plan had been kept. For example, Hwang et al. (2017) assesses that the plan to strengthen the science-society linkages in various ways fell short of the promise made by the goal.

[Table 1-8] Government R&D on Future strategic technology (6T)

Category	2002	2003	2004	2005	2006
Expenditures (KRW billion) (a)	2,484.0	2,357.8	2,686.7	3,165.8	3,101.3
Govt R&D budget (KRW billion) (b)	6,141.7	6,515.4	7,082.7	7,799.6	8,909.6
a/b (%)	40.4	36.2	37.9	40.6	34.8

Source: NTIS

[Table 1-9] Composition of Government R&D

Category	2002	2003	2004	2005	2006
Basic	18.7	19.5	19.9	23.0	23.4
Applied	29.7	27.9	25.3	24.8	24.4
Development	51.6	52.6	54.7	52.2	52.2

Source: NTIS

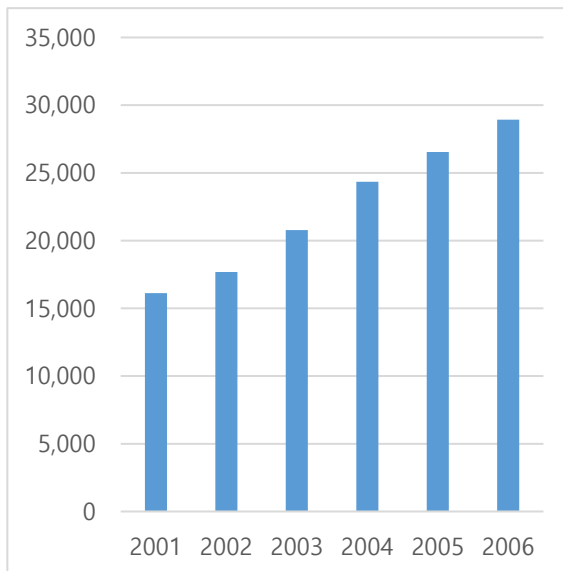
[Table 1-10] Government R&D on future strategic technology (6T)

Category	2002	2003	2004	2005	2006
Expenditures (KRW billion) (a)	2,484.0	2,357.8	2,686.7	3,165.8	3,101.3
Govt R&D budget (KRW billion) (b)	6,141.7	6,515.4	7,082.7	7,799.6	8,909.6
a/b (%)	40.4	36.2	37.9	40.6	34.8

Source: NTIS

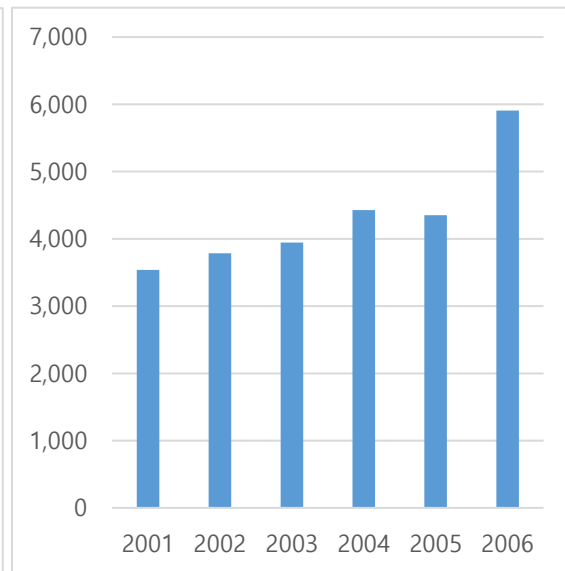
The outputs generated by the R&D activities seem to justify the investments during the plan period, quantitatively at least. During the period from 2001, the year before the plan started, to 2006, the last year of the plan period, GERD increased 1.69 times and the government R&D expenditures 1.55 times, which resulted in the 1.79 times increase in the number of scientific publications and 1.66 times increase in the number of US patents registered during the same period. The phenomenal growth of the US patents owes very much to the areas of information and communications technology, the number of patent applications of which grew from 668 in 2001 to and bio-technology 2,634 in 2006, which in turn is due to the R&D strategy focused on the area.

[Figure 1-5] Number of scientific publications



Source: NTIS

[Figure 1-6] Number of US patents granted to Koreans

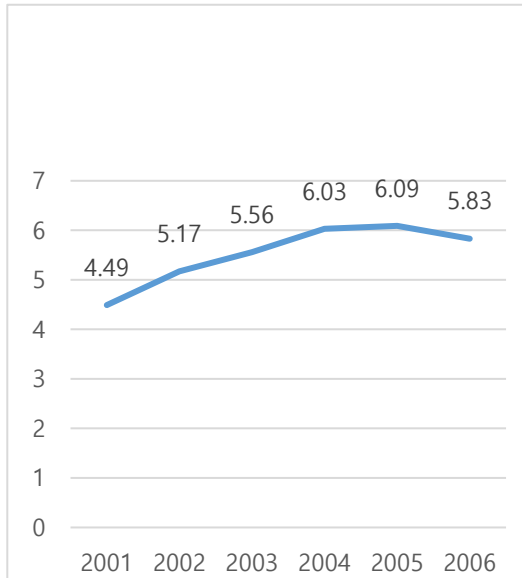


Source: NTIS

The scientific and technological outputs are translated into social and economic development, even though not instantaneously. It is practically impossible to numerically explain which outputs affect what industries to what extent and the time required for the impacts to take place. We may guess very roughly the impacts by looking at the changes of some economic indicators during the plan period. For example, During the plan period, Korea's share in the world exports of high-technology products increased from 4.49% in 2001 to 6.09% in 2005 and then fell slightly to 5.83% in 2006 (NTIS), indicating that Korean high-tech industries' competitiveness was enhanced significantly during the plan period. This may be in part due to the focused investments in the development of strategic technologies during the plan period.

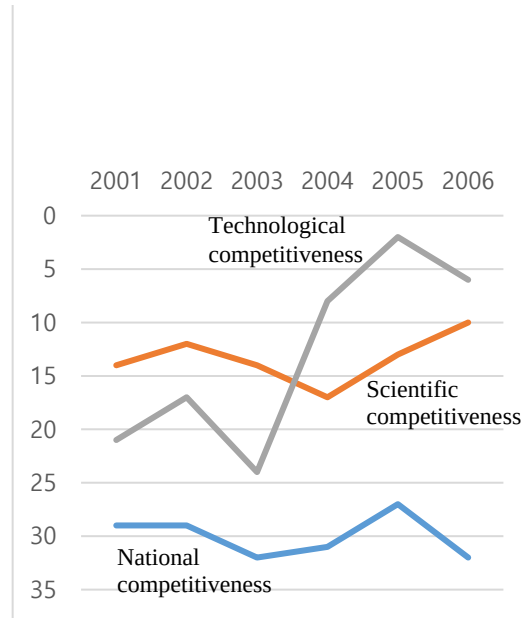
According to the IMD World Competitiveness Report, Korea's scientific and technological competitiveness also was visibly improved during the plan period – Korea's world rank in scientific competitiveness rose from 14th place in 2001 to 10th place in 2006, and technological competitiveness from 21st place in 2001 to 6th place in 2006 (IMD, various years). This means that the plan's goal of elevating Korea's S&T competitiveness to the level of world top ten countries was attained. In addition, the vision of the plan to attain a GDP per capita of \$15,000 by the year 2006 was realized as Korea's GDP per capita rose well over the target to \$17,690 in 2006. Yet, however, the overall national competitiveness of Korea fell a little from 29th place in 2001 to 32nd place in final year of the plan period (IMD, various years), suggesting that the improvements in S&T capability had not been properly complemented by improvements in other factors that determine the national competitiveness together with S&T capability.

[Figure 1-7] Korea's share in the World high-tech product exports: %



Source: NTIS

[Figure 1-8] Changes in Korea's international competitiveness: Rank in the world



Source: NTIS, IMD (various years)

4.2 Assessment and Policy Lessons

The plan's policy goal was achieved to a certain extent – Growth of R&D investments, enhanced technological capability of innovation players, in particular private industries made possible the structural transformation of the industries and economic growth. Korea has been able to attain the development goals based on the S&T Framework Plan, owing very much to:

- (1) Strong policy capacity of the government: Effective intervention in S&T affairs
 - Information and analytical capacity
 - Operational capacity of the government to translate policy ideas into actionable policy measures and to put the measures into action
 - Political capacity to set the vision for development, to put the society together toward the set goal, and to mobilize resources required to implement the policy plans.

- (2) Hard-working, well-educated human resources with strong desire for change for the better
- (3) Heavy investments in R&D and advanced education and concentration of the limited resources on the development of strategic areas of technology
- (4) Consistent linkage of S&T policy to industrial development: R&D priority dictated by industrial development

Even though Korea was able to realize the vision and goal of the plan, various challenges remain there. The fundamental problems of the Korean innovation system still remain uncorrected.

- (1) Dominance of the government in S&T and innovation policy process is still there. Despite the growth of the private sectors' capability in R&D and innovation, the influence of the government still remains largely unchanged.
- (2) The imbalance in the innovation system has not been corrected – the imbalance between scientific research and industrial technology development, the imbalance between large enterprises and SMEs, the regional imbalance in innovation performances are still there.
- (3) To nurture creative researchers, universities' role in R&D and innovation has yet to be further strengthened. Korea is one of the few countries in the world where universities are minor players in R&D.
- (4) As far as science, technology and innovation system is concerned, Korea is still one of the least internationalized countries among OECD members.
- (5) Many of the above problems have something to do with the dominance of government in STI. A paradox -- as long as the government succeeds in planning, we can hardly expect the government to change. Very often, success is used as an excuse to stay unchanged.

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The achievements of the S&T Framework Plan can be assessed using several criteria – resource

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CHAPTER 2

Technology Commercialization in Korea

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 2. Technology Commercialization in Korea

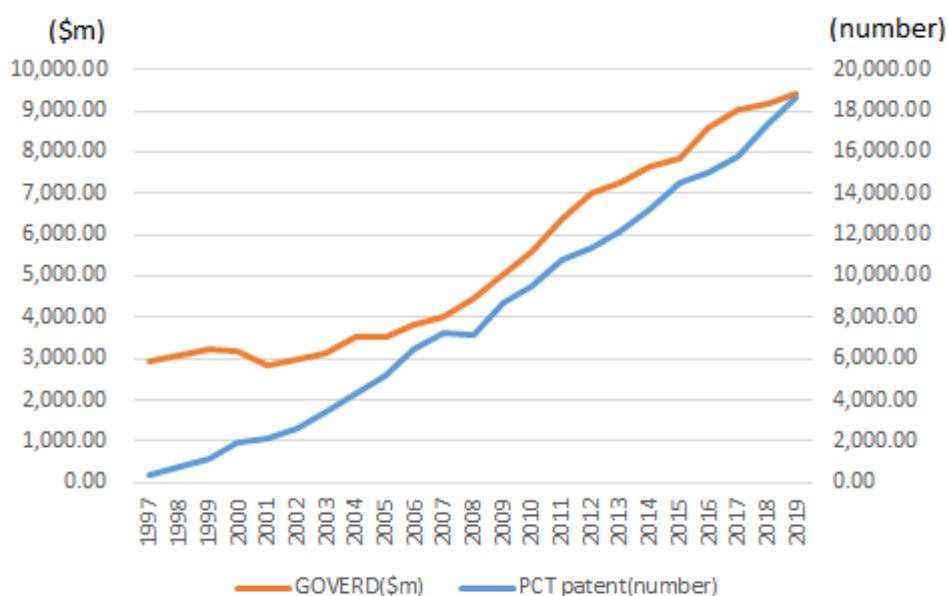
Dr. Soo J. SOHN, Senior Research Fellow, STEPI

1. Background

A typical growth model is a function based on labor(L) and capital(K). But, a growth model based on innovation considers technology, A , as a main factor. That is, the growth model changes $Y = f(L, K)$ to $Y = f(L, K, A)$. Thus many innovative subjects invest big R&D resources into R&D activity to obtain qualified technologies. And these technologies need to have legal property rights to prevent their efforts and investments. This is the reason why many major advanced countries and industrialized countries started to pay attention to IPRs after the industrial revolution. The legal system provides holders who create intellectual property, that is, creators who engage in creative activities and inventors who engage in invention activities the right to creative activities. It is basic institution to prevent technologies that made through R&D from abusing or misusing. It will be incentive for innovators to invest their time and effort to create such IPRs. Therefore, many innovative countries organized their insitutions for granting IPRs in general to give incentives to engage in more creative activities or invention activities. Therefore, it can be said that the more advanced science and technology, the more the importance of innovation is emphasized, and thus the more the importance of intellectual property is emphasized.

Since 1990, Korean government budget for R&D has been radically increased, and IPRs such as patents resulted from R&D was also extended. Based on this situation, the Korean government was interested in creation, dissemination, and utilization of IPRs. As industrialization evolves, the granting of rights, IPRs, to that R&D outputs and IPRs input for industrial revitalization become issues.

[Figure 2-1] Government R&D Budget and PCT patent application of Korea



note: GOVERD(Government expenditure on R&D)

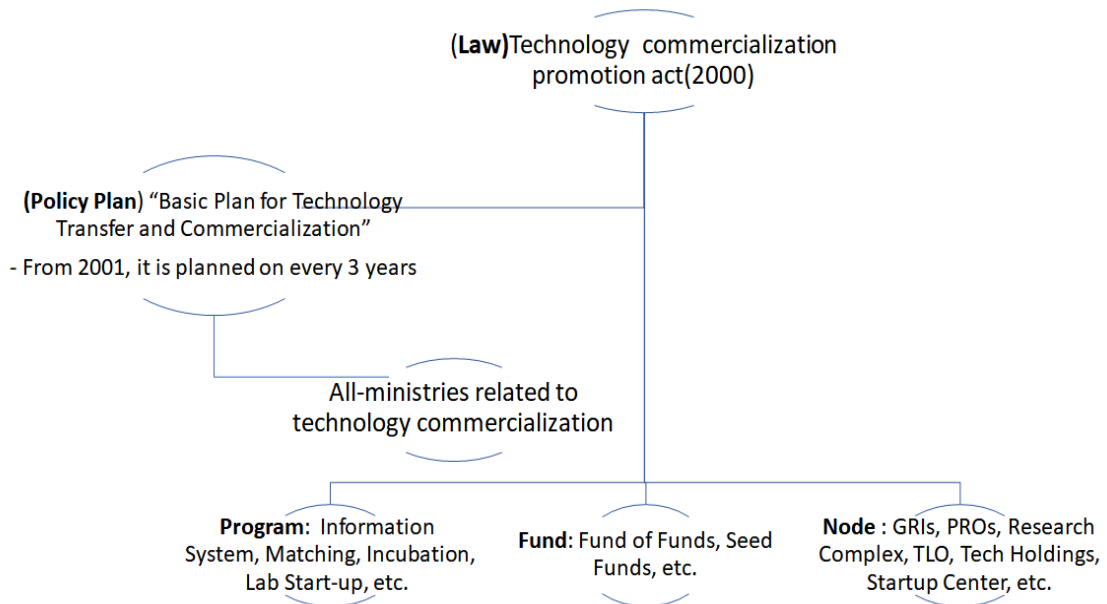
As seen in [Figure 2-1], Korea can be called a country that is currently creating a lot of R&D investments and therefore a lot of patent results. Korea has been trying to overcome its resources limitation, such as small nature resources, small markets, and short history for industrialization, etc. This is a main reason why Korea is more interested in innovation. That is, it is because Korea is weak in all areas including the labor force, natural resources, and literally capital power required for growth. So it had to make a very important effort in technological development and innovation. Therefore, Korea increased the R&D input has been quite quickly compared to its history of economic scale and growth, and made remarkable achievements in the end.

And now Korea has faced with another challenge and needs to shift from catch-up innovation policies to leading innovation policies. Therefore many innovation policies focus on technology commercialization.

2. Policies for Technology Commercialization

The law, 'The Technology Transfer Act', to promote knowledge transfer was launched in January 2000. This law provided a legal basis for the country's technology commercialization policies. Based on this law, many policies for hardware infrastructure (organizations, spaces, facilities, etc.), incubation programmes, information networks, training programmes, technology finance and technology valuation, etc. have been introduced and extended for encouraging technology commercialization (technology transfer, technology-based startup, etc.). And the pan-department has been designed the basic plans to secure national competitiveness in technology commercialization. Policies such as training experts, establishing information networks, strengthening technology scale-up, and creating commercialization funds were planned. Especially regulation, demonstration, connection and tech finance are considered as the most important factors. And policies for establishment of technology-based venture ecosystem has been increased.

[Figure 2-2] Structure of policy for technology commercialization



Source: author

Many support projects are implementing by government ministries including key players like the Ministry of Trade, Industry, and Energy, the Ministry of Science and ICT, and the Ministry of SMEs and Startups. Each ministry has designed a representative program. For example, the Ministry of SMEs and Startups launched TIPs(the Tech Incubator Program for Start-ups) programmes for incubating startups. It's a co-incubation program in which the government matches start-ups discovered by private accelerators. The Ministry of Trade, Industry, and Energy carries out technology development projects linked to commercialization. This is to bridge and incubate the gap between supply technology and demand technology by connecting public research institute and companies (start-ups, SMEs, etc.). The Ministry of Science and ICT operates the R&D special zone system, Innopolis. This is a platform for the transfer of public technologies and start-ups based on public technologies. It supports many useful programmes, such as regulatory exceptions, post-R&D for scale-up, and incubation, etc. As such, many ministries are operating technology commercialization programmes tailored to their characteristics.

And the main frame has changed over time. In the 2000s, policies related to technology commercialization began to be introduced. In particular, manpower training, establishment and support of dedicated organizations such as TLOs. And the technology information network, NTB (National Technology Bank) was established and the technology evaluation system was developed.

In the 2010s, main policy support was focused on building infrastructures and platforms, for example incubation centers, Innopolises(special zone), Technoparks, CCEI, etc. Technoparks have now been established in 19 regions. And Innopolises support matching tech supply-demand, tech-based start-up companies/products/ services under the law 'Innopolis Act'. Innopolises have been established in main 5 regions(Daedeok, Daegu, Busan, Gwangju, Jeonbok) and since 2017, small and high-density intensive area(Town) have been designated and nurtured as R&D special zone, Innotown. Now, there are 14 Innotowns nationwide. CCEI(Center for Creative Economy Innovation) serves as a hub for innovative startups. CCEIs were opened in 19 regions. They provide programmes for the vitalization of local start-ups and the promotion of entrepreneurship. And it was also focused on bridging gaps or steps. In particular,

consensus for connecting was formed. In addition, local governments also established their own institutions dedicated to supporting activation of technology-based regional industries and these institutions are now leading the TC support efforts of the local governments.

[Table 2-1] The flow of key policies by times

Times	2000s	2010s	2020s
Main issues	- building infrastructure - strengthening expertise	- connecting nodes - bridging gaps/ steps.	- Changing leader/ planner 'public subject' to 'private players'

For the past two decades, policy has been placed on introducing and maturing the technology commercialization system, building infrastructure for it, supporting the capacity building of key stakeholders and building technological competences, supporting the dissemination and application of public technologies in the field, and establishing and strengthening institutions dedicated to implementing all these technology commercialization support projects.

3. The Next Step in Korea's Technology Commercialization

Korea has created a government-led technology commercialization environment. Support policies for commercialization include direct measures such as investment and R&D support, and indirect measures, such as support of entrepreneurship education, mentoring and consulting, workspace supply, and networking events, seed funding, matching investment. But despite these support, there are still big gaps between labs and markets, suppliers and demanders, technologies and capitals. And there is also the issue of rent seeking by agents acting in encouraging technology commercialization. As such, some of the past policies have been successful and some have acted as obstacles.

It is important to understand the environmental changes and limitations surrounding technology commercialization to be able to design the next steps. All countries are the same. Accurately diagnosing the current level of each country is a very important process for designing the next step. External shock is also a big variable. Especially, digital transformation will have a great impact on technology commercialization. Above all, we can use a virtual space like the metaverse. And as a demonstration tool, the digital twin can be used. From this point of view, the two keywords, 'speed' and 'transformation', will be important triggers. Thus the next step of commercialization policy will be to build a system that can speed up the flexible transformation and lead by the private sector.

Korea is in the vortex of change, such as the global pandemic, technological hegemony, Russia-Ukraine war, and digitalization, convergence, inflation, especially dramatic demographical changes, etc. Therefore, it is important to obtain technological competitiveness. In particular, technological prowess is important to be competitive in the international market and to have a comparative advantage over other countries. Even with this understanding, an accurate forecasting of the impacts of these changes is still a challenging task to complete. Therefore, an optimal policy would be to lay down the framework for technology commercialization, so that the country will be able to respond to changes quickly and flexibly. Realistically, a feasible next step will be to

design an optimal path in areas where government policy intervention is needed and plan a structure that can ensure policy intervention through efficient intervention models.

South Korea's sustainable innovation efforts and related experience have made it an innovative country in a relatively short period of time. In order for this growth to continue, a transformative commercialization policy is needed. Above all, it is to create an environment centered on the private sector. And it is to set excellence as an important policy criterion. In other words, it is not a distributive support, but a reliable support for an excellent subject. In addition, it is necessary to create a global value chain for commercialization. There are many fields that require global cooperation, such as demonstration or technology packages.

Commercialization requires a long cycle from securing good technology to creating added value. Above all, the uncertainty of technology and market is high. Therefore, there are areas of market failure that cannot be achieved only by the private sector, and this is the reason for government intervention. Appropriate and efficient government intervention is required to strengthen technology commercialization capabilities.

4. References

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CHAPTER 3

Technology Commercialization in Lao PDR

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 3. Technology Commercialization in Lao PDR

Dr. Sengpasith HOUNGALOUNE, National University of Lao PDR

1. Introduction

1.1 Backgrounds

In Lao PDR, it is essential to develop the National Master Plan for Science, Technology and Innovation Development and Application as well as the National Policy on Science and Technology Research. The plans should focus on survey and study development indicator of relevant sectors, human resource, science and technology research innovation networking and infrastructure development. It is specially to increase competitiveness of trade, capability of research, and industrial development to be more innovative.

Recently, there are several areas that are important for developing of science and technology such as technology commercialization, small and medium-sized enterprises (SME) innovation, international science, technology and innovation (STI) Collaboration, etc.

Technology commercialization seems a complicated term at first sight, but its meaning is an easy one; it is the process of transforming knowledge into services or products. Simple, exceptional ideas are analyzed in labs, and then converted into services and products to improve our everyday life (Ghazinoori, 2005). They are then launched to the wider international community.

The main purpose of technology commercialization is to add more value to new products and services. Technology commercialization is the most effective and efficient way to turn ideas into real services and products, that will eventually satisfy market's needs. The process also looks at whether the inherent strategies have a long-term economic advantage (Bahreini and Shadnam, 2007).

Nowadays, technology commercialization is attracting worldwide attention and plays an important role in the integrated national approach for innovation and research. Likewise, businesses are always looking for new strategies to improve their market status, increase their business awareness and develop successful products and services. Thus, by implementing the right ideas and converting them into products and services, they are successful (Dehghani, 2015).

However, technology commercialization is not an easy process. It requires the understanding of some basic concepts which all institutions, universities and corporations should be aware of in order to successfully achieve this process. Similarly, the fundamental concepts of intellectual property (IP) protection represent an important issue related to technology commercialization, and it is helpful to be informed regarding the ways through which potential intellectual property conflicts can be avoided (Kim and Cho, 2022; Dehghani, 2015).

Therefore, it is necessary to include technology commercialization in a national master plan science and technology (S&T) development to compete with the growth of S&T, and commercialization system with the needs of the market and society. It is critical that research and development (R&D) capacity needs to be aligned with the market needs, a system with developed S&T and R&D capacity and process for the practical policy decision-making need to be established by applying available opportunities and potentials in line with the growth of economic and social needs.

1.2 International environment

In today's world, technology commercialization is one of the most controversial issues (Lee and Lee, 2015). A number of companies and countries give first priority to this issue

and allocate their attention among a number of activities (Dehghani, 2015). Nevertheless, there are some obstacles to technology commercialization even within most of the industrial countries i.e., financial issues, inefficiency of organizational government and administration, lack of interaction between research teams, lack of mass production, evaluation of research findings, implementation of reforms, and product optimization which are being gradually eliminated (Kim and Cho, 2022; Dehghani, 2015). Park and Glenn (2020) have warned that companies failing to adopt accelerating technological changes might go out of business, leading to personal bankruptcy and even a national crisis.

Developed countries have a long history in commercialization of ideas and research outputs. For instance, university-industry collaboration in the U.S., concerning linkages between universities and industries in commercialization of innovations and research findings, dates back to decades ago, to the Morrill Act of 1862, also known as the Morrill Land-Grant acts, defense R&D during World War II, and competition between two Cold War rivals, the Soviet Union and the U.S. (Dehghani, 2015). These are U.S. statutes that allowed for creation of land-grant colleges. In addition, the Bayh-Dole Act, enacted in law in 1980, encouraged universities to acquire patent government-funded research inventions and to engage with industries in technology transfer and research commercialization (Wu et al., 2015).

The Japanese government has been formulating various measures to strengthen industry-university collaboration. For example, the Japanese government has been sending professional advisors on industry-university collaborative intellectual property to universities since 2016 and intellectual property strategy designers to universities since 2019. They have also implemented measures to reduce and exempt patent fees and examination application fees for universities. These measures are expected to improve the weakness of the patent strategy at Japanese universities and the situation where there is a lack of expertise in intellectual property licensing strategies. It is also necessary to focus on cultivation of human resources who are responsible for consistent practice of technology transfer from acquisition of IP rights, financing arrangements, and development of the new markets (Masato, 2020).

In Germany, the transfer of knowledge and technology from higher education institutions and higher education-related institutions to industry and vice versa is based on a highly differentiated system of different stakeholders. In addition to the universities themselves, there are three large clusters of non-university research institutions: Helmholtz Association, the Max Planck Society and the Leibniz Association with their respective institutes (Czychowski, 2019). Germany has also made strategic efforts to strengthen the competitiveness of SMEs through public technology transfer (Kim and Cho, 2022). The government-funded research institutes (GRI) of Germany is operated by four large research groups. Among them, the Fraunhofer Society was established in 1949 to support the research needs of industries (Kim and Cho, 2022). The model agreements for R&D cooperation at the Federal Ministry of Economics and Technology prevailed in the 2010s. These model agreements, which are now in their third edition, attempt to reconcile the interests of all parties involved, universities and research institutions as well as SMEs and large industry, which were all involved in the negotiations on these models at the Federal Ministry of Economics and Technology. At this point, only the essential contents of these model agreements are reproduced in order to give a feeling of which topics in Germany play a role again and again in the transfer of technology under contract law (Czychowski, 2019).

Meanwhile, the Singaporean government attaches great importance to the linkage between R&D and business. Therefore, Singapore universities have a strong link with industry to commercialize research results or contribute to society (Kim and Cho, 2022).

South Korea enacted the Technology Transfer Promotion Act in 2000 (Kim and Cho, 2022). Currently, various policies are being implemented, such as the seventh Technology Transfer and Commercialization Promotion Plan and the fourth Basic Plan for Research Results Management and Utilization in Korea. Accordingly, the quantitative growth of technology transfer is increasing at a rate of 7% per year (KISTEP, 2020).

However, the success rate of public technology commercialization remains less than 15% (Kim and Cho, 2022). Thus, qualitative growth is disappointing when compared to quantitative growth. Small- and medium-sized enterprises (SMEs) in Korea account for 99.9% of all enterprises and 83.1% of total employment and are closely related to the

national economy (Kim and Cho, 2022). However, these enterprises find it challenging to develop core technologies independently. Therefore, SMEs must find other ways to reinforce their technological capabilities within a short period of time and escape the crisis (Kim and Cho, 2022).

It has been reported that almost all research universities in the world established technology transfer offices (TTOs) to manage new technology commercialization (Wahyudi et al., 2019). Generally, a technology transfer office (TTO) is established to support the research organizations to manage their intellectual assets that provides a benefit for society with their transformation. The general roles of TTO include establishing relationship with firms and community actors, generating new funding support from sponsored research or consulting opportunities, providing assistance on all areas related to entrepreneurship and intellectual property, facilitating the formation of university-connected companies utilizing university's technology (start-up) and/or university resource (spin-off) to enhance prospect or further development and generating net royalties for the university's technology and collaborating partner (Wahyudi et al., 2019).

Several researchers have identified success factors for technology transfer and commercialization based on analyses of several studies as summarized in [Table 3-1] below.

[Table 3-1] Success factors for technology transfer and technology commercialization

Factor Classification		Technology Transfer and Commercialization
Technology provider	R&D capability	number of papers and patents, research capacity (HR, R&D investment/fund), research facilities, etc.
	Management ability	R&D budget, resource allocation capacity, administrative capability, etc.
	Technology diffusion capability	Professional status of a technology licensing organization, technology transfer experience, technology marketing activities.
Technology adopter	Technology absorption capacity	number of developers, ability to utilize technology, dedicated technology adoption team, etc.
	Technology commercialization willingness	willingness and support of owner, entrepreneurship, efforts for technology transfer, etc.
	Technology Commercialization capability	marketing capability, commercialization capability, understanding customer needs, etc.
Technology	Technology characteristics	compatibility, complexity, maturity, etc.
	Intellectual property rights	patent application/registration status, industrial applicability
	Technology trading market	transaction brokerage, consulting, legal service, technology valuation, etc.
Environment	Government support	R&D projects, funding support, related laws and institutions
	Commercialization environment	Technology financing, industrial characteristics, market size, market volatility, competitive situation, etc.
	Cooperation partnership	collaborative relationships, communications, trust, etc.

1.3 Domestic environment

The Lao government established the Ministry of Science and Technology (MOST) in 2011, in regard with the decree number 309/PM, officially issued on September 28, 2011. It indicated the government's interest in fostering R&D. However, Lao PDR sought to reduce the overlap of R&D policies. In February 2021, the government settled the MST and transferred its roles and responsibilities to Ministry of Education and Sports (MOES), Ministry of Post and Telecommunications (MPT), and other relevant line ministries. In October 2021, the government established the Ministry of Technology and Communications (MTC) under the Decree on Organization and Activities of Ministry of Technology and Communications, No.625/PM, officially issued on October 22, 2021. It is responsible mostly for MOST's responsibilities and roles in terms of governance and management of technology, innovation, telecommunication, internet, post, and cyber security.

Lao PDR's 7th Socio-Economic Development Plan (2011-2015) included the development technology and promotion of technology transfer was one of the significant elements in such development plan. The technology and technology development for SMEs was supported under the Development Strategy of Science and Technology (2013-2020) of the Ministry of Science and Technology (MOST).

- A legal and policy framework supporting development of technology in universities, R&D labs and support centers on IP rights exist has also been established by the Government (ERIA and OECD, 2014).

There is limitation on the financial support schemes funded by the government, donors, and other organizations. The pilot public funds supporting R&D activities are also provided with a commercial orientation and limited allocation.

2. Policy Analysis

2.1 Institutions related technology commercialization in Lao PDR

2.1.1 Technology Transfer and commercialization in Lao Context

The concepts of “technology transfer” and “technology commercialization” have been used interchangeably depending on the subject and research topic (Kim and Cho, 2022). However, some studies suggest the definition of technology transfer as “the transfer of intellectual property such as accumulated technology, knowledge, and know-how developed for the achievement of a specific purpose through contract and negotiation between the parties to the technology transfer”. While technology commercialization was defined as “developing, producing, and selling products using technology, or improving related technology in the process” (Kim and Cho, 2022; Roh et al., 2017).

With respect to Lao context, since the technology transfer and technology commercialization may be used interchangeably as there is no specific law and regulations on technology commercialization. Regarding to the law on Technology Transfer No. 33/NA dated November 8, 2017, the technology transfer defines as “transferring of right and ownership partly and wholly on the technology such as know-hows, experiences, processes, methods, equipment, tools, and vehicle from a transfer person to a receive person”.

2.1.2 Laws Related to technology commercialization in Lao PDR

The law that directly related to technology commercialization in Lao PDR includes the law on Technology Transfer No. 33/NA dated November 8, 2017 (consists of 11 Chapters and a total of 58 Articles). The law defines the principle, regulations, practices, measures, restrictions, and standards for management and monitoring technology transferring to promote the research, development, creativity, trade, investment, production, service, and industry to ensure that the technology applied is effectively and contribute to sustainable growth of the national economy.

Based on the laws, it emphasized directions and policies for technology transfer development as follows:

- The Government of Lao PDR (GoL) gives priority for the technology transfer by facilitating the R&D, innovation, local wisdom, production, management of technology transfer and innovation application to ensure for raising of living standards of people, and to set the foundations for steady industrialization and modernization supporting to the national economy growth sustainably;
- GoL supports and promotes activities and investments for the technology transfer by providing infrastructures, human resources development, budgets, equipment, accessible to development funds, and provide information related to science and technology; and
- GoL supports and promotes individuals, legal entities, and organizations domestically and internationally to invest in the development of advance technology, modern, suitable, and environmental friendly by providing special credit, exemption or reduction of taxes according to the law.

Promotion of the technology transfer include:

- Promotion on the policy related to the technology transfer;
- Promotion on human resources development; and
- Promotion on marketing related the technology transfer.

Besides, the Law on Intellectual Property, No. 38/NA dated November 15, 2017 is another law that could be related to technology commercialization in Lao PDR (consists of 10 Chapters and a total of 170 Articles). This law determines principles, regulations and measures relating to the management and protection of intellectual property rights [in order] to support and promote inventions, creativities, local knowledge-based economy, development of science and technology, the transfer of technology within the country and from abroad, to ensure legitimate interests of the owner of the intellectual property and the interests of state, society aiming to promote trade, investment and the competitiveness according to market based economy mechanism effectively and efficiently and integrate into regional and international levels and contribute in socio-

economic development of the nation and industrialization and modernization of the country.

2.2 Analysis of Lao PDR current situations related technology commercialization

2.2.1 Descriptive Statistics

2.2.1.1 Intellectual property Performance

The National Assembly passed a revision of the Law on Intellectual Property comprehensively in 2011, and an amended version in 2017 (though changes are not considered significant). After the dissolution of the Ministry of Science and Technology (MOST), the Ministry of Industry and Commerce (MOIC) now controls the issuance of patents, copyrights, and trademarks. The consolidation of responsibility for intellectual property rights (IPRs) is a positive development, but the enforcement capacity is insufficient and lacks qualified patent examiners.

In Lao PDR, the intellectual property protection is weak, but steadily improving. The country has been a member of several regional and international organizations as follows:

- Member of the World Intellectual Property Organization (WIPO) in 1995;
- Joined Paris Convention for the protection of industrial property in 1998;
- Joined Patent Cooperation Treaty (PCT) in 2006;
- Protocol Relating to the Madrid Agreement Concerning the International Registration of Marks in 2016; and
- Joined the Association of Southeast Asian Nations (ASEAN) and ASEAN working group on IP cooperation in 1997.

■ Patent

A “Patent” means the official certificate issued by the state organization to protect inventions that they are novel, include an inventive step and are practicable for

industrial application. Whereas “Invention” means the technical solution to create new product or process of production to resolve a specific problem.

Measuring the link between publications and patents has been a subject of academic research in recent decades. It helps us to understand the intensity and orientation of research as well as the relation between science and technology (UNESCO, 2018). A measurable relation lets us to examine knowledge transfers and potential spillovers; to describe the knowledge base of particular technologies; to disclose the technological neighborhood of scientific themes or research fronts; to reveal an ongoing innovation process (from research to technology then on to commercialization).

Patents include a unique source for technological knowledge, which they are regarded a good proxy for R&D activities, invention skills, and for the scope of technological innovation of sectors and firms, countries, and regions. The use of patent statistics makes it possible to track technological changes and to examine knowledge transfer and R&D cooperation between various sectors and countries (UNESCO, 2018).

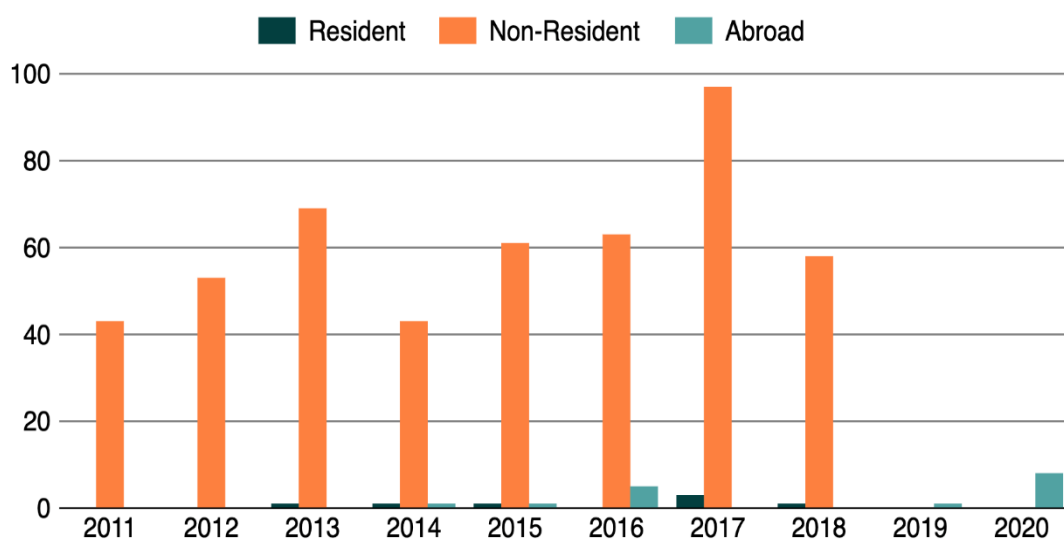
According to requirement of the Government represented by Department of Intellectual Property (MOIC), in order to obtain a patent (with a Term of Protection of Patents 20 years), an invention shall meet all the following requirements:

- (1) shall be new: meaning such invention has not existed, not been disclosed to the public by publication, or by use or in any other means in the Lao PDR or any place in the world prior to the date of filing the application for registration or where priority is claimed prior to the priority date of the application for a patent;
- (2) shall involve an inventive step: meaning that would not be obvious to a person having ordinary knowledge in that field of technology;
- (3) shall be industrially applicable in industry, handicraft, agriculture, fishery, services, etc.

Exclusion from patent protections include discoveries, scientific theories, mathematical methods, schemes, rules or methods for performing mental acts, methods for doing business, methods for treatment of human and animal body and inventions which are contrary to the public order or the national cultural morality.[Figure 3-1] presents the number of patent applications filed in the Lao PDR by origin of the applicant (residents

and non-residents) between 2011 and 2020. The number of applications ranged between 43-97 patents annually. Most of the patent applied are from non-resident applicants. The highest applications of 97 patents were recorded in 2017. The number of patent applications were significantly decreased in 2019 and 2020, which this may due to the Covid-19 pandemic that affected all socio-economic system including the IP sector.

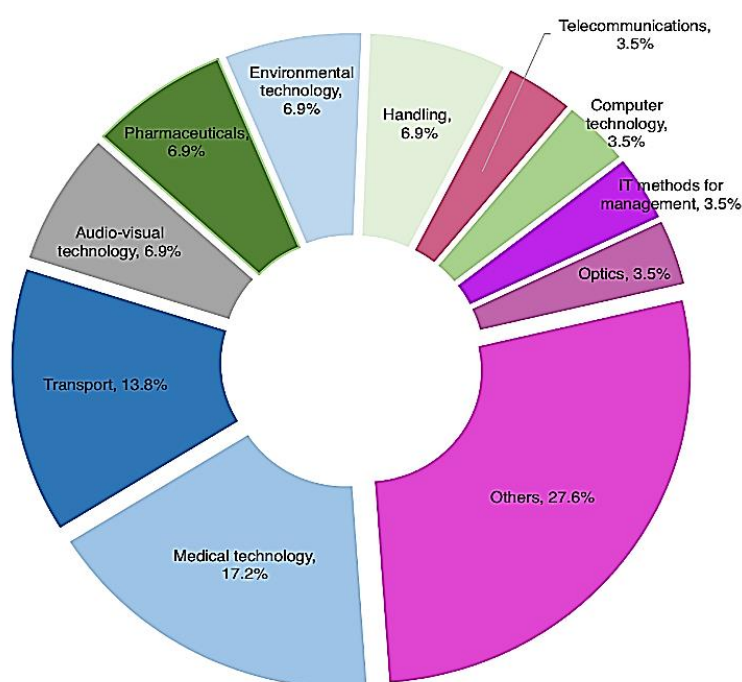
[Figure 3-1] Number of patent applications in Lao PDR (2011-2020)



Source: WIPO statistics database; last updated: 11/2021

[Figure 3-2] shows the distribution of patents (2001–2015) by top field of technology. Medical technologies (17.2%), transport (13.8%), audiovisual technologies (6.9%), pharmaceuticals (6.9%), environmental technologies (6.9%) and handling (6.9%) are the main technological fields.

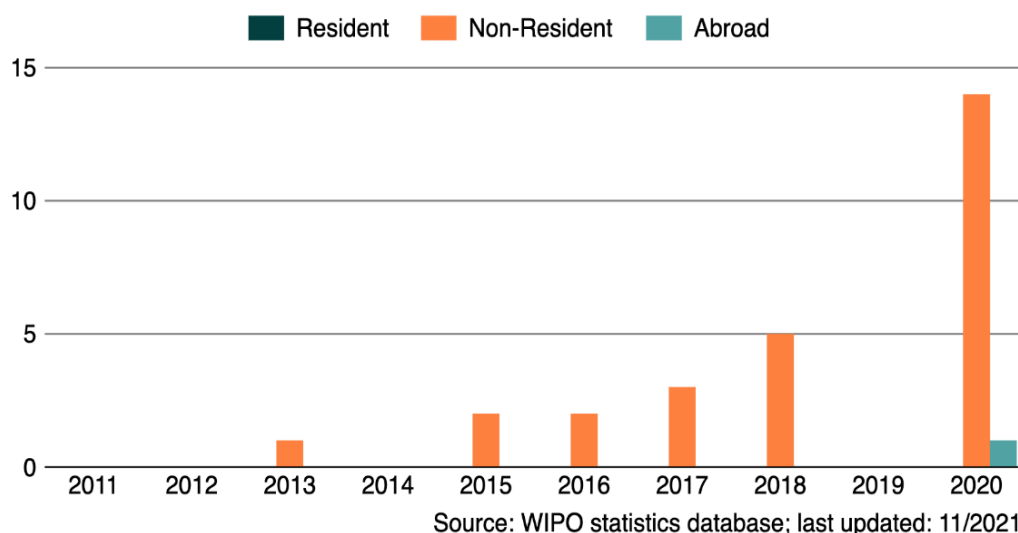
[Figure 3-2] Patent applications in Lao PDR by top field of technology, 2001–2015



Source: UNESCO, 2018

Department of Intellectual Property (DIP), MOIC is responsible for patent application process and approval in Lao PDR. The successful rate of patent grants is relatively low as shown in [Figure 3-3] Only 1 patent was granted in 2013; 2-5 patents were granted between 2015-2018; and the highest patents granted of 14 patents was recorded in 2020. The small number of patent applications and granted patents in Lao is due to lacking of a suitable legal system on IPR until current years.

[Figure 3-3] Patent Grants in Lao PDR (2011-2020)



■ Trademark

A trademark is defined as a sign capable of differentiating the goods or services of one enterprise from those of other enterprises, which it is protected by IPRs. Principally, an exclusive right on the use of the registered trademark is conferred by a trademark registration. It is implied that the trademark can be used entirely by its owner or licensed to another party for use in return for payment.

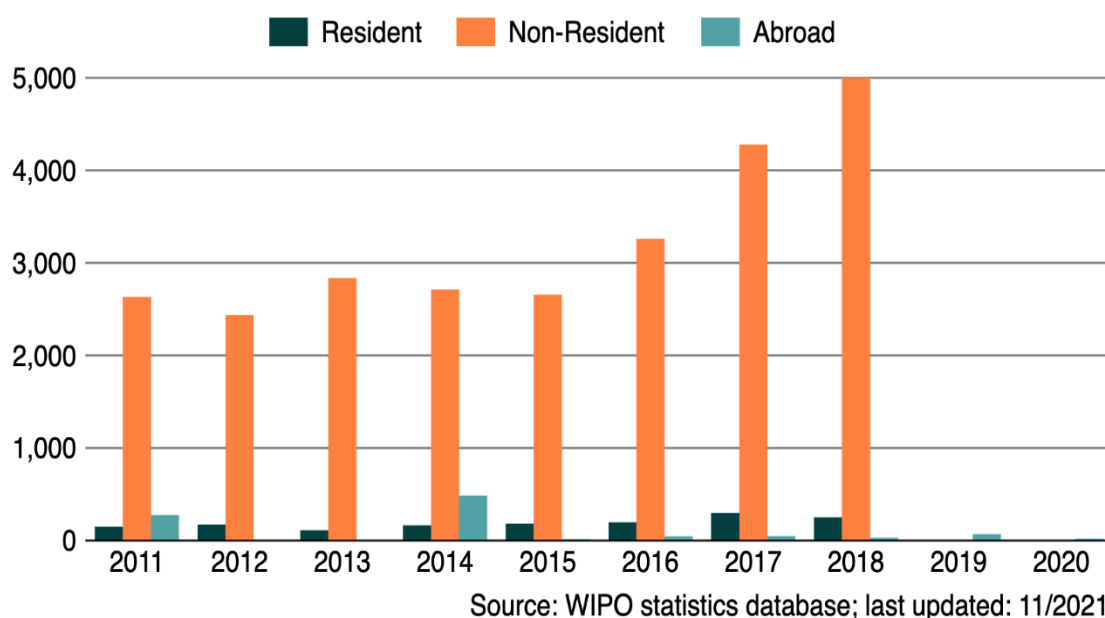
Recently trademark data have also been used to convey information on two key aspects of innovation, which are not usually covered by traditional indicators: marketing innovation and innovation in the services sector (Milot, 2009).

The relation among trademark deposits and product innovation is fairly straightforward: the commercialization of novel products is occasionally linked with the creation of a new trademark to connect about the innovation and subsequent possibly become the reference on the market for the product, which in turn enables companies to suitable the benefits of their innovation. When it comes to marketing innovation, the link with trademark deposits is more complex (WIPO, 2021b).

[Figure 3-4] shows the evolution in trademark applications by residents, non-residents, and abroad between 2011 and 2020 in Lao PDR.

Besides, other IP covered by the Department of Intellectual Property (MOIC) including copyright, industrial designs, geographical indicators, new plant variety, layout-design of integrated circuits, and IP disputes resolution.

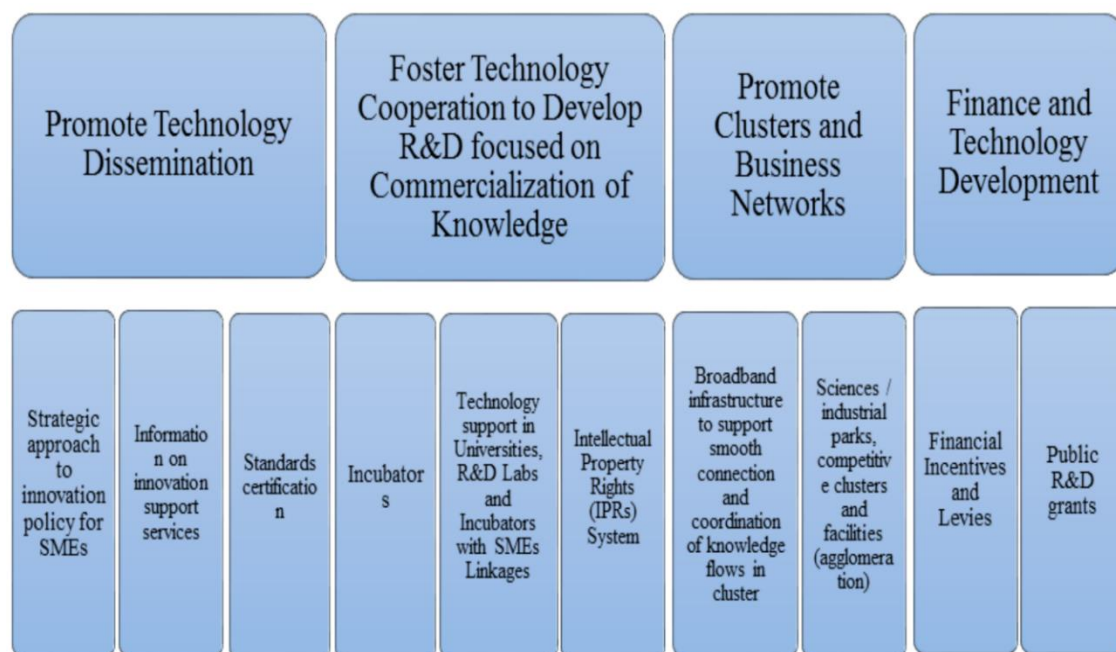
[Figure 3-4] Trademarks applications in Lao PDR (2011-2020)



2.2.1.2 Technology Transfer Performance

ERIA and OECD (2014) conducted an assessment framework for technology and technology transfer for ASEAN countries. There are four main policy sub-dimensions to measure the upgrade of technological capability and transfer indicating in [Figure 3-5]:

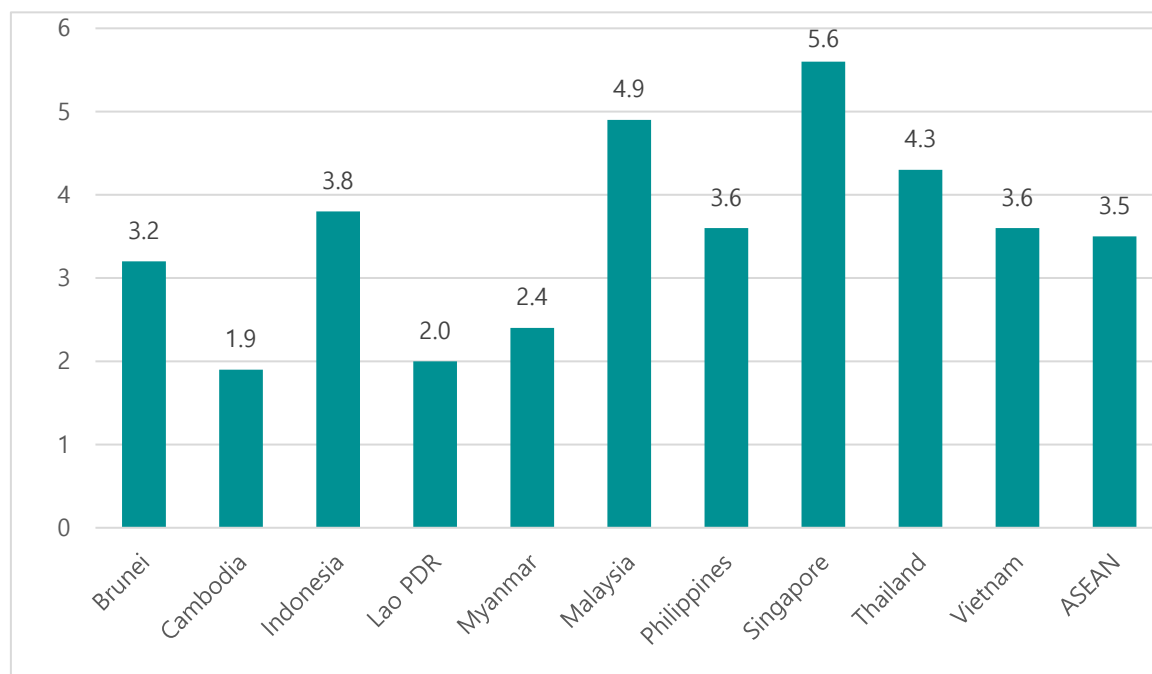
[Figure 3-5] Assessment Framework for Technology and Technology Transfer



Source: ERIA and OECD, 2014

The results from ERIA and OECD, 2014 study recommended that the main gap is in the policy to promote technology and technology transfer among the poorer ASEAN Member States (AMSs) and their advanced counter parts. The gap includes lacking of a strategic approach to SMEs innovation policy, limited access to standard certification services, lack of technology provision in universities, poor provision of information on innovation support services, R&D labs, incubators, and little linkages with SMEs. Lack of broadband infrastructure support for IPRs, insufficient financial incentives for technology development and R&D activities, poor protection and low promotion of IPRs, and, underdeveloped science/industrial parks and competitive clusters are also the causes of the gap as inferred from the scores listed in [Figure 3-6] and [Table 3-2].

[Figure 3-6] Overall Scores for Technology and Technology Transfer



Source: ERIA and OECD (2014)

Overall Scores for Technology and Technology Transfer of ASEAN countries is 3.5 (average), while three countries including Cambodia, Lao PDR, and Myanmar showed the overall score lower than an average of other countries in ASEAN. Singapore has highest score for Technology and Technology Transfer, following by Malaysia and Thailand.

Lao PDR had low scores for in all four key policy sub-dimensions: (i) Promote technology dissemination (2.3), (ii) Foster technology cooperation to improve R&D focused on commercialization of knowledge (2.0), (iii) Promote clusters and business networks (2.3), and (IV) Financial incentives for technology development through levies, public R&D grants (1.5). Therefore, it is very important for Lao PDR to consider the promotion and policy development for technology and technology transfer.

[Table 3-2] Technology and Technology Transfer

	BRN	CAM	IND	LAO	MMR	MYS	PHL	SGP	THA	VNM	ASEAN
Promote Technology Dissemination	3.0	2.5	4.0	2.3	2.7	5.0	3.5	5.0	4.5	3.5	3.6
Strategic approach to innovation policy for SMEs	3.0	2.0	4.0	2.0	1.0	5.0	4.5	5.0	4.0	4.0	3.5
Information on innovation support services	3.0	2.5	3.0	2.0	2.0	4.0	1.0	5.0	4.5	2.0	2.9
Standards certification	3.0	3.0	5.0	3.0	5.0	6.0	5.0	5.0	5.0	4.5	4.5
Foster Technology Cooperation to Develop R&D focused on Commercialization of Knowledge	4.3	1.7	4.8	2.0	3.3	4.5	4.2	6.0	4.3	3.8	3.9
Incubators	4.0	1.0	4.5	2.0	4.0	4.5	4.0	6.0	4.5	3.5	3.8
Technology support in Universities, R&D Labs and Incubators with SME Linkages	5.0	1.0	4.0	1.5	3.0	3.5	3.5	6.0	4.5	3.5	3.6
Intellectual Property Rights (IPRs) System	4.0	3.0	6.0	2.5	3.0	5.5	5.0	6.0	4.0	4.5	4.4
Promote Clusters and Business Networks	3.0	1.8	3.5	2.3	2.5	5.5	3.8	6.0	5.0	4.0	3.7
Broadband infrastructure to support smooth connection and coordination of knowledge flows in clusters	4.0	2.5	5.0	3.0	2.0	6.0	4.5	6.0	5.0	3.5	4.2
Science / industrial parks, competitive clusters and facilities (agglomeration)	2.0	1.0	2.0	1.5	3.0	5.0	3.0	6.0	5.0	4.5	3.3
Finance and Technology Development	2.5	1.8	3.0	1.5	1.0	4.8	3.0	5.5	3.5	3.0	3.0
Financial Incentives	3.0	2.5	3.0	2.0	1.0	5.0	3.0	5.0	3.5	3.0	3.1
Public R&D grants	2.0	1.0	3.0	1.0	1.0	4.5	3.0	6.0	3.5	3.0	2.8
Average	3.2	1.9	3.8	2.0	2.4	4.9	3.6	5.6	4.3	3.6	3.5

Note: BRN = Brunei, CAM = Cambodia, IND = Indonesia, LAO = LAO PDR, MMR = Myanmar, PHL = Philippines, SGP = Singapore, THA = Thailand, VNM = Vietnam (Source: ERIA and OECD, 2014)

2.2.2 Governance

With respect to technology transfer and commercialization in Lao PDR, the Technology Promotion and Transfer Center (TPTC) and DTI of MTC play important roles for technology and technology transfer. The DTI comprises of 6 divisions: (1) Administration Division, (2) Policy and Strategy Division, (3) Technology Management Division, (4) Innovation Division, (5) Advanced Technology Division, and (6) Satellite Imagery Information Technology Division. The TPTC also comprises of 6 divisions: (1) General Affair Division, (2) Star-up Division, (3) Technology Promotion Division, (4) Digital Transfer Division, (5) Technology Transfer Division, and (6) Technology Zone Division.

Besides, Department of Intellectual Property (DIP), MOIC responsible for intellectual property promotion and development that includes facilitating for applications and grants of patent, trademark, copyright, industrial designs, geographical indicators, new

plant variety, layout-design of integrated circuits, and IP disputes resolution, which can support technology and technology transfer. The MOIC was reformed and restructured since its establishment. It was recently legislated as stipulated in the Prime Minister's Decree on Organization and Activities of Ministry of Industry and Commerce, No. 604/PM, issued on October 15, 2021. The MOIC acts as a secretariat to the government in responding to governance and management at the macro level. Its responsibilities cover various fields, such as industry, handicrafts, industrial zones, chemical management, internal and international trade, and electronic commerce. The MOIC is also responsible for protecting consumers, adjusting prices, managing the market, and enhancing businesses' competition. The DIP comprises of 9 divisions:

1) Copyright Division, 2) IP Promotion and Development Division, 3) Plant Variety Protection Division, 4) General Affair Division, 5) Policy and International Affair Division, 6) IP Service Center, 7) Patent Division, 8) Trademark and Geographical Indication Division, and 9) IP Dispute Resolution Division.

2.2.3 Stakeholders

Regard to stakeholders for technology and technology transfer in Lao PDR, Technology Promotion and Transfer Center (TPTC) and Department of Technology and Innovation (DTI) of MTC play important roles in promoting, development, implementation and management for technology and technology transfer. These two departments also work together with other line ministries related to technology transfer such as: Department of Intellectual Property (DIP) of MOIC, Research Institute for Technology and Innovation of MOES, etc.

2.2.4 Resources

2.2.4.1 Human resource

The Human Development Index (HDI) has been measuring by the United Nations Development Programme (UNDP) since 1989. It describes in a summary way the achievements of a country in relation to the three-dimensions of human development including access to knowledge, a long and healthy life, and a decent standard of living.

The global rank of countries is obtained by representing each individual HDI value in descending order. Countries are classified as achieving very high (>0.800), high (0.700-0.799), medium (0.550-0.699) or low (<0.550) human development, according to their ranking (UNESCO, 2018).

In 2015, Lao PDR got a HDI of 0.586, which corresponds to a medium human development category positioning it at 138 out of the 188 countries and territories covered by the last report (UNDP, 2016). In 2019, Lao PDR got a HDI of 0.613, which still corresponds to a medium human development category (UNDP, 2019).

Lack of qualified academic staffs is one of the main reasons for the low quality of education and R&D in Lao PDR. The education institutes have difficulty retaining skilled and experienced academic personnel due to low salaries paid and unattractive benefits to academic staffs, so it is leaving institutes with only many young, underqualified, and inexperienced academic staffs (Onphanhdala & Philavong, 2022). The key constraint on undertaking R&D in innovation is lack of academic staff with advanced degrees.

[Table 3-3] Human resources of each relevant stakeholder

Stakeholders	Body	R&D Budget	Manpower			
			PhD	Master	Bachelor & Diploma	Total
TPTC	MTC	N/A	0	24 (10*)	39 (23*)	63 (33*)
DTI	MTC	N/A	0	12 (5*)	23 (10*)	35 (15*)
DIP	MOIC	N/A	0	4 (2*)	24 (12*)	28 (14*)

Note: * is the number of female staffs.

2.2.5 Issues

Generally, transfer of technologies and innovations is affected by numerous barriers understood as “any kind of limitations and features that hamper the effective functioning of a technology transfer and research commercialization system, and, as a result, block interactions between the R&D sector and enterprises, therefore impeding the development of innovative entrepreneurship” (Matusiak and Guliński, 2010; and Mazurkiewicz and Poteralska, 2017). The application of technological innovations is acknowledged as a driver for economic and social development considering the significance of the issue from the scientific and practical points of view.

With respect to technology transfer (TT) and technology commercialization (TC) in Lao PDR, there are numerous issues facing for the country as follows:

- Institutions responsible for technology transfer and innovation management are lack of experiences and skills;
- Limited financial supports for innovation and technology development at universities, research institutes, and lack of private funds at enterprises;
- Lack of infrastructure and facilities to support innovation and technology including technology transfer and commercialization;
- Lack of human capacities for STI at central and local government levels as well as at private companies;
- Lack of disseminate the knowledge experiences and best practice on TT;
- Technologies of universities are embryonic and not mature, therefore highly risky to develop and invest. TT from universities to industries in Lao PDR is still limited, desynchronized, and weak;
- Practical utility is not proven: Technology is not developed in response to market need (i.e., only at Lab.-level, lack of collaboration with businesses and lack of networking or linkage among public and private sectors);
- Weak protection of IPRs at universities, which affected University Technology Transfer (UTT) and creativity within universities;

- Poor utilization of patent information for research, and IP promotion policy and act need developments;
- Lack of innovation promotion and technology commercialization for technology makers and transfers;
- Lack of offices or bodies which specialize in the commercialization of research results or patents like models of Technology Licensing Organization (TLO);
- The size of Lao PDR enterprises is still small with a proportion of MSMEs covers more than 99% of total enterprises. A proportion of micro, small, and medium enterprises are 94%, 5%, and 0.7%, respectively according to the 3rd national economic census (2019-2020).

2.3 Challenges and Opportunities

SWOT analysis of the technology transfer and technology commercialization for Lao PDR is provided as follows:

[Table 3-4] SWOT analysis of the technology transfer and technology commercialization for Lao PDR

Strengths	Weaknesses
- Several regulations related to technology and technology transfer are available such as: law on Technology Transfer No. 33/NA, Law on Intellectual Property No. 38/NA, Lao on Science and Technology No. 30 / NA, etc. - There is an increase for Human Development Index (HDI) e.g., Lao PDR got a HDI of 0.586 in 2015, which increased to 0.613 in 2019 corresponding to a medium human development category. - Lao PDR become members of several international organizations (i.e., WIPO). - Five national universities and few research institutes available. - Stakeholders (MTC, MOIC, MOES, etc.) support for STI.	- Institutions responsible for technology transfer and innovation management are lack of experiences and skills. - In-sufficient human capacities in STI. - Lack of infrastructure and facilities to support STI development. - Lack of the financial supports from public and private sectors. - Weak awareness on STI including TT, IPRs knowledge for both public and private sectors. - Lack of corporation with universities on TT to connect with private sectors in Laos. - Technologies of universities are embryonic and not mature, therefore highly risky to develop and invest. - Low Global Innovation Index (GII): Laos ranks the 117th among 132 economies in 2021.
Opportunities	Threats
- Legal statutes can be strengthened to protect, promote, and develop IP, TT and develop national innovation systems. - Learn from developed countries who have experiences and skills related to STI including TT and TC. - GoL supports or invests more on budget of R&D for STI development. - Number of patent applications filed between 2011 and 2020 ranged between 43-97 patents annually. - Over 2000 trademark applications annually recorded between 2011-2018. It has potential to get higher applications on IP, with a good promotion strategy.	- Technology is not developed in response to market need. Gaps between public and private sectors. - The size of Lao PDR enterprises is considered small with a proportion of MSMEs covers more than 99% of total enterprises. - Lack of Technology Licensing Organization (TLO) and Technology Transfer Offices (TTOs). - Lack of innovation promotion and technology commercialization for technology makers and transfers. -

2.4 Policy Recommendations

- (1) Firstly, the legislative framework needs to be improved, where the main role should be assigned to intellectual property rights (IPRs) protection. Strengthened IPRs facilitate open innovation adoption. Thus, robust and certain IP protection and collaboration are complements in supporting the innovation process.
- (2) A strategy is developed from each side of technology transfer, and to achieve a successful result of innovation, a significant exchange of knowledge is required, on the one hand, and some ways and mechanisms to limit the exchange of other knowledge, on the other hand. Intellectual property rights help to deal with this controversy.
- (3) A specialized program appears necessary to be developed to support the technology transfer network to bring market demand closer to the capabilities of domestic scientific and industrial complex, and create an active circulation of intellectual rights and innovative developments.
- (4) It is necessary to stimulate and accelerate the development of technology markets. It is necessary to develop regional and sectoral technology transaction markets connected to the nationwide network of technology transactions, to promote linkages and mergers between technology and capital markets, and to expand investment channels.
- (5) It is necessary to improve the quality of technology transfer services by developing standards for technology transfer services, improving market pricing mechanisms, creating a reliable specialized statistical system for the technology transfer services sector, improving the rules for identifying technology contracts and registration rules. Obviously, widespread use of common standards allows connectivity among diverse technology transfer participants.
- (6) Lastly, there is a need to improve the infrastructure in the field of telecommunications with the outside world and within the country. Scientific centers will be able to gain access to global knowledge, interact with foreign scientific centers and carry out joint work in real time. The exchange of data in an international format will greatly speed up the process of innovation generation, and consequently, it will lead to an improvement in the state of Lao PDR technology transfer.

3. National STI Master Plan

3.1 Direction of the Plan

3.1.1 Vision

“Transforming Lao PDR to a Sustainable Lower Income Economy by 2030 through Strengthening Science Technology and Innovation Capability”.

3.1.2 Strategy

To achieve the proposed vision, several strategies are proposed as follows:

- Strategy 1: Technology Commercialization – Promote the linkage of the supply and demand for technologies.
- Strategy 2: MSME Innovation – Promote the development MSMEs to generate demands for technologies.
- Strategy 3: International STI Cooperation – Take advantages for the strengthening of S&T capacity.

3.2 Objectives for “Stagey 1: Technology Commercialization”

Several objectives as below are proposed in order to achieve the Strategy 1: Technology Commercialization – Promote the linkage of the supply and demand for technologies.

- Objective 1-1: (Infrastructure Creation) Development of human capacity and facilities related to technology commercialization and technology transfer.
- Objective 1-2: (Consensus Creation) Promotion the awareness on technology commercialization, technology transfer, and IPRs knowledge for both public and private sectors.
- Objective 1-3: (Relationship Creation) Establishment of the cooperation mechanisms between universities or research institutions with industries.

- Objective 1-4: (Value-Chain Creation) Strengthening cooperation, integration and interconnection related to technology regionally and internationally.

3.3 Actions to achieve each objective

The table below illustrates the proposed actions to achieve each objective under Strategy 1: Technology Commercialization – Promote the linkage of the supply and demand for technologies. The table presents details actions, responsible stakeholders, monitoring process, indicators for each objective. To achieve the proposed vision, several strategies are proposed as follows:

- **Strategy 1:** Technology Commercialization – Promote the linkage of the supply and demand for technologies.
- **Strategy 2:** MSME Innovation – Promote the development MSMEs to generate demands for technologies.
- **Strategy 3:** International STI Cooperation – Take advantages for the strengthening of S&T capacity.

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- **Strategy 1:** Technology Commercialization – Promote the linkage of the supply and demand for technologies.
- **Strategy 2:** MSME Innovation – Promote the development MSMEs to generate demands for technologies.
- **Strategy 3:** International STI Cooperation – Take advantages for the strengthening of S&T capacity. The table below illustrates the proposed actions to achieve each objective under Strategy 1: Technology Commercialization – Promote the linkage of the supply and demand for technologies. The table presents details actions, responsible stakeholders, monitoring process, indicators for each objective.

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CHAPTER 4

SME Innovation Support in Korea

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022



Chapter 4. SME Innovation Support in Korea

Prof. Soogwan DOH, University of Ulsan

1. Introduction

Sustainable economic development and growth for many countries have become an important policy agenda, with innovation from each sector of society serving a key role. This is especially true in today's knowledge-based society where knowledge and technology-based innovations are critical for promoting economic development (Collinson, 2000; Doh and Acs, 2010; Doh and Kim, 2014). Not only is innovation one of the most important aspects of knowledge creation (Collinson, 2000), but the future of prosperity depends to a large extent on successfully promoting innovation (Baumol, 2004). Therefore, much research has aimed to understand the drivers of innovation as well as how innovation promotes sustainable economic development and growth. These studies mostly emphasize innovation spurred by high levels of human capital, R&D investment and expansion of entrepreneurship (Grossman and Helpman, 1994; Kirzner, 1997; Baumol, 2002; Audretsh and Keilbach, 2004; Acs, 2006; Doh and Kim, 2014). Specifically, investment in human capital and R&D will lead to entrepreneurship based on new knowledge and technology, generating various Small and Medium-sized enterprises (SMEs). These innovative SMEs, according to prior studies, significantly contribute towards economic development and growth (OECD, 2004a, 2004b; Kramer et al., 2007).

However, SMEs, due to comparative disadvantage in various aspects such as labor, capital, information, management skills, and government regulations, face a steep challenge competing with large sized enterprises. Nevertheless, SMEs have positively contributed to sustainable economic development and growth through notable

achievements such as innovation, job creation and export expansion. Thus, governments from various countries, including South Korea, have supported SMEs in various ways. In 1960s, South Korea, a poor agrarian society based on agriculture and light industry, achieved rapid economic growth in 1970s by transitioning to an industrial society based on heavy and chemical industries (Doh and Kim, 2014). South Korea successfully transitioned by investing in capital-intensive heavy chemical industries. The needed capital came from exporting goods from labor-intensive manufacturing industry. Export expansion coupled with development of chemical industries – shipbuilding, cars, and chemicals – allowed South Korea opportunities to transition from an underdeveloped to a developing country.

Although South Korea's economic development and growth continued until ICT industry developed in early and middle of 1990s, South Korea faced Asian financial crisis in 1997. At that time, President Kim Dae Jung, who had just been elected as a president, inherited a nationally bankrupted economy. In addition, he also inherited a host of other problems caused by rapid economic growth such as imbalanced relations between central and local governments and economic and social inequality. These problems raised the need for Kim Dae Jung administration to develop a balanced economic development plan for regions and promote a new industry that can serve as an engine of growth and enhance labor flexibility.

To resolve these problems, President Kim's administration first reformed each economic sector's vulnerability by implementing restructuring policies to enhance their competitiveness. These measures included big business (Chaebol) reform, banks sector reform, and corporate restructuring based on 'industrial development act' (February 1999). To expedite the restructuring process and maximize its effect, various restructuring policies were promoted, including financial support for finance sector, protection and tax breaks for the unemployed and depositor and monetary and credit policy (total loan limit control policy, interest rate control policy, foreign exchange policy, maintenance of foreign exchange management system, currency cooperation with

foreign countries, foreign investment attraction policy, SME financial support policy, fostering venture business, e-commerce activation, etc.).⁸

One of the economic policies Kim Dae Jung administration prioritized in the midst of the crisis was the policy measures supporting SMEs. Because President Kim Dae Jung perceived SME reform as critical, SME policy measures were highly prioritized among Kim Dae Jung administration's various economic policies. Most of these policies were introduced during the first year of the administration, as President Kim aggressively promoted venture capital initiatives to overcome economic difficulties through job creation. Such policy effort was part of a larger plan by the administration to enhance competitiveness of enterprises and create new jobs. These efforts paid off, contributing to overcoming financial crisis and improving sovereign ratings. Of course, some policies adopted by the administration justified wrongdoings of big businesses (Chaebols), collapsed middle class through large scale restructuring and escalated labor repression and social conflict. Despite these side effects, there is a consensus that the administration's policies promoting venture corporations based on technological innovation and financial support were critical to ending national bankruptcy early and normalizing economic difficulties.

This research analyzes the historical context that enabled South Korea to promote SME innovation and employment expansion during the financial crisis and in the larger scheme of things, achieve sustainable economic development and growth. In this line of argument, author of this research examines and reviews the functions and the role of the Korean government in promoting SME innovation, in addition to how relevant governance and institutions were implemented. To this end, this study seeks to provide policy implications for promoting their economic development and growth and supporting SMEs for underdeveloped or developing countries. Although country context may differ by countries, Korean government's diverse policies to support particular industrial base and group of corporations in the midst of the financial crisis can provide important policy implications.

⁸ President Archives, Ministry of the Interior and Safety
(<https://www.pa.go.kr/research/contents/policy/index020602.jsp>).

2. Historical Background of Korea's SME Policy

2.1 SME Definition

In South Korea, as of December 1st, 2022, SME is defined as enterprises eligible to receive support according to the Framework Act on Small and Medium Enterprises⁹, which establishes and stipulates legal basis for fostering and promoting SMEs. According to the Ministry of SMEs and Startups (2018), the criteria for being SME rests on economic scale of SME and its degree of independence regards to its subsidiaries. The first criteria, economic scale of SME, is based on the size and upper limit of each industry. The economic scale criteria for each industry applies different revenue criteria based on standard industry classification (main category, middle category for manufacturing). For the upper-limit criteria, corporations with more than 500 billion won of total assets (total capital + total liability) is not an SME regardless of industries they belong to. Overall, if a corporation qualifies the size criteria but not the upper limit criteria, the corporation is not an SME.

Next, in many cases, an individual company can grow or diversify its business by acquiring a subsidiary. It can be seen as unreasonable to financially support these companies that combined with their parent company, exceed SME's economic scale criteria. Therefore, for companies 1) belonging to business groups subject to restrictions on cross-shareholding under the Monopoly Regulation and Fair Trade Act, 2) with total assets of KRW 500 billion or more (including foreign corporations, excluding non-profit corporations, etc.) with a stake of 30% or more directly and indirectly as the largest investor, their revenue is calculated proportional to their total investments. Even if their total revenue qualifies them as SMEs, based on the degree of independence criteria, cannot be labeled as SMEs.

SMEs are classified into small and medium-sized enterprises. Businesses are classified as small enterprises if their average sale falls within the criteria stipulated under the Table 3 Enforcement Ordinance of the Framework Act on Small and Medium Business. Middle enterprises refer to businesses excluding small businesses in SMEs.

⁹ See https://elaw.klri.re.kr/kor_service/lawTotalSearchList.do

2.2 SME and Innovation

Conventional understanding of the innovation process assumes that innovation lies in the domain of large enterprises (Acs and Audretsch, 2005). However, in the new perspective (i.e. entrepreneurship theory) on innovation process, entrepreneurial SMEs as well as large established enterprises play an important role in innovation (Rothwell, 1989; Link and Bozeman, 1991; Lee et al., 2010; Lausen and Salter, 2004; Acs and Audretsch, 2005; Doh and Kim, 2014). SMEs play an important role contributing to regional development and enhancing economic performance by innovating technology and strengthening their capacity (Jones and Tilley, 2003; Doh and Kim 2014). SMEs rely on innovative dynamics more so than large enterprises in terms of product innovations, and critically depend on production flexibility in terms of process innovations (Hall and Harvie, 2003). For example, prior studies suggest that SMEs do a better job capturing the benefits of networking for innovation in the knowledge-based environment (i.e. Rogers, 2004; Edwards et al., 2005). Link and Scott (2005) and Mowery and Sampat (2005) also maintain that in a more innovative environment, SMEs do a better job exploiting external economies through their proximity to the R&D centers of large firms and universities.

All in all, SMEs' performance implies productivity growth, employment creation and export growth. These findings suggest that SMEs' performance is an important driver of innovation as well as economic growth. For instance, SMEs' high productivity could mean SMEs' high innovation, given that productivity is an important function of innovation. In terms of economic growth, raising employment and creating jobs within SMEs imply their competitiveness in the market, whereas export growth implies SMEs' international competitiveness in global market. Therefore, government should prioritize its policy orientation on promoting job creation, mitigating unemployment rates and increase international competitiveness.

Despite SME's contribution to innovation, productivity, economic development and growth, SMEs face vulnerability achieving economy growth and development on its own compared to its counterpart, large enterprises (OECD, 2004b). For this reason, governments have been promoting various policies to support and foster SMEs. Korean

government has also prepared and implemented various projects such as start-up, funding, management, sales channels, technology development and mutual growth to strengthen fostering SME's.

2.3 Status and Development of the Korean SMEs

SMEs play an important role in Korean Economy (Nugent and Yhee, 2002). According to business statistics, SMEs account for ninety-nine percent of total enterprises, eighty-two percent of total employment and forty-eight percent of export (Official webpage of the Korean Ministry of SMEs and Startups).¹⁰ Table 1 shows the status and trends of Korean SMEs from 2016 to 2020. As these data suggests, SMEs play a particularly important role in the Korean economy due to their large share of the workforce dependent on SMEs (Nugent and Yhee, 2002).

[Table 4-1] The Number of SMEs and their Employee in South Korea from 2016 to 2020

All Industry (one person or above)		Year 2016	Year 2017	Year 2018	Year 2019	Year 2020
No. of Firms	Total	6,085,434	6,301,013	6,643,756	6,893,706	7,295,393
	No. of SMEs (%)	6,080,914 (99.9)	6,296,210 (99.9)	6,638,694 (99.9)	6,888,435 (99.9)	7,286,023 (99.9)
	No. of Others (%)	4,520 (0.1)	4,803 (0.1)	5,062 (0.1)	5,271 (0.1)	9,370 (0.1)
No. of Employees	Total	19,635,071	20,094,913	20,591,641	21,076,582	21,580,496
	No. of SMEs (%)	16,361,595 (83.3)	16,689,525 (83.1)	17,103,938 (83.1)	17,439,595 (82.7)	17,541,182 (81.3)
	No. of Others (%)	3,273,476 (16.7)	3,405,388 (16.9)	3,487,703 (16.9)	3,636,987 (17.3)	4,039,314 (18.7)

Source: Official webpage of the Korean Ministry of SMEs and Startups (accessed in November 13, 2022); Also, see Appendix.

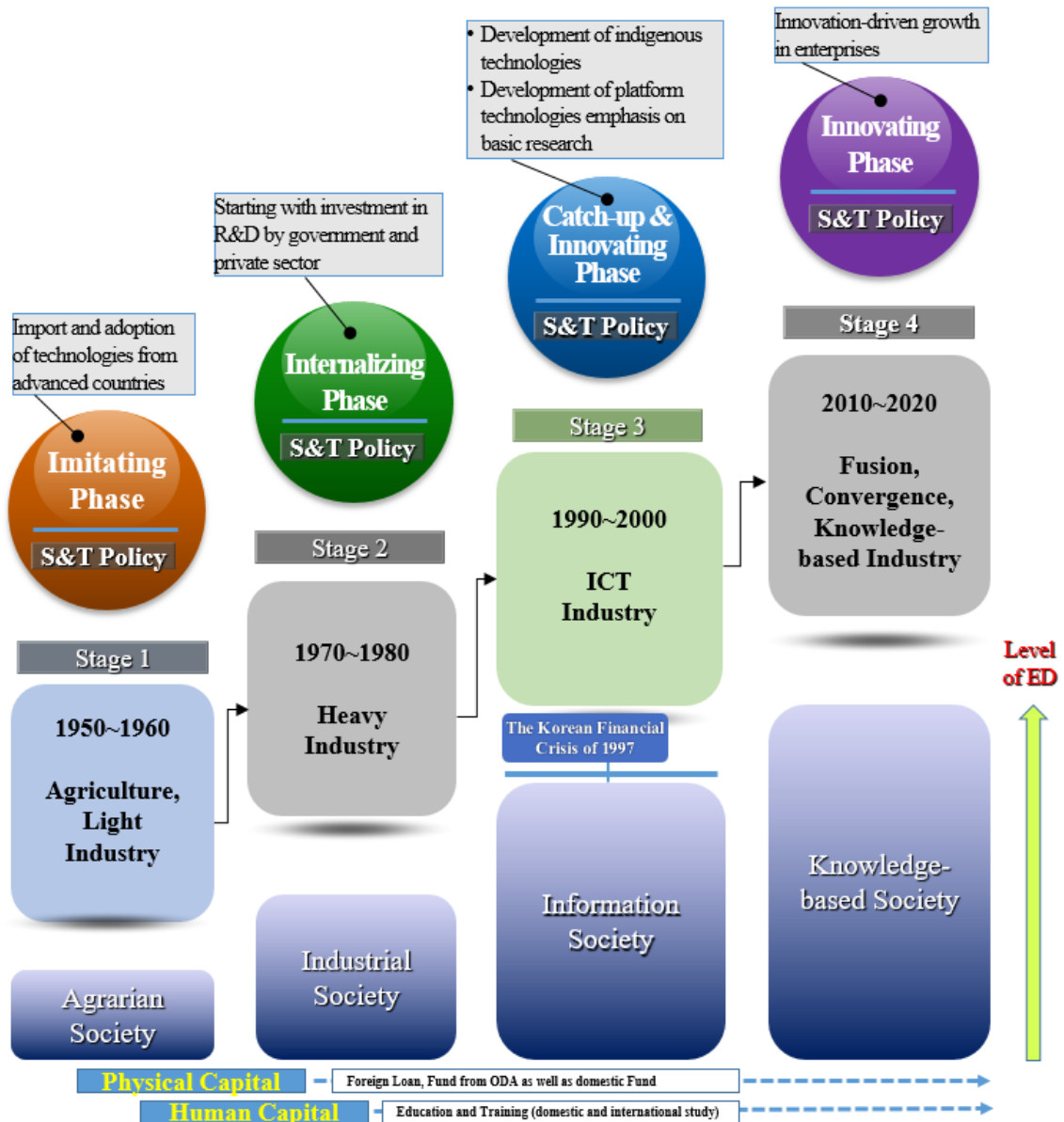
¹⁰ <https://www.mss.go.kr/site/eng/02/20202000000002019110610.jsp>

2.4 Asian Financial Crisis and Transition of SME Innovation Policies in Korea

As of 2020, South Korea has 7,286,023 SMEs with more than 17 million employees (17.54 million). South Korea's successful SME performance can be attributed to various government policies intended to promote SME innovations during the Asian Financial Crisis. Until late 1980s, South Korea's economy mostly relied on heavy and chemical industries but since early 1990s, various efforts have been made by the government to transition into information technology economy. However, spread of financial liberalization in 1990s coupled with South Korea's weak financial risk management system severely weakened South Korean economy's international competitiveness. As a result, South Korean economy eventually fell prey to the Asian Financial Crisis 1997, one of the most severe crises in Asia's history. To capitalize the crisis into an opportunity, South Korean government strived to transform South Korea into a market-oriented economy and knowledge-based economy.

As [Figure 4-1] suggests, South Korea was struck with the Asian Financial Crisis at the early stage of information society. During the crisis, Kim Dae Jung administration assessed that promoting ICT industry – a foundation to information society – can facilitate recovery of Korean economy. Thus, the administration aggressively sought to promote policies supporting SMEs, expecting such innovative approach would bring job creation and economic growth.

[Figure 4-1] Stage of Social Development and Evolution of S&T Policy in Korea



Source: Park (2019: 390) and revised by author

After the financial crisis 1997-1998, South Korean government strived to reduce the influence of big businesses (chaebol) through promotion of technology-based SMEs, while directly intervening in the venture capital market in 1998 for its revitalization. The intervention enabled innovative small firms with technology but needing startup funds

to obtain investment from the venture capital firms (VCFs) and limited partnerships funds (LPFs). To recap, [Figure 4-2] compares South Korean government's SME support policies before and after the Asian Financial Crisis.

[Figure 4-2] Korean SME Innovation Policies before and after the Asian Financial Crisis

1960s: Created SME Policies · Established a legal framework for assisting and fostering SMEs · Small and Medium Enterprise Association Act('61), Private Contract with SME Associations('65), Basic Act on Small and Medium Enterprises ('66)
1970s - 1980s : Protected and fostering SMEs · Fostered SMEs as a player of supplying parts to the heavy chemical industries · SME-Exclusive Industry('75), Interrelated SMEs to Large Companies('80) · Created Credit Guarantee Fund('76), Small Business Corporation('79), Technology Credit Guarantee fund('89)
1990's: Pursued structural improvement of SMEs · Shifted policy focus to autonomy, opening and competition with the inauguration of WTO (Jan.'95.) · Announced removal of SME-Exclusive Industry('94), Reduced Private Contract with SME Associations('95), Enacted the Act on Structural Improvement and Management Stabilization of Small and Medium Business('95), Established Small and Medium Business Administration(SMBA)('96)
Financial Crisis - 2002: Fostered venture businesses and promoted start-ups · Emphasized venture policy as a growth engine such as IT and NT and established measures to complement weak areas such as Small and Women's business · Opened KOSDAQ('96), Special Act on Venture Business('97), Small Business Act('97), Women's Business Act('99)
2003 -: Strengthen Self-Sustaining and Innovation Capacities · Implemented the strategy to secure SME competitiveness suitable for the innovation-driven era · Comprehensive plan to enhance SME competitiveness('04.7), Plan to vitalize venture businesses('04.12), 12 tasks for policy innovation('05.1), Comprehensive plan for self-employment('05.5), Revision of financing policy('05.6)

Source: Kim (2007: 133) and revised by author

As shown in the [Figure 4-2], South Korean government enacted SME related laws in 1960s and promoted various policies intended to support, protect and foster SMEs in 1970s and 80s. But in the late 1990s, the Asian Financial Crisis 1997-98 had tremendous adverse effect on South Korean economy and brought significant changes to various sectors. Further, not only did the Asian Financial Crisis trigger structural reform to Korean economy in the late 1990s but brought changes to South Korea's SME policies. Specifically, to reduce influence of big businesses (Chaebol) on Korean economy, the Korean government aggressively promoted policies supporting innovative venture corporations and technology based SMEs. In this spirit, the government pursued various supportive policies aimed to promote SME's innovation such as promoting large scale policies supporting SMEs. In addition, the government support policies for SMEs also served as means to resolve unbalanced growth problem between central and local government.

3. Government Support Policy for SME Innovation in South Korea

3.1 SME Technology Innovation Policy

To promote SME innovation, the Korean government emphasized policies promoting technological innovation. These policies include SME Technology Innovation Program, Industry-University-Research Institute Consortium Program, New Technology Purchasing Assurance Program, and the Korean Small Business Innovation Research (KOSBIR) Program (Kim, 2007: 135-137). First, SME Technology Innovation Program aimed to boost technological innovation of SMEs through R&D and build R&D capability. To this end, the program was launched in 1997 to strengthen technological competitiveness by supporting SME's development of new products and its process. Second, Industry-University-Research Institute Consortium Program aimed to support SMEs with inadequate innovative capacity to resolve technological problems and to strengthen their technological innovation by utilizing capable human capital and equipment from universities and research institutes. Through the program, performances of patent, prototype, and process improved. Third, New Technology Purchasing Assurance Program, to assist SMEs in commercializing their newly developed technologies, guaranteed that the government agencies and public institutions would purchase SME's new technology products. Fourth, the Korean Small Business Innovation Research (KOSBIR) Program was launched in 1998 to support SME's activities promoting technological innovation. The program makes recommendations to 16 public agencies – 10 government agencies and 6 government-funded research institutes with largescale R&D budget – to provide more than 5% of their R&D budget to SMEs. In most SME technology innovation policies, government supports more than 50% of the budget, which implies that SME technology innovation requires large-scale government investment and support.

3.2 Government Investment in the ICT Industry

It would not have been an exaggeration to suggest that South Korea's IT capability before 1990s lagged behind advanced countries by at least ten years and at most forty years. Today, scholars generally agree that South Korea's IT miracle that surprised the world started with the launch of Ministry of Information and Communication in 1994, which brought and sustained intensive investment policies on IT industry (Lee, 2008). It is worth noting that such policy measures took place during the Asian Financial Crisis 1997-98, one of the most devastating crises in Asia's history. The crisis raised number of unemployed to 2 million and Kim Dae Jung administration aggressively promoted IT policies with hope that ICT industry would serve as a cure. [Table 4-2] summarizes Kim Dae Jung administration's ICT policies.

[Table 4-2] Government Support Policy for the ICT Industry during the Asian Financial Crisis

Year	Policy Contents
1998	- Provided emergency support of KRW 300 billion in postal finance funds to SMEs in IT industry to foster venture companies. - Allocated 135 billion won to the public labor business funds as an additional budget to utilize highly educated but unemployed population for informatization projects. - Promised to grant tax cuts and exemptions to custom-made semiconductor companies. - Supplemented the special military service system to secure manpower in ICT industry - Established corporation to construct techno parks in Incheon, Ansan, Daegu, Gyeongsan, Gwangju, Cheonan, Asan and Gongju.
1999	- Established joint project promotion plan for e-money - Supported 83 billion won to develop human resources in the IT field - The Ministry of Information and Communication announced the 'Five Year Plan for Information and Communication Technology Development' from 2000 to 2004, which plan to invest 4 trillion won to both private and public sectors through 6 key technology development projects such as next generation internet, optical communication, digital broadcasting, wireless communication, software and computer.

Year	Policy Contents
	- Revealed to develop internet technology 1,000 times faster than the internet speed at that time through technology development.
2000	- Established optical cable network in 144 major regions nationwide. - Provided free high-speed internet service to all elementary, middle and high schools nationwide. - Enacted the Information and Communications Infrastructure Protection Act from July 1st, 2001 to establish protection of information infrastructure. - Promoted investment for the ICT industry by absorbing telephone tax into value-added tax. - Provided high-speed internet service to towns and villages nationwide and implemented high speed technology verification pilot project to underdeveloped rural areas. - Framework Act on Electronic Commerce, Comprehensive Measures for Vitalization of Electronic Commerce (2000).
2001	- Invested 600 billion won in addition to 400 billion won in venture investment funds to prepare for new industry development plans such as IT, BT, NT, CT, ET. - Established Cultural Contents Agency (2001) and legally incorporated as Korea Cultural Contents Agency in October 2002. - Completed early repayment of full IMF debt on August 23, 2021.

Source: Lee (2008), President Archives of the Korean Ministry of the Interior and Safety and revised by author.

As the [Table 4-2] suggests, to recover from the Asian Financial Crisis 1997-98, Kim Dae Jung administration invested 400 billion won in the IT, BT, NT, CT and ET sectors and additionally invested 600 billion won in venture investment funds to promote new industry development plan. In addition, various policies supporting SMEs were promoted to actively foster venture companies, such as the establishment of the Cultural Contents Agency for the promotion of ICT-based cultural and artistic contents. Among these active government efforts, also included are the 2000 Comprehensive Unemployment Measure to solve massive unemployment caused by the IMF economic crisis and the promotion of the ICT industry as a part of this measure for job creation.

Specifically, the government comprehensively supported spaces needed for start-up, high priced advanced equipment and technology and market information to SMEs with

excellent ideas and technologies but without start-up funds. The government also promoted policies expanding domestic software market through strengthening crackdowns, supporting domestic software companies to enter overseas markets and fostering component software industry – an industry that emerged as a core sector of software industry. Such government efforts brought a rapid increase of venture companies, localization rate of computer parts, new employment, GDP and IT environment conducive for IT development and growth.

3.3 Entrepreneurial Venture Business Growth in the ICT Industry through Korean Government Financial Support

In South Korea, entrepreneurship has been an important driver of SME innovation in the ICT industry. The notion of entrepreneurship not only includes creation of new firms, but also the economic activities of newly established corporations. In essence, entrepreneurship is about recognizing and exploiting opportunities (Kirzner, 1997). Not surprisingly, entrepreneurship is associated with innovation and other entrepreneurial features such as risk-taking and proactiveness (Covin and Slevin, 1989; Lumpkin and Dess, 1996; Hessels, 2008). Further, entrepreneurship also facilitates spread of knowledge. According to knowledge spillover theorists, entrepreneurship is the knowledge filter serving as a conduit for knowledge spillovers (Acs, 2008; Don and Kim, 2014). SMEs are critical for entrepreneurship because it can bring technological innovation and strengthen the capacity to do so. As a result, performance of SMEs is a significant factor for building a competitive economy in a global market. In sum, SMEs play an important role enhancing innovation and economic performance.

During the Asian Financial Crisis 1997-98, Kim Dae Jung administration paid attention to the newly growing ICT industry. To overcome the economic crisis, the administration tried to expand employment and job creation through fostering venture business that required new knowledge and technology. As land formation plan for techno park – a research complex promoted for ICT and venture corporations since February of 1997 – began to materialize, a corporate body responsible for constructing techno parks in Gongju-si, Asan-si, Cheonan-si, Gwangju-si, Gyeongsan-si, Daegu-si, Ansan-si and

Incheon Metropolitan City was established in 1998. Furthermore, various policy efforts such as Venture Business Financial Support Expansion Plan (September 1998), Venture Business Startup Fund Support Plan (February 1999), Job Creation and Unemployed Protection Plan (March 1999), Venture Investment Association Investment Management Plan (2001) were put forth. Such atmosphere generated numerous venture corporations and served as an opportunity to attract talented individuals to ICT industry. However, when venture capital funds became easily available from government's supportive policies, it also generated problems for the IT industry. With abundant venture capital funds available, many of the talented prospective college graduates chose to start their own venture firms rather than seek employment in the IT industry. Such trend brought labor shortage and became a problem for the IT industry, as demands for computer and internet related businesses were high but supply of needed labor low. This imbalance of labor brought a heated competition between corporations in ICT industry to scout for IT professionals.

[Table 4-3] summarizes the data of venture firms from times of the Asian Financial Crisis 1997-98 to recent years. The data suggests that the number of venture corporations have increased from 2,042 in 1998 to 39,101 in 2020.

[Table 4-3] The status and development of Venture Firms in Korea since 1998 year

Year	No. of Venture Firms	Average Sales Amount (Unit: 100 Million Won)	Average Number of Employees	Financial Situation		Current Status of Patent Rights (Mean)
				Equity Capital Ratio (%)	Debt Ratio (%)	
1998	2,042	-	-	-	-	-
1999	4,934	46.9	39.60	-	-	0.90
2000	8,798	53.0	39.11	-	-	2.90
2001	11,392	65.6	31.60	50.4	97.6	3.20
2002	8,778	68.0	37.50	42.3	128.2	3.00
2003	7,702	73.4	39.10	38.3	142.0	3.30
2004	7,967	78.5	33.50	50.1	99.4	3.70
2005	9,732	68.8	34.80	47.4	111.2	4.10
2006	12,218	74.2	29.10	45.3	120.5	5.00

Year	No. of Venture Firms	Average Sales Amount (Unit: 100 Million Won)	Average Number of Employees	Financial Situation		Current Status of Patent Rights (Mean)
				Equity Capital Ratio (%)	Debt Ratio (%)	
2007	14,015	53.1	30.20	44.6	124.1	3.20
2008	15,401	60.1	27.30	39.1	154.9	2.80
2009	18,893	64.5	27.00	39.1	153.7	3.00
2010	24,645	72.2	27.30	41.2	142.7	2.90
2011	26,148	70.3	25.50	40.4	147.5	3.00
2012	28,193	67.2	24.70	40.6	146.1	3.50
2013	29,067	68.4	24.70	40.5	146.7	3.90
2014	29,844	71.9	24.00	42.6	134.5	4.20
2015	31,189	69.2	23.30	39.2	155.4	4.50
2016	33,289	68.5	22.90	40.9	144.6	5.40
2017	36,065	49.0	18.80	45.7	118.7	5.80
2018	36,065	53.0	19.80	46.9	113.4	3.30
2019	36,503	53.0	22.20	43.8	128.3	5.50
2020	39,101	52.9	20.90	46.8	113.7	5.30

Source: Ministry of SMEs and Startups' Survey of Korea Venture Firms. Each Year.¹¹

Ultimately, the government's ICT policies introduced to overcome the economic crisis through job creation successfully fostered ICT industry. Specifically, the policies increased labor force with related skills in ICT and the ratio of localized ICT production while generating new jobs that made enormous contribution to South Korea's economic growth (Lee, 2008). It is particularly worth noting that a handful of Korean corporations – LG Electronics' display sector, Samsung Electronics' semiconductor and memory sectors, and SK telecom's communication sector – were able to develop a world-class competitiveness due to Korean government's decision to concentrate resources in fostering ICT industry.

¹¹ e-National indicator(https://www.index.go.kr/potal/main/EachDtlPageDetail.do?idx_cd=1195)

3.4 Financial Policy

Government's financial support – both direct and indirect – can facilitate SME innovation by assisting innovative SMEs with financial problems. Therefore, Korean government has also provided both direct and indirect financial support for SMEs. For instance, SMEs with potential for growth in the future has received various benefits such as tax reduction, loan support and special listing in stocks market. In addition, the government offered financial support to SMEs with low security solvency that could not obtain loan credit guarantee services. In particular, Korean government operated various funding programs to support SMEs with technological capacity but faced financial constraints.

While Korean government operated various funds such as Small and Medium Business Fund (SMBF), Informatization Promotion Fund (IPF), Cultural Industry Promotion Fund (CIPF), Science and Technology Promotion Fund (STPF), Credit Guarantee Fund (CGF), Technology Credit Guarantee Fund (TCGF), the government also supported SMEs through various other measures such as the Venture Investment Fund. [Figure 4-3] summarizes the financial support system for SMEs operated by the government.

- Small and Medium Business Fund (SMBF), a fund supported by Small and Medium Business Corporation, provides funds needed for start-up support and structural advancement of small and medium-sized enterprises. The Korean government has established this fund to secure financial resources necessary for the promotion of small and medium-sized enterprises' start-ups, balanced industrial development and the establishment of industrial infrastructure and expansion of management infrastructure and advancement of structures.
- Informatization Promotion Fund (IPF) is a fund established with the goal of improving citizen's quality of life and promoting economic development by promoting informatization, establishing an infrastructure for the information and communication industry, advancing information and communication infrastructure and supporting information and communication R&D (Article 33 of the Framework Act on Promotion of Informatization). South Korea, through audacious and flexible investment of the Informatization Promotion Fund, has laid a foundation for the knowledge and information society by establishing 1) a high-speed information and communication

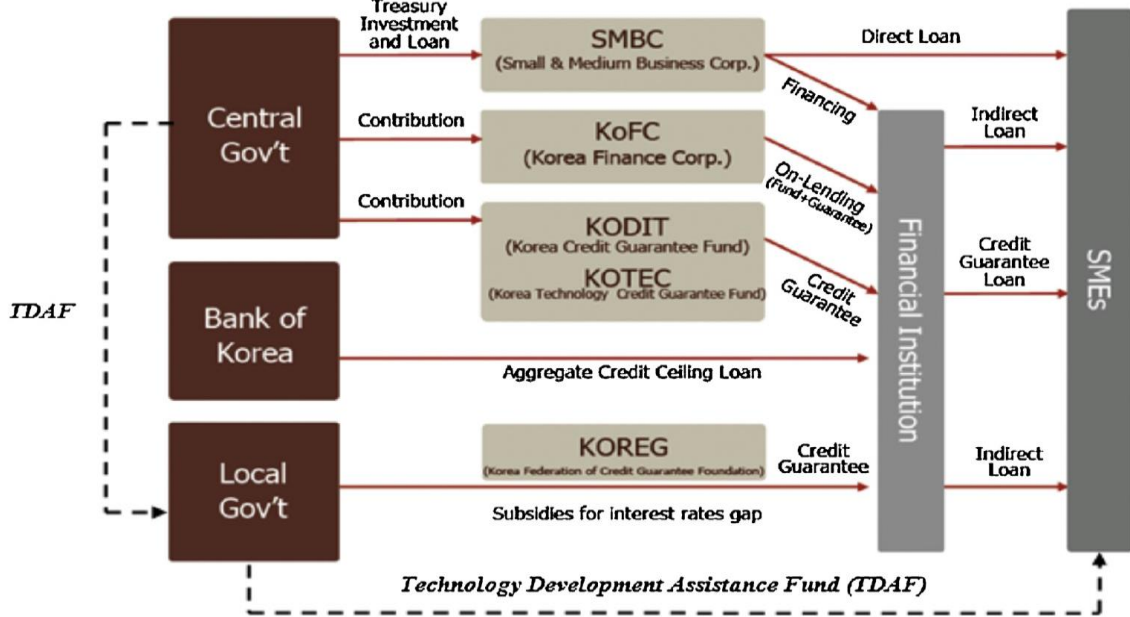
network that provides access to high-speed internet for everyone 2) an infrastructure to advance e-government by creating an electronic document distribution environment for all central administrative agencies and the expanding G4C base and the national tax database. In addition, with the IT industry emerging as a key growth industry for the Korean economy, the IPF generated an environment conducive for IT venture companies such as software support centers and investment associations to actively venture and grow. Furthermore, with the technological development support through IPF, it became possible to advance information technology research and foster ICT human resource. These findings suggest that IPF, as a financial vehicle, enabled the IT industry to grow and become central to Korean economy after the Asian Financial Crisis (Yoon et al., 2002; Special Committee on e-Government, 2002).

- The Cultural Industry Promotion Fund (CIPF) provides loans to SMEs related to cultural industry, invests in new technology business investment associations or SME venture investment association to financially support and foster SMEs as well as venture businesses related to cultural industry, and manage contributions and subsidies related to promoting cultural industry.
- The Science and Technology Fund (STF) is a fund established to support promotion of science and technology and development of science and technology culture. STF supports projects promoting science and technology such as international exchange, human resources and research and academic activities related to science technology. The fund also supports companies, educational institutions, research institutions and science and technology related organizations seeking to use research outputs into practical use. In sum, the fund is used as an investment on new technology operators or venture businesses, loans and investments prescribed by Presidential Decree.
- Korea Credit Guarantee Fund (KCGF) was established to establish a sound credit order and facilitate corporate financing by guaranteeing the debts of SMEs. As a fund management type quasi-governmental institution under the Financial Services Commission, it is the most specialized institution providing various types of support for SMEs in Korea.
- Korea Technology Appraisal Guarantee System (KTAGS) is an institution established to enhance technological competitiveness of companies by focusing on technology assurance and evaluation. KTAGS, a fund management type quasi-governmental

institution under the ministry of SMEs and Startups, changed its name to Korea Technology Finance Corporation in 2016. The KCGF and KTAGS are funds created to support SMEs and technology innovation company and provides a very important government credit guarantee service to finance SMEs.

- In addition to direct/indirect financial support through fund management and credit guarantee services, the Korean government also provided technology development subsidies to SMEs for the fostering of SMEs and technological innovation. These subsidies provided financial support necessary for regional SMEs to promote technology development.

[Figure 4-3] Financing Policy for SME Innovation in Korea



Source: Yi (2012); Doh and Kim (2014: 1562)

3.5 Laws Related to the Korean SMEs

Korean Ministry of SMEs and Startups reports various laws related to Korean SMEs on its website.¹² The laws include 'Framework act on small and medium enterprises', 'Framework act on small and medium enterprises', 'Act on facilitation of purchase of

¹² Ministry of SMEs and Startups webpage (<https://www.mss.go.kr/site/eng/02/20203000000002019110618.jsp>).

small and medium enterprise-manufactured products and support for development of their markets', 'Act on special cases concerning the regulation of the special economic zones for specialized regional development', 'Act on the protection of and support for micro enterprises', 'Act on special measures for the promotion of venture businesses', 'Act on support for female-owned businesses', 'Act on the fostering of self-employed creative enterprises', 'Act on the promotion of collaborative cooperation between large enterprises and small-medium enterprises', 'Act on the promotion of technology innovation of small and medium enterprises', 'Promotion of disabled persons' enterprise activities act', 'Special act on support for small urban manufacturers', 'Small and medium enterprise cooperatives act', 'Small and medium enterprises promotion act', 'Special act on support for human resources of small and medium enterprises', 'Special act on the development of traditional markets and shopping districts', 'Special act on the promotion of business conversion in small and medium enterprises', 'Support for small and medium enterprise establishment act', 'Regional credit guarantee foundation act', 'Act on Support for Protection of Technologies of Small and Medium Enterprises', 'Act on Special Cases Concerning Support for Technoparks', and 'Korea Technology Finance Corporation Act'.

As these laws suggest, South Korean government uses various laws to support and foster SMEs. These laws, in the process of promoting government's SME policies, have become an important legal basis. [Table 4-4] summarizes these laws.

[Table 4-4] Laws Related to Korean SMEs

Laws	The Purpose of Each Law
Framework act on small and medium enterprises	The purpose of this Act is to provide for basic matters concerning the direction-setting for small and medium enterprises and measures for fostering small and medium enterprises with the aim of supporting their creative and independent growth, improving their industrial structure, and facilitating the balanced development of the national economy.
Act on facilitation of purchase of small and medium enterprise-manufactured products and support for development of their markets	The purpose of this Act is to contribute to the enhancement of the competitiveness of small and medium enterprises and to the management stability thereof by facilitating the purchase of small and medium enterprise-manufactured products and supporting the development of markets therefor.
Act on special cases concerning the regulation of the special economic zones for specialized regional development	The purpose of this Act is to systematically support specialized regional development and to help stimulate regional economies and the growth of the national economy by selectively applying special cases for regulations in accordance with the characteristics of each region through the designation and operation of special economic zones for specialized regional development.
Act on the protection of and support for micro enterprises	The purpose of this Act is to contribute to improving the social and economic status of micro enterprises as well as to the balanced development of the national economy by promoting independent business activities of micro enterprises and furthering the management stability and growth thereof.
Act on special measures for the promotion of venture businesses	The purpose of this Act is to contribute to the facilitation of the structural adjustment of the industry as well as to the enhancement of the competitiveness thereof, by promoting the conversion of existing enterprises into venture businesses and the establishment of venture businesses.
Act on support for female-owned businesses	The purpose of this Act is to seek actual gender equality in the economic realm by actively supporting the activities of female-owned businesses and women's establishment of business and to contribute to the development of the national economy by facilitating women's economic activities and elevating the status of businesswomen.

Laws	The Purpose of Each Law
Act on the fostering of self-employed creative enterprises	The purpose of this Act is to contribute to the growth of the national economy by fostering self-employed creative enterprises through the promotion of the establishment of self-employed creative enterprises by the people who have creativity and specialty and through the creation of the foundation for the growth thereof.
Act on the promotion of collaborative cooperation between large enterprises and small-medium enterprises	The purpose of this Act is to sharpen the competitiveness of large enterprises and small-medium enterprises by consolidating win-win cooperation between them and to attain their shared growth by resolving the polarization between large enterprises and small-medium enterprises with the aim of laying the foundation for sustainable growth of the national economy.
Act on the promotion of technology innovation of small and medium enterprises	The purpose of this Act is to contribute to the development of the national economy through the strengthening of the technological competitiveness of small and medium enterprises by expanding infrastructure to promote technological innovation of small and medium enterprises and establishing and implementing policies related thereto.
Promotion of disabled persons' enterprise activities act	The purpose of this Act is to contribute to the growth of the national economy by proactively facilitating business start-ups and entrepreneurial activities of disabled persons, improving the economic and social status of disabled persons, and endeavoring to enhance their economic strength.
Special act on support for small urban manufacturers	The purpose of this Act is to contribute to the development of the national economy by establishing a support system for the growth and development of small urban manufacturers and encouraging their economic activity.
Small and medium enterprise cooperatives act	The purpose of this Act is to provide for the matters concerning the establishment, management, and development of cooperative organizations, through which small and medium enterprisers consolidate and promote collaborative projects, with aims of providing equal economic opportunities to small and medium enterprisers and encouraging their independent economic activities to improve the economic status of small and medium enterprisers and accomplish the balanced development of the national economy.

Laws	The Purpose of Each Law
Small and medium enterprises promotion act	The purpose of this Act is to strengthen the competitiveness of small and medium enterprises through their structural advancement, and expand the business sphere for small and medium enterprises, thereby contributing to the balanced development of the national economy. <Amended by Act No. 9683, May 21, 2009]
Special act on support for human resources of small and medium enterprises	The purpose of this Act is to contribute to the balanced development of the national economy and society by enhancing the competitiveness of small and medium enterprises and promoting the employment thereof with support for programs facilitating the supply and demand of human resources, upgrading human resource structure and improving awareness of small and medium enterprises.
Special act on the development of traditional markets and shopping districts	The purpose of this Act is to seek the revitalization of local business districts and balanced development of the distribution industry through facilitating the modernization of facilities and business management of traditional markets and shopping districts in order to contribute to the growth of the national economy.
Special act on the promotion of business conversion in small and medium enterprises	The purpose of this Act is to promote the business conversion of small and medium enterprises that suffer difficulties due to changes in the economic environment, which will enhance the competitiveness of small and medium enterprises and upgrade the industrial structure, thereby contributing to the healthy development of the national economy.
Support for small and medium enterprise establishment act	The purpose of this Act is to contribute to the establishment of a solid industrial structure through sound development of small and medium enterprises by facilitating the setting-up of small and medium businesses and developing a firm basis for their growth.
Regional credit guarantee foundation act	The purpose of this Act is to facilitate financial accommodation, revitalize the regional economy and contribute to the promotion of welfare of the populace by establishing credit guarantee foundations and the Korean Federation of Credit Guarantee Foundations to enable them to guarantee obligations of small enterprises, micro enterprise, etc., and individuals short of security solvency in the district.

Laws	The Purpose of Each Law
Act on Support for Protection of Technologies of Small and Medium Enterprises	The purpose of this Act is to strengthen the capability of small and medium enterprises to protect technologies and their technical competitiveness, by expanding infrastructure to protect technologies of small and medium enterprises and by formulating and implementing policies related thereto, thereby contributing to the development of national economy.
Act on Special Cases Concerning Support for Technoparks	The purpose of this Act is to cluster human and physical resources of enterprises, universities, research institutes, etc. in certain places to jointly develop technology and to bring about regional innovation based on close connections and collaboration among such entities, thereby contributing to creating jobs, invigorating regional economies, and promoting national competitiveness.
Korea Technology Finance Corporation Act	The purpose of this Act is to facilitate the financing of new technology business by stabilizing and developing technology guarantee systems through the establishment of the Korea Technology Finance Corporation, and further contribute to the development of the national economy.

Source: Webpage of the Korean Ministry of SMEs and Startups.¹³

¹³ Ministry of SMEs and Startups webpage (<https://www.mss.go.kr/site/eng/02/20203000000002019110618.jsp>).

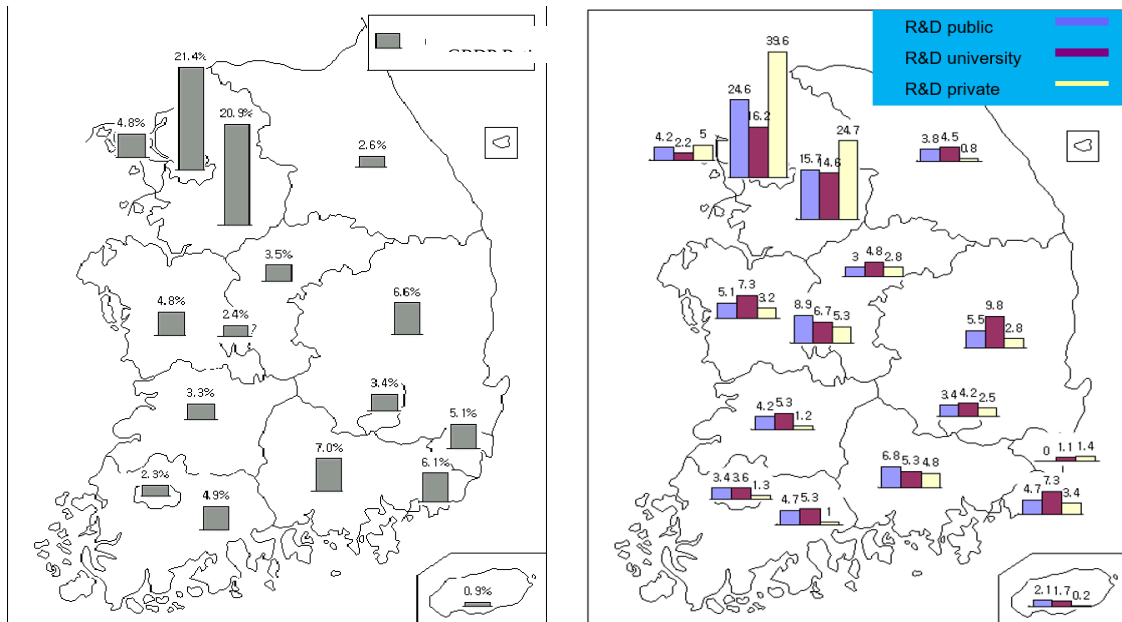
4. Government Financial Support for Regional SME Innovation

4.1 Disparity between Large Capital Area and Local Areas

Amid the economic crisis, South Korean government achieved notable results through its SME and venture corporation policies. However, the problem of unbalanced growth between capital and local areas remains a problem waiting for a solution. The disparity between Large Capital Area (LCA) and local cities and provinces have exacerbated. In 2001, GRDP of LCA comprised 47.1% of the total areas and the local tax revenue of LCA 58.7% of the total tax revenues. In terms of fiscal independence, LCA scored 84.1% compared to the national average at 62.9%.

The regional disparity problem is also serious in manpower and R&D fund. In 2001, 62.7% of R&D funds, 58.6% of labor force, and 66.7% of R&D institutions were located in LCA. Further, 67% of universities related R&D institutions and 60.1% of students reside outside of the LCA, but funds and quality of manpower remain an issue in local areas.

[Figure 4-4] GRDP Ratio and R&D Fund of each Region in Korea: Year 2001



4.2 Necessity of Regional Balance for Competitiveness

The issue of regional balance was one of the pivotal campaign pledges of President Roh, who became president of South Korea in 2003. In the campaign, President Roh emphasized that the inefficiency of concentration in LCA will eventually reach a critical point and that investment in local area will reap profits in the long run. After being elected, President Roh launched a powerful Presidential Commission. The commission suggested that local industries, combined with the knowledge and skill base of neighboring colleges and universities, should be the engine for regional development. Based on this recommendation, the commission designed the Regional Innovation System (RIS).

Included in the RIS were various strategies for regional industrial development, including development via SME innovation. However, because regional SMEs faced various limitations, the government's financial support was a priority above all. For instance, to promote innovation, regional SMEs need technological skills to introduce new and effective technologies. However, regional SMEs lack adequate financial resources to invest or develop such technologies. What also makes such investment and development challenging is that the profit made from the technology is not guaranteed. More importantly, most regional SMEs simply lacked technological infrastructure to develop next stage of advanced technology. Due to these limitations, Korean government began to provide a large-scale financial support to regional SMEs, especially for the technological innovation of regional SMEs.

In effort to promote regional industries, the Korean government has invested huge amounts of public funds conducive to regional SME innovation with following assumptions. First, the investment and support of regional SMEs may strengthen their market competitiveness. Second, the investment and support would considerably impact their local competitiveness and in turn, promote regional economic growth. According to Doh and Kim (2014), the government's financial support has positively contributed to the innovation of regional SMEs. Specifically, a positive relationship has been observed between 1) the technological development assistance of government and patent acquisitions of regional SMEs 2) the technological development assistance

and new design registrations 3) networks with universities and regional SMEs' patent acquisitions and new design registrations 4) R&D expenditures and patent acquisitions.

In this line of argument, role of local government is critical, given their proximity to regional SMEs. Specifically, local governments are, in particular, very active in the design and delivery of programs for SME innovation. Therefore, the national government should empower local government to promote SMEs innovation. Ultimately, the aim of government support for SMEs is establishing a competitive and innovative SMEs not needing government's financial aid. However, when the government is involved, its role should be facilitative, not dictatorial.

By facilitating, it means the local government should provide a variety of support services for regional SME innovation. This includes taking initiatives in provisions for targeted and quality business support services; immediate, technical, and managerial training programs; lowering administrative costs and burdens of regional SMEs; building network cross sectors; provisions for financial incentives and assistance; legal framework reinforcement.

In South Korea's case, among local government's various policies supporting SMEs, technology development fund played a critical role enhancing regional SME's technological innovation. Specifically, the government intervention in the form of payment such as the technology development fund helped overcome the economic crisis and the limitations of regional SMEs in ameliorating the regional growth problem.

4.3 Government Financial Support for Regional SME Innovation in Korea: Technological Development Assistance Funds (TDAF)

Regional SMEs face severe constraints in the competitive market. In such market, technological development and innovation are a necessary condition for regional economic growth. Therefore, to enhance SMEs' performance, SMEs need to acquire technological skills and introduce effective and new technologies into their firms. However, SMEs face several constraints such as lacking adequate financial resources and technical infrastructure to support and develop advanced technology. Furthermore, the

profit brought to SMEs through the technology is not guaranteed. All in all, these conditions hamper and limit performance of SMEs.

It is in this context government interventions are required to overcome the constraints of SMEs. Government interventions by financial assistance (technology development assistance fund) can enhance SME performance such as technology development and innovation and management capacity. Korean government's industrial policies can be divided into two categories. First, applying national (central) industrial policies to the local (regional) level and second, promoting local industrial policies to achieve their own regional economic prosperity. To achieve these goals, government technology development assistance policy plays a critical role. A part of national industrial policies devolved from the Department of Knowledge and Economy to local government level in Korea, the policy aims to apply national industrial policies to local level.

In this regards, South Korea's local industrial policies have primarily tried to reduce the gap between regional economies and increase GRDP by vitalizing local firms, particularly the SMEs. To this end, the Korean government has concentrated its investment in expanding local infrastructure, growth of self-sufficient corporations through SMEs' technology development assistance and through strengthening industrial competitiveness at the local level since 1999 (Doh and Kim, 2014)

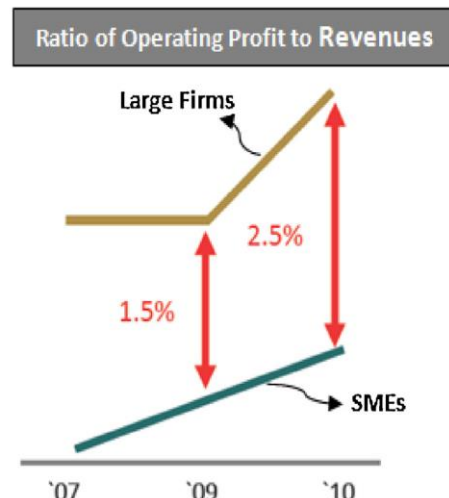
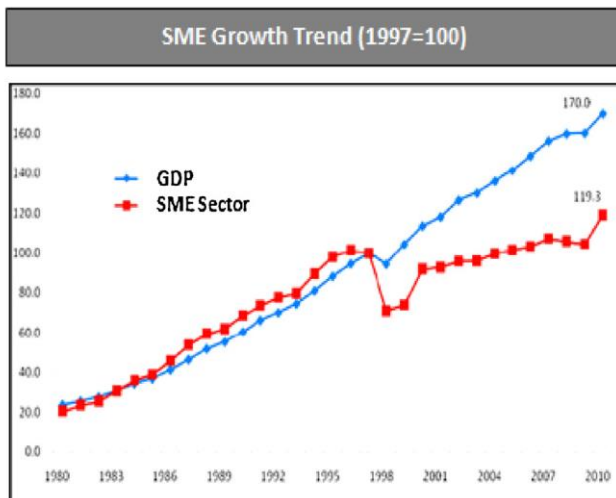
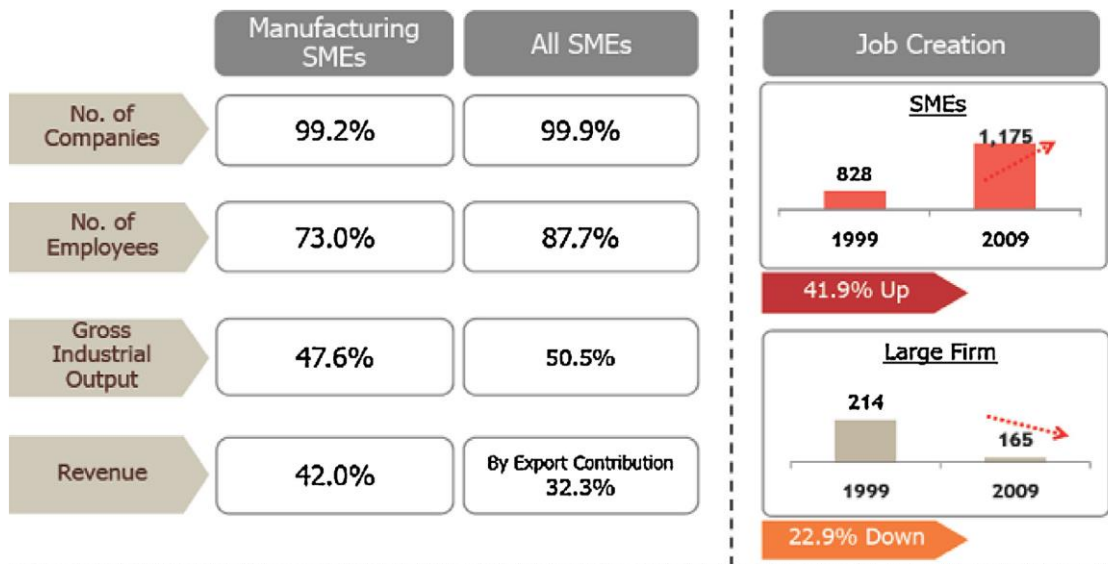
As such, government intervention in South Korea is critical to overcome regional SMEs' constraints. Specifically, Technology Development Assistance Fund (TDAF) has been implemented at the provincial government level. As a part of local industrial policy, it aims to strengthen the competitiveness of local firms and the economy by supporting common technology, core technology and fundamental technological innovations.

In addition, considering the importance of regional SME innovation in the Korean economy, the Korean governmental authorities have, for many years, carried out a variety of programs to support SMEs. Specifically, the Korean government has implemented various strategies for a balanced regional development to decrease income disparity between the industrial and agricultural sectors since DJ Administration brought policy initiatives to promote regional industries. As a means of promoting regional industries, the Korean government has invested huge amounts of public funds

conducive to SME innovation on the assumption that innovative SMEs may strengthen their competitiveness in markets and make a considerable impact on local competitiveness.

Since the Korean government provided financial support for SME innovation in regional industries, more attention has been given to the design and implementation of such programs. Compared to 1999, when faced with the Asian Financial Crisis, in 2009, 10 years later, the number of SMEs and new jobs have increased. Such result in large part can be credited to government's SME support policies.

[Figure 4-5] The status and development of SMEs in 1999 and 2009 in Korea



Source: Adapted from Yi (2012); Doh and Kim(2014: 1559)

As a result of the consideration of government, South Korea is regarded as having set a successful model for SME development. According to Small and Medium Business Administration (SMBA) of Korea's internal report, Korean SMEs have developed steadily as a backbone of the national economy and are making important contributions to nation's economy, creating new job opportunities, the leverage necessary to strong, sustainable and balanced growth within the frame of the global economic system.

5. Policy Implications for SME Innovation

Today, South Korea invests high share of GDP on technologically innovative SMEs and venture corporations. Among OECD countries, South Korea's level of investment in innovative SMEs and venture corporations ranks among the top. These investments in particular have been used as an important financial resource for venture companies. Only about twenty-five years ago South Korea faced one of the economic crises in Asia's history, Asian Financial Crisis 1997-98 with an underdeveloped economic structure. But the Asian Financial Crisis 1997-98 South Korea experienced in late 1990s brought an opportunity to transform South Korea's economic structure into an advanced one. Further, the SME support policy such as the financial support for fostering venture companies has been evaluated positively for overcoming the crisis through job creation. In particular, active venture business fostering policies based on ICT industry and SME actively promoted during Kim Dae Jung administration's early stages – expansion of venture business financial support, venture business start-up fund support, venture investment association fund management, support for purchase of export bills for SMEs, repayment delay for SMEs' foreign loan, and financial support for SMEs – were mostly intended to resolve financial difficulties of SMEs. However, South Korean government's success in overcoming the Asian Financial Crisis early and in fostering technologically innovative SMEs were possible not only due to their financial support for businesses but also due to various other measures by the government such as legal foundation for SMEs, actively fostering venture companies and technologically innovative SMEs in the ICT field, providing technology development subsidies for technologically innovative SMEs and promoting balanced growth policy for metropolitan and non-metropolitan areas.

In this research, we analyzed South Korea's historical experience that created and expanded jobs through SME innovation in facing the Asian Financial Crisis that lead to sustainable economic development and growth. According to the analysis of this study, the Korean government has performed various functions and roles and promoted establishment of related institutions and governance. Therefore, the functions and roles of the government displayed by the Korean government so far provides important policy implications for underdeveloped and developing nations promoting SME innovation for

economic performance. Although the context of other countries may not match South Korea's, the process of the government promoting diverse support policies for specific industries or business groups in the midst of Asian financial crisis provides important policy implications as follows.

First, for SME innovation, it is necessary for the government to establish SME Innovation Plan. SME Innovation Plan is a formal document that contains visions and goals of SME Innovation and strategies to attain vision and goals, comprehensive guideline on resource allocation and based on an assessment of the nation's needs and capabilities as well as internal and external environments surrounding SME innovation in each country. The plan also includes a plan focusing on SME innovation policy and a sectoral plan focusing on SME innovation based on the guidelines of the planning authority or a higher-level plan. It is also necessary to consider the period for SME Innovation Plan (ex. Five-Years, Ten-Year Plan).

Second, for SMEs to successfully innovate, human resources with high expertise and skills (high skilled people) need to flow into SMEs. Thus, the government needs to establish and promote human resource development plan appropriate for SME innovation. Korean government during the early stage of ICT industry began to actively support ICT industry's venture enterprises and SMEs as a means to overcome the Asian financial crisis, and in the process the Korean government invested financial resources to develop human resources needed for ICT industry.

Third, the government needs to prepare legal and institutional foundations to promote necessary policies supporting SME innovation. The central government needs to prepare the necessary finances at a national level for SME innovation and enact legislation that can support SMEs innovation both directly and indirectly. Moreover, based on these legislations, the government needs to establish and operate organization responsible for financially supporting SME innovation and a governance system including cooperation between central government departments and between central and local governments as well as a cooperative governance system among various organizations related to SME innovation.

Fourth, a government support policy for regional SME innovation is necessary. In particular, since most SMEs are distributed across regions, promoting technological innovation through local SME's technology development support is needed. Further, it is necessary to strengthen the role of local governments, the main body with the role of providing financial support to SMEs. Local governments must develop and operate various cooperative programs so that local techno parks, local universities, and local SMEs can cooperate to achieve synergy effects through the establishment of industry-academia-research networks.

Fifth, government needs to manage basic data related to SMEs operating in each industry. Establishing such data of SMEs in each industry serves as a basic data for understanding their performance and activities, and analysis of these data allows us to evaluate effectiveness of government support policies for SME's innovation.

Finally, SME innovation cannot be accomplished quickly nor by SMEs alone. Therefore, government needs to make efforts to promote policies fostering and supporting SME innovation in a mid to long-term perspective and ensure that government's support policies for SME innovation can focus through the establishment of SME innovation clusters. Ultimately, efficient government policies will eventually accelerate SME innovation both in terms of process and outcome, an important aspect underdeveloped and developing countries with limited financial resources to consider.

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CHAPTER 5

SME Innovation Support in Lao PDR

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 5. SME Innovation Support in Lao PDR

Dr. Phanpakit OUPHANHDALA, Lao People's Revolutionary Party

1. Introduction

Research and Development (R&D) is a key factor to improve the national economy, particularly R&D on science and technology (S&T). Policymakers have taken R&D and S&T for innovation to strengthen commercial productivity. It aims to transform economic structure to industrialization and modernization in achieving the sustainable development goals (SDGs) by 2030. Watson et al. (2003) revealed that S&T helps the government to enhance poverty alleviation and economic development. However, development would increasingly depend on an ability of each country to comprehend, interpret, select, adapt, practice, transfer, diffuse, produce, and commercialize scientific and technological knowledge to match its culture, aims, and development's level. In South Korea, the government has implemented the 5-year technology promotion plan since the 1960s, which became the foundation of science and technology development, particularly the establishment of Korea Institute of Science and Technology (KIST) (IDN, 2011).

Governments around the world are urging to broaden and deepen their support for innovation development in the private sector and the economy. In the case of Korea, the innovation of SMEs is measures by technological innovation, such as patent, utility model, trademark, and new design registrations. Doh and Kim (2014) suggested that the assistance of the Korean government on technological development and universities' researches play a major role in regional MSME innovations. In addition, regarding a study on external relationships in the organizational innovation carried out by Simao et al. (2016), it revealed that relationships with business partners contribute to positive

performance of organizational innovation, in particular through new organizational practices, rewards to employees and communication that allow introduce organizational innovations along the supply chain. Despite this, based on results from emerging and developed economies, PwC (2010) reported that innovation depends on well-functioning markets. It also emphasized that the rule of law and arm's length transaction are essential for such markets to function well, which need to be initiated before tax or other incentives can be effective.

In order to ensure the achievement of SDGs and innovation, most countries have developed national development planning and their implications for the sustainable development agenda, particularly the national master plan for technology and innovation development. Plan production requires greater ownership and political control of the processes. By examining 107 national development plans, Chimhowu et al. (2019) revealed that the plans vary in terms of evidence used, consistency's degree between different parts of the plan, plan developing's process, and the extent of funding. The World Bank Group also released a practitioner's guide to innovation policy in developing countries. The guide aims to reduce the large information gap about what works in developing countries to improve the design and implementation of innovation policy instruments and increase their effectiveness. It includes 21 instruments grouped in 10 sets of related instruments: (1) grants for innovation or R&D projects, (2) vouchers for innovation and collaboration, (3) loans and guarantees for innovation, (4) tax incentives for R&D, (5) demand-pull instruments, (6) technology adoption and generation instruments, (7) early-stage support for innovation ventures, (8) inducement instruments, (9) quality infrastructure, including standards, metrology, and testing, (10) clusters and networks for innovation (Cirera et al., 2020).

As Lao PDR (hereafter referred to as Laos) has been struggling to improve its economy since its establishment in 1975, it has made considerable progress in economic development. However, this country does not rapidly shift from agrarian economy to industrial one. It is because more than half of labor force still engages in the agricultural sector, whereas its output is less than a quarter in GDP contribution. There are several challenges of the transformation because Laos does not have any heavy industry. Only mineral ores exploitation, assembly plants, and light manufacturing in Special Economic

Zones (SEZs) that are available. In recent year, Laos has relatively advanced its manufacturing. For example, many provinces initiate to exploit potash mines, which is form as small and medium industries. In addition, there are feasibility studies of iron exploitation project in different parts of the country. However, by setting the Republic of Korea's state of development as a benchmark¹⁴, Laos is still at an early stage of transformation from agrarian to industrial economy.

With these respects, the 9th Five-Year National Socio-Economic Development Plan (2021-2025) has been set out as the fundamental direction for creating a new turning point in socio-economic development. It aims to ensure quality establishment of the political-ideological factors, the economic system and the material and technical basis, which align with the green and sustainable development direction. The 9th NSEDP also aspires to ensure comprehensive preparation to support the country's firm graduation from the Least Developed Country status. There are 5 prior focusing areas, such as agricultural, mining, tourism, logistics sectors, and digital economy.

One of the most important drivers of the 9th NSEDP achievements is a support from Science and Technology Innovation (STI). In order to implement the plan, it is essential to focus on a development of enterprises, particularly micro, small, and medium enterprises (MSMEs). Because MSMEs dominate economic activity and accounts for substantial employment in Laos. According to the latest economic census (2019-2020), there are more than 130,000 enterprises of which 99.3% are MSMEs. As Lao MSMEs struggle with many constraints, such as primarily level of management and organization, limited use of technology, insufficient skilled labor, credits access constraints, and market access limitations. It is necessary to improve MSMEs' capacity to compete, particularly an enabling environment for them to grow.

It is essential to develop the National Master Plan for Technology and Innovation Development and Application in Lao PDR (NMPSTR) and the National Policy on Science and Technology Research (NMPSTR) as a fundamental of MSME innovation. The plans

¹⁴ There are 4 stages of development: (i) 1950s-1960s: Agricultural incentives, (ii) 1960s-1970s: Focused on heavy industry development, (iii) 1980s-1990s: ICT concentration, (iv) 2000s-Present: The age of innovation, AI and big-data.

should focus on survey and study development indicator of relevant sectors, human resource, science and technology research innovation networking and infrastructure development. It is specially to increase competitiveness of trade, capability of research, and industrial development to be more innovative.

The NMPSTR could be a foundation of innovation development for Lao MSMEs. It should contribute to promote R&D and application of STI to all sectors by 2030. The setting goal is to enhance innovative utilization for Lao SMEs and build its own technological innovation in the future. It implies that it should aim to improve human resources in science and technology R&D, create effective policy ecosystem to facilitate MSMEs, and strengthen funding mechanism to support MSMEs' operation and development. In order to achieve these, technology transferring could help Laos to rapid improve innovative SMEs, particularly management, production, value-added, etc. To support this, it is important to ensure internal and external markets for Lao products. The strategy also focuses on implementing priority sectors for SME development in accordance with the green growth. In addition, the government should invest in R&D to increase the efficiency of production and services, create value-added, and upgrade agricultural production to advance the industry that uses modern S&T. The strategy includes a promotion of scientific research and develop new socio-economic developments, particularly R&D institutes, and human resources.

2. Vision

2.1 Background

2.1.1 Changes in International Surrounding MSME Innovation

2.1.1.1 A Brief History of Innovation

The term “innovation” is simply defined as doing something different. It can also be explained as a new idea, product, device, or novelty. However, throughout its history, innovation has been broadly conceived and defined. It concerns any kind of novelty, such as artistic, scientific, technological, organizational, cultural, social, or individual (Godin, 2008). It first appeared in the thirteenth century in law texts as “novation”, which referred as a term for renewing contracts or a new version of a contract. In the particularly entrenched religious atmosphere of the sixteenth and the seventeenth century in Europe, innovation gathered negative associations thrown at people who sought to rewrite religious texts, resulting doctrinal innovation was anathema (Godin, 2010).

With respect to the Industrial Revolution in the nineteenth century, a term innovation became associated with science and industry. It focused more on invention, particularly technical invention. It was because of the rise of consumer culture, increased number of patents, and the support of the government on building labs for R&D. The twentieth century representation of innovation had a more positive value, and it owes to ‘usefulness’ (Godin, 2011). In 1939, Austrian economist Joseph Schumpeter also defined invention as an act of intellectual creativity, which implied to a new product. Meanwhile, he also defined innovation as the adoption of that new thing or bringing a new product to market. To make innovation’s meaning clearer, Godin (2010) emphasized that innovation term between the 1950s-1980s could be defined as a process of theoretical research in labs, applications of research, and commercialized products.

To sum up, regarding Canadian Historian Benoît Godin’s extent researches, innovation is also an incarnation of terms like “imitation” and “invention”. It implies that innovation was developed through three concepts from imitation (literary theory and art theory),

invention (history, sociology, management, and economics of technology), and innovation of creativity.

2.1.1.2 MSME Innovation Policy in Asia

Based on the work of Park & Kim (2020), it offers an essential summary of innovation policies in selected Asian countries, including necessity, definition, instruments, and governance. They categorized these selected Asian countries into four groups in accordance with their income level and market size: 1) high-income countries (Japan and Korea), 2) big-market countries (China and India), 3) middle and upper-middle income countries (Malaysia and Thailand), 4) low-income countries (Bangladesh).

1) High-Income Countries

Japan is the first Asian country to succeed in industrialization and catch up to advanced economies. Japanese innovation policies have been developed since the late 1800s, which referred to the Meiji Restoration period. Its early innovation policies focused on expanding access to international knowledge and accumulating indigenous human capital. From the mid-1960s, the focus shifted to the development of in-house indigenous technologies. The economic and technological gap between Japan and the US narrowed as industrial productivity increased sharply in the late 1980s. In the early 1990s, Japan expended research apparatus and promoted a new R&D system for the country through the Science and Technology Basic Plan (STBP) by continuing strengthening university and industry linkages, commercialization of intellectual assets, increase funding of competitive research grants, and encourage four priority domains (life sciences, IT, environmental sciences, and nanotechnology). With respect to governance of innovation policies, the Comprehensive Council for Science, Technology, and Innovation is in charge of the function of S&T budgeting, linkages among academia, public institutions and industry, and R&D evaluation to support innovative research investments beyond the existing framework – as well as the S&T master plan.

Korea experienced dramatic growth in innovation input-output and economic growth. During the 1960s and 1970s, S&T were needed for technology evolution. The government aspired to enhance S&T capabilities through various instruments. For instance, the

establishment of the Korea Institute of Science and Technology (KIST) and the Ministry of Science and Technology (MoST) in 1966 and 1967, respectively (Figure 2). The government also enforced the S&T Promotion Act to promote technological innovation in the country. Since 1971, the Korea Advanced Institute of Science and Technology (KAIST) has been referred as the first national research university, particularly research-oriented science, and engineering. As the Korean government launched the first national R&D program, promoted private firms' research institutes, and reformed tax incentives to stimulate private firms' R&D investment, Korea became a country where companies led innovation, with private R&D expenditure surpassing public R&D spending since the mid-1980s. Currently, the "Korean Vision 2040 STBP" is a foundation of innovation policies, which includes issues for improved human society and economic and scientific development. It implies that innovation policies that address social problems and S&T issues are recently emerging.

2) Big-Market Countries

China has successfully acquired the market economy in the socialist planned economy, and is expected to overtake the US in terms of GDP by 2030. After the foundation in 1949, the government established the Chinese Academy of Sciences and launched the First Five-year Plan (1953-1957) in respond to the demands of S&T development. It aimed to modernize agriculture, industry, and defense. In the particularly entrenched atmosphere of the Cultural Revolution (1966-1976), the accumulated research base was undermined. From the late 1970s, China experienced a rapid change in planned economy. The government increased corporate productivity by providing incentives while ensuring managerial autonomy and participation to the global value chain. In addition, the Technology Spreading Program was introduced and Innovation Fund for Technology-based SMEs was established in 1990 and 1999, respectively. The government also improved the mid- to long-term S&T development plan since the 2000s. From the 2010s, China transformed from public and government-driven to private-driven, which showed a significant transition to digital convergence and urbanization. With respect to governance, the key ministries in charge of innovation policy are the State Council, Ministry of Science and Technology, and Ministry of Industry and Information Technology.

India is the top ten largest economy in the world, but its GDP per capita remain insufficient to reach the level of the upper-middle-income country. During 1947-1955, India established many public laboratories and research institutions, and built the political foundation for reflecting S&T. However, an attempt to link S&T with national development plan was failed. In 1958, the linking S&T with the national development plan was finally realized. As a result, the overall direction and resource allocation for the S&T policies came from the Five-year Economic and Social Development Plan. With the Science Policy Resolution (SPR) in 1958, India published the Technology Policy Statement (TPS) in 1983, Science and Technology Policy (STP) in 2003, and Science Technology and Innovation Policy (STIP) in 2013, which were foundation of India's innovation policies. With respect to governance, the advisory and coordinator bodies consist of the National Innovation Council (NInC), the Science Advisory Council to the Prime Minister (SAC-PM), the Science and Technology Advisory Committees (STAC), and the Inter-Sectoral STAC (IS-STAC).

3) Middle and Upper-Middle Income Countries

Malaysia did not have a distinct science, technology, and innovation (STI) policy until the mid-1980s. Although the first ministry, the Ministry of Technology, Research, and Local Government (formerly MOSTI) was established in 1973. After that the government launched the New Economic Policy (NEP) to focus on industrialization. The significant STI-driven policy was the First National Science and Technology Policy, which also had a separate chapter for S&T in the Fifth Five-year Plan since 1986. Decades later, with the government assistance, Malaysia successfully transferred R&D capabilities from public institutions to private sectors as other high-income countries. However, the number of patent applications is limited for residents, which innovation activities depend on foreign companies or foreigners. With respect to governance, STI policies overlap among various ministries, such as the Ministry of Higher Education, the Ministry of International Trade and Industry, the Ministry of Energy, Green Technology and Water, the National Innovation Agency of Malaysia, et cetera.

Thailand's import substitution policy shifted to an export-oriented strategy from the early 1970s. In the early stages of industrialization, productivity was achieved through

labor-intensive manufacturing. In order to improve labor's capacity, the government expended school and healthcare facilities as a part of poverty eradication agenda. In the early 1990s, holistic approach for innovation and SME Promotion Master were launched, reflecting the first and the second SME Promotion Master Plan for 2002-2006 and 2007-2011, respectively. In the particularly entrenched of 20 years strategy, Thailand 4.0 was designed to promote and support innovation, creativity, R&D, advanced technology, and green technology. Currently, Thailand is already classified as an upper-middle-income country. With respect to governance, STI policy operations is somewhat fragmented, complex, and partially overlapped.

4) Low-Income Countries

After struggling from the war, *Bangladesh* became independent in 1971. But the government was unsettled for around 20 years, which slow down the initiatives for innovation policies. At the early stages of development, it focused on industrial policies for economic development. In the late 1990s, the first policy envisioned industrial development. Its objectives outlined in the statement had clear sense of direction. In the 2010s, the government formulated the National Science, Technology, and Innovation (NST&I) Policy considering innovation-based development. The ICT Act 2006 (as amended in 2013) was also formulated to achieve Vision 2021 for Digital Bangladesh. Several innovative projects have been executed by the information and communications technology division since then. Despite these, the government attracted and deepened FDI, making the GDP continued to grow and Bangladesh became a lower-middle-income country in 2015. With respect to governance, the Ministry of Posts, Telecommunications, and Information Technology was established in 2014.

To sum up, MSME innovation policy in high-income and big-market countries is integrated and systematic. There is specific ministry or institutes responsible for R&D and S&T that reduce overlap plans and implementation. In addition, research institutes also play a key role in R&D and innovation. At the beginning, SME policy trends begin with an expansion access to international knowledge and accumulating indigenous human capital. Then its focus shifted to the development of in-house indigenous technologies by establishing and improving research institutes. With strong political and

social stability, it also plays an important part for investment and arm's length of business dealing. On the other hand, the governments have offered support to improve technological innovation for the private sector. As the government leaves a room for technological development to the private sector to be more autonomy, they also legislate regulations to enforce and protect rights of businesses and consumers in order to utilize technology sustainably.

2.1.1.3 Current Situation

Currently, the COVID-19 pandemic and its measures to control the outbreak entails economic performance. The global economic turned down to -3.1% in 2020 from 3.0% in 2019 (IMF, 2022a). An attempt to lift mobility restrictions and enhance economic activities encouraged the global economy gradually recovers. It was estimated to grow around 6.1% in 2021. However, a reimposition of mobility restrictions and energy crisis have enforced a weaker position of the global growth in 2022. The growth rate is expected to decline to 3.2% in 2022 (0.4% lower than in the April 2022 World Economic Outlook) (IMF, 2022b). On the other hand, increasing food and energy prices squeeze living standard and revise up global inflation, which worsen monetary and fiscal situations worldwide. As a result, global output would decline to 2.9% in 2023 (IMF, 2022b).

Despite a contraction of global economy, digitalization and innovation play an important role in SME development and survival. SMEs face innovation challenges compared to large enterprises due to limited resources. It is by far worse when it comes to developing countries. They are constrained in their innovative activities due to a lack of financing and a shortage of skilled labor. In contrast, SMEs seem to have a fast-decision-making process when react to technological opportunities compared to larger companies, particularly information technology. For instance, German SMEs use digitalization to advance their position in the market and expand niche strategies. As SMEs want to accelerate their digital transformation processes after hit by the pandemic, many EU SMEs have turned to new technologies to meet customer expectations by empower their employees and ensure a safe workplace. Moreover, by using technology, they find

new ways to communicate and improve digital transaction to leverage their competitiveness.

2.1.2 Changes in Domestic Surrounding MSME Innovation

2.1.2.1 Innovation in Lao Context

The term innovation in Lao context was complex due to language interpretation and its development process from a term of invention. It first appeared as a term “invention” on Section 1, Article 3, item 5 of Law on Intellectual Property (Amended), No.01/NA, officially issued on December 20, 2011. Invention was defined as the technical solution to create new product or process of production to resolve a specific problem. The law also defined a term “utility innovation” as a new innovative work derived through technical improvements, which involve simpler steps than with inventions (Section 1, Article 3, item 7). According to the intellectual property law, it sets a foundation for production development that any product of the human intellect is protected by the law, which enhances production to be more innovative.

In 2013, a term innovation was first appeared on section 1, Article 3, item 1 of the Law on Science and Technology, No.30/NA, officially issued on July 19, 2013. The law defined it as the use of knowledge, capacity, production methods and scientific and technological experience to research, invent, serve [and] develop new forms of quality products and goods which are to be introduced to the society. With respect to R&D, it was noticed on Decree on Science and Technology Development Fund, No.174/GOL, 23 May 2014. R&D was defined as science research and develop existing technology to create new knowledge, application, technology systematically in order to design new products to serve a society.

The term innovation and R&D are also emphasized on related legislations, whereas a term research is generally used at research institutes. For example, the Law on Technology Transfer, No.33/NA, officially issued on November 8, 2017, the Law on Intellectual Property, No.38/NA, officially issued on November 15, 2017, the Law on Advanced Technology, No.07/NA, officially issued on November 16, 2021, and the Law on the Promotion of Micro, Small, and Medium Enterprises (Amended), officially issued

in July 2022. In order to ensure the implementation of R&D, the government also legislated several decrees, such as the Prime Minister's Decree on Organization and Activities of Ministry of Post and Telecommunications, No.22/PM, official issued on January 16, 2017, and the Prime Minister's Decree on Organization and Activities of Ministry of Science and Technology, No.314/PM, officially issued on September 29, 2017.

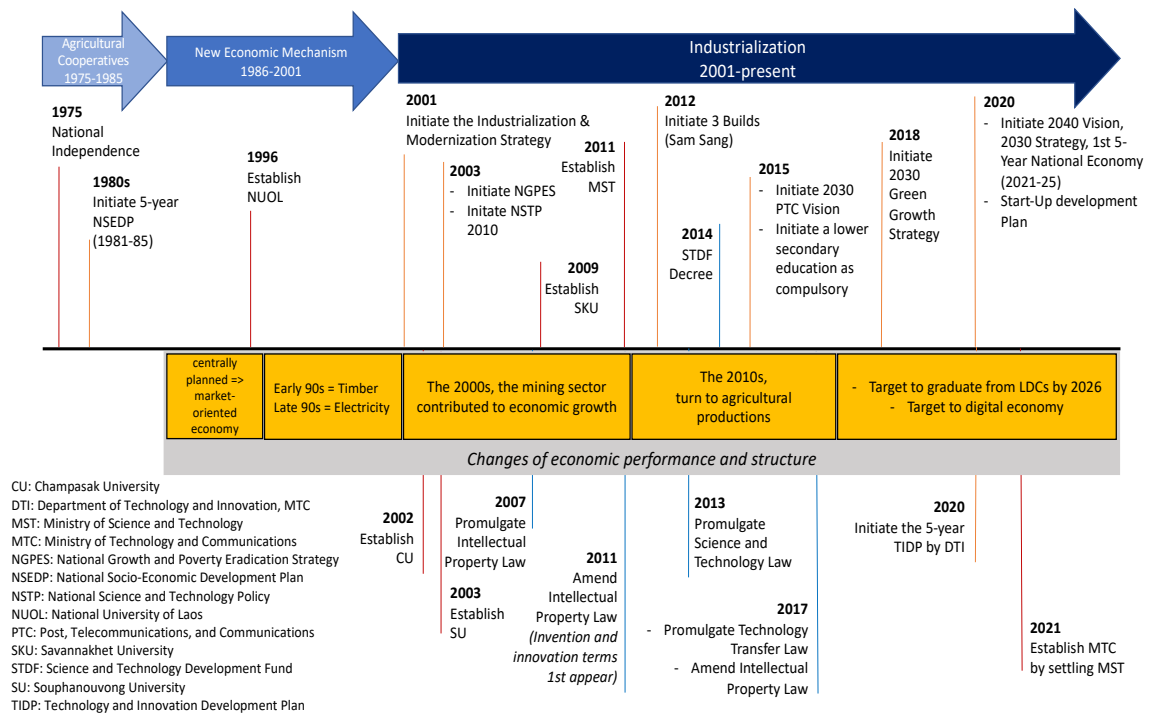
With respect to governance, in 2011, the Lao government established the Ministry of Science and Technology (MST) in regard with the decree number 309/PM, officially issued on September 28, 2011. It indicated the government's interest in fostering R&D. However, Lao PDR sought to reduce the overlap of R&D policies. In February 2021, the government settled the MST and transferred its roles and responsibilities to the Ministry of Education and Sports (MOES), the Ministry of Post and Telecommunications (MPT), and other relevant line ministries. In October 2021, the government established the Ministry of Technology and Communications (MTC) under the Decree on Organization and Activities of Ministry of Technology and Communications, No.625/PM, officially issued on October 22, 2021. It is responsible mostly for MST's responsibilities and roles in terms of governance and management of technology, innovation, telecommunication, internet, post, and cyber security.

2.1.2.2 Innovation Policy in Laos

Since the country graduation in 1975, Lao economy has experienced substantial structural change by transforming from a central-planned economy to a market-oriented driven after a major reform in 1986. The reform showed significant attempts to reduce state control over Lao economy and enhance more openness to international trade and foreign investment. As the Lao government interested to foster the industrial sector for economic growth contribution, they initiated the 20-Year Industrialization and Modernization Strategy in 2001 and the 2030 Green Growth Strategy in 2018. The aims of strategies focus on helping Laos to graduate from the Least Developed Country (LDC) by 2020, but failed. In 2020, the government also initiated the 20-Year Visions (2040), 10-Year Strategy (2030), first 5-Year National Economy (2021-2025), and the Startup Development Plan. These strategies were aimed to strengthen a continuation of

industrialization and modernization direction in order to achieve LDC graduation by 2026. A summary of changes of innovation policies in Laos is displayed in [Figure 5-1].

[Figure 5-1] Changes of Lao PDR's Innovation Policy



Source: Author's compilation

The National Strategy on Economic Digital Development Plan:

- (1) Offering policy and trade ecosystem which aims E-commerce, electronic payments, logistics, productivities, and services to growth sustainably.
- (2) Collaborate with MSMEs, Startup sectors to adapt into modernization services and access digitalization to help their business.
- (3) Completed E-Government to mobilize and manage government services which reduces process to entrepreneurship.
- (4) Turn 50% of government services into online to increase indicators of incomes through digital services and develop Lao ICT to be better.
- (5) Completed one-stop central and local government services.

- (6) Developing one special digital economic zone to attract foreign investment.
- (7) Developing digital infrastructure via internet connecting in the pilot areas of China-Laos railway.
- (8) Developing internet services in rural areas of Laos, at least 80% could access the internet and developing 5G services in specific economic zone.
- (9) The contribution of digital economy could raise up 3% to 5% of GDP¹⁵.

National Master Plan for Start-up:

- (1) Raise awareness of the significant Startup can contribute the national economy.
- (2) Formulating regulations, policies, and laws by ensuring its development on the fair competition, equality and beneficial to all.
- (3) Human resource development and training for young entrepreneurs.
- (4) Development digital infrastructure.
- (5) Creating source of funding and facilitate access to funding resource.
- (6) Creating Startup market, particularly exchanging platforms.
- (7) Strengthening collaboration and cooperation of Startup Ecosystem.

2.1.2.3 SME in Lao Context

The definition for SMEs in various countries is different. In general, SMEs are defined by both qualitative and quantitative definitions regarding economic development and other circumstances. In terms of quantitative definition, the criteria for classifying enterprises into different categories are usually number of employees, value of assets and sales.

With respect to Lao context, definition of SMEs has been revised since 1996, which solely based on number of employees. In 2004, SMEs were also categorized by the value of assets and the annual average turnover. It stated that: “Small and Medium-Sized Enterprises are independent establishments which are legally registered and are running their activities in accordance with the laws of Lao PDR.” However, this definition

¹⁵ Some documents presented at the 3rd Ordinary Session of the IX National Assembly in 2022 showed the contribution of digital economy to GDP could rise to 10% by 2040.

was not updated for over a decade. There were many critical concerns on its application, especially for the terms of total assets and annual turnover, which tend to be under-evaluated. As a result, many commercial banks established their own criteria of SME for the use of loan assessment (Onphanhdala, 2013).

[Table 5-1] Definition of Enterprises in Lao Context

		1996 ^a	2000 ^b	2004 ^c	2017 ^d		
		All Sectors			Manufacturing	Commerce	Services
Employee	Micro	-	-	-	1 - 5	1 - 5	1 - 5
	Small	1-9	1-9	1-19	6 – 50	6 – 50	6 – 50
	Medium	10-29	10-99	20-99	51 – 99	51 – 99	51 – 99
	Large	≥ 30	≥ 100				
Assets	Micro	-	-	-	≤ 100	≤ 150	≤ 200
	Small	-	-	≤ 250	≤ 1,000	≤ 1,000	≤ 1,500
	Medium	-	-	≤ 1,200	≤ 4,000	≤ 4,000	≤ 6,000
	Large	-	-				
Turnovers	Micro	-	-	-	≤ 400	≤ 400	≤ 400
	Small	-	-	≤ 400	≤ 2,000	≤ 3,000	≤ 1,500
	Medium	-	-	≤ 1,000	≤ 4,000	≤ 6,000	≤ 4,000
	Large	-	-				

Source:^a The enterprise survey by the Ministry of Industry and Handicraft (currently, it is Ministry of Industry and Commerce) together with Germany development agency (former GTZ and currently, it is GIZ).

^b The National Statistical Center (currently, it is National Statistics Bureau).

^c The Prime Minister's Decree, No. 42 officially issued in 2004.

^d The Prime Minister's Decree on MSME categorize, No. 25/GOV, officially issued on 16 January 2017.

Regarding the Prime Minister's Decree on Categories of Micro, Small and Medium Enterprises, number 25/GOL, issued on 16 January 2017, SMEs are categorized into 3 different sizes, such as micro, small, and medium enterprises (MSMEs). MSMEs are defined by annual average number of employees, total assets, and turnovers. The decree also categorizes MSMEs by sectors, which includes manufacturing, commercial,

and services sectors as shown in Table 1. In 2022, the administration reconsidered the classification in accordance with internal and external standards.

2.1.2.4 Laws Related to SME Development in Lao PDR

Doing business in Lao PDR had to comply with the new Enterprise Law No. 11/NA, issued date on November 09, 2005, which replaces Business Law No. 03/94/NA, issued date on July 18, 1994. The legislation that directly relevant to SMEs was the Decree on the Promotion and Development of Small and Medium Sized Enterprises No.42/PM (consists of 6 Chapters and a total of 29 Articles), which the government has completed the process of converting it into the Law on Promotion of Small and Medium Enterprises, No.011/NA, issued date on December 21, 2011. It was amended and promulgated the latest version in 2022.

The latest draft of SME law was just presented at the 3rd Ordinary Session of the IX National Assembly on June 29, 2022. It was amended and promulgated in July 2022 as the *Law on the Promotion of Micro, Small and Medium Enterprises*. The law consists of 11 Parts, 5 Chapters, and 72 Articles. It determines principles, regulations, and measures to implement, manage, and monitor MSMEs in order to align with current economic performance and ease possible constraints of MSME development. The law also emphasizes input promotion to reduce business costs for industrial and agricultural MSMEs. It aims to enhance invention and innovation adaptation to lift quality of MSMEs and start-up. Moreover, the law defines directions and policies to implement the SME Promotion and Development Fund, which ensure a consistency with the Law on State Budget (amended version). The revised context could help to solve various constraints of previous SME fund implementation, which attempts to contribute to job creation, income generation, living standards rising, and to lay the foundations for gradual industrialization and modernization.

Based on the law (2022 version), it emphasized directions and policies for MSME promotion and development: (1) creating an enabling regulatory and administrative environment; (2) accessing to financial sources; (3) customs, tax, and accounting; (4) building and developing entrepreneurs; (5) business Development Services; (6) MSME

and large enterprises cooperation; (7) increasing productivity and quality; (8) accessing and expanding markets; (9) business group/partnership; (10) business place development; (11) using advanced and eco-friendly technology; (12) patent and intellectual property right; (13) information providing and accessing; (14) enhancing inputs; (15) enhancing invention and innovative adaptation; (16) Start-Up promotion; (17) accessing to procurement; (18) other related policies.

In order to implement the Action Plan for SME promotion and development, all measures should be planned and implemented in the form of projects and programs. Those should be aligned with socio-economic development plans and the laws. Based on Part VI of the law, it emphasizes MSME promotion organization as follows¹⁶:

- (1) MSME Promotion Committee
- (2) MSME Promotion Office
- (3) MSME Promotion for the Private Sector

2.1.3 Current Situation

Over the past decade, Laos was one of the fastest-growing economies in ASEAN. The GDP growth increased steadily since 1990, even its trends turned down in recent years, but still ranging from around 6.3% to 7.2% on average regarding the mining sector. As the pandemic was reported, Lao economic performance sharply dropped to 3.3% and 2.3% in 2020 and 2021, respectively (NIER, 2021a, 2021b); World Bank, 2022a). However, the projection by international agencies – World Bank (0.2% and 2.5%), ADB (-0.5% and 2.3%), and World Bank (-1.8% and 2.5%) (ADB, 2022; IMF, 2022b; World Bank, 2021; 2022a). In 2022, Lao economy was expected to place in a stronger position due to natural resources exports between 3.2% to 3.8% (ADB, 2022; IMF, 2022b; World Bank, 2021; 2022a). However, increased prices suppress living standards and employment, which worsen fiscal and monetary situations, particularly high inflation, local currency depreciation, and defaults.

¹⁶ More details can see in Annex 1

Since the COVID-19 outbreak was reported, it forced people to change household consumption habits, mainly food consumption. Affordable households preferred cooking and eating at home by using grocery and food delivery services or advance ordering for takeaway. Delivery and takeout possibilities increased significantly to serve this change. Food delivery vendors expand their businesses, especially in large cities. In addition, online transaction becomes more popular, but limited among food, groceries, and small parcel delivery. Buyers and sellers use electronic methods for payment via mobile phone. For example, Sengkhambhouthavong et al. (2021) revealed that more than half of goods and services payment at micro and small shops in large cities is made via mobile application. Their empirical results also confirmed that businesses relevant to restaurants, entertainment venues, accommodations, retailers, and wholesalers are more likely to adopt e-payment, which is statistically significant at a 10% level. Moreover, several platforms that perform electronic transactions have been launched with an increasing demand for online services. However, it is limited in large cities and capable enterprises whom can access such resources.

2.2 Future Vision 2030

To promote R&D and application of STI into all sectors. The setting goal is to enhance innovative utilization for Lao SMEs and build its own technological innovation in the future. It implies that it should aim to improve human resources in science and technology R&D, create effective policy ecosystem to facilitate MSMEs, and strengthen funding mechanism to support MSMEs' operation and development. Generally, we set our goals to improve properly operation of MSMEs. It aims to strengthen capacity to compete of Lao MSMEs in terms of novelty or originality of technology and/or business model, marketability and growth potential of products or services, and financial stability.

2.3 Roles of STI for MSME Innovation

2.3.1 R&D Performance

Although Laos has made considerable progress in lifting access to education in response to increasing social demand and setting a lower secondary education for compulsory

education regarding the education law 2015. It further needs to improve the quality and relevance of the education system, R&D, and institutions. Asian Development Bank (ADB), in its recent analysis of the Lao economy and human resources, identified an increasing lack of educated and skilled human resources as a critical constraint to inclusive growth in Laos's transforming economy and a constraint on private sector growth and economic diversification (ADB, 2011, & ADB, 2012). In addition, human development index shows that Laos ranks at the medium level of human development (HDI, 2021).

The Global Innovation Index (GII) ranks world economies according to their innovation capabilities. The GI consists of seven pillars: (1) institutes, (2) human capital and research, (3) infrastructure, (4) market sophistication, (5) business sophistication, (knowledge and technology outputs), (7) creative outputs. To put Laos' benchmarking against other lower middle-income economies, Laos' innovation performance is weaker in most GI pillars, excepting business sophistication pillar with a relatively difference. Regarding to these indicators, Laos ranks the 117th among 132 economies in 2021, with only 123 innovation inputs and 112 innovation outputs (WIPO, 2021).

[Table 5-2] GI 2021 Rankings and Innovation Performance at Different Income Levels

Performance	High-Income	Upper Middle-Income	Lower Middle-Income
Above expectations for level of development	Korea (5 th)	China (12 th)	Vietnam (44 th)
	Singapore (8 th)	Thailand (43 rd)	India (46 th)
	Japan (13 th)		
In line with level of development		Malaysia (36 th)	
Other economies			Cambodia (109 th)
			Bangladesh (116 th)
			Laos (117 th)

Source: WIPO (2021)

As shown in [Table 5-2], Laos has a weak innovation performance, which is categorized in lower middle-income group as well as Cambodia and Bangladesh. On the other hand, among top 10 of high-income economies, Korea and Singapore rank the 5th and the 8th in the best innovation economies, whereas Korea and Singapore rank the 1st place in human capital and research and institutes pillars, respectively.

2.3.2 Startup Innovative ICT products

The national master plan for startup was initiated and the startup promotion was recently emphasized in the promulgated MSME law in 2020 and 2022, respectively. Before this initiation, the startup has been operated under the 2011 SME promotion law. There are relatively numbers of startup in Laos, particularly foreign own and joint-venture companies. This kind of company tend to adapts technology for operation, as a result, startups at the early growth stage are those online platform businesses, such as Delivery Hero (known as Food Panda delivery), Loca (taxi services), M-Money (electronic wallet by the Lao Telecom company), etc. Business Start-ups in Laos seem to focus on online platform development for e-payment, logistics services, digital currency exploitation. For example, the Loca provides electric taxi services with Loca Pay that accept international cards such as Visa for payment. Despite this, with respect to business ventures, foreign developers plan to build the high technology park in Vientiane Capital. It could contribute to digital economic development by bringing new technologies and accelerate start-ups in Laos.

Whereas, manufacturing and production businesses are barely developed as Start-ups, particularly agricultural production and other primarily manufacturing, etc. However, agricultural producers aim to produce for exporting, particularly to China. It is because of Lao producers could utilize the Laos-China Railway and gain more from agricultural export preferences that granted by the Chinese government.

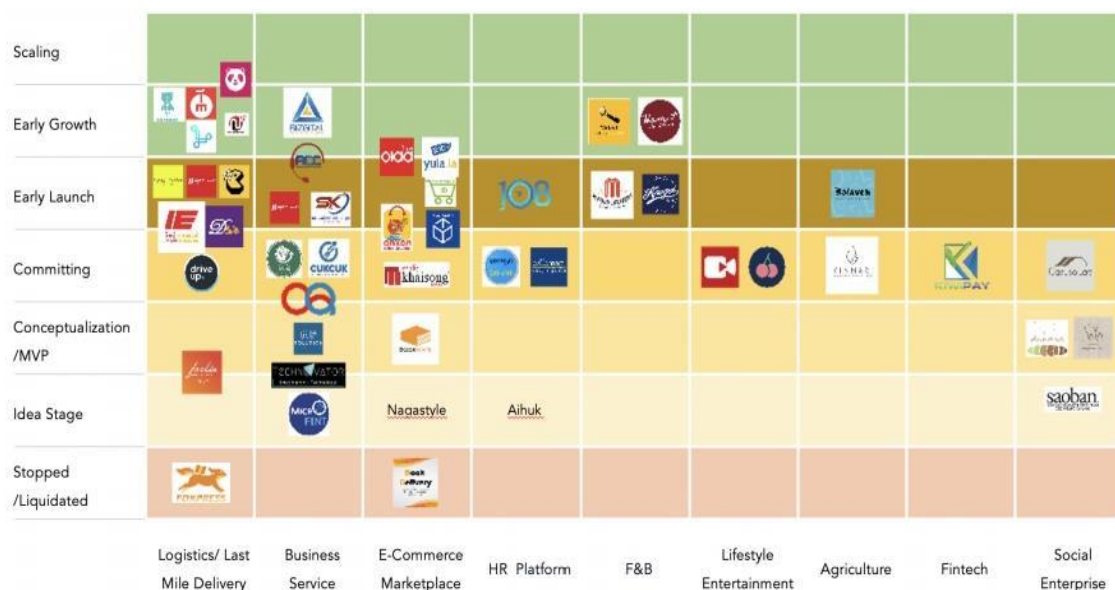
[Table 5-3] Startup Innovative ICT Products Evaluation

Category	Qualitative Description
Scaling	The solution has proven to be able to acquire clients at an optimistic rate. A level of “market-fit” in a sense that the product/solution can be seen as addressing clients’ expectations with a clear value proposition. The founders are usually in search of more resources and assets to grow their company.
Early Growth	The solution is gaining tractions and we start to see adoption from customers. The product is, at this stage, not growing exponentially but market validation is becoming more obvious.
Early Launch	The solution is launched to the market and made available to customers. It is yet to see positive business metrics and established customer adoption.
Committing	The founders have committed time and assets to the execution of the startup.
Conceptualization /MVP	The solution is brought alive with a pilot, a mock-up, a prototype that is usually ready to be launched with basic features.
Idea Stage	The solution is not yet materialized by any pilot or concrete products. It usually takes the shape of a “pitch deck” or concept note.
Stopped /Liquidated	The company has ceased to operate or has left Laos.

Source: DOSMEP¹⁷, MOIC

¹⁷ Currently, the Department of SME Promotion becomes the MSME Promotion Office in regard to the Law on MSME Promotion (amended), issued in July 2022.

[Figure 5-2] Startup Innovative ICT products



Source: DOSMEP, MOIC

2.3.3 Governance

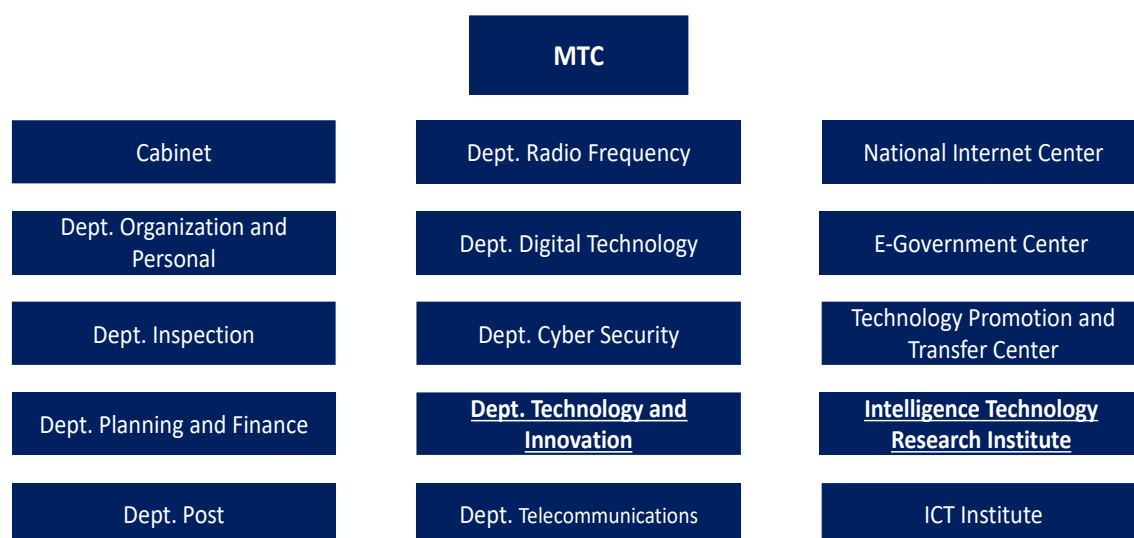
2.3.3.1 Science, Technology and Innovation

The MTC is the line ministry primarily responsible for governance and management of technology, innovation, telecommunication, internet, post, and cyber security of Laos as stipulated in the Prime Minister's Decree No.625/PM, officially issued on October 22, 2021. It is also responsible mostly for MST's responsibilities and roles in terms of governance and management of technology and innovation, as the government sought to reduce the overlap of R&D and S&T governance. MST's roles also went to MOES and MPT (as mentioned earlier in Section III, Subsection 1). The MTC consists of 15 departments, centers, and institutes, as shown in [Figure 5-3].

Department of Technology and Innovation (DTI) plays a major role in S&T for innovation. It acts as a secretariat to the MTC, which is responsible for setting up national master plans, policies, and strategies for R&D, and formulating legislation on the management of technology and innovation. The department is in charge of promoting research on

S&T and managing the interaction between science sectors and related organizations at the central and the local government to implement and certify S&T research projects. The DTI comprises with six divisions: (1) Administration Division, (2) Policy and Strategy Division, (3) Technology Management Division, (4) Innovation Division, (5) Advanced Technology Division, (6) Satellite Imagery Information Technology Division.

[Figure 5-3] Governance Structure of Ministry of Technology and Communications



Source: The Prime Minister's Decree on Organization and Activities of Ministry of Technology and Communications, No.625/PM, officially issued on October 22, 2021.

To strengthen international cooperation in terms of knowledge and technology transferring, the Department of Planning and Cooperation (DPC) acts as a secretariat to the MTC, serving as an interconnector between the MTC and other ministries. For instance, it coordinates directly with the Ministry of Planning and Investment (MPI) for using government budget and ODA in the science, technology, and innovation (STI) sector; whereas it coordinates with the Ministry of Foreign Affairs (MOFA) for bilateral, multilateral, and international organization cooperation.

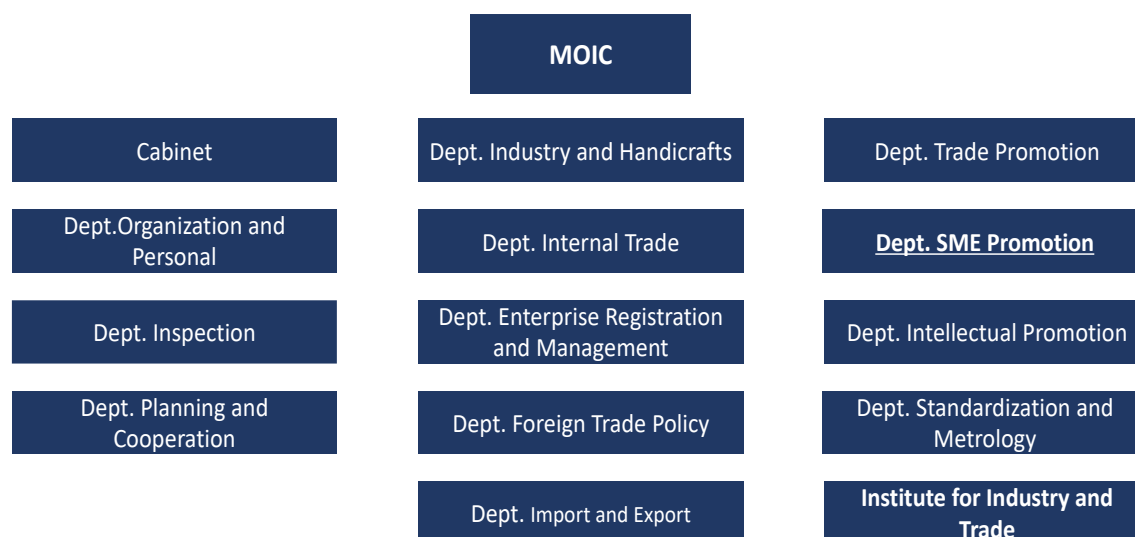
With respect to coordination, the Cabinet of Academy of Science and Technology (CAST) plays a role in functioning and coordinating with the Sectoral Scientific Committee of Ministries, Equity and Establishment of the MTC. It is also responsible for preparing and

presenting related proposal in consideration of scientific and technological researches nationwide. Currently, it was settled to work under a supervision of the MOES.

2.3.3.2 SME Promotion and Development

The MOIC was reformed and restructured since its establishment. It was recently legislated in 2021 as stipulated in the Prime Minister's Decree on Organization and Activities of Ministry of Industry and Commerce, No. 604/PM, officially issued on October 15, 2021. The MOIC acts as a secretariat to the government in responding to governance and management at the macro level. Its responsibilities cover various fields, such as industry, handicrafts, industrial zones, chemical management, internal and international trade, and electronic commerce. The MOIC is also responsible for protecting consumers, adjusting prices, managing the market, and enhancing businesses' competition. Enterprise registration, promotion, intellectual property, measures, trade facilitation, and regional and international integration are responsibilities of the MOIC. Its governance structure consists of 14 departments, cabinet, and institutes, as shown in [Figure 5-4].

[Figure 5-4] Governance Structure of Ministry of Industry and Trade



Source: The Prime Minister's Decree on Organization and Activities of Ministry of Industry and Commerce, No. 604/PM, officially issued on October 15, 2021.

Department of Small and Medium Enterprise Promotion (DOSMEP) was established under the Prime Minister's Decree No. 42/PM, 20 April 2004, entitled "Decree on the Promotion and Development of Small and Medium Sized Enterprises". Its main objective is the promotion of sustainable development of Lao SMEs. Currently, based on the Law on the Promotion of Micro, Small, and Medium Enterprises (amended), issued in July 2022, DOSMEP changed to the name 'MSME Promotion Office' and its responsibilities and duties remain the same (more details can be seen in Annex 1).

It comprises with six divisions and fund: (1) General Affairs Division, (2) Entrepreneur Development Division, (3) SME Support Division, (4) Productivity and Quality Division, (5) SME Planning and Statistics Division, (6) SME Promotion Fund. The SME Promotion Fund plays an important role in financing and providing supports to SME promotion and development programs.

2.3.4 Stakeholders

R&D stakeholders consist of the public and the private sectors. The public sector includes line ministries, local government, and research institutions. The MOES and the MTC are two primarily pillars of R&D for innovation. Regarding MOES's roles, higher education institutes are deepening their responsibilities for R&D, particularly the NUOL. In addition, the MOES established the Research Institute for Technology and Innovation, in the hope to strengthen R&D and STI. Meanwhile MTC offers an assistance for connecting to science sectors and related organization within central and local to implement and certify S&T research. Both ministries also work together with other line ministries in order to enhance R&D and STI development in Laos. Each line ministry has its independent education and research institutes, as shown in [Table 5-4].

There is a relatively small number of the private sector as R&D stakeholders. However, the LNCCI as an association takes a lead role in the private sector's stakeholder. Enterprises that have their own R&D section probably are Phu Bia Mining, DPS Exploration and Mining Company Limited, HSMM Group of Companies, ALS, etc. Most of them are mining exploitation companies. On the other hand, other enterprises usually utilize research outputs from the public sector or their business partners rather than

operate their original labs or R&D centers – outsourcing bases, such as CP Group. For example, agricultural enterprises utilize research results from the NAFRI for agricultural production, etc.

International stakeholders providing supports to innovation and technology development, particularly MSME innovation, are Korea, China, and Japan. More details can be seen in Annex 2.

[Table 5-4] R&D Stakeholders

	Line Ministries	Education and Research Institutes/ Associations	Relevant Degree
Central	MOES	<i>Public and private higher education institutes</i> <i>Public: NUOL, CU, SU, SKU</i> <i>Vocational Education Development Institute (VEDI)</i> Research Institute Research Institute for Technology and Innovation Research Institute for Educational Sciences	RD1: R&D pillars
	MTC	Intelligence Technology Research Institute (ITRI)	
	MOIC	Institute for Industry and Trade (IIT)	RD2: Major research institutes
	MPI	Institute for Research Development (IDR)	
	MAF	National Agriculture and Forestry Research Institute (NAFRI)	
	MEM	Research Institute for Energy and Mines (RIEM)	RD3: Relevant research institutes
	MONRE	Natural Resource and Environment Research Institute (NRERI)	
	MPWT	Public Works and Transport Research Institute (PTRI)	
	BOL	<i>Banking Institute</i>	RD4: Education institutes that are also provide research works
	MOF	<i>Academy of Finance and Accounting (AFA)</i>	
	MOFA	<i>Institute of Foreign Affairs</i>	
	MOJ	<i>National Judicial Institute</i>	
	MOLSW	<i>Lao-Korea Skills Development Institute</i>	

	Line Ministries	Education and Research Institutes/ Associations	Relevant Degree
	MPT	<i>Institute of Posts and Telecommunications (IPT)</i>	
	MICT	<i>Institute of Mass Media, Culture and Tourism (IMCT)</i>	RD5: Relevant education institutes that are also provide research works in media, healthcare, and administrative fields
	MOH	<i>Lao Tropical and Public Health Institute (Lao TPHI)</i> <i>Institute of Traditional Medicine</i> <i>Institut Pasteur du Laos (IPL)</i> ¹⁸	
	MOHA	<i>Public Administration Research and Training Institute (PARTI)</i>	
Local		Research centers at the provincial level (Not all provinces, only major provinces)	
Association	LNCCI	Lao National Chamber of Commerce and Industry	

Source: The Prime Ministry's Decree No. 22/PM, issued date on January 16, 2017 (MPT)

The PM's Decree No. 96/PM, issued date on March 9, 2017 (MOH)

The PM's Decree No. 573/PM, issued date on September 20, 2021 (MONRE)

The PM's Decree No. 575/PM, issued date on September 28, 2021 (MOHA)

The PM's Decree No. 600/PM, issued date on October 14, 2021 (MOF)

The PM's Decree No. 601/PM, issued date on October 14, 2021 (MOJ)

The PM's Decree No. 602/PM, issued date on October 14, 2021 (MOLSW)

The PM's Decree No. 603/PM, issued date on October 15, 2021 (MAF)

The PM's Decree No. 604/PM, issued date on October 15, 2021 (MOIC)

The PM's Decree No. 623/PM, issued date on October 22, 2021 (MOF)

The PM's Decree No. 624/PM, issued date on October 22, 2021 (MEM)

The PM's Decree No. 625/PM, issued date on October 22, 2021 (MTC)

The PM's Decree No. 633/PM, issued date on November 18, 2021 (MOES)

The PM's Decree No. 636/PM, issued date on November 26, 2021 (MPI)

The PM's Decree No. 677/PM, issued date on December 21, 2021 (MPWT)

The PM's Decree No. 678/PM, issued date on December 21, 2021 (MICT)

Remarks: - R&D stakeholders at the central level are arranged by relevant degree to R&D and S&T for MSME innovation from the most relevant as RD1 to the least relevant as RD5.

- Research institutes that are also responsible for human resource development and providing training are shown in *Italic*.

¹⁸ Translation to English as *Pasteur Institute of Laos*, which is a long term joint between Lao Ministry of Health and Institut Pasteur Paris for infectious diseases research and training, etc.

BOL	Bank of Lao PDR	MOIC	Ministry of Industry and Commerce
MAF	Ministry of Finance	MOJ	Ministry of Justice
MEM	Ministry of Energy and Mines	MOLSW	Ministry of Labour and Social Welfare
MICT	Ministry of Information, Culture and Tourism	MONRE	Ministry of Natural Resource and Environment
MOES	Ministry of Education and Sports	MPI	Ministry of Planning and Investment
MOF	Ministry of Finance	MPT	Ministry of Post and Telecommunications
MOFA	Ministry of Foreign Affairs	MPWT	Ministry of Public Works and Transport
MOH	Ministry of Health	MTC	Ministry of Technology and Communications
MOHA	Ministry of Home Affairs		

2.3.5 Resources

2.3.5.1 Human Resources and Qualifications

The Ministry of Home Affairs (MOHA) reported that there are 176,151 civil servants in 2021. Teacher staffs' coverage reach more than 40% of total government staffs. Since teacher staffs have the largest proportion of government staffs, we could consider teacher staffs at a university level as a key proxy on undertake R&D for innovation.

Four public universities confront with a degree level program and continuing programs for general and higher education. There are National University of Laos (NUOL), Champasak University (CU), Souphanouvong University (SU), and Savannakhet University (SKU) since 1996, 2002, 2003, and 2009, respectively. In 2007, the University of Health Sciences (UHS) was formed following the separation from a faculty of NUOL and the Ministry of Health (MOH) administers the UHS. All 5 public universities offer bachelor-degree programs, and only NUOL and UHS offer master degree programs and PhD programs, while SKU offers master degree programs since 2019.

[Table 5-5] shows the qualifications of faculty and staff members in two selected universities in accordance to the latest data availability. It is obvious that NUOL's most members hold master's degree, which accounts for approximately 54.2% (increases from 44.8% in 2020). With respect to higher educational level, PhD holders of NUOL and SKU account for approximately 10.5% and 3.3%, respectively. It implies that teaching staff's qualifications in the other universities remain low and insufficient. These higher education institutes tend to have a lower qualification of academic staff comparing to the MOES strategy that requires 30% of PhD staffs (actual is 9.4%), 60% of master degree staffs (actual is 52.3%), and 10% of bachelor degree staffs (actual is 38.0%). Moreover, there is an insufficient academic staff who has an advanced academic title. For instance, at the NUOL, only 5 lecturers hold *professor* title, and 87 lecturers hold *associate professor* title.

One of the key reasons for the low quality of education and R&D in Laos is the lack of qualified academic staff. Due to low salaries paid to academic staffs, education institutes have difficulty retaining skilled and experienced academic personnel. It is leaving institutes with too many young, underqualified, and inexperienced academic staff. Lack of academic staff with advanced degrees is a key constraint on undertaking R&D in innovation (Onphanhdala & Philavong, 2022).

[Table 5-5] Qualifications of Faculty and Staff Members in Selected Public Universities

	PhD	Master	Bachelor	Others	Total
NUOL	186 (41)	956 (408)	572 (343)	53* (33)	1,763 (825)
SKU	11 (2)	145 (50)	176 (93)		332 (145)
Total	197 (42)	1,101 (458)	797 (470)		2,095 (970)

Source: NUOL's Report, issued in July 2022 and SKU's information in educational year 2020/21.

Remarks: * 4 of them holds a post-graduate and none of them are females.
Females in parentheses.

2.3.5.2 Financial Resources

1) R&D Fund

The impact of R&D was mentioned earlier in the introduction part, which is a key factor to improve the national economy. According to its importance and necessity, governments around the world are broadening supports into R&D and STI. In 2020, public expenditure on R&D accounts for approximately 2.63% of global GDP. Whereas Korea and Japan's R&D spendings are greater than the global level, accounting for approximately 4.81% and 3.26%, respectively. For Laos, public expenditure on R&D accounts for approximately 0.04% in 2002, which is the lowest in Southeast Asian nations (Figure 5-9).

Since the government interested in fostering S&T, the CAST was established together with the setting of the Science and Technology Development Fund (STDF) that established since 2014 under the MTC (former MST). Based on a regulation, the amended Decision on STDF Implementation that stipulates in 2022, it also categorized 3 funding types for S&T development: (1) ODA to support STI research projects, STI activities, research subsidiary, and rewards; (2) joint donation between STDF and relevant sectors; (3) loans from STDF referred to MOF and Fund Board's conditions.

However, based on actual implementation, the Survey on Capability of Science and Technology Organization on Science and Technology Implementation in Lao PDR in 2020 supported by the STDF, there are 4 funding sources: (1) government budget, (2) fees, (3) external sources, (4) ODA. The government budget source for R&D and S&T refers to annual 1% of total public expenditure. The spending is limited due to the fiscal deficit and budget constraints. As a result, at the beginning, the budget only went to natural science research field. Currently, it splits to both natural and social science research fields.

Findings in Table 6 show that a major funding source for S&T is from the government budget, which accounts for around 69% of total sources. The ODA also plays an important role in supporting S&T implementation. When categorizing sources by implementation, higher education institutes received the largest proportion of funding. They obtained funds from the central budget. Meanwhile, R&D implementation

(projects) received most supports from the ODA, which accounted for approximately 55% of total funding for R&D implementation. On the other hand, MTC implantation share a similarly proportion of sources between the central budget (45%) and ODA (44%).

[Table 5-6] Structure of Fund Sources for S&T During 2015-2019

	Source	Total	By Implementation		
			R&D	Higher Education Institutes*	MTC (Former MST)
		100%	8%	59%	33%
1	Government Budget	69%	38%	83%	52%
1.1	Central	60%	37%	72%	45%
1.2	Local	9%	1%	11%	7%
2.	Fees	5%	6%	7%	2%
3.	External Sources	3%	0%	4%	2%
3.1	S&T Activities and Technology Transfers	0%	0%	0%	0%
3.2	Production, Business, and Services	3%	0%	3%	2%
4.	ODA	23%	55%	7%	44%

Source: The Survey on Capability of Science and Technology Organization on Science and Technology Implementation in Lao PDR in 2020. STDF, MTC, Vientiane Capital, 2021.

Remarks: * at university level

With respect to expenditure on STDF decision, the fund can be categorized into 3 types: (1) R&D and STI promotion fund for developing products and services, by using 50% of STDF's annual budget, (2) S&T research fund shall use 40% of STDF's annual budget, (3) fund for policy development and capacity building of STI shall use 10% of STDF's annual budget. For administrative expenditure shall not exceed 5% of total values of research project.

[Table 5-7] also displays criteria of research project by size, budget, partner/sector, and implementation duration.

[Table 5-7] Size, Budget, and Implementation Period of Research Project

	Research Project	Budget (Million LAK)	Relevant Sector	Period
1	Small	< 100	≥ 1 sector	At least 1 year
2	Medium	100 – 500	At least 1 research partner/sector	At least 2 years
3	Large	500 – 1,500	At least 2 research partners/sector	At least 2-3 years
4	Govt. special research project for emergency issues: disasters, pandemic, etc.	Referred to actual activities		

Source: the amended Decision on STDF Implementation that stipulates in 2022.

2) SME Promotion Fund

The government authorized to establish the SME Promotion and Development Fund to provide support to SME promotion and development programs under the Prime Minister's Decree No. 123/PM, officially issued on March 3, 2010. The SME Promotion and Development Fund shall be managed by the National SME Promotion and Development Office (former the Department of SME Promotion and currently the MSME Promotion Office, Ministry of Industry and Commerce). Separate regulations for the management of the SME Promotion and Development Fund will be issued. The latest regulation in regarding to the implementation of SME development fund refers to the Government Decree on SME Promotion Fund, No. 299/GOL, officially issued on September 4, 2019, which amended from the Decree No. 123/PM promulgated in 2010. Currently, the SME Promotion Fund was emphasized on the latest MSME law (2022 version).

[Table 5-8] displays significant activities of SME promotion fund, whereas budgetary of the fund is shown in [Table 5-9].

[Table 5-8] Milestones of the SME Promotion and Development Fund

Year	Contents	Remarks	Amount
2010	Decree on SME Promotion and Development Fund, No. 123/PM, March 03, 2010	The SME Promotion and Development Fund (SPDF) establishment	-
2011	SME Development Plan 2011-2015	Revised conditions to access to the fund	
	Budget from the government to SPDF	Funding	32 billion kip
2012	Lao Development Bank provides credits for SMEs		
2014	Loans and ODA from World Bank		111.68 billion kip
2016	SME Development Plan 2016-2020	Credit access promotion policy	
	Commercial banks provide credits for SMEs		
2019	Decree on SME Promotion Fund, No. 123/PM, March 03, 2010	Amended decree No. 123/PM	
2020	Budget from the government to SPDF	Funding	200 billion kip

Source: 2020 Annual Report, SME Promotion Fund, MOIC

[Table 5-9] Income and Expenditure of SME Promotion Fund

Income Sources (Part V, Article 36)	Expenditure (Part V, Article 39)
- The government; - Grants; - Internal or external loans; - Voluntary donations and contributions by individuals and organizations; - Fund revenue; - Other sources that are not violent against the law.	- Credits; - Fund for academic projects; - Capital and interest payment; - Administrative expenditure.

2.3.6 MSME Performance

There are three economic censuses implementing since 2006 by the Lao Statistics Bureau (LSB), Ministry of Planning and Investment. The 2nd and 3rd Economic Censuses were conducted in 2013 and 2019, respectively. [Table 5-10] displays number of enterprises by employee. Regarding the census, SMEs or hereafter referred as MSMEs in the latest context (legislation) dominate economic activity and accounts for substantial employment in Laos, covering more than 99% of total enterprises. It implies that a competitiveness of Lao SME is essential for economic development.

[Table 5-10] Number of Enterprises by Employee

Number of Employee		1 st Census 2006	2 nd Census 2013		3 rd Census 2019
< 5 persons	Small Enterprise	118,592 (93.4%)	107,270 (85.9%)	Micro Enterprise	(90.6%)
5-9 persons		5,268 (4.2%)	11,520 (9.2%)	Small Enterprise	(9.0%)
10-99 persons	Medium Enterprise	2,835 (2.2%)	5,777 (4.6%)	10-50 is Small Enterprise	(0.2%)
				51-99 is Medium Enterprise	
> 100 persons	Large Enterprise	198 (0.2%)	306 (0.3%)	Large Enterprise	(0.2%)
Total		126,913	124,873		133,977

Source: The Economic Census by Lao Statistics Bureau, Ministry of Planning and Investment.

The 3rd national economic census (2019-2020) reported that there are 133,997 enterprises in Laos, based on size, of which a proportion of micro, small, and medium enterprises are 94%, 5%, and 0.7%, respectively. Since a survey result is categorized as MSME size, we can consider it as the nation-wide MSME in Laos. Regarding the establishment year, most enterprises started their businesses in the 2010s and the 2000s, respectively, as shown in [Table 5-11]. Among these enterprises, 914 firms are inactive, which account for approximately 0.7% of total enterprises.

[Table 5-11] Number of Enterprises by Period in 2019

No.	Year	Numbers of firm	
		Numbers	Percentage
1	1970-1979	228	0.2
2	1980-1989	1,042	0.8
3	1990-1999	5,757	4.3
4	2000-2009	26,873	20.1
5	2010-2019	100,097	74.7
	Total	133,997	100

Source: The 3rd National Economic Census (2019-2020) conducted by Lao Statistics Bureau, Ministry of Planning and Investment.

According to three economic censuses, the largest proportion of enterprise by sector is wholesales and retails trade. It accounts for around 64.5%, 62.8%, and 63.8% regarding the first, the second, and the third census, respectively. As shown in [Table 5-12], other major subsectors include manufacturing, hotels and restaurants, and other services. Enterprises that locate in urban areas account for approximately 59.5%, while firms locate in rural with road areas and without road areas account for approximately 38.8% and 1.7%, respectively. [Table 5-13] displays numbers of non-agricultural firms categorized by province (as agricultural enterprises cover only 0.7%, we could consider a survey result as numbers of non-agricultural enterprises). Most enterprises locate in large cities, such as Vientiane capital, Savannakhet and Champasak.

[Table 5-12] Proportion of Enterprises by Sector

Sector	1 st Census 2006	2 nd Census 2013	3 rd Census 2019
No. of Enterprises	126,913	124,873	133,997
Wholesale and retail trade; repair of motor vehicles and motorcycles	64.5	62.8	63.8
Manufacturing	19.0	12.5	14.1
Accommodation and food service activities	2.8	11.7	10.8
Other services	2.7	3.9	2.4
Transportation and storage	2.9	2.8	2.4
Real estate	0.5	0.6	1.5
Construction	0.5	0.5	0.8
Other	7.1	5.2	4.2

Source: The Economic Census by Lao Statistics Bureau, Ministry of Planning and Investment.

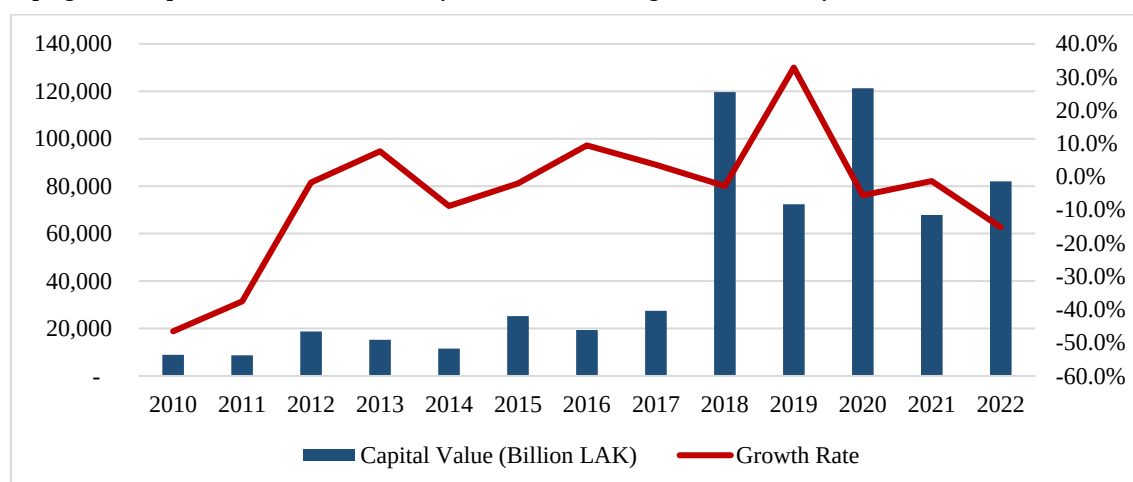
[Table 5-13] Number of Enterprises by Province in 2019

No.	Provinces	Numbers of firm	
		Numbers	Percentage
1	Vientiane capital	29,874	22.3
2	Phongsaly	1,699	1.3
3	Luang Namtha	3,802	2.8
4	Oudomxay	6,254	4.7
5	Bokeo	2,825	2.1
6	Luang Prabang	9,450	7.1
7	Houaphanh	3,775	2.8
8	Xayabouly	6,605	4.9
9	Xiengkhouang	5,903	4.4
10	Vientiane province	12,803	9.6
11	Bolikhamxay	5,765	4.3
12	Khammouan	4,405	3.3
13	Savannakhet	15,411	11.4

No.	Provinces	Numbers of firm	
		Numbers	Percentage
14	Salavan	4,406	3.3
15	Sekong	1,683	1.3
16	Champasak	15,791	11.8
17	Attapeu	2,726	2.0
18	Xaysomboun	1,020	0.8
	Total	133,997	100.0

Source: The 3rd National Economic Census (2019-2020) conducted by Lao Statistics Bureau, Ministry of Planning and Investment.

[Figure 5-5] Growth Rate and Capital Value of Registered Enterprises from 2010 – 2022



Source: National Enterprise Database, Enterprise Registration and Management Department, Ministry of Industry and Commerce. Data as of August 2022.

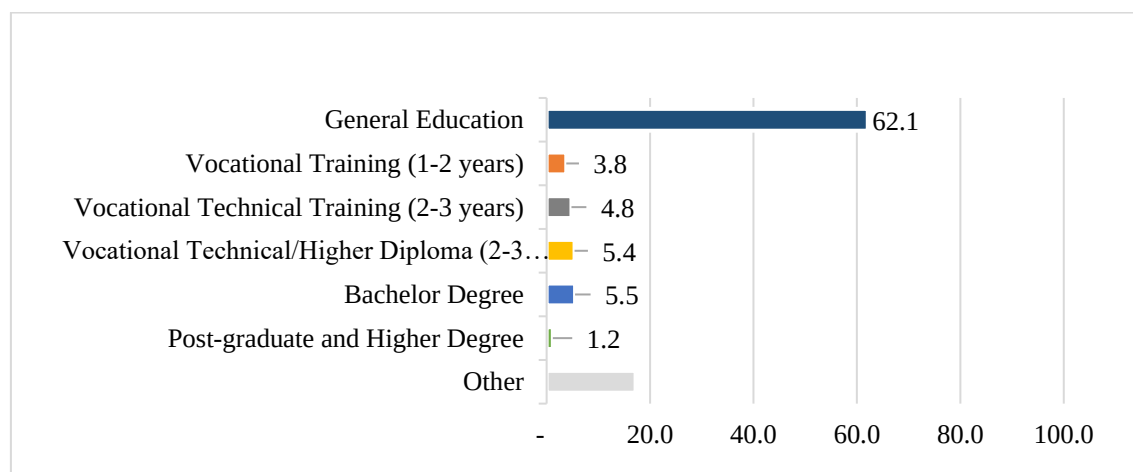
Despite the economic census, accessible database of MSMEs is the National Enterprise Database (NED)¹⁹, which provides essential information of registered enterprises. Major registered enterprises emerged since 2010. [Figure 5-5] displays a fluctuation of growth rate of registered enterprises in Laos. The growth rate shows positive sign from 2016 to 2019, whereas the growth rate shows a declined trend since 2020. Please note that the

¹⁹ Enterprise Registration and Management Department, Ministry of Industry and Commerce. Link: <http://www.ned.moic.gov.la/index.php/en/>

data show in the figure is total registered enterprises by year. It includes both active and inactive enterprises, which its total number is probably bigger than the 3rd National Economic Census.

With respect to operating types, most firms are family business owned by individual operation rather than professional business operation. Joint venture and foreign-owned firms remain very limited. With respect to nationality, 95% of enterprises are run by Laotian owners, who mostly hold general education degree, as shown in [Figure 5-6]. They also tend to have a small amount of physical and financial assets as well as low level of management skills. Most SME are primarily focused on domestic markets. Only a small proportion of enterprise exports to foreign markets. The low level of foreign export is due to the low level of productivity, products do not meet international standards and lack of information about overseas market.

[Figure 5-6] Proportion of Enterprises by Educational Level in 2019 (%)



Source: The 3rd National Economic Census (2019-2020) conducted by Lao Statistics Bureau, Ministry of Planning and Investment.

Remarks: - General education consists of 5-years primary education + 4-years lower secondary education + 3-years upper secondary education.

- Vocational training consists of 5-years primary education + 1-2-years vocational programs.

- Vocational technical training consists of 5-years primary education + 4-years lower secondary education + 2-3-years vocational programs.

- Vocational technical/higher diploma consists of 12-years general education + 2-3-years vocational programs.

- Bachelor degree consists of 12-years general education + 4-years bachelor degree curriculums.

Enterprises face several internal and external constraints. The external constraints of SMEs are divided into four subgroups: (1) competition and taxes, (2) infrastructure, (3) operational costs, (4) regulation issues. On the other hand, shortage of human resource, low level of technology and lack of capital are three major internal constraints of Lao SMEs. Based on the enterprise survey²⁰ by the GIZ in 2013 revealed that the level of technology as a perceived constraint of enterprises in Laos. Many medium enterprises rated low level of technology (GIZ, 2014). Lack of capital was also considered a very big issue for medium enterprises and a big issue for a similar proportion of micro enterprise.

Several SME surveys by using the CIS local assessment methods²¹ in 5 main dimensions: (1) Management Strength (MS), (2) Technical Capability (TC), (3) Sales Capability (SC), (4) Human Resource Capability (HR), (5) Organization Strength (OS) were conducted to evaluate an operational credibility of SMEs in different sectors. By using this method, there are several SME surveys in Vientiane capital, Luang Prabang, Savannakhet, Champasak provinces, Luang Namtha, and Oudomxay.²² In Vientiane capital, a survey was conducted to assess manufacturing sector in 2017. Whereas, surveys conducting in Luang Prabang, Savannakhet, and Champasak provinces focused on services sector, particularly tourism sector in 2018. Its subsectors include accommodations, restaurants, and souvenir retails and travel agents. In 2019, CIS assessment surveys were conducted in Luang Namtha and Oudomxay provinces, which also focused on tourism sector.

Regarding survey results, Lao SMEs barely use technology to improve their operations due to a small investment of human resource development and R&D. Whereas, joint venture or foreign own enterprises seem to show better performance in all dimensions

²⁰ They are small scale surveys since 2005 by the GIZ (German organization) every two years covering five major provinces, namely Vientiane Capital, Luang Namtha, Luang Prabang, Savannakhet, and Champasak. All the surveyed reports are accessible online both in Lao and English.

²¹ Credibility Index for SMEs (CIS) is a tool to evaluate the operational credibility of SMEs and will be applied to SMEs in ASEAN Member States (AMSs) to enhance competitiveness of SMEs in the region. The CIS base model was developed by Japanese expert team from Advanced Institute of Industrial Technology, which comprises a set of checklists formed by six appraisal dimensions including Management Strength, Technical Capability, Sales Capability, Human Resource Capability, Organization Strength and Financial Strength covering 478 questions. The CIS base model was developed from the base model since 2017 (Dr. Phanhpakit Onphanhdala led the production at that time). The assessment covers manufacturing and services sectors.

²² More details can be seen in Annex 3.

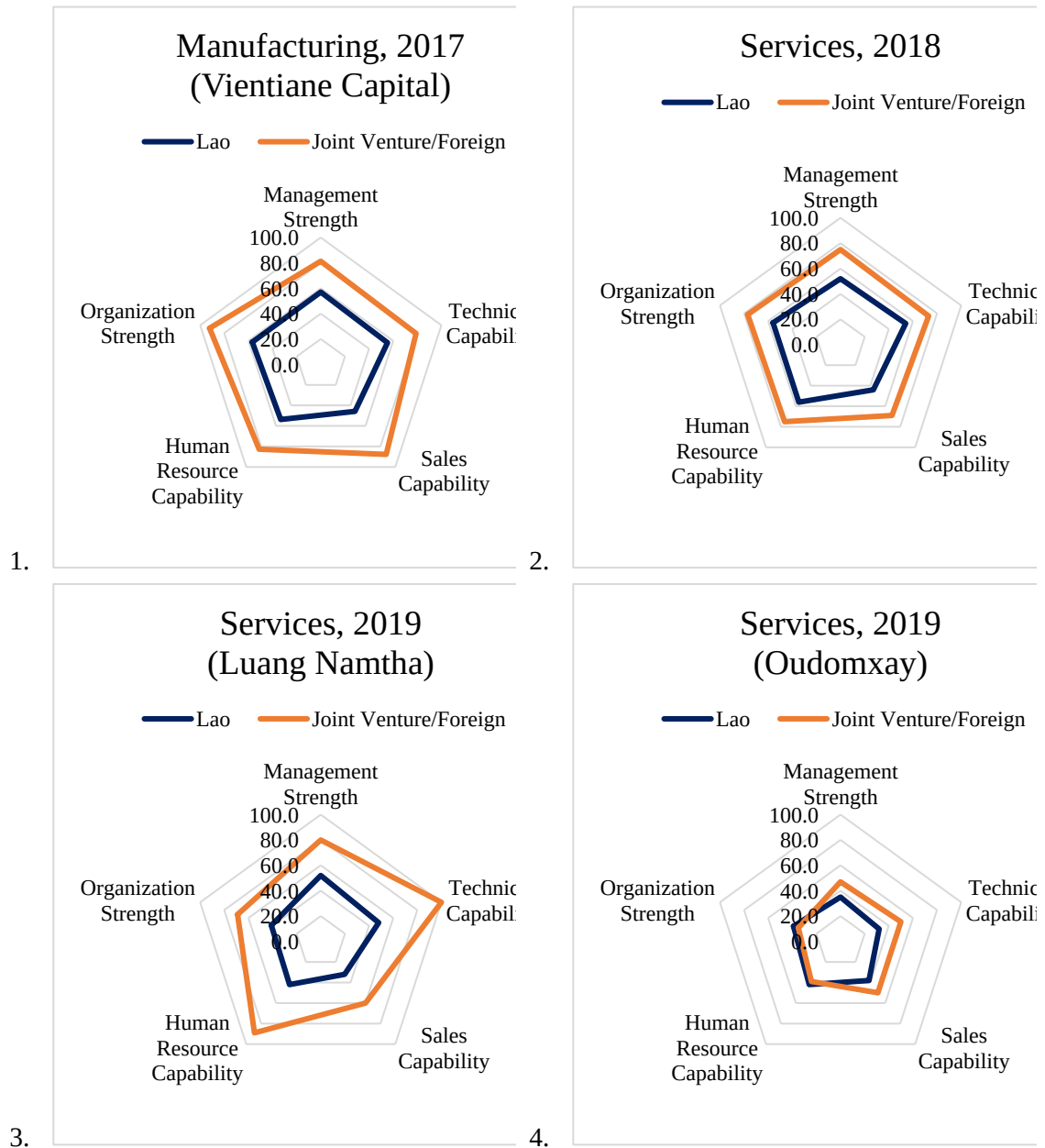
of CIS assessment, as shown in [Figure 5-7] (APEN, 2017; Onphanhdala and Philavong, 2017; Onphanhdala et al., 2018, Onphanhdala et al., 2020).

Despite the CIS assessment, there is also a working paper on private sector development under the Lao-China Economic Corridor (LCEC) that investigated impacts of the LCEC or the Laos-China Railway (LCR) on the Lao private sector.²³ Results of Key Informant Interview in Oudomxay, Luang Prabang, and Xiengkhouang provinces reveal that the private sector expected lower cost of transport and increasing number of tourists. This could offer an opportunity to generate more income if Lao enterprises would be able to access Chinese market. However, Lao firms struggle with several constraints in terms of readiness when the LCR operates. It is due to difficulties to access to finance, lack of skilled labor and technique, and out of date machinery (Bannalath and Insisienmay, 2021).

Most Lao MSMEs are primarily management and organization. They barely use technology into production and services, except some business start-ups. For these reasons, the MSME innovation Master Plan should focus on building capability of MSMEs by initiating technological utilization and further extend to create own technology.

²³ Surveys focused on tourism, manufacturing, and agricultural sectors.

[Figure 5-7] CIS Assessment Results in Manufacturing and Services Sectors



Source: APEN (2017); Onphanhdala and Philavong (2017); Onphanhdala et al. (2018), Onphanhdala et at. (2020)

2.3.7 Issues, Challenges, and Opportunities

2.3.7.1 Issues

With respect to resources, the government invests in human resource management by establishing higher education institutes, vocational institutes, and skill development institutes over the past decades. They have been increased in number, but limited qualification. There are scholarships for excellent students to study abroad from the government and international grants. But due to low salaries paid to academic staffs, education institutes have difficulty retaining skilled and experienced academic personnel. On the other hand, although, there is SME Promotion Fund providing financial resources for MSMEs, it seems MSMEs find it is difficult to access to those credits comparing with large enterprises. According to these obstacles, the MSME innovation Plan should emphasize public expenditure on human resource development. Further expanding to oversea scholarships is to accumulate human capital and high paid to returned students is to retain advanced degree personnel. Moreover, it is essential to ensure accessing to credits that is provided to MSMEs.

With respect to institutions, Lao SMEs as well as R&D implementation are at the early stage of innovative development, which focus on technology utilization rather than invention or creation. There are numbers of research and education institutes, but applied research to improve products and services is poor. Research based course is insufficient to build research base and human capital. Research fund is limited. The government should enhance and focus on more on basic research and innovation. The MSME innovation Plan should also emphasize a consistency between various sectors and a cooperation between the public and the private sectors.

With respect to governance, administrations that responsible for SME and innovation were assigned to avoid overlap responsibilities. The government also lifts the SME promotion department to more autonomous MSME Promotion Office. Ministry of Technology and Communications was established and other responsibilities were assigned to related line ministries. However, collaboration mechanism should be strengthened and systematically to avoid sophisticated procedures and burdens.

With respect to policies, this year, the government has legislated the latest law of MSME promotion and taken an initiation to resolve previous challenges of MSME development process. But the law should emphasize that loans for MSMEs should be allowed to invest overseas to provide more room for doing businesses internationally. In addition, as the government also attempts to ease procedures for facilitate doing business environments, but the effort is insufficient to catch up with demands and global trends.

Despite these, in terms of R&D, policies lack of a term development, particularly in science and technology research for SMEs' innovation. Definition, conditions, and instrument are not clear and complex. Implementation periods of R&D and S&T projects are not appropriate to support a continuation of researches, which obstruct to accumulate knowledge and human capital. Therefore, it is essential to modify research term. For example, short-term, mid-term, and long-term projects could be determined as 1 year, 2-3 years, and 4-5 years, respectively. Besides this, it is necessary to strengthen a continuation of researches and implementation after project completion. In addition, regulations related research, such as terms, intensive, cooperation, public governance, and MSME development should be reconsidered and reformed, particularly the government assistance to support resources for R&D and S&T implementation.

2.3.7.2 SWOT Analysis

[Table 5-14] shows SWOT analysis for MSME innovation. Strength, Weakness, Opportunity, and Threat align with MSMEs, Start-ups, and R&D aspects.

[Table 5-14] SWOT Analysis for MSME innovation

Strengths	Weaknesses
- There are numbers of national policies and strategies to promote modernization, including digital technology utilization. - Strong and stable politic and economic performance foster doing business in Laos and enhance international cooperation. - The government broadens its support to education and research institutes. - A growing of ICT lays a foundation for MSMEs and Start-ups. - Geographical advantage to connect the region. - Natural resource rich to reduce defaults risks. - Markets for eco-friendly businesses (agriculture, tourism, and logistics). - Stable fertility rate with expected more young population in the future.	- Complex and sophisticated regulations and procedures for doing businesses. - Lack of qualified academic staffs and limited research studies. - Insufficient skilled labor in quantity and quality due to poor human resource investment from both the public and the private sectors. - Poor language communication skills. - Entrepreneurs have limited knowledge about laws, regulations, and financial technology. - Lack of capital to operate businesses. - Limited fund to support R&D and poor infrastructure. - Entrepreneurs in rural areas have limited access to infrastructure, facilitation, and credits. - Subsidiary to collapse businesses is limited making entrepreneurs barely to take a risk. - Law and regulation enforcement is weak.
Opportunities	Threats
- Economic corridors provide opportunities to expand markets – EWEC, NSEC, LCEC, etc. - Digital technology development offers low cost to utilize affordable technology into businesses. - There are more young entrepreneurs and more creativities. - Large numbers of mobile phone and internet coverage nationwide.	- Global economic and energy crises. - Inflation, local currency depreciation. - Credit rating for Laos is poor to attract foreign investment. - Climate change threats. - Foreign competitors in markets.

Source: Author's compilation.

3. Strategies and Objectives

3.1 Strategy 2. MSME Innovation:

The MSME Innovation Plan is basically set as the five years plan that align with the Five-Years SME Development Plan (2021-2025), the Five-Years Manufacturing and Handicraft Sector Development (2021-2025), and the 9th NSEDP.

It aims to promote the development MSMEs to generate demands for technologies. Main objectives include:

3.1.1 Objective 2-1: Improved human resources in Science and Technology R&D.

3.1.1.1 Action 2-1-1: Promotion of scientific R&D with a concentration of basic research

1) Directions

- Offering policies and improve infrastructure for strengthening human capital, building human resource, and enhancing R&D, particularly via education institute, research institutes, and broadening access to international knowledge and technology.
- Providing fund research projects.
- Enhancing international cooperation to support researches.
- Reconsider and reform regulations related researches to strengthen a continuation of researches and implementation after project completion, and extend various fields of research projects.

2) Implementations

- The MOES and the MTC shall be responsible for promote scientific R&D with a concentration of basic researches, allowing local and international researchers to participate academic institutes for enhancing more research projects.

- Both ministries shall be responsible for review and legislate relevant regulations to manage and enhance R&D, particularly among education institutes.
- Trainings and capacity builds are essential to improve human resources, particularly academic staffs, government staffs, entrepreneurs, etc.
- The MOES and the MTC shall work with the MOIC to strengthen cooperation between the public and the private sectors, providing opportunities to accelerate innovation and R&D.
- The MOF shall be responsible for reallocating national budgetary for R&D fundings. In addition, international sources of funding should be managed appropriately to support R&D.
- International cooperation is essential for exchanging and transferring knowledges, lesson learnt, and technologies.

3) Evaluation

- To evaluate R&D performance, R&D public expenditure should be taken into account. It includes a total expenditure, expenditure by size of research project, by research field, a percentage of GDP, etc.
- Quality and quantity of academic and research works are required to evaluate R&D performance. It should focus more on accepted academic journals and other publications internationally. One of indicators could be Global Innovation Index (GII). The GI consists of seven pillars: (1) institutes, (2) human capital and research, (3) infrastructure, (4) market sophistication, (5) business sophistication, (knowledge and technology outputs), (7) creative outputs.
- To evaluate human resource development, Human Development Index (HDI) could be one of indicators, particularly the knowledge dimension.

3.1.1.2 Action 2-1-2: Qualified human resources and skilled labor to enhance entrepreneurship and improvement of MSMEs' competitiveness

1) Directions

- Investing in human resources to offer more opportunities to become entrepreneurs.
- Creating opportunities for people to access sources of capital, production techniques, and modern technology.

2) Implementations

- The MOES and the MOIC should continue their roles to develop human resources and provide advises to strengthen potential entrepreneurs.
- The MOIC shall take a lead to promote and develop MSMEs, particularly the MSME Promotion Office in order to offering low interest loans for MSMEs.
- In order to ensure funding, the fund, BOL, and MOF should harmonically collaborate in accordance with regulations, policies, and implementation to be consistency.
- International cooperation is essential for exchanging and transferring knowledges, lesson learnt, and technologies.

3) Evaluation

- To evaluate human resource development, Human Development Index (HDI) could be one of indicators, particularly the knowledge dimension.
- The growth of local business registration and capital investment.
- Local divestment trends to ensure the growth of businesses and the improvement of local entrepreneurs.

3.1.1.3 Action 2-1-3: Strong and modern cooperation and integration with the region and internationally

1) Directions

- Strengthening cooperation to receive attention from external assistance agencies or foreign universities.

- Strengthening and consolidating research capacity development activities with universities or research institutes at the subregional and the international levels.
- Enhancing harmonization among member universities or research institutes in terms of R&D and credit transfer systems, etc.

2) Implementations

- The MOES shall build a closer partnership in science, technology, and innovation including through jointly formulating and implementing the masterplan is an aim to connect our strategies of innovative development and foster new impetus for cooperation.
- The MTC and the MOES take a lead to strengthen international cooperation for exchanging and transferring knowledges, lesson learnt, and technologies.
- In order to ensure funding, the fund, BOL, and MOF should harmonically collaborate in accordance with regulations, policies, and implementation to be consistency.

3) Evaluation

- To evaluate Action 2-1-3, numbers of annual co-research projects/outputs shall be considered as well as research findings' utilization.
- A process of scholarship/credit transferring regionally and internationally.

3.1.2 Objective 2-2: Effective policy ecosystem to facilitate MSMEs.

3.1.2.1 Actions 2-2-1: Effective technology and innovation utilization in accordance with green and environmentally friendly

1) Directions

- Offer opportunities to obtain technology transfer and adapt into businesses.
- Promote patent, trademarks, and R&D investment to develop products.
- Helping MSMEs identify, acquire, assimilate and utilize existing technologies from foreign or domestic sources.
- Creating an agency and/or organization that plays a linkage role between technology owners and MSMEs.

- Focus on implementing priority sectors in accordance with the green growth strategy to bring harmony between the various sectors, including science and technology.
- Increase the efficiency of production and services, create added value, upgrade agricultural production to advance to the industry that uses modern science and technology.
 - The agricultural sector:
 - To raises the productivity and use of modern tools, application of new techniques and sciences and innovations.
 - To create access to credit and markets; strong agricultural infrastructure and natural disaster resilience.
 - Industry sector: Technical-Science-Innovation-Wisdom-Technology, with packaging and labeling to enhance product quality and competitiveness.
 - Service sector: Upgrading telecommunication, logistics and postal transportation to be agile and modern in order to support commercial transactions, clearing and other activities efficiently.

2) Implementations

- The MTC and the MOIC take a lead to promote and development MSMEs and enhance their competitiveness and shall harmonically collaborate in accordance with regulations, policies, and implementation to be consistency.
- MTC, as key stakeholders together with MOIC and MOES, responsible for technological innovation and R&D, which include patent, trademarks, research outputs, technological transferring, and innovation endorsement.
- As a key stakeholders, the MTC shall work with line ministries to utilize technological innovation improve production.

3) Evaluation

- Adoption of technology into production checklists and spending.
- The growth of patent and trademarks registration.
- Business spending on R&D investment.

- Value-added products and market extension.

3.1.2.2 Actions 2-2-2: Efficient public administration in accordance with the direction of public administration by law

1) Directions

- To formulating regulations, policies, and laws by ensuring its development on the fair competition, equality and beneficial to all, particularly MSMEs.
- To offer policy and trade ecosystem to facilitate MSMEs, such as to enhance E-commerce, electronic payments, logistics, productivities, and services to growth sustainably.
- To be an interconnector to collaborate with MSMEs, Startup sectors to support them adapt into modernization services and access digitalization.
- To improve the E-government for reducing doing business procedures and enhance transparency by mobilize and manage government services. For example, 50% of government services should be done via online to rise indicators of incomes through digital services.
- To strengthen one-stop central and local government services.
- Strengthen strong and modern cooperation, integration, and interconnection within, with the region and internationally.

2) Implementations

- The MOIC takes a lead to promote and development MSMEs and enhance their competitiveness, particularly the MSME Promotion Office. The ministry should continue its role and ensure an enforcement of the latest MSMEs promotion law.
- The MOIC shall work with the BOL to improve and accelerate electronic payment to facilitate business operations.
- The PWT shall responsible for logistics and infrastructure improvement.
- The MTC shall responsible for improve the E-government in accordance with relevant ministries for reducing doing business procedures and enhance transparency by mobilize and manage government services.

3) Evaluation

- A proportion of Lao, joint-venture, and foreign own MSMEs – numbers, assets, markets, etc.
- Improved doing business indicators and rankings.
- A Frequency and amount of value of electronic transaction.
- The growth of business productivities.
- 50% of government services should be done via online to rise indicators of incomes through digital services.

3.1.2.3 Action 2-2-3: Strong and efficient cooperation between the public and the private sectors

1) Directions

- Enhancing cooperation mechanism between the public and the private sectors efficiently and verifiably.
- Development of technology/industrial parks to facilitate interactions among enterprises, entrepreneurs, and researchers.

2) Implementations

- The MOES and the MTC shall work with the MOIC to strengthen cooperation between the public and the private sectors, providing opportunities to accelerate innovation and R&D.
- The MOIC shall responsible for expanding international markets for exports by advancing bilateral and multilateral cooperation with major partners.

3) Evaluation

- Improved quality and reduced quantity of cooperation procedures.

3.1.3 Objective 2-3: Strengthening funding mechanism to support MSMEs' operation and development.

3.1.3.1 Actions 2-3-1: Finance reliable to sources to support MSMEs' operation and development

1) Directions

- Strengthen funding mechanism to help MSMEs access to credits and improve their liabilities.
 - Conditions, collateral, terms, interest rate, etc.
 - Enhance MSMEs to use technology for financial management and transaction for improve its capacity to compete in terms of payment facilitation and transparency.
 - Encourage MSMEs to proper manage financial account documents, bookkeeping, tax payment document, installation daily maintenance and real estate owners.
 - Ensure cooperation mechanism between line ministries relevant MSME promotion funding in accordance with efficiency, transparency– MSME Promotion Fund (MOIC), BOL, MOF, etc.

2) Implementations

- The MOIC takes a lead to promote and development MSMEs and enhance their competitiveness, particularly the MSME Promotion Office.
- MSME Promotion Office/Fund plays a key role to support an operation and development of MSMEs by offering low interest loans to businesses.
- The BOL and the MOF shall be responsible for offering loans for MSMEs and reallocating national budgetary for MSME promotion.
- The BOL and the MOF should harmonically collaborate in accordance with regulations, policies, and implementation to be consistency.

3) Evaluation

- Credits and loans for MSMEs in terms of conditions, amounts, etc.
- Reduced NPLs.

- The growth of value added of MSMEs.
- Improved E-commerce and online transaction methods.
- Bookkeeping ownership and financial status checklists.

3.1.3.2 Actions 2-3-2: Well-round MSMEs in terms of management and organization systematically

1) Directions

- Create opportunities to access sources of capital, production techniques, modern technology in adopting to management and organization of businesses.
- To develop digital and ICT infrastructures and exchanging platforms for MSMEs.
- To assist MSMEs to create plan, set up management, and business direction.

2) Implementations

- With respect to MSME innovation Plan, the MOIC takes a lead to promote and development MSMEs and enhance their competitiveness, particularly the MSME Promotion Office.
- The ministry should continue its role and ensure an enforcement of the latest MSMEs promotion law.
- Under the MOIC's supervision, the MSME Promotion Fund plays a key role to support an operation and development of MSMEs by offering low interest loans to businesses.
- In order to ensure funding, the fund, BOL, and MOF should harmonically collaborate in accordance with regulations, policies, and implementation to be consistency.
- The MTC shall responsible for ITC infrastructure improvement to ensure ICT coverage, which in accordance with the PWT's infrastructure plans.
- The MOF shall be responsible for reallocating national budgetary for infrastructure expenditure.

3) Evaluation

- ITC infrastructure coverage.

- Management and organization checklists.
- Utilization of business development services.

3.1.3.3 Actions 2-3-3: Efficient administration to ensure funding source stability

1) Directions

- Ensure transparency of fund providing to improve Lao MSMEs and entrepreneurs.
- Strengthen financial stability of funding sources.
- Monitoring credit providing and ensure it goes to targeted MSMEs.

2) Implementations

- the MSME Promotion Fund plays a key role to support an operation and development of MSMEs by offering low interest loans to businesses.
- The BOL and the MOF should harmonically collaborate in accordance with regulations, policies, and implementation to be consistency in terms of loans and credits.

3) Evaluation

- Improved funding mechanism.
- Numbers of MSME access to credits and MSME's productivity.
- Financial status of MSME Promotion Fund.

[Table 5-15] displays the metrics for evaluation of the achievement of the MSME innovation Plan. The subsequent planning is to lay a solid foundation for creating own technology and extend digital knowledge further, which requires international and partnership supports.

[Table 5-15] Metrics for Ex-Post Evaluation

	Key Actions	Evaluation
Objective 2-1		
Action 2-1-1	Promote scientific research and develop new socio-economic developments with a concentration of basic research	- R&D public expenditure - Academic and research works - Global Innovation Index (GII) - Human Development Index (HII)
Action 2-1-2	Qualified human resources and skilled labor	- Human Development Index (HII) - Local business registration and capital investment - Local divestment
Action 2-1-3	Strong and modern cooperation and integration with the region and internationally	- Co-research projects/outputs - Scholarship/credit transfer
Objective 2-2		
Action 2-2-1	Effective technology and innovation utilization in accordance with green and environmentally friendly	- Adoption of technology into production checklists and spending - Numbers of patent and trademarks registration - Business spending on R&D investment - Value-added products and market extension -
Action 2-2-2	Efficient public administration in accordance with the direction of public administration by law	- A proportion of Lao, joint-venture, and foreign own MSMEs – numbers, assets, markets, etc. - Doing business indicators - Electronic transaction value and frequency - Business productivities - E-government (%)
Action 2-2-3	Strong and efficient cooperation between the public and the private sectors	- Cooperation procedures

	Key Actions	Evaluation
Objective 2-3		
Action 2-3-1	Finance reliable to sources to support MSMEs' operation and development	- Number of MSMEs get loans - Amount of credits/loans - NPLs - Interest rate, reformed legislations - E-commerce methods - Online transaction methods - Bookkeeping ownership
Action 2-3-2	Well-round MSMEs in terms of management and organization systematically	- ITC infrastructure coverage - Management and organization checklists - Utilization of business development services
Action 2-3-3	Efficient administration to ensure funding source stability	- Funding mechanism - MSME access to credits - Financial status of MSME Promotion Fund

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5. Annex

Annex 1

Part VI: Organization of Micro, Small, and Medium Enterprises Promotion

	Organization	Description
1	MSME Promotion Committee (It is a non-permanent committee, which to be appointed by the Prime Minister upon the proposal of MOIC)	- Deputy Prime Minister who supervises economic field as Chairman; - Minister for Industry and Commerce as Vice Chairman; - Vice Minister for Industry and Commerce as member; - Vice Minister for Finance as member; - Vice Governor of Bank of Lao PDR as member; - Vice Minister Planning and Investment as member; - Vice Minister for Agriculture and Forestry as member; - Vice Minister for Education and Sports as member; - Vice Minister for Information, Culture and Tourism as member; - Vice Minister for Natural Resources and Environment as member; - Vice Minister for Labor and Social Welfare as member; - Vice Minister for Public Works and Transport as member; - Vice Minister of Technology and Communications as member; - Vice President of National University of Laos as member; - President of Lao National Chamber of Commerce and Industry as member; - President of Association of MSME Promotion at central level as member; - Director of MSME Promotion Office as member and secretary.
2	MSME Promotion Office ²⁴	- It was changed from Department of SME Promotion and under a supervision of MOIC.
3	MSME Promotion of the Private Sector	- Lao National Chamber of Commerce and Industry (LNCCI); - Provincial Chamber of Commerce and Industry; - Association of MSME Promotion; - Other business organizations

²⁴ Former the SME Promotion and Development Office

Annex 2**S&T International Cooperation**

Partner	Assistance Program	Fields
Korea	- Science and Technology Innovation (STI) - Other projects	ICT Education Healthcare Energy
China	- Digital Silk Road (DSR) - Belt and Road Initiative (BRI)	Digital Infrastructure Logistics Education Healthcare Energy
Japan	- Innovation Asia - Other projects	ICT Education Healthcare Infrastructure Energy Finance

Annex 3

1) Survey Respondents for Manufacturing Sector Assessment in Vientiane Capital in 2017

Interviewees	Lao SME	Joint Venture/Foreign Owner
Owner	22	-
Top Manager	1	3
Middle Manager	6	4
Head of Sector	1	2
None Specify	2	-
Total	32	9

Source: APEN (2017); Onphanhdala and Philavong (2017)

2) Survey Respondents for Tourism Sector Assessment in Luang Prabang, Savannakhet, Champasak provinces in 2018

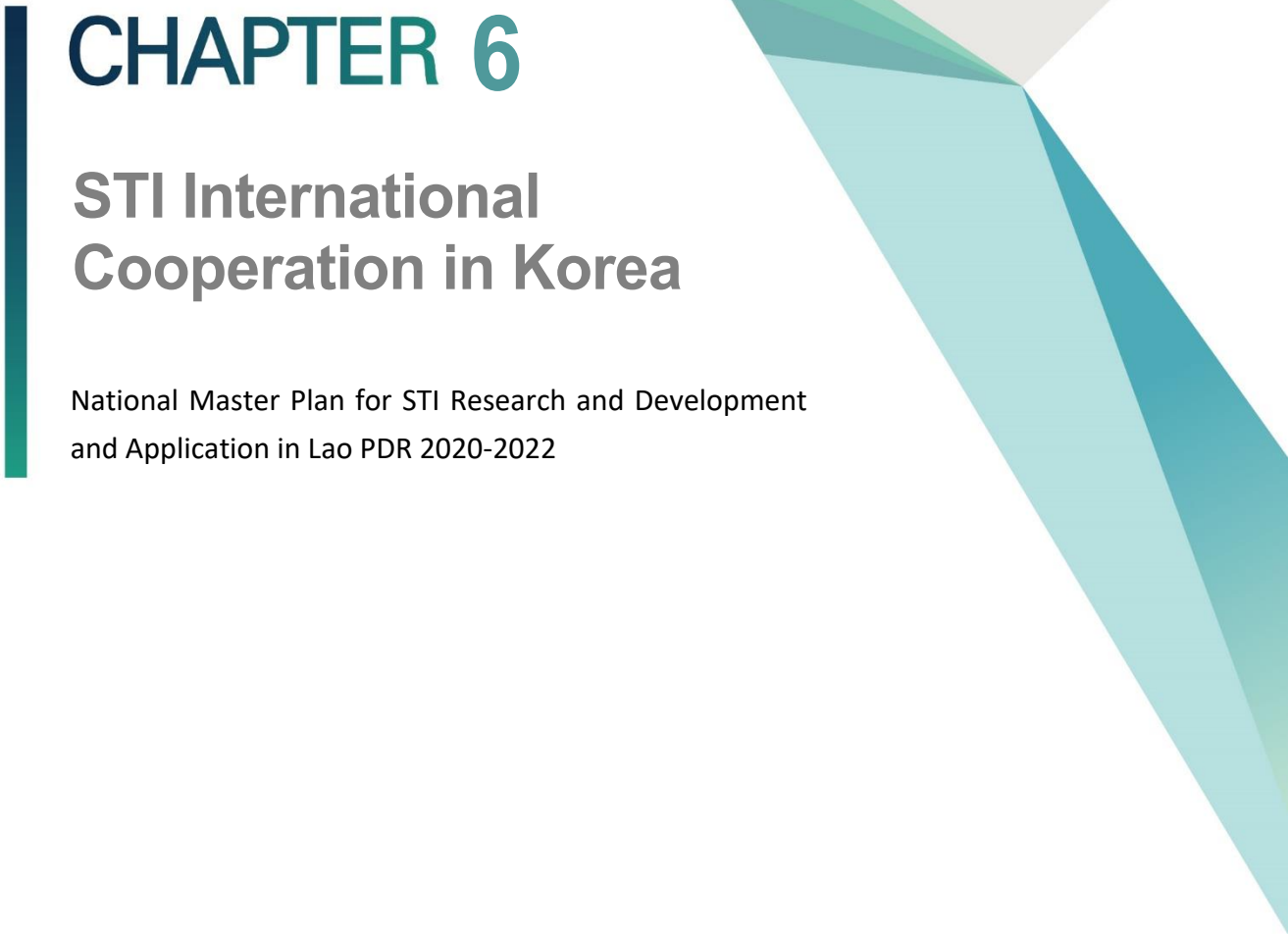
Interviewees	Luang Prabang		Savannakhet	Champasak
	Joint Venture/Foreign	Lao	Lao	Lao
Owner and Top Manager	4	16	14	17
Middle Manager and Other	14	11	12	11
Total	18	27	26	28

Source: Onphanhdala et al. (2018)

3) Survey Respondents for Tourism Sector Assessment in Luang Namtha and Oudomxay provinces in 2019

Interviewees	Luang Namtha		Oudomxay	
	Joint Venture/Foreign	Lao	Joint Venture/Foreign	Lao
Owner and Top Manager	1	17	2	10
Middle Manager and Other	0	1	0	3
Total	1	18	2	13

Source: Onphanhdala et al. (2020)

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and green, also pointing downwards.

CHAPTER 6

STI International Cooperation in Korea

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 6. STI International Cooperation in Korea

Dr. Jingwon KANG, KISTEP

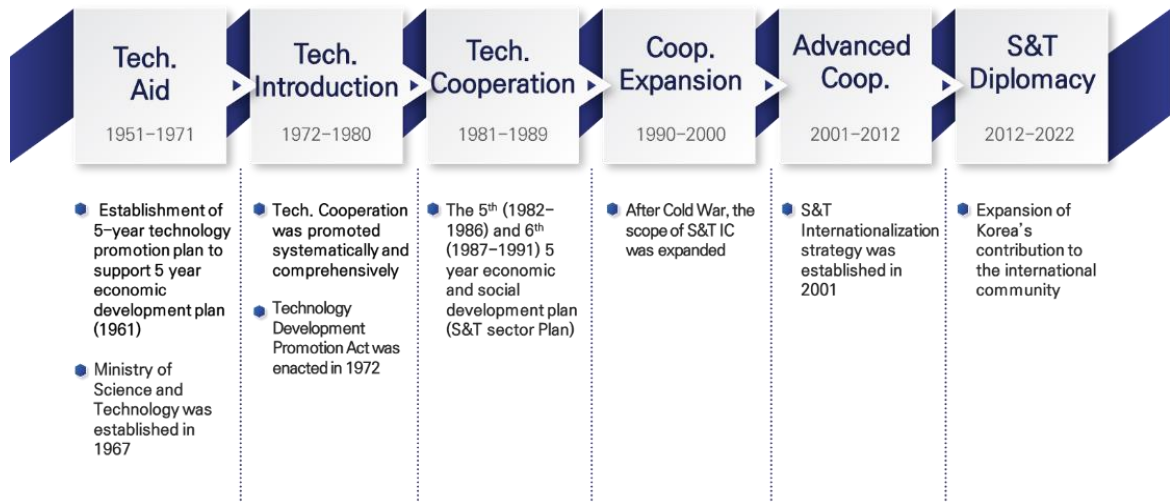
1. Change of S&T IC Strategy²⁵

1.1 Periods of S&T IC in Korea and their Evolution

The periods of international S&T cooperation can be divided from the Korean war to the present into five sub-periods: period of technology aids, technology introduction, technology cooperation, cooperation expansion, advanced cooperation and S&T diplomacy, respectively. This division is not exclusive, but it can be seen that although overlapping characteristics exist, they represent the most important characteristics as it is shown in the [Figure 6-1].

²⁵ The history of international cooperation in science and technology in Korea was written based on 'Institutional evolution of Korea's IC of S&T from the viewpoint of the historical institutionalism' (Se-in Park, 2011)

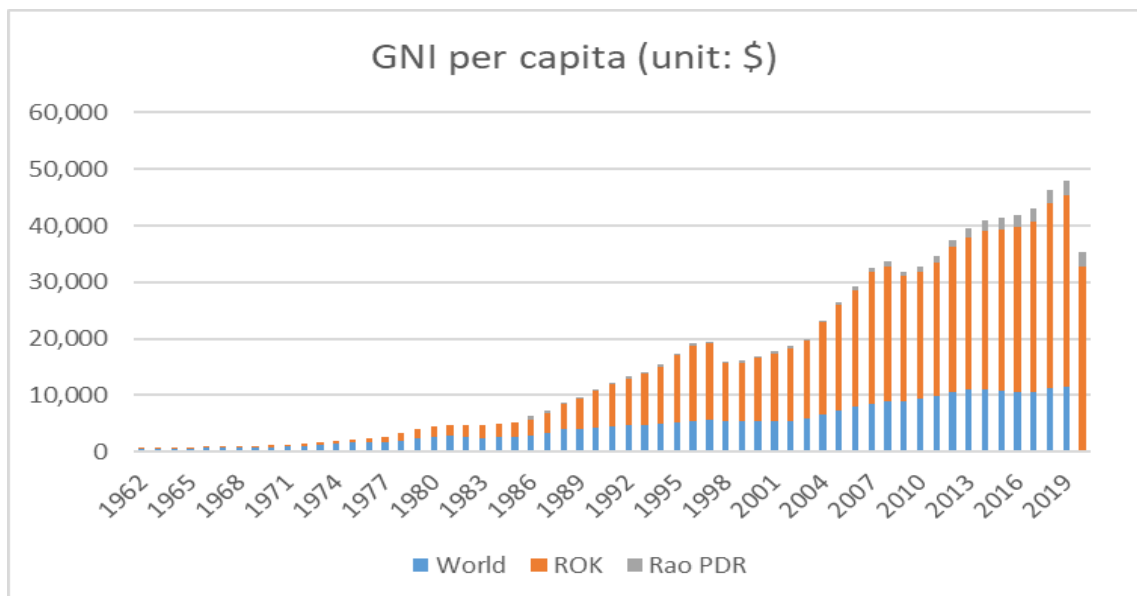
[Figure 6-1] The periods of S&T IC



Source: author summarized the main characteristics of each period

When comparing the current national income of Laos in this period of Korea, it can be seen that Laos is in the period of technological cooperation with Korea (refer to the [Figure 6-2] below).

[Figure 6-2] GNI per capita in World, ROK and Rao PDR



Source: https://kosis.kr/statHtml/statHtml.do?orgId=101&tblId=DT_2AS017 &conn_path=I3

The S& T IC strategy was changed in terms of organization, volume, type counter-part, program and so on over the periods. The main changes are as follows; organization: from ministry of Economy to ministry of Science & Technology; volume: \$ 1 million ('51-'55) for Tech. Aid to \$ 2.86 billion ('21) for ODA; joint research: KRW 1.6 billion ('85) to 17 billion ('16); type: bilateral for assistance to bilateral for cooperation with multilateral relations; technology assistance, technology adoption, technology cooperation, human exchange, joint research, hosting research institutes, etc; counter-part: advanced countries centered to diverse counter parts such as international organizations, developing countries, northern countries (Russia, Eastern Europe, etc.); R&D Program: joint research ('85) and cooperation foundation ('96).

1.2 Technology Aid Period (1951-1971)

In this period, there was the ideological confrontation between capitalism and communism after the end of World War II. Due to the Korean war (1950), the economic foundation of Korea was devastated, and this became a social background in which Korea had no choice but to pursue economic development with foreign aid. Therefore, in the 1950s, Korea had no choice but to rely on aid from advanced countries including the United States for capital and materials for economic activities. In the early 1960s, per capita GNP was only about \$100, and dependence on the primary industry represented by agriculture and fisheries was absolute, and industrialization was not progressing.

Against this background, from the early 1960s, the government established an economic development plan to achieve independent economic development and began to promote it in earnest. At that time, the domestic industrial base was weak, so the technology needed to foster strategic industries was imported from advanced countries and then digested. In the early 1960s, the government tried to resolve the legitimacy of the unstable regime through economic development. And this interest in economic development leads to the scientific and technological promotion, and the technical promotion five-year plan is established to support the first five-year economic development plan that started in 1961. This plan has the meaning of being the first

comprehensive plan to promote the development of science and technology as a whole, and shows that the government recognized that science and technology policy is one of the important driving forces for national economic development beyond the existing educational policy. Ministry of Science and Technology, 2008).

At the same time, at the Korea-US summit in May 1965, the Korea Institute of Science and Technology (KIST) was established with the support of the US to establish a research institute. In January, the Science and Technology Scarlet Act was passed by the National Assembly, and on April 21, 1967, the Ministry of Science and Technology was established. The 1st Five-Year Technology Promotion Plan is significant in that it is the first comprehensive plan to promote the overall science and technology. It shows that science and technology policy was recognized as one of the important driving forces for national economic development beyond the existing educational policy. With the establishment of this plan, the development of science and technology was recognized as a national task and promoted as a part of the national long-term development plan. included in the plan.

International cooperation in science and technology from immediately after the Korean War to the early 1970s was a period in which Korea mainly received aid from advanced countries and international organizations. In 1963, a technology grant project that invited trainees from developing countries was also started, but the amount of funding received from the United States and international organizations was larger than Korea's own funds. In addition, this period can be defined as the period when Korea received technical assistance mainly from advanced countries or international organizations because it was insignificant compared to the amount of support received from advanced countries or international organizations.

Technical assistance came from advanced countries such as the United States and Japan, international organizations of the United Nations and UNDP, and multilateral cooperation organizations such as the Colombo Project, and most of them were grant aid. Among these, aid from the United States was the largest, followed by the United Nations and the Colombo Program.

[Table 6-1] Technical assistance in technology aid period (1951-1971)

Classification	Invitation of Exp.	Dispatch of trainees	Service contract	Supplied Goods	Sum	Ratio
AID/DG	35,444(1,535)	12,075(3,348)	46,964	21,052	115,535	64.4
UN	5,673(968)	4,739(1,851)	19,255	14,529	44,196	24.7
Colombo Plan	857(124)	4,292(1,538)	-	1,934	7,083	4.0
ETC	731(45)	8,638(2,261)	973	2,171	12,513	7.0
Sum	42,705(2,672)	29,745(8,998)	67,192	39,686	179,328	100

Source: Science and Technology Yearbook (1968, 1978, 1980)

1.3 Technology Introduction Period (1972-1980)

In the 1970s, a new international technological cooperation plan was raised with the UN centering on it. As a representative example, the Country Program has been adopted since 1972, centering on the UNDP, the leading organization for technical cooperation among UN organizations. In the past, the support method based on the unilateral decision of the donor country was changed to a new technology cooperation method centered on the recipient country that provides support in accordance with the long-term development plan of the recipient country.

Meanwhile, with the advent of the GATT system, the protectionism policies of advanced countries have strengthened and the international free trade environment has deteriorated significantly. In addition, resource nationalism emerged internationally as the Organization of Petroleum Exporting Countries (OPEC) was established in 1972, and the Middle East War broke out the following year, resulting in an oil shock. Such rapid changes in the international environment were adversely affected by Korea, which was pursuing export-oriented economic development.

As the basic direction of economic policy in the 1970s was set to advance the industrial structure by fostering the heavy and chemical industry, science and technology policies were also promoted to support it. Heavy and chemical industrialization requires

enormous capital and technology. Accordingly, the government pursued a bold foreign capital and technology introduction policy, unlike the 1960s, when foreign capital was regulated. The government's response to changes in the international environment in the 1970s was to establish the foundation of industrial technology to support heavy and chemical industrialization led by the government. At that time, most of the technologies required by private companies were imported from abroad or embodied in capital goods, so the ability to absorb and digest them was accumulated. To this end, the Ministry of Science and Technology enacted the Technology Development Promotion Act in 1972 to promote the independent development of industrial technology and the introduction of appropriate advanced technology, and to support digestion and improvement. This Act stipulated both direct support such as subsidy payment and policy finance, and indirect support such as tax and customs support and market creation. In addition, government-funded research institutes for each specialized field established under the leadership of the Ministry of Science and Technology were mainly responsible for the development of independent industrial technology and support for the introduction and digestion of advanced technologies by companies.

Korea's international cooperation in science and technology initially started as a form of economic cooperation, but with the creation of the Ministry of Science and Technology in 1967, it became possible to systematically and comprehensively promote technological cooperation that had been sporadically promoted by each ministry. In 1972, in order to support the 3rd Five-Year Economic Development Plan and the Third Five-Year Science and Technology Plan, the manpower supply and demand plan and R&D strategy, the Five-Year Plan for International Technology Cooperation was announced. The significance of the Five-Year International Technical Cooperation Plan for international cooperation in science and technology is that science and technology are not a means of economic growth, but an important factor leading the way, and advanced technology even if a price is paid rather than passive technical cooperation. In 1979, the second liberalization of technology introduction was carried out, and a technology introduction consultation center was established at KIST to introduce and mediate appropriate technology following the liberalization of technology introduction. In addition, information support for technology introduction contracts was strengthened, and tax and loan support were provided in accordance with the

Technology Development Promotion Act for cases where the technology was digested/ improved after technology was introduced.

[Table 6-2] The change of Tech. Introduction in 1960-1980 (unit: \$ Million)

Classification	Amount (No.)	Ratio of Increase	Amount / No.
1962-1966	0.8 (31)	-	0.03
1967-1972	26.5 (326)	-	0.08
1974	17.8 (86)	55%	0.21
1976	30.4 (131)	15%	0.23
1978	85.1 (296)	46%	0.29
1980	107.2 (222)	14%	0.48

Source: Ministry of Science and Technology (2008)

It started to participate in the international science and technology network from the technology source period. Some examples are the invitation of Korean scientists and engineers from abroad, which began in 1968, the dispatch of domestic scientists and engineers to international conferences, and the invitation of eminent foreign scientists. These projects are meaningful in that Korea wanted to participate in international R&D networks beyond a one-sided technology source and promote leading international cooperation.

[Table 6-3] The performance of invitation and dispatch

Classification	'68	'69	'70	'71	'72	'73	'74	'75	'76	'77	'78	'79	Sum
hosting overseas Korean	7	22	21	30	28	50	56	38	54	51	53	83	493
Dispatch to International meeting	32	26	33	23	20	16	11	6	18	13	13	11	232
Invitation of eminent foreign scientists	2	12	23	21	20	21	31	17	16	27	10	5	205

Source: Science and Technology Yearbook (1980)

In this period, the main changes of S&T IC are as follows; organization: international cooperation bureau to technology cooperation bureau; volume of technology introduction: \$ 107.2 million ('80), technology cooperation: \$ 8.9 million ('71) to 13.7 million ('80); type: bilateral for aid to bilateral for cooperation (Tech. Aid, Tech. Introduction, Tech. Donation); counter-part: including developing countries as well as international organizations, developed countries, developing countries.

1.4 Technology Cooperation Period (1981-1990)

In the 1980s, major advanced countries, including the United States, invested huge amounts of money and manpower in the development of advanced technologies while seeking a transition to a new development strategy in order to escape the swamp of a long economic recession. As a result, the development of advanced technologies has been rapidly developed, centering on advanced countries. Technology protectionism began to intensify as competition for technology development between advanced countries developed fiercely.

Developed countries have provided protection for new intellectual property rights, including software and material patents, while preventing leakage of their advanced technological information to competitors. In addition, through trade negotiations, etc., measures to effectively protect the country's industry and technology have been strengthened. Meanwhile, Korea achieved rapid economic growth with the development of heavy and chemical industries in the 1970s, but in the 1980s, the Korean economy reached a limit where it could no longer maintain its competitiveness in the past, which relied on cheap labor. Accordingly, the demand for economic advancement and industrial structure reorganization increased, and a new approach that reflected this situation was required in terms of science and technology policy.

In the 1980s, as Korea's export competitiveness, which relied on cheap labor, reached its limit, the government took note of the need to actively respond to changes in the international economic environment and technological environment, so that Korea's science and technology must enter the level of advanced countries. Therefore, in the

1980s, the 5th Five-Year Economic and Social Development Plan: Science and Technology Sector Plan (1982-1986) and the 6th Five-Year Economic and Social Development Plan: Science and Technology Sector Plan (1987-1991) were established.

In addition, in 1986, the Ministry of Science and Technology established a long-term plan for science and technology development for the 2000s (1987 - 2001) in order to systematically promote science and technology policies from a long-term perspective.

In order to continuously pursue economic development while actively responding to the above-mentioned changes in the domestic and overseas environment, it was important to secure independent scientific and technological research and development capabilities to develop cutting-edge technologies. With this shift in perception, technology development was emphasized as a national task, leading to the conversion of export drive policy to technology drive policy.

In addition, a project was developed to actively attract excellent scientific brains from abroad and to support the dispatch of excellent domestic manpower abroad.

In the 1980s, Korea tried to develop cutting-edge technology in Korea beyond simple technology introduction from advanced countries, but international cooperation in science and technology with major advanced countries was still important. However, as Korea's economy grew, it tried to reestablish international cooperation as a mutually beneficial relationship on an equal footing with advanced countries, away from the position of receiving support unilaterally. Therefore, it actively promoted the signing of science and technology cooperation agreements for equal cooperation with major advanced countries that had been promoting scientific and technological cooperation, and based on this, various cooperative projects represented by international joint research were promoted.

Until the early 1980s, Korea's technology introduction was limited to major advanced countries, and among them, the United States and Japan accounted for most. From 1962 to 1985, 78% of Korea's technology introductions and 74.5% of technology introductions were made to the United States and Japan (International Science and Technology Cooperation Statistics, 1997).

If technology introduction is concentrated in a specific country, if that country's technology protectionism is strengthened, it may become difficult to acquire core technology or necessary technology, and there are also concerns about weakening technological competitiveness and technology dependence. Therefore, it is necessary to secure technological diversity through technological cooperation with various countries, strengthen its own R&D capabilities, and secure new markets. As a result, the number of countries that signed the Science and Technology Cooperation Agreement, which can be said to be the basis of scientific and technological cooperation, increased from 5 countries in 1979 to 36 countries in 1989.

[Table 6-4] Status of Science and Technology Cooperation Agreements in the 1980s

Classification	Agreement signed countries
before 1979	Spain ('75), Paraguay ('75), India ('76), USA ('76), Costa ('79)
1981	France, Colombia, Zaire, Peru, Philippines
1982	Nigeria, Dominica
1983	Iraq
1984	Italy, Haiti
1985	Sweden, Gambia, UK, Thailand, Japan, Pakistan, Malaysia, Uruguay
1986	West Germany
1987	Guinea-Bissau, Greece
1988	Egypt
1989	Finland, Hungary, Mexico

Source: Ministry of Foreign Affairs website

Since the late 1970s, as industrialization progressed, the number of technology introductions and payments for technology introduction have gradually increased. In this situation, the international joint research project was promoted to meet the increasing demand for technology and to strengthen its own research field (Science and Technology Yearbook, 1986).

After the international joint research project was promoted in earnest in 1985, looking at the performance of the international joint research project, a total of 354 tasks were promoted from 1985 to 1990, and a total of 16.8 billion won was invested.

[Table 6-5] Performance of joint research (1985-1990)

Classification	1985	1986	1987	1988	1989	1990	Sum
Budget	16	23	27	27	26	50	169
No. of Project	39	49	69	66	55	76	355

Source: Science and Technology Yearbook (1991)

By signing a science and technology cooperation agreement with advanced countries, it was intended to transform into a reciprocal relationship unlike the past, where support was only unilateral. Technology diplomacy-type international cooperation in science and technology, which had been weak in the past, was strengthened. The implementation of the international joint research project has the meaning of a full-fledged international cooperation policy for science and technology. This is a different type of international cooperation in science and technology from the past.

1.5 Cooperation Expansion Period (1991-1999)

With the beginning of the 1990s, the international community underwent major changes. With the collapse of the communist regime, the collapse of the Cold War system and the dissolution of international relations based on ideology and military power, international relations centered on the economy began to take on the core importance.

From January 1995, the WTO system, which set new standards for international trade and investment, was launched. In science and technology policy, a situation in which policy must be planned/implemented according to the rules stipulated by international conventions has begun to develop. The development of globalization following the collapse of the Cold War system was similar to Korea. In particular, the Kim Young-sam

government presented globalization as a national goal and actively promoted globalization. However, the opening brought up along with globalization has cost Korea a great deal. The rapid opening, which was not sufficiently managed, became a major cause of the foreign exchange crisis, which eventually led to a situation in which the IMF requested a bailout and implemented a restructuring policy at the request of the IMF.

Until the mid-1980s, the issue of comprehensive coordination was not discussed much because the Ministry of Science and Technology had been in charge of the functions of the department in charge of science and technology in the indifference of other ministries. However, as mentioned above, as the government's R&D budget continued to increase in the 1990s and various ministries and ministries promoted R&D projects in various forms, the need for comprehensive coordination of science and technology policies increased.

To this end, in February 1998, the Ministry of Science and Technology was elevated to the "Ministry of Science and Technology", and in January 1999, the National Science and Technology Committee was established with the president as its chairperson under the 'Special Act for Science and Technology Innovation'. After the establishment of the National Science and Technology Committee, through the establishment of the 「Science and Technology Long-Term Vision for 2025」 in 1999 and the 「Basic Plan for Science and Technology」 based on the Framework Act on Science and Technology in December 2001, the 'medium-term vision/goal for the development of science and technology in Korea' and A 'promotion strategy for achieving the goal' was presented.

The key change in science and technology policy in the 1990s was that the strategy to create new technologies by breaking away from the imitation strategy that was dependent on foreign sources of science and technology began to be discussed in earnest. Against this background, a large-scale national R&D project that intensively supports national R&D resources was started in order to secure strategic competitiveness and technological competitiveness to compete with advanced countries. These projects prioritized areas where competitiveness could be secured through mid- to long-term technology forecasting and demand surveys, while clearing the strategic nature to achieve specific goals.

The 1990s was a time when the Cold War system, which had continued for half a century, ended, and exchanges with Russia, Eastern European countries and China began and rapidly expanded. In the field of international cooperation in science and technology, cooperation with these countries was more focused than other countries. In particular, at a time when Korea was experiencing difficulties due to the strengthening of technological barriers due to technology protectionism despite the need to narrow the technological gap with advanced countries, scientific and technological cooperation with Russia or China was found to be a new cooperative partner, not a western advanced country. Therefore, international cooperation in science and technology with northern countries such as Russia and China was newly added during this period in addition to the existing international cooperation in science and technology with advanced western countries.

In Korea, information gathering activities on the latest scientific and technological trends abroad have been promoted since the 1970s. However, in the 1990s, as our international activities expanded to various regions such as Eastern Europe, Latin America, Africa, and ASEAN in addition to the existing developed countries in the West, infrastructure-building projects were strengthened. In particular, while promoting international cooperation with Russia, it was necessary to establish a foundation in advance to facilitate joint research and attraction of manpower. Accordingly, the cooperation-based project started with the Korea-Russia manpower exchange project in 1992, and in 1996, it was separated from the international joint research project due to differences in the task selection procedure, and was newly established as a foundation for international cooperation.

International cooperation in science and technology was promoted not only by government organizations but also by public institutions. As the scale of IC S&T increased, the scope of cooperation became more diversified, and professionalism was required, the government began to entrust work to public institutions. The scope of science and technology cooperation has been expanded. IC S&T was also promoted with northern countries such as Russia, China, Eastern Europe, and North Korea, where cooperation was restricted due to ideological barriers in the past. The science and technology international cooperation program has become more diverse. In addition to the existing

joint research-type projects, human resource exchange and foundation-building type projects were added.

1.6 Advanced Cooperation Period (2000-2012)

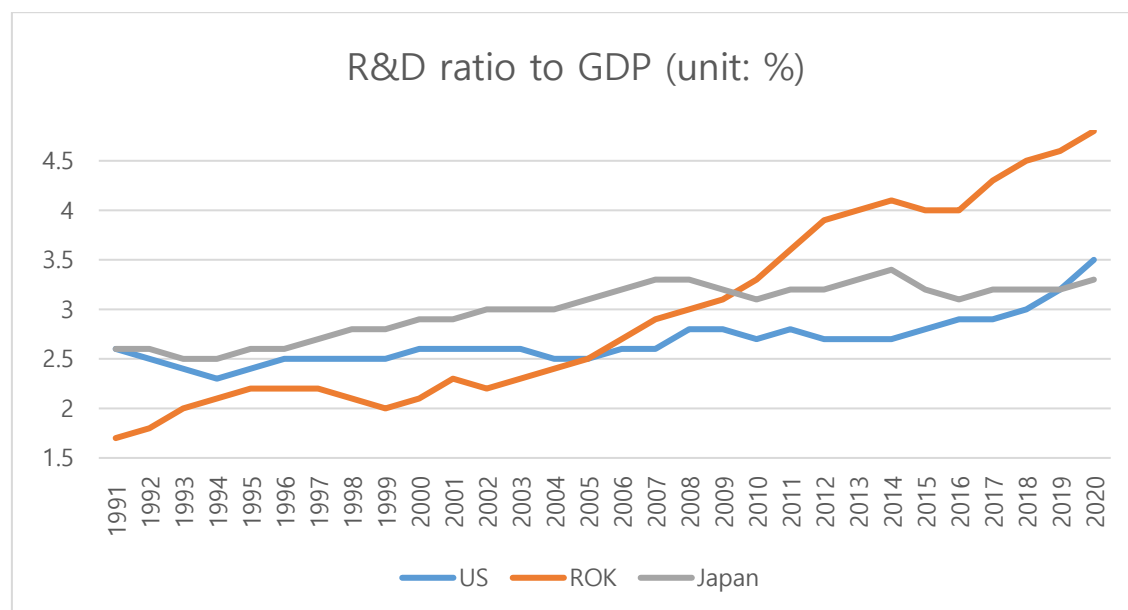
In the 21st century, with the rapid development of information and communication technology, knowledge has emerged as the biggest factor of production. Meanwhile, with the development of information technology, economic and scientific and technological development activities around the world are gradually becoming more open, globalized and networked. In Korea, the limitations of the catch-up innovation system, such as a lack of creative manpower, a lack of ability to develop source technology, sluggish industrialization of R&D results, and a focus on mature technology that allow latecomers to catch up, were pointed out. Therefore, overall strategy revision through innovation was inevitable, and the need for new growth engines to be discovered and a transition to an innovation system that went beyond imitation and pursuit was raised (Ministry of Science and Technology, 2008).

The Roh Moo-hyun administration has presented “building a science and technology-oriented society” as one of its major national tasks, and has been advocating 'entering the second science and technology' and 'establishing a Northeast Asian R&D hub' as key national tasks. To this end, the 「Science and Technology Centered Society Promotion Planning Group」 was formed in January 2004, and the position of information science and technology assistant was newly established within the Blue House

In order to prepare for the above environmental changes, Korea has expanded its investment in R&D as well as in promising new technologies such as biotechnology (BT), nanotechnology (NT), environmental technology (ET), aerospace technology (ST), which have high added value and are the basis of next-generation growth industries. This is an essential strategy for inducing high growth by discovering future growth engines centered on high value-added next-generation industries in Korea, where natural resources are limited.

In 2002, efforts were made to enhance the ability to plan science and technology development at the national level, such as creating a national technology road map for 99 key technologies and a vision for science and technology 10 years later. As a result, national R&D projects have also been focused on leading the industry and discovering specialized technologies in the public welfare field, which the government should focus on nurturing. As one of these efforts, the Roh Moo-hyun administration has made efforts to continuously increase the government R&D budget with the goal of increasing government R&D investment by 7% of the government budget by 2008.

[Figure 6-3] R&D ratio to GDP



Source: OECD Main Science and Technology Indicators data base

Korea also needed to systematically and efficiently utilize overseas scientific and technological resources to overcome the domestic poor scientific and technological research resources, and to participate in solving global problems such as global warming, environmental pollution, and energy depletion as a member of the international community. Against this background, in 2001, the 'Science and Technology Internationalization Promotion Strategy' was announced. According to this science and technology internationalization promotion strategy, the basic direction to promote the whole government's international cooperation in science and technology more

systematically was set, and each project was more systematically coordinated and promoted.

On the other hand, in the 2000s, there was a shift in the internationalization base construction project for international science and technology cooperation that Korea had promoted so far. In the 2000s, a new type of international cooperation project called the project to attract excellent overseas research institutes was added. In the 2000s, as the size of the Korean economy grew and the role that the international community demanded of Korea increased, international cooperation with regional multilateral councils in science and technology international cooperation expanded. In 2003, the EU and the EU agreed to sign a science and technology cooperation agreement, and promoted various multilateral scientific and technological international cooperation, such as participation in the International Thermonuclear Experimental Reactor (ITER) project and the 6th EU Framework Program. In order to promote the exchange and utilization of excellent overseas scientific and technological manpower, an incentive system was prepared, such as simplifying the procedure for obtaining nationality for foreign highly skilled manpower.

With the establishment of a science and technology internationalization promotion strategy, an annual science and technology internationalization plan was established and executed, thereby promoting international science and technology cooperation more stably. Partner countries have been expanded to include Central and South America, Africa, Southeast Asia, as well as Western advanced countries and northern countries, and international organizations and regional cooperation organizations (APEC, ASEM, etc.) have been added. It is participating in various multilateral science and technology cooperation programs such as the EU Framework, EUREKA, CERN (Conseil Européen pour la Recherche Nucléaire), and HFSP (Human Frontier Science Program).

1.7 Science Diplomacy Period (2013-2022)

According to the World Economic Outlook data of the International Monetary Fund (IMF) on the 26th of December 2021, Korea's nominal gross domestic product (GDP) for this year was estimated to be 1823.9 trillion dollars (about 216.6 trillion won) in US dollars.

This is the 10th largest among 191 countries in the world. Korea's GDP next year is expected to be 1.97.7 trillion dollars (about 226.6 trillion won), and it is also expected to rank 10th in the world. If the IMF's forecast is realized, Korea will maintain the world's 10th largest economy for three consecutive years from last year to next year.²⁶ At the 68th Trade and Development Council held at the Geneva headquarters in Switzerland on the 2nd of July (2021, local time) of the United Nations Conference on Trade and Development (UNCTAD), 195 member countries unanimously changed Korea's status from 'developing country group' to 'developed country group' changed The developed countries group has increased to 32 countries with Korea included. Since the establishment of UNCTAD in 1964, Korea is the first country to change its status from a developing country to a developed country.

Korea has already been classified as an advanced country in the international community in terms of economy and diplomacy. According to the International Monetary Fund (IMF) report, Korea ranked 10th in the world with a gross domestic product (GDP) of \$1.55 trillion in 2020. According to the World Trade Organization (WTO), Korea's exports reached \$512.5 billion last year, ranking 7th in the world. Last year, GDP per capita was \$31,497, ranking 26th in the world²⁷

As Korea's contribution to the international community is requested, Global problems such as climate change, biodiversity and Covid-19 are getting serious, and the role of science and technology innovation is becoming more important in solving these problems. Global protectionism is intensifying and trade conflicts and competition for hegemony between the US and China are intensifying. In this period, pursuing strategies to create new industries and markets by combining science and technology and ICT, and to create good jobs by strengthening existing industries with three trends; Leading the achievement of the national income of 30,000 dollars through the role of science and technology, Pursuing investments to improve the quality of life such as culture, welfare, safety, and environment, focusing on R&D investment for the purpose of economic

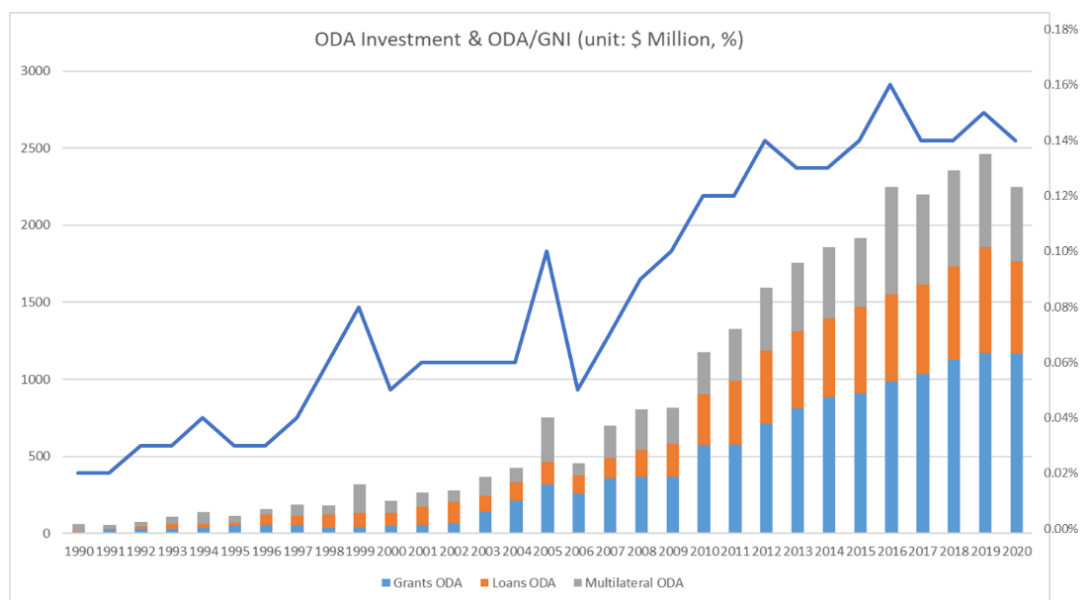
²⁶ <https://www.yna.co.kr/view/AKR20211224134000009>

²⁷ <https://www.hani.co.kr/arti/opinion/editorial/1002068.html>

growth and Paradigm shift to a leading R&D system such as activation of open technological innovation, creating an innovative ecosystem where challenges and growth occur actively.

It corresponds to the globalization of science and technology and the strengthening of the strategic nature of the 3rd (2013-2017) and 4th (2018-2022) Science and Technology Basic Plan. It intends to expand its contribution to the international community by expanding ODA, establishing the Science and Technology Diplomacy Center, and promoting the Asian research framework for resolving common issues in Asia. In this time, Korea secured donor country from recipient country with joining the Development Assistance Committee (DAC) (2009) and loan donor country (2016). In addition, Korea abandoned WTO developing country status (2019) as a result of index showing the developed country such as the 12th in terms of gross domestic product (GDP), 6th in export, and US\$30,000 in national income (joined the OECD in 1999) and so on. In this circumstance, Korea has strengthened international cooperation and promoting joint research that contributes to solving common human problems such as infectious diseases, energy, and fine dust, etc. The contribution to the ODA shows the increase of ODA investment in [Figure 6-4] below.

[Figure 6-4] ODA investment & ODA/GNI



Source: OCED Stat

The concept of science and technology diplomacy has not been established, but it is used to include an active role rather than international cooperation in science and technology. The efforts to secure international leadership in line with the national status based on quantitative expansion of international science and technology cooperation and a strategic approach. In this period, Korea reinforced the role of international cooperation in science and technology such as support for national diplomacy and entry into the global market.

2. S&T IC vs. S&T Diplomacy

2.1 R&D Program (S&T IC R&D Budget)

It is not easy work to draw the data of S&T IC R&D budget. Because we do not have the index of S&T IC in budget system in Korea as other countries and just can draw the approximation of data through the summation of R&D programs related S&T IC. In 2021, 9 Ministries, 32 programs are related to the S&T international cooperation with MSIT (19 programs, KRW 285 billion) and MOTIE (3 programs, KRW 165 billion). The S&T IC R&D budget and trends from 2017 to 2021 is shown in [Table 6-6] below.

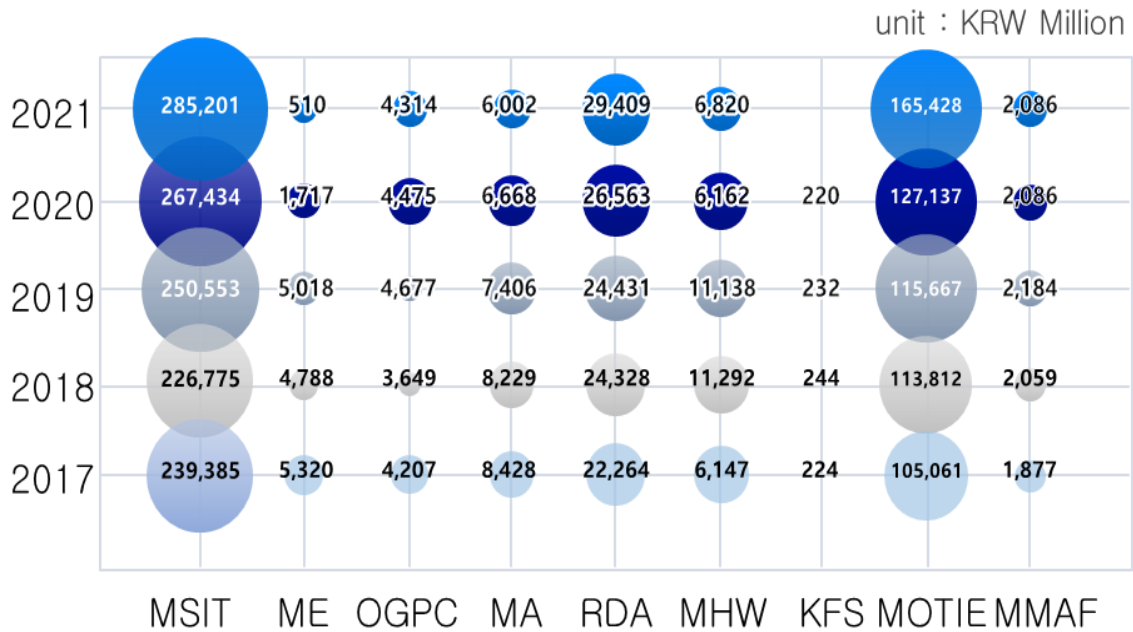
[Table 6-6] The budget of S&T IC R&D

category	2017	2018	2019	2020	2021	Increase rate
S&T International Cooperation (ratio)	0.39 (2.0%)	0.40 (2.0%)	0.42 (2.1%)	0.44 (1.8%)	0.50 (1.8%)	7.5%
Government R&D budget	19.46	19.67	20.53	24.22	27.20	9.2%

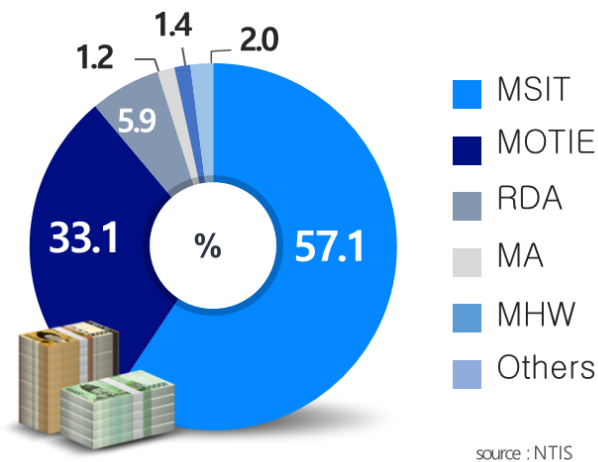
source : NTIS

The budget of S&T IC R&D can be divided into ministries and years as shown in figure below. The majority of this budget is consisted of ministry of Science and IT and ministry of Trade Industry and Energy as shown in [Figure 6-5] and <6-6].

[Figure 6-5] The budget of S&T IC in terms of ministries

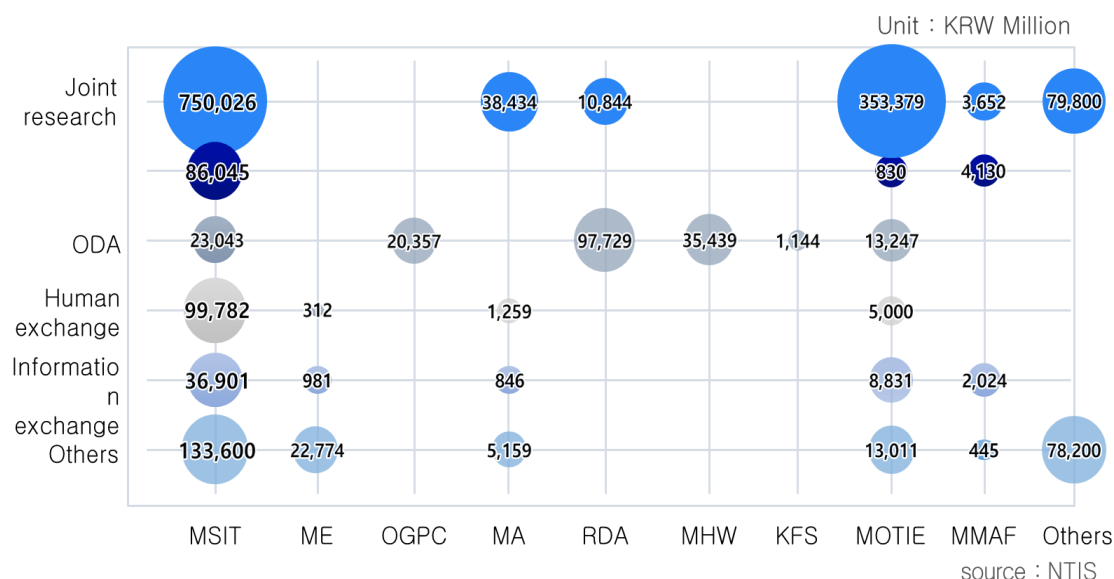


[Figure 6-6] The ratio of budget of S&T IC in terms of ministries



According to the type of S&T IC, the main types are joint research, ODA, human exchange, infrastructure, information exchange in that order in the base of international cooperation (2016~2020) as shown in [Figure 6-7] below.

[Figure 6-7] The budget of S&T IC in terms of IC types



The S&T areas of IC can be divided into construction / transportation, disaster and safety, aerospace / maritime, defense, machinery / manufacturing, materials / nanotechnology, agriculture / forestry / fishery / food, life / welfare / medicine, energy / resources, environment / weather, ICT / SW and others. The main S&T areas (2019) are energy/resource, aerospace/maritime, life/welfare/medicine in that order as shown in [Table 6-7] below.

[Table 6-7] S&T Areas of international cooperation

S&T area	International cooperation (2019)	
	Funding, KRW Billion (%)	No. of projects (%)
Construction, Transportation	3.8 (1.0)	22 (1.8)
Disaster and Safety	3.9 (1.0)	16 (1.3)
Aerospace, Maritime	45.8 (12.0)	26 (2.1)
Defense	5.3 (1.4)	7 (0.6)
Machinery, Manufacturing	12.1 (3.2)	50 (4.0)
Materials, Nano	19.5 (5.1)	156 (12.6)

S&T area	International cooperation (2019)	
	Funding, KRW Billion (%)	No. of projects (%)
Agriculture, Forestry, Fishery, Food	7.0 (1.9)	97 (7.8)
Life, Welfare, Medicine	39.8 (10.5)	217 (17.6)
Energy, Resources	94.9 (24.9)	147 (11.9)
Environment, Weather	14.5 (3.8)	59 (4.8)
ICT-SW	20.4 (5.4)	119 (9.6)
Others	113.4 (29.8)	320 (25.9)
Total	380.6 (100.0)	1,236(100.0)

Source: Ministry of Science and ICT internal document

The typical program for IC is S&T IC Program, which is one of National R&D Programs and performed by ministry of science and ICT. The [Table 6-8] below shows the ministry of science and ICT's R&D in 2022 and the location of S&T IC program in the structure of R&D.

[Table 6-8] 2022 Ministry of Science and ICT's R&D

Classification	S&T (KRW Billion)	ICT (KRW Billion)
Basic Research	- Basic Research Program(2,013.6)	-
Original Research	- Original Technology Development Program(1,287.1)	- ICT Research Development Program (1,042.1)
	- Space Technology Development program(412.5)	
	- Nuclear Research Development Program(247.1)	
	- Nuclear Fusion Research Program(73.9)	
	- Light Source Research Program (130)	

Classification	S&T (KRW Billion)	ICT (KRW Billion)
	- Social Problem Solving Research Program(51)	- Standardization Research Program(41.2)
	Sum (2,201.6)	Sum (1,083.3)
Commercialization	- Industry-Academia-Research Institution Cooperation / Technology Commercialization Program (288.5)	- ICT Commercialization Program(76.9)
Human Training	- S&T Human Training Program(181.9)	- ICT Human Training Program(117.2)
Foundation	- S&T IC Program(51.4)	- ICT Foundation Program(96.2)
	- International Science Business Belt Program(312.1)	
	Sum (363.5)	
Total	(5,049.1)	(1,373.6)

Source: Ministry of Science and ICT internal document

The S&T IC program belongs to category foundation of R&D and occupied very small volume of budget in the total R&D of ministry of Science and ICT. This means the ratio of S&T IC program's volume should be increased and sub-divided into specific areas such as regional division and cooperation type and so on.

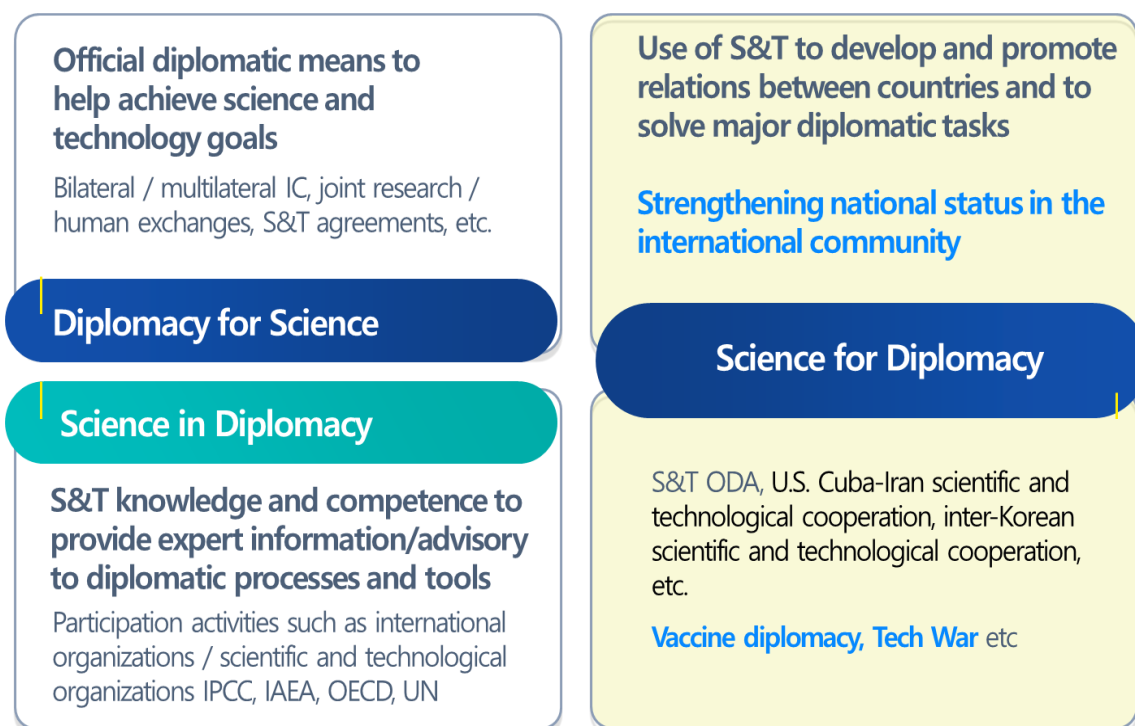
Current Issues can be divided two categories the definition of S&T diplomacy and other issues related to the competition of emerging technology between US and China.

2.2 Definition of S&T diplomacy

Nowadays there are a lot of mentions about science diplomacy in the S&T IC area, foreign affairs, public diplomacy, international organization and so on. But it is impossible to find the definition of science diplomacy in official document and there are many similar words such as digital diplomacy, technology diplomacy, cyber security diplomacy, etc. Even though we do not have official definition of science diplomacy,

the royal society and AAAS divided science diplomacy into 3 dimensions of activities as shown in [Figure 6-8] below.

[Figure 6-8] Three dimensions of S&T Diplomacy



Source: New Frontiers in Science Diplomacy (2010)

First, the diplomacy for science is official diplomatic means to help achieve science and technology goals such as bilateral and multilateral IC, joint research, human exchanges, S&T agreements etc. Second, science in diplomacy means S&T knowledge and competence to provide expert information and advisory to diplomatic processes and tools such as participation activities in the international organizations and scientific and technological organizations (IPCC, IAEA, OECD, UN, etc.). In these three dimension has limitation to classify the current issue, reshaping of the global supply chain, which seems to be close to the hard power rather than soft power. Compared to the science for diplomacy as soft power, the new category is put into the dimensions of S&T diplomacy. The four dimensions of S&T diplomacy are shown in [Table 6-9] below.

[Table 6-9] Four Dimensions of S&T Diplomacy

Classification	S&T as hard power	S&T as soft power
Command-based leadership	Diplomacy in Science: A type of competition for S&T resources (economic power: hard power) - S&T diplomacy to secure a stable supply chain	Science in Diplomacy: Controlling S&T standards, institutions, norms, etc. (e.g. COP26) to make others do things they don't want them to do - S&T serve as scientific advice (soft power)
Persuasion-based leadership	Diplomacy for Science: A type of cooperation through persuasion and argument based on scientific and technological achievements (hard power) - S&T IC	Science for Diplomacy: A type of using S&T as a persuasive resource (soft power) for incentives and rewards - Active use of various forms of S&T (ODA, CERN, Track 2, etc.)

The existing three dimensions are corresponded to command-based leadership (science in diplomacy) and persuasion-based leadership (science for diplomacy) as soft power and command-based leadership (diplomacy in science) and persuasion-based leadership (diplomacy for science) as hard power. The division between command and persuasion is related to the relative strength of cooperation and competition, while the difference of S&T between hard power and soft power influence is based on the relative closeness to the economic values and common human values. As the importance of S&T diplomacy is increasing and the realms of science and technology and diplomacy are converging, the boundary of dimensions of S&T diplomacy is getting more blurred.

2.3 Conflict between US and China

As mentioned in 2.2, the conflict between US and China made the new definition of S&T diplomacy emerging and us face the drawing science and technology diplomacy strategies. As the era of free trade and cooperation has passed, and the era of

protectionism and competition has arrived, China and the United States are trying to build their own empire by uniting friendly forces.

Countries such as Korea, which pursued cooperation between China and the United States at the same time, faced a moment of choice, and for a considerable period of time had concerns about making a strategic choice in a conflict situation. In particular, in the case of the semiconductor supply chain, economy and security are converging, and science and technology are directly affecting the era of economic security like technological sovereignty. Therefore, setting a direction based on international norms is important, and rather than choosing one of the two, it is the direction that the United States and China will seek to achieve the necessary cooperation based on principles.

2.4 Implication from Korea's case

In addition to Korea's economic development (in connection with the economic development plan), international cooperation in science and technology has also been developed. First, the external environmental influence such as world economy, end of cold war, climate change, etc. should be carefully considered. In the case of Korea, international cooperation is relatively low importance (investment ratio) compared to other fields such as emerging technology, basic research, infrastructure and so on. Implementation of international cooperation is not highly feasible due to separation of plan and its budget, which is applied to other fields as well. Now, as an official advanced country, enhancing international leadership by strengthening science and technology diplomacy is a big task.

Before the planning of master plan, the necessary data should be secured. In the absence of data, policy can be established on the base of survey or text mining. Based on the data, current status of S&T IC needs to be evaluated and this is the start of planning. After that, the goal of S&T IC should be set up regarding the external environment and internal other plan and schemes. According to the Lee and Lim (2001), they suggested three types of technological pursuits of latecomers such as path-following catch-up, path-skipping catch-up and path-creating catch-up. Lao PDR needs to select the way to obtain the goal.

3. References

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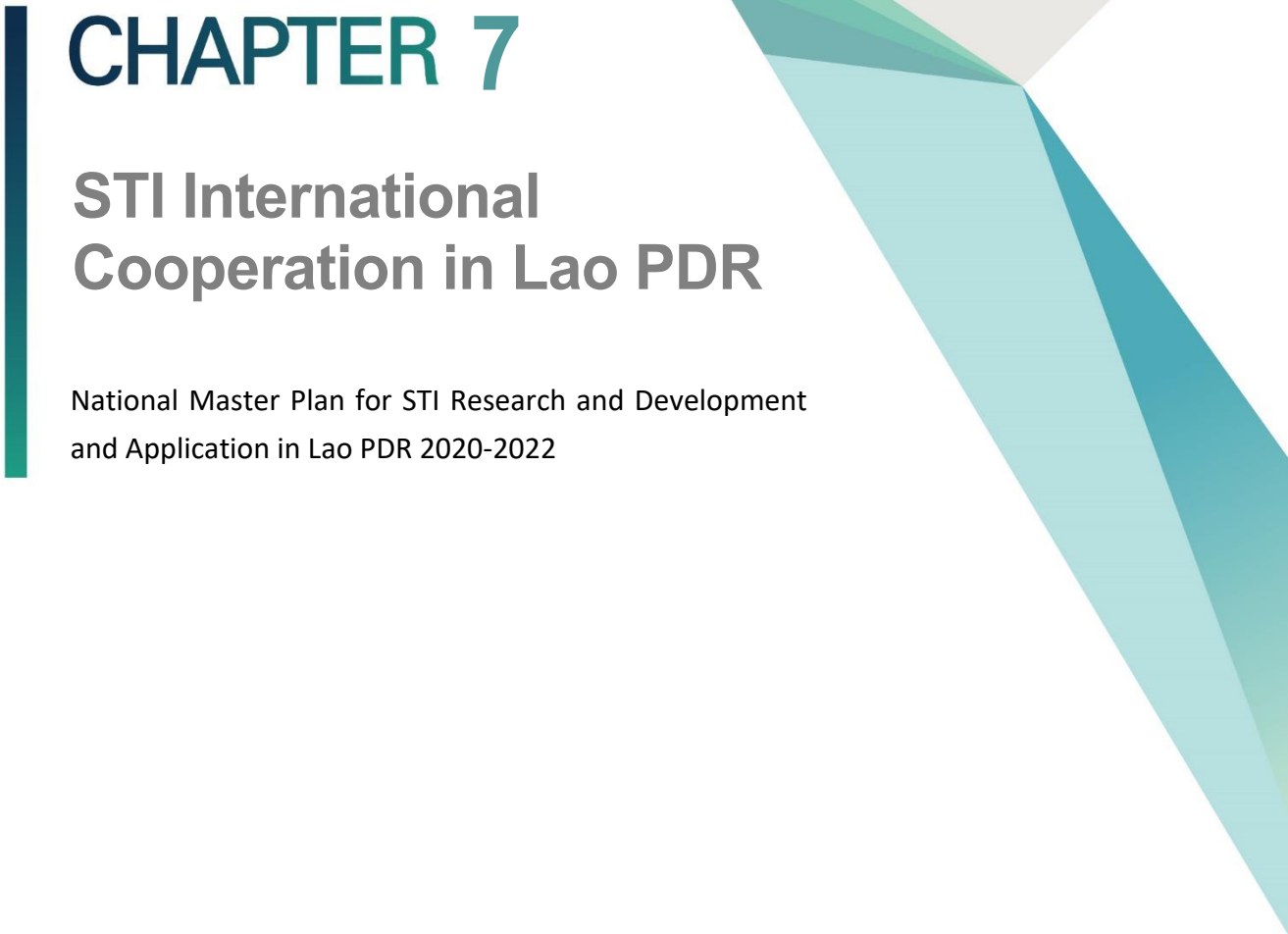
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CHAPTER 7

STI International Cooperation in Lao PDR

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 7. STI International Cooperation in Lao PDR

Prof. Vimontha KHIEOVONG, National University of Lao PDR

1. Introduction

1.1 Background

Lao People's Democratic Republic is the landlocked country in Southeast Asia. Currently, Lao PDR has a total population of 7,379,358(2021)^[1]. with a GDP per capita of USD2,568(current US\$,2018), Lao PDR is classified among the group of lower middle-income countries. The Human Development Index (HDI) 2018 ranked Lao PDR as 139 out of 189 countries, which is on the category of "medium human development", Laos' human development improves slightly as the ranked country in 137 of 189 counties. Over the past twenty years, Lao PDR has made great progress in social and economic development, including halving the absolute poverty rate, reducing hunger, and improving education and health outcomes ^[2].

In recent years, science and technology development has advanced rapidly. Therefore, Lao PDR should seize this opportunity for socio-economic development, especially the application of science and technology in production, services, and human resource development to achieve higher quality growth and to be able to compete at the regional and international levels, in addition to improving public governance and administration to be more efficient in revenue collection and expenditure.

According to the implementation report of 9th Socio Economic Development Plan for (2021-2025) is taking place at a critical moment, both domestically and international, offering important opportunities for Lao PDR to achieve its development goals. The key factors that will drive development in the coming five years are 1) young population, 2)

rich natural resources, 3) geographical location of Lao PDR and growth of countries in Asia, including neighboring economies and 4) application of modern science and technology. There are 6 goals relates to technology and communications sectors 1) The national economy is growing in a quality, balanced and sustainable manner. 2) Improved human resources, with the ability to research and apply science and technology and wisdom to promote diversified production and services with higher value. 3) The material and mental well-being of the people has been gradually improved. 4) Development in the direction of green and environmentally friendly. 5) Strong and modern cooperation, integration, and interconnection within, with the region and internationally. 6) Effective public administration and society are sacredly protected in accordance with the direction of public administration by law.

The purpose is to focus on international Science, Technology, and Innovation (STI) collaboration that relate to of 9th Socio Economic Development Plan for (2021-2025), the key factor 4 which is application of modern science and technology as Science, Technology, and Innovation. In addition, follow the goal5 to strong and modern cooperation, integration, and interconnection within, with the region and internationally. Therefore, this is to report the status of international STI collaboration in Lao PDR.

International STI collaboration can improve STI systems of country and build STI capabilities by learning from international experiences and benefiting from external sources of knowledge, expertise, and technology. Improvements can help them learn better about existing technologies as well as creating new knowledge and innovations

1.2 International environment

Developed and developing countries, in addition to developing science and technology to be a force for national development, also invest in new innovations, increase the production of various modern technologies and cooperation aimed at sharing lessons and technology, and foster the development of human resources in terms of STI.

In the ASEAN Plan of Action on Science, Technology and Innovation (APASTI), [3] implementation plan includes activities that are supportive of both social and economic developments and consistent with the relevant characteristics of the ASEAN Economic Community (AEC) 2025 as well as the ASEAN-Socio-Cultural Community (ASCC) Blueprints 2025 . The strategic thrusts (1) to collaborate between public and private sector to strength strategic collaboration among academia, research institutions, networks of centers of excellence, and the private sector to create an effective ecosystem for capability development. (2) to enhance mobility of scientist and researchers, connecting people to people and enhancing the participation of women and youth in STI. (3) to Establish innovative system and smart partnership with dialogue and other partners to nurture STI enterprises to support MSMEs, nurture knowledge creation and STI applications to raise competitiveness. (4) Raise public awareness and strengthen STI enculturation to enhance ASEAN science and technology cooperation.

ASEAN-China STI Cooperation [4], STI cooperation is further enhancement between ASEAN and China in area of mutual interest and promote innovation driven development. ASEAN-China STI cooperation , to explores new and sustainable cooperation mechanism, to enhance exchanges of STI talents to promote regional S&T development and capacity building, by continuing the Talented Young Scientist Program, training courses of advanced applicable technology and S&T management, to identify key priority to further enhance the innovation agenda through exchanges of information, experience and expertise in research and development, technology transfer and commercialization. Collaboration among science park to share the best practices and experiences while promoting and intensifying the usage of existing STI parks to enhance innovation and networking. collaboration in technology transfer through the sharing of experiences and best practices, and further enhancing collaboration among research institutes, higher learning institutions and enterprises among all ASEAN Member States and China. Utilize the ASEAN-China Cooperation Fund to implement all related activities under ASEAN-China STI initiatives.

Laos and the Republic of Korea have been developing bilateral relations and cooperation more and more over the past two decades since they established diplomatic relations on October 25, 1995 and in 2019, Laos and the Republic of Korea [5] had signed three

memorandums of understanding (MOUs) to further the strengthen the cooperation which had three cooperation agreements on intellectual property, technical cooperation, and vocational cooperation. In bilateral meeting, two sides highly valued friendship and reviewed cooperation of the past, and to enhance the bilateral cooperation in the fields of politics, diplomacy, trade and investment, agriculture, energy, and infrastructure development, and witnessed the signing of four cooperation documents in the areas of ICT, industry-trade, agriculture, and economic development.

1.3 Domestic environment

History of Science and Technology in Lao PDR ^[7,8], In August 1988, the Council of Ministers changed the name of the National Scientific Technique Committee to establish the Ministry of Science and Technology. So that the new Ministry of Science and Technology support economic-social development, the government designated that it should work in four areas: 1. Scientific and technological services, 2. Scientific research and technological development, 3. Science and technology management, 4. International co-operation in science and technology. During this period, the role of science and technology was focused on international co-operation activities with foreign countries and international organizations. By June 1991, the First Workshop on the Formulation of Basic Sciences and Technology Policy was organized in Vientiane.

On 23 July 1997, Lao PDR participate the Association of Southeast Asian Nations (ASEAN). This was an important step forward to work co-operate on science and technology at regional level.

In 2003 the First National Science and Technology Policy was released and was implemented up until the year 2010. The policy outlined five main goals related to science, technology and innovation. The first goal improved R&D to ensure the mobilization of STI resources. The second goal is to upgrade education for higher STI knowledge, the aimed to provide agricultural, industrial and services development work for the poverty reduction strategic program and to the industrialization and modernization of the country. The third goal increased the coordination and cooperation between various sectors and research institutes to ensure the cooperation

of the production sector and the scientific community to obtain the maximum benefit of the country from STI activities. The fourth goal upgraded and strengthen the public's awareness on science and engineering knowledge. The fifth goal promoted the technological innovation and adaptation system. These policy goals intent to be accomplished through increased funding and international cooperation.

The 2003 STI policy called for a reorganization of government agencies and international cooperation to compensate for weak capacity. In 2007, due to the expansion of science, technology and environmental activities beyond Vientiane to the provinces, the government transformed the Science, Technology and Environment Organization into the National Organization of Science, Technology and the Environment. With the purpose of improving the efficiency implementation of its activities, there are two new organizations: Resources and Natural Environment (RNE) and The National Authority of Science and Technology (NAST).

In 2007, the National Authority for Science & Technology (NAST) was under the Prime Minister's Office. NAST acted as a secretary to central government and managed all the science and technology activities in the country, intellectual property rights, standardization and metrology, even formulating legislation.

On 28 September 2011, a decree of the Prime Minister of Lao PDR No.309/PM the government of Lao PDR established the Ministry of Science and Technology officially, which is (re) created the Ministry of Science and Technology (MOST) from the original operation between 1988 and 1993. MOST is addressing important challenges such as: (i) Improvement of the organizational structure for the design and implementation of STI policies; (ii) development of the legal framework related to science, technology and innovation; (iii) capacity building and human resources development; (iv) infrastructure development; and (v) normalization and standardization functions, in the priority fields of Science and Technology are : (a) Biotechnology and Ecology,(b)Renewable Energy and Materials, (c) Information Communication and Technology(ICT).

Since MOST was establishment, several laws have submitted to National Assemble and first proposed laws were on intellectual property and electronic transactions. These laws are placed to encourage socio-economic development and facilitate integration into the

regional and international cooperation, especially in the relation to the ASEAN Economic Community.

Policy on Science and Technology within the framework of Socio-Economic Development Plan, the regional and international aspect of the STI policy is to link with international organizations to carry out scientific research and development, using a wide scope, especially in the priority areas of development.

The ASEAN Plan of Action on Science, Technology and Innovation (APASTI) in the period of 2016-2025, it opened the opportunity for the Lao PDR to improve the quality of education through exchange and collaboration with high-standard educational institutions in advanced ASEAN countries such as Singapore and Malaysia.

The Ministries relate to Science and Technology has collaborated with regional, international to promote and develop technology as well as human resource development.

Ministry of Technology and Communication (former Ministry of Science and Technology and former Ministry of Post and Telecommunication). The cooperation on ICT has taken step forward especially the cooperation on mutual benefits:

- (1) Cooperation with China and Vietnam: signed a Memorandum of understanding (MOU) at both ministry and department levels in various areas. Sending 19 staffs to study master's degree in the field of ICT. In 2019 Lao staff received master and PhD 5 scholarships per year and continually in next 3 years and received short term and long-term training grants in the framework of cooperation with China more than 219 grants. Received 4 aid projects:
 - -The project on upgrading the institution of Information Communication and Technology (ICT) for 10million US Dollar.
 - -The project on Digital TV in 100 villages for 8 million US Dollar.
 - -The project on information linkage together with lower Mekong River countries, the project value is 139,500 US Dollar.

- The project to organize seminar/meeting to share the knowledge and experience in ICT sustainable development with lower Mekong River countries, the project value is 40,900 US Dollar.

(2) Cooperation with ASEAN and subregion

Completed the preparation and participated in the annual meeting in the ASEAN and sub-regional cooperation framework. Successfully hosted and chairmanship of the 19th ICT Ministers' Meeting (TELMIN19) and the 20th ASEAN ICT Senior Officials' Meeting (TELMIN20) as part of the ASEAN cycle in Vientiane.

received fund in the ASEAN and sub-regional cooperation framework for 10 activities worth 340,924 US dollars, which contribute to develop the ICT of the Lao PDR.

(3) Bilateral cooperation and other international organization

Signed a memorandum of understanding (MOU) on cooperation at the ministerial level in various areas with the Republic of Korea. Vietnam, China, S. Korea and Thailand; At the same time, the Ministry received a total of 30 grants for seminars, trainings/activities from APT and ITU; Signed a Statement of Intent agreement between the Ministry and the Microsoft Company on the subject of Youth Online Learning Platform.

(4) Lao-China Cooperation project to improve Science, Technology and Innovation.

Ministry of Technology and Communication (MTC) have development plan for five year (2021-2025) as the goal9: Strengthen cooperation with foreign countries within the framework of bilateral, ASEAN, sub-regional and international organizations, pursuant to the foreign policy guidelines of the Lao PDR, peace, independence, friendship, and cooperation, and implement multi-directional, multilateral and multifaceted cooperation to implement the work plans and enhance the role of the MTC. MTC has plan to cooperate with ASEAN, sub-regional, the framework of bilateral, international organization.

Ministry of Industry and Commerce integrated with the region and international according to develop plan for 5years (2021-2015). International cooperation: (1) with Russia for mining, (2) with Republic of Korea to develop on commercial and investment, (3) with Japan to transfer technical on industrial and handicraft.

Organization cooperation: (1) Mekong Lancang Cooperation (MLC) work on industrial, handicraft, agriculture and establish the national chemical database, (2) GIZ for wood industrial and commercial, (3) EU cooperate the field of wood processing, (4) ASEAN for wood product to endorse the ISO on wood-based product.

The law relates to international collaboration:

(1) Law on the promotion of small and medium sized enterprises, No 011/N.A
21December2011

- Article8: International Cooperation

The government promotes the relationship and cooperation with foreign counties, regional and internationally on SME promotion activities, especially on the field of productivity, by exchange knowledge and information; the enhancement of their human resources' knowledge capacity, transferring technology acquiring assistance and others, basing on the independence, equal and benefits of both sides, in accordance to the international agreements and conventions, that the Lao People's Democratic Republic is a signatory members

- Article19: The cooperation between SMEs and large sized enterprises and foreign investment enterprises

The government promotes the cooperation between the SMEs and the large sized enterprises and foreign investment enterprise, by establishing an enabling policy and favorable mechanism and enacting necessary legislation and measures.

The large size enterprises and foreign investment enterprises shall give opportunity to the SME, on supplying raw materials, semi products, commodities a service as their sub-contractor network, and to enhance the strengthen for SMEs by transferring advances technology and skills on management.

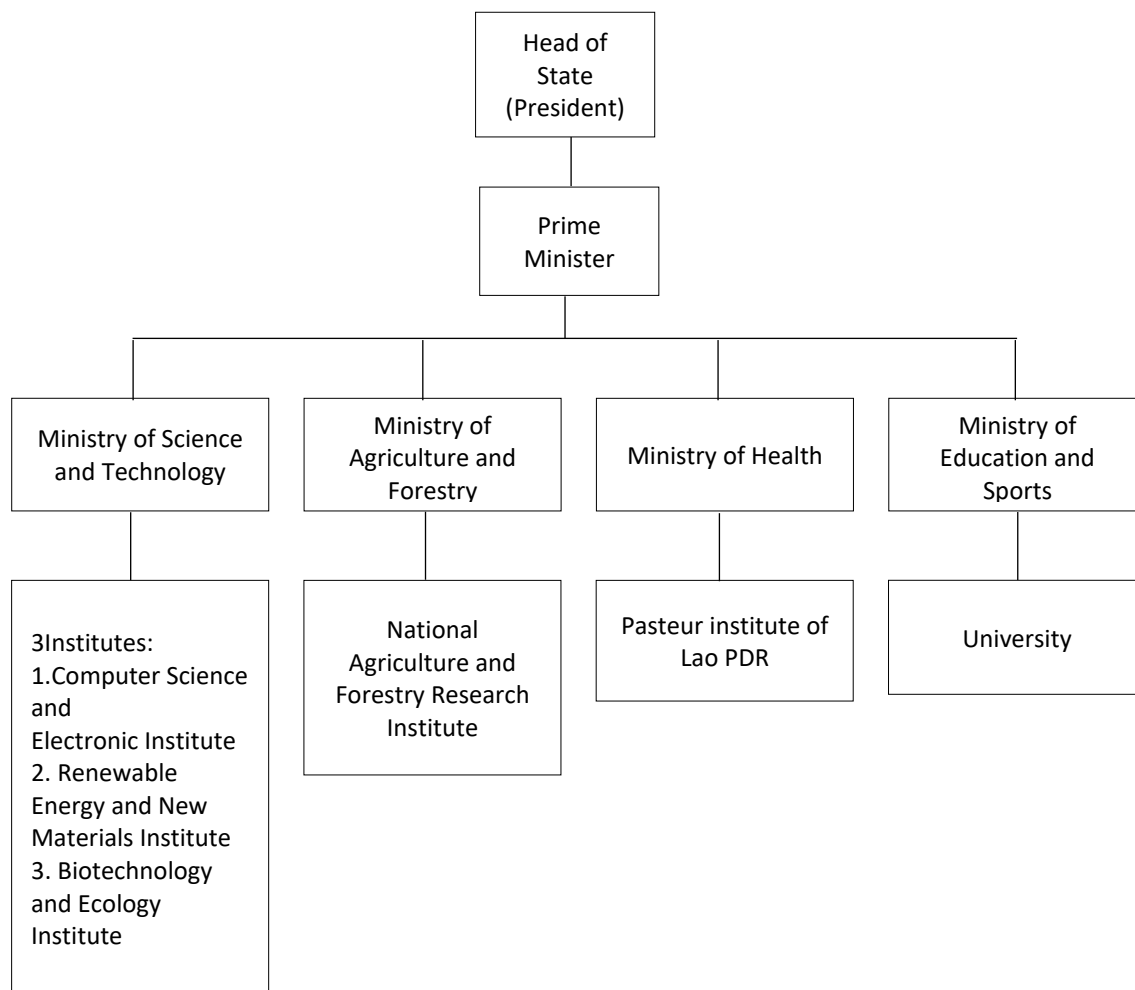
The SMEs shall improve ability on management and technology development to ensure the effective provision of quantity and quality of commodities and services suitable to the requirements of the large sized enterprise and foreign investment enterprises.

2. Policy Analysis

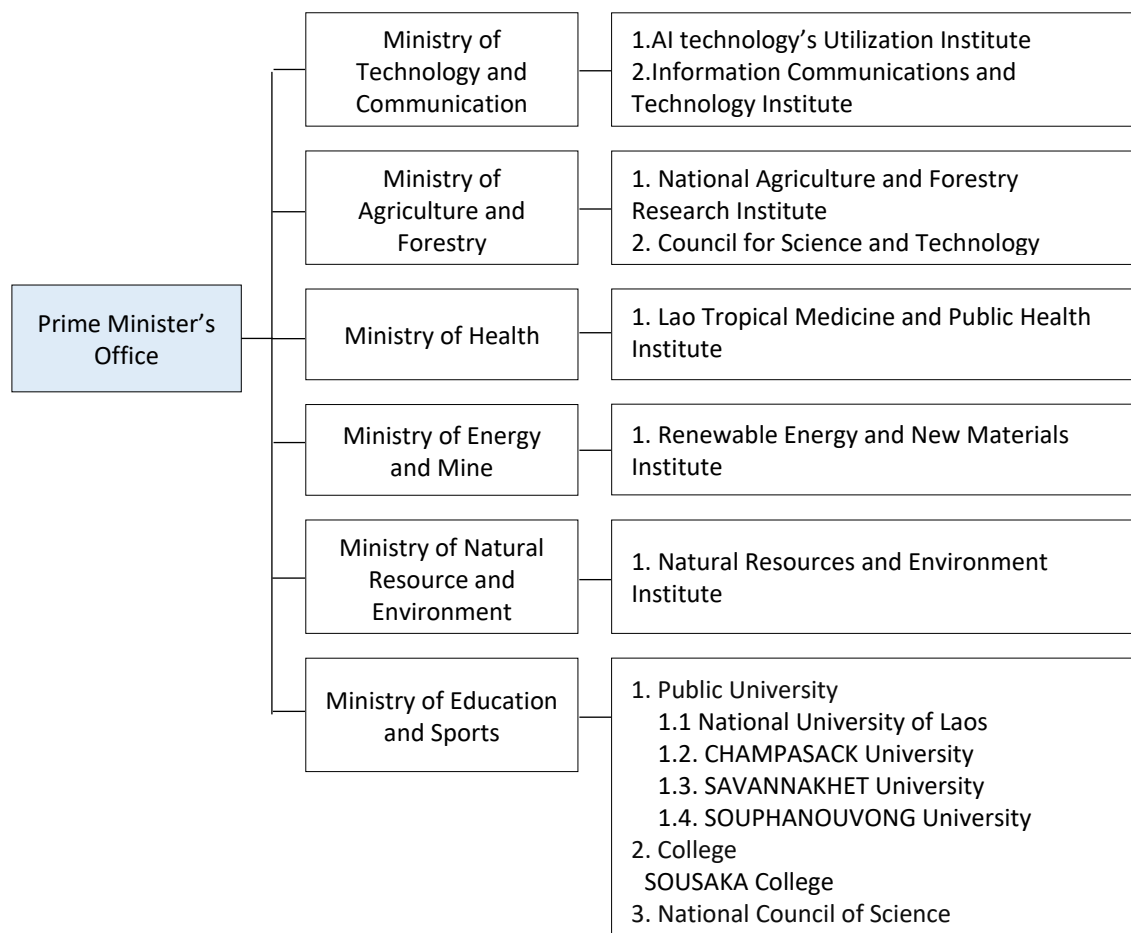
2.1 Institutions

In the 2014, The institutions related to STI which is under Ministry of Science and Technology, Ministry of Agriculture and Forestry, Ministry of Health and higher education is under Ministry of Education and Sport. The organization chart is shown in [Figure 7-1], currently institute reorganize and under new ministry as show in [Figure 7-2].

[Figure 7-1] Organization Chart Related to S&T, 2014



[Figure 7-2] The institutes related to STI Organization chart, 2022



- AI Technology's Utilization Institute

The institute is reformed from Computer Science and Electronic Institute (CSEI) and under Ministry of Technology and Communication which has role in research, development software, hardware, network, automation, transfer and providing technical services related to computer science, electronics and AI technology. The institute conducted range of research and development projects such as development of Lao Fonts, E-book and E-dictionary.

- Information Communications and Technology Institute

The institute play role in researching, creating institutional curriculum, and developing personnel. The vision is to develop the institute as a leading institute in the field of posts

and telecommunications to reach the national- regional an international standard on ICT education, research and development.

- Renewable Energy and New Materials Institute (REMI)

REMI has role in research, development, transfer, promotion, applications and services related to renewable energy and new materials. REMI conducted several research and demonstration projects. In addition, REMI organized training courses on renewable energy technology and applied.

- Biotechnology and Ecology Institute (BEI)

BEI has role in research, development, application and service in biotechnology and ecology. BEI was set up as an herbarium and a natural history museum and has a function to disseminate and raise awareness on science and ecological systems. BEI conducts some research activities to apply biotechnology to identify plant and animal diseases.

- Office of National Council of Science

Referring to Prime Minister's decree number 04, dated 28 January 2002, the National Council of Science (NCS) is responsible for setting up the Cabinet Office of NSC that will assist NSC and Ministry of Education and Sports in coordinating with research institutes, universities, ministries and equivalent agencies across the country for research and development on Science and Technology, and for promoting NCS activities. * In the past, National Council of Science was under Ministry of Science and Technology*.

- Council for Science and Technology (CST)

The core mandate is to facilitate science and technology research and innovation in the Ministry of Agriculture and Forestry (MAF). CST contributes to the achievement of the MAF research outcomes.

- National Agriculture and Forestry Research Institute (NAFRI)

Modernized agriculture technologies are developed for improvement of agricultural production systems which support national food security and nutrition, commercialized production, and the use of resources is managed at a sustainable level. The main research themes are the following: (i) promotion of bio-diversity to create a gene bank, while deepening the understanding of native plants and wild animals; (ii) improvement of agricultural productivity, including a programme for each rice-plant

and livestock species and other programmes which focus on maize, cassava, and coffee; (iii) response to climate change is given priority, such as research on protecting crops from floods; and (iv) information provision to farmers (CRDS, 2015)

- Lao Tropical Medicine and Public Health Institute (Lao TPHI)

The role and functions of Lao TPHI are (a) serving as a central focal point for promoting, supporting, consulting, advising, coordinating and implementing health research; (b) developing regulation and standard measures to protect benefits for human subjects participating in research; (c) providing evidence based scientific research to decision makers for developing, improving health policy, strategies, action plans, and health work plans of the Ministry of Health(MOH) to enhance the quality of health services for better health of the population; (d) developing and improving health research data bases as a center of excellence, providing the evidence base and research outputs that can be synthesizes so as to offer policy recommendations; (e) providing in-country short-term training on Public Health management for health professionals at all levels, according to each five year socio-economic plan of the country; (f) providing regional long-term training on Public Health and tropical medicine in order to: (i) enhance the competence of junior medical doctors and public health professionals committed to improve ASEAN populations' health, (ii) deliver an international master degree, focusing on practice in priority health issues: Public Health, epidemiology, clinical medicine, therapeutics, research, and management, (iii) to foster and monitor health development projects based on students' initiatives and implementations, (iv) to create a forum for academic exchange and research, with an international perspective, and to expand scientific networks in the region; (g) co-ordinate for the Council of Medical Sciences and serve as secretariat for the National Ethics Committee for Health Research; and (h) other roles assigned by MOH.

- Institute Pasteur of Laos (IPL)

The IPL has a mandate from Lao Ministry of Health to fulfil activities of public services: (1) Research and diagnostic on emerging infectious diseases and vector borne diseases; (2) training, Education and Capacity building and (3) technical assistance to National Centre for Laboratory and Epidemiology (NCLE) for investigation of epidemics.

- Natural Resources and Environment Research Institute

The NREI play a role as secretariat to Ministry of Natural Resources and Environment

in studying, research on natural resources and analysis environment quality and provide technical services in the area of natural resources and environment. Its mandate includes implementing the policy, strategy, regulation, action plan on natural resources and environment; to promote research, development, and the utilization of prototype on natural resources and environment which best suitable for Lao context; and other tasks.

- National University of Laos (NUOL)

NUOL provide higher education in areas required for the socio-economic development of Lao PDR; performing research in natural and social sciences; preserving the arts, culture and traditions of the nation; and providing academic services to society. There is the degree of bachelor in all and some master's degree programs in some of its academic disciplines are available, and it is preparing to offer Doctoral degree programs. The university is part of the following regional networks: (a) ASEAN University Network (AUN); (b) ASEAN University Network/Southeast Asia Engineering Education Development Network (AUN/SEED-NET); (c) Agence Universitaire de la Francophonie (AUF) and (d) Greater Mekong Sub-Region Academic and Research Network (GMSARN).

- CHAMPASACK University

The main activities of CHAMPASACK University are (1) Providing Teaching-learning at the higher education level; (2) developing scientific, interdisciplinary and applied research; (3) providing capacity building and academic services base on communities-base in accordance to government policy for better well-being of its people. CHAMPASACK University has focused on agriculture and will be a center of excellence in Agriculture.

- SAVANNAKHET University

The mission of SAVANNAKHET University are (1) SKU Development is aligned with the socio-economic development strategy of the three provinces (Savannakhet, Khammouane and Bolikhamxay); (2) provide equal access to higher education for both the urban and rural people, in particular, underprivileged people from rural and remote area; (3) development of SKU must be closely related to national strategy of education development so as to gradually approach the national and international standard and (4) Centre for human resources development, in the central region.

- SOUPHANOUVONG University

The University conducts the educational functions to prepare the specialists, researchers and multidisciplinary scientific scholars; organizing scientific research; protecting and promoting Lao's unique national and multiethnic arts and culture; and to provide academic services to society.

- SOUSAKA College

SOUSAKA College has role to develop human resource in business administration and information technology fields to support the need of national socio and economic devilmment. SOUSAKA College promotes human resources at post diploma level, offering bachelor's degrees in business administration, accounting, IT and computer engineering.

- University of Health Sciences

The University of Health Sciences (UHS) is a unique health institution that produces health professionals at higher level (bachelor's degree) in Lao PDR. The University has three main functioning roles: training, service and research.

- The survey on STI organization in 2020, it can be classified into three categories as below

- Research and development organization

Scientific research organizations, research and technology development organizations, the organizations are created under the form of institutions, research centers which performs the work relate to the goal of national socio-economic development

- Higher education

University, Institute, College

- The Sector under (former) Ministry of Science and Technology

the amount of organization is shown in table 1, the main role and the most important on the STI is higher education, research and development respectively

[Table 7-1] Organization relate to STI based on the survey ,87 organizations

Description	Amount of organization	Percentage (%)
Research and Development	13	14.9%
Higher education	44	50.6%
Sector	30	34.5%
Total	87	100%

[Table 7-2] STI organization in public and private sector

Sector	Amount of organization	Percentage (%)
Public sector	69	79.31%
Private sector	18	20.69%
Total	87	100%

[Table 7-3] The research area in STI

Research area	Amount of organization	Percentage (%)
Natural Science	6	6.90%
Technical Science and Technology	21	25.29%
Medical Science and Medicine	6	6.90%
Agriculture Science	10	11.49%
Social Science and Humanities	38	43.68%
Other under (former) Ministry of Science and Technology	5	5.75%
Total	87	100%

[Table 7-4] Staff in STI

Number of staff	Organization			
	Research and Development	Higher education	Sector	Total
Staffs<30	7	14	12	33
30<Staffs <50	4	5	13	22
50<Staffs <100	1	10	4	15
Staffs>100	1	15	1	17
Total				87

[Table 7-5] Staff in public university

University	PhD	Master	Bachelor	Pre-bachelor	Total
National University of Laos	186	956	572	53	1,763
CHAMPASACK University	22	161	236	8	419
SAVANNAKHET University	11	145	170	6	332
SOUPHANOUVONG University	18	218	178	14	428

*[update2022]

This table6 represents the faculty staff in the National University of Laos (the Faculty of Engineering and the Faculty of Natural Sciences, these two faculties are related to STI which is shown in employee statistics, educational qualifications and academic position.

■ Faculty of Natural Science, NUOL:

The Faculty of Natural Sciences is one of 13 faculties in the National University of Laos which is leader of Science and Technology, consisting of the department of Mathematics and Statistics, department of Physics, department of Chemistry, department of Biology ,department of Computer Science and also provides academic services upon request from government sector and private sector to develop the socio economic beside to maintain and to promote cultural along with education, learning, exchange and adopting international culture selectively and also conducting research for the benefit of socio economic development. (<https://fns.nuol.edu.la/home/>)

■ Faculty of Engineering, NUOL:

The faculty has the role to equip engineers with solid scientific knowledge and technical skills to keep up with rapidly changing technology in the world place. Moreover, the faculty maintains an excellent reputation in higher education, fosters intellectual growth, provides a rich learning environment, supports and encourages the development of student's potentials. To further enhance quality training, keeping up with international development, and to expand research activities, FE/NUOL has also established collaborations and exchange programs with overseas universities. The administrator attaches great importance to the reform of education as well as the reform of learning to adapt to the current situation as much as possible by collecting and organizing academic guideline and planning work in educational institutions to provide student

with knowledge that is effective and can be used in the most practical applications. The range teaching program for bachelor's and master's degree have been launched by faculty to address the needs of national Socio-Economic development. Faculty of Engineering consist of the department: Civil Engineering, Mechanical Engineering, Electrical Engineering, Electronic and Telecommunication Engineering, Road and Bridge Engineering, Mining Engineering, Environmental Engineering, Transportation and Logistics Engineering, Chemical Engineering, Computer Engineering and Information Technology. Student in faculty of engineering will focus on theory and apply principles in conjunction with practical laboratory practice, they study and experiment under the guidance of professors with specific expertise, student will also be able to internship in companies or organizations related to their field study. (<https://www.fe-nuol.edu.la/>) (<https://www.youtube.com/watch?v=VZPKQUUGd1o&t=2s>)

[Table 7-6] STI staffs in National University of Laos

- Work type:

Faculty of	Lecturer	Administrator	Study abroad	Total(Female)
Engineering	156(37)	25(24)	3(1)	184(62)
Natural Science	94(41)	6(2)	8(4)	108(47)

- Academic Position:

Faculty of	Professor	Associate Professor	Teacher	Assistant teacher	Total(Female)
Engineering	0(0)	15(1)	39(7)	3(0)	57(8)
Natural Science	0(0)	4(0)	31(10)	33(20)	68(30)

- Educational qualification:

Faculty of	PhD	Master	Bachelor	Pre-bachelor	Total(Female)
Engineering	38(3)	98(31)	44(24)	4(4)	184(62)
Natural Science	23(7)	75(36)	10(4)	0(0)	108(47)

* annual report of faculty those faculties relate to STI

* number of staff (number of Female) [update2022]

2.2 Analysis of current situations

2.2.1 Descriptive Statistics

The primary mechanism for policy implementation was setting the target for the Gross Domestic Expenditures on R&D (GERD) between 4% and 5%. This was approximately 100 times the expenditures on R&D estimated by the 2002 R&D survey.

2.2.2 Governance

Ministry of Science and Technology (Currently Ministry of Technology and Communication) was established in 2011, this is new step to develop Science, Technology and Innovation. Ministry is one of the main actors with Science and Technology(S&T) governance structure in Lao PDR[10]. The ministry consists of seven departments (including Science, Technology and Innovation, Intellectual Property, Standardization, IT, human resources, Inspection and Planning and has three institutes: the Institute of Ecology and Biological Technology, the Institute of Renewable Energy and New materials and the Institute of Computer Science and Electronics. It hosts two offices and 18 science and technology bureaus at the provincial level. The ministry particularly focuses on three thematic areas: biotechnology and ecology, renewable energy and materials, and ICT. The main mission of the ministry is to apply scientific knowledge in order to reduce poverty and to overcome the status as “least developed country” by 2020. In order to implement this mission, five goals have been identified: Improvement of the organizational structure to promote policies, Development of the legal system related to S&T, establishing of sustainable innovation infrastructure, Development of human resources related to S&T, Standardization of local functions.

The government of Laos has dissolved the Ministry of Science and Technology, transferring its departments to various ministries. Departments under the ministry will be transferred, along with all public servants, to five departments under existing ministries including the Ministry of Education and Sports, Ministry of Post, Telecom and Communications (recently Ministry of Technology and Communication), Ministry of Industry and Commerce, Ministry of Agriculture and Forestry, and the Ministry of Energy and Mines.

Ministry of Technology and Communication (MTC) is a number of key functions related to the research, development, management, promotion and promotion of technology and communication (Prime Minister's Decree No. 625 / PM, October 22, 2021) are the following: 1) study on policies, strategies, laws and legislations of the technology and communications sector to submit to the Government. 2) Create, improve technological and communication infrastructure across the country to facilitate technology, innovation, telecommunications, internet, post and cyber security. 3) Develop, manage and promote the use of public administration systems and services with digital technology to the same standard as modern urban systems, public sector networks, public and integrated information systems, national database systems, digital identity systems, administrative systems, administrative systems, administrative systems. 4) Promote the transformation of the digital economy, industrialization and modernization of management, production, trade and services as the core for all parties in the development of digital technology. 5) Create, develop and manage cyber security management across the country to ensure the Internet network. 6) Manage, administer, promote, develop and provide infrastructure services and information center systems of the country to provide e-information services for the government and society. 7) Encourage the intelligence, innovation of individuals and social organizations in research, development, transfer, use of various techniques and technologies and use the results of research, innovation and application of technology in various forms, research on business ventures, production, import-export, investment in the development of the hardware, software, digital and high-tech industries and advanced and modern technologies.

2.2.3 Stakeholders

- Ministries relate to Science Technology
- Institutes
- Higher education (University, College)
- Business sector

2.2.4 Resources

In Lao PDR, the government's STI budget for R&D is allocated by the Ministry of Finance and approved by the National Assembly each year.

2.2.5 Issues

Science, Technology and Innovation (STI) has become one of the important potentials for economic growth and social development. It plays an important role in rapidly promoting the growth economy and encouraging the movement of the economic structure in a more advanced and modern direction. Through changing the production elements and raising the competitiveness of the products of enterprises and the economic. Laos have international STI collaboration with various countries and international organizations to encourage STI through seminars, training, study abroad and research development.

2.3 Challenges and Opportunities

2.3.1 Strengths

- collaboration in research and development, promotion of the use of modern technology with countries, international organizations, including technical assistance, funding are carried out according to the foreign policy guidelines of the government that have been laid out.
- The government play roles importance to science and technology by setting policy guidelines, formulate national socio-economic development 5-year plan in each phase to develop all legal acts related to the protection and promotion of research, development and use of advanced and modern technology.
- Political stability is strong, there is democracy, unity and society are peaceful.
- The national economic is expanding continuously and steadily, including the Gross domestic product (GDP) increasing each year. There are abundant natural resources. The terrain is conducive to science and technology, research and development. moreover, connecting with neighboring countries on the regional and internationally.

- Cooperation in research and development, promoting the use of modern technology with countries and international organizations technical assistance and grants are broad in nature according to the government's foreign policy.

2.3.2 Weakness

- The main fund to develop the STI in Laos still come from the FDI, Loans, ODA and private sector is very less. The lack of human resource in STI.
- related legal system on science and technology is not complete system. the coordination mechanism between central and local is not systematic.
- Personnel with knowledge, ability, experience and conduct research in science and technology are limited in terms of quantity and quality. human resource development is not exactly plan and it is not prioritized focus area.
- Science and Technology infrastructure is still limit and unable to use the modern equipment which the existing equipment is old and out of date.
- government budget and involved sector for science and technology is not enough.
- Increased energy consumption has led to a shortage of energy and research on using of renewable energy and materials Lack of food supplies and the need for safe food to consume, Management of natural resources and the environment change, The weather that may be affected by various natural disasters. therefore, people need to develop knowledge of science and technology, there are enough human resource with knowledge and ability in science and technology both in quantity and in quality to develop country.
- establish the infrastructure of Science and Technology has significant need fund to ensure the development in each phase.

2.3.3 Opportunities

- Government open cooperation with foreign countries.
- The rate of investment, foreign assistance, international organizations, business and enterprises sector including personal businesses to the socio-economic development and science and technology development tends to increase.

- Human resource potential, the natural resources and topography of a country are the facilitating conditions for scientific research and technology development.
- In the era of globalization, regional development of the ASEAN Community, Lao PDR 's socio-economic development, competitiveness and international economic cooperation, international economic linkages which require modern science, technology and innovation.

2.3.4 Threats

- Demographic and social changes are increasing rapidly, Aging society and population mobility, social and cultural infiltration, Receiving and distributing news from multi direction, The distribution of prosperity is unstable characterized that affect the socio-economic development of Lao PDR.
- region development into the ASEAN Community, Membership of the World Trade Organization, Competition and cooperation for advance economic development which the government have to well manage mechanisms and strongly promote the coordination in science and technology. There is a complete set of systematic protection regulations to be comparable and equal to such developments. adoption of modern technology is not affected to environment.

2.4 Policy Recommendations

- Policy to facilitate research and development collaboration.
- Policy to promote research and innovation.

3. National STI Master Plan

3.1 Direction of the Plan

3.1.1 Vision

“Transforming Lao PDR to a Sustainable Lower Income Economy by 2030 through Strengthening Science Technology and Innovation Capability”

3.1.2 Strategy

■ Strategy3: International STI Cooperation

“Take advantages of S&T International cooperation for the strengthening of S&T capability”

3.2 Objects and Actions

The International Cooperation strategy regarding to the cooperation such as Bilateral, multilateral, small multinational and International Organization.

The investment and technology attract FDI, development of R&D capability utilize ODA for research manpower, R&D equipment, policy capacity building.

■ Objective3-1: Capacity building and human resource development

Action3-1: “Development of human resources in science and technology by enhancing the development of skills and knowledge for employees of the relevant ministries and universities, institutions, including those of the industrial zone”

Strategic Plan for development of Science and Technology, human resource development is necessary to drive Socio Economic development. In Lao PDR, the number of human resources is growing rapidly. In 2006, the number of staff who graduated PhD degree is 255, master’s degree 2,017, bachelor’s degree is 10,833 and

other 70,461. In 2010, the number of PhD graduated increases to 495, master graduated is 4,212 bachelors graduated is 22,185 and other is 85,639. In 2006 to 2010, higher education increased at an average rate of 46 percent per year. In 2009-2012 higher education enrollment ratio of 1,898 students per 100,000 population. In 2011 to 2012, the ratio increased by 2,015 per 100,000 population.

Government sent Lao student to study abroad in various filed in Vietnam is 61.2% of all studying abroad, Thailand is 12.9%, Republic of China is 9,7% and study other country is 15,2%

According to data, the increasing of human resource is the quantity but low quality because the mechanism of human resource development is not consistent with the field of study that needs to be developed in each phase. The distribution of knowledge-based resources does not meet the actual needs, the most of resource work at institutions rather than working in the business sector and local area. higher education level has only been initiated the master's degree in some field area, but it has not yet had enough capabilities and conditions to implement doctoral degree in various fields.

Competent and highly educated employees working in science and technology development are insufficient to meet the demands.

Human resource development in science and technology is still very small compared to ASEAN country, in 2010 data show that there are only 1.1 researchers per 10,000 population in science and technology. Therefore, the strategy of science and technology development focuses on human resource development at various levels to have quantity and quality, it can be equivalent and suitable to join the ASEAN Science and Technology Community and connecting internationally.

Ministry of Technology and Communications, Ministry of Education and Sports will be responsible for monitoring and evaluating.

The funding: Government budget, ODA, FDI

The monitor progresses: adopt the knowledge to local area which expand the knowledge and collaboration, sending staff to attend the training, seminar and study trip, the assessment of attending.

The indicators of evaluation: Number of participants, roadmap, number of researchers, number of organize trainings, seminar and workshop.

■ **Objective3-2: Introduction of effective technology and utilization of innovation**

Action3-2: Promotion of Technology Transfer

Technology plays an important role in the development of socio-economic as technology is developing rapidly, we must develop and apply modern technology to our country.

Vision to 2030 is to promote the import and transfer of science and technology will be international, to stimulate the national socio-economic development rapidly, bringing the country's science and technology level closer to international. The import of science and technology requires constructive practice, guarantee of national independence, safety, does not affect the environment and to be sure for sustainable development of the country and equality, both sides have mutual benefits.

Technology plays an important role in the development of socio-economic as technology is developing rapidly, we must develop and apply modern technology to our country.

Vision to 2030 is to promote the import and transfer of science and technology will be international, to stimulate the national socio-economic development rapidly, bringing the country's science and technology level closer to international. Promote the transfer and dissemination of research results and development of science, technology and innovation to society.

The import of science and technology requires constructive practice, guarantee of national independence, safety, does not affect the environment and to be sure for sustainable development of the country and equality, both sides have mutual benefits.

Ministry of Technology and Communications inline with Department of Technology and Innovation will be responsible.

The funding: Government budget, ODA, FDI and International Organization.

The monitor progresses: Monitor the technology transfer by registration, implement project to strengthen academic competence in the management of technology transfer both in the central and local areas, Improve and upgrade services, transfer of science, technology and innovation (the focus plan 7 of DTI), Cooperation project for the development of a Lao-China high-tech zone. to take the advance technology from other countries and can get the technology from ODA or loan, government must survey necessary technology base on the SME or universities need , government find necessary technology from industries in foreign countries, utilize the technology and development base on resources.

The indicators of evaluation: Number of Contact, Number of registrations as number of technology transfer, Questionnaire of satisfaction and usage, Number of project and service relate to technology transfer.

Action3-3: Promotion of joint R&D

Countries in the region and in the world are increasingly invested in research and development opposite to Least developed countries still have limited budget and capabilities of research and development. The adoption of science and technology is still very low compared to developed countries and developing countries. Lao PDR invests small amount in research and development than other ASEAN countries (in 2012 it was 0.04% of the Gross Domestic Product (GDP) or 1% of the government's investment budget.

In the past year, the National Budget Plan did not invest in research, and development much. A science and technology development fund has been established. Attracting international aid and cooperation is still limited and inefficient. Science and technology research and development projects that the government pay particular attention to do

not have a specific budget plan or mechanism for budget management and investment plans.

The vision to 2030 will strengthen the framework of science, technology and innovation in Laos in term of advance, modern and equivalent to other countries in the region and to support potential sector that can adapt, invent new technologies to develop the country with an economy of wisdom, encourage establishing infrastructure such as equipment's and laboratory that necessary for scientific research and technological development. And also cooperated with various research institutes and universities, both domestically and internationally to strengthen the framework for research and development of science, technology and innovation.

According to Strategic Plan⁶, Cooperation of Science and Technology Development with the global community is very necessary, therefore, it must be carried out in a competitive, motivated, creative, autonomy, national security and equally efficient manner.

Plan and implement research projects in development of Science and Technology between business units, universities, and research institutes in Lao PDR , increase international scientific seminars, organize an exhibition of research in science and new technology, attract foreign scientists to participate in research projects at different levels for human resource potential and encourage freshly researcher ,create the National Institute of Science and Technology by cooperating with countries that have Scientific and technological expertise ,increase regional and international cooperation in science and technology, including neighboring countries to attract more investment and experience, continue to mobilize Lao scientists who live abroad to participate scientific and technological research to develop country.

Responsible for promotion of joint R&D: relevance ministries, higher education, Business /Industries sector.

The funding: Government budget, ODA, FDI, International Collaboration, Science and Technology development fund

The monitor progresses: roadmap, government need to allocate the fund for joint R&D, to find R&D issue which researchers in universities and research institutes are interesting, to find proper partners from ASEAN countries or neighboring countries, to promote and evaluate the achievement of research.

The indicators of evaluation: the number of Project on R&D, a policy to facilitate research and development collaboration.

■ Objectiv3-3: Infrastructure in Science and Technology

Action3-4: Organization structure and management of Science and Technology

Lao PDR has scientific and technological research organizations such science and technology council, research and development institute, educational institute, research and development center, laboratory, research and experimental institute, Science and Technology Service Center, companies relate to science and technology, however, the coordination between the service and consulting sectors is not good executed and rigorous.

The vision is to improve the organizational structure and management of Science and Technology in line with the approach of strength infrastructure with a focus on economic development thar can be linked to the international economy, improve Science and Technology policies and laws to be a system, to support the advancement of Science, Technology and Innovation, conduct Research and Development in Science and Technology in closely cooperation with universities to create human resource for carry out research and development and use of modern technologies. Learn the informative system from well-developed countries and adaptation to develop the informative system in Lao PDR.

The funding and responsible is Government of Lao PDR

The monitor progresses: the effective cooperation between governments, universities, institutions and the private sector.

The indicators of evaluation: a policy to promote research and innovation, the Laws on organization and management of science, technology and innovation to be complete and able to connect with ASEAN countries, a standard statistical database system for STI to be able to report and link domestically and internationally, the Mechanisms for managing technology and innovation both centrally and locally

Action3-5: Strengthening infrastructure on STI

research laboratories in institute are still scattered and lack equipment for research and education, including measuring instruments, testing equipment, and research tools. Infrastructure is necessary and need to set up laboratory and all facility for research and development which have to match with the well-developed countries.

The funding and responsible is Government of Lao PDR

The monitor progresses: increasing and joint R&D

The indicators of evaluation: a national institute of R&D in Science and Technology, deployment, using and service in three main provinces, a state-of-the-art scientific research institute equivalent to ASEAN countries, infrastructure on Science and Technology.

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CHAPTER 8

STI Brain Drain and S&T University: A Study on the Struggle of KAIST

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 8. STI Brain Drain and S&T University: A Study on the Struggle of KAIST

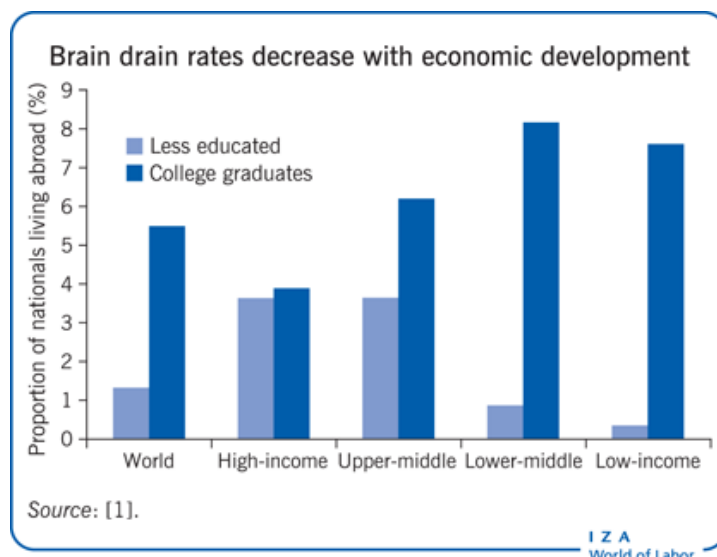
Prof. Youjung Shin, Jeonbuk National University

1. Introduction

The brain, i.e. the educated people, is an important resource for developing countries. Yet, the brain who has earned social and economic capital with higher education tends to look for better living condition. So, they often move to different places or countries instead of staying in their own countries. In result, developing countries which urgently need the brain and their contributions used to struggle to attract and capture the brain – not to mention to raise them. This is called as “brain drain problem.” According to the United Nations Global Migration Database, about 34% of global migration occurred from developing countries to developed countries and this number has been rapidly increasing.²⁸ [Figure 8-1] shows the brain drain rate depending on the level of economic development in each country. Not surprisingly, low-income countries showed highest brain drain rate compared to high-income countries; interestingly, it is not low-income countries but lower-middle income countries that showed the highest brain drain rate, which this paper will revisit again later. What’s notable in Figure 1 is that, if we compare the less educated and college graduates, it was college graduates which showed the greatest gap in leaving the country depending on its economic size.

²⁸ Frédéric Docquier, “The Brain Drain from Developing Countries,” *IZA World of Labor*, May 2014, <https://wol.iza.org/articles/brain-drain-from-developing-countries/long>.

[Figure 8-1] The Brain Drain Problem



Source: Frédéric Docquier (2014)

South Korea was one of the countries which suffered from the brain drain problem. After the Korean War (1950-1953), most of the national infrastructure including educational system and infrastructure was devastated.²⁹ In the late 1950s, the so-called Minnesota Project (1955-1961) was carried out for the reconstruction of the Seoul National University. In the name of the Minnesota Project, the University of Minnesota in the U.S supported the Seoul National University to restore its basic material and intellectual foundation by, for example, giving aids for equipment, instruments, books, and building reconstruction.³⁰ Yet, it was far from enough to revive the educational system and culture in South Korea. Particularly, the graduate school system remained as weak; even within Seoul National University as the Minnesota Project mostly concerned about their undergraduate system. In this context, those who wanted higher education or those who wanted to continue their study as researchers used to go abroad and rarely came

²⁹ Geun-Bae Kim, *The Structure of Scientific and Technological Revolution in Korea* (Paju: Dulnyouk, 2016); KyongHee Han, *The Making of Korean Engineers: Historical Formation and Development* (Paju: Dulnyouk, 2021); for more about the impact of the Korean War on the Korean Economy, see Jong-Won Lee, "The Impact of the Korean War on the Korean Economy," *International Journal of Korean Studies* 5.1 (2001): 97-118.

³⁰ For more about the Minnesota Project, see Ock-Joo Kim and Sang-Ik Hwang, "The Minnesota Project: The Infuence of American Medicine on the Development of Medical Education and Medical Research in Post-War Korea," *Korean Journal of Medical History* 9.1 (2000): 112-123.

back to the country. From 1962 to 1976, the annual average of non-return rate among those who left the country was about 62.62% in South Korea which was really high; higher than for example Jordan, Iran, Iraq, India, and Lebanon.³¹

Not always did the brain leave their country for economic reason. Scholars have pointed out diverse motivations and reasons – from economic reasons to political reasons, individual reasons, and social reasons – for brain drain.³² In South Korea, a continuous political conflict with North Korea as well as the dominance of authoritarian regime made the brain to concern about the political stability in South Korea. It played a role to a certain extent in draining the brain out of the country. Also, individual and social reasons such as health-care issues, family education, and living condition were also important factors in motivating the brain to leave the country. Several Korean researchers have periodically conducted a survey on searching the major reasons of brain drain.³³ Some pointed out the importance of economic reason and others pointed out the role of political factors. For example, Ha-Joong Song conducted a survey on the way Korean scientists and engineers in the U.S. decided their career path – either staying in the U.S. or returning to South Korea – in the 1980s and 1990s.³⁴ However, what is important here than the result of survey per se is the fact that the major reasons of brain drain have been changing even within one country. Then, the question is how to keep monitoring the changing reasons of brain drain and appropriately dealing with this problem.

³¹ Manyong Moon, “Korea’s ‘Brain-Drain’ and the Role of the Korean Institute of Science and Technology,” *Korean Culture* 37 (2006): 229-261; depending on the survey method, the number of non-return rate of South Korea was fluctuated but it was high enough to make the government concern.

³² For example, see Andreas Ette and Nils Witte, “Brain Drain or Brain Circulation?: Economic and Non-Economic Factors Driving the International Migration of German Citizens,” in *The Global Lives of German Migrants: Consequences of International Migration Across the Life Course*, eds., Marcel Erlinghagen et al (Springer Cham, 2021).

³³ Hyun-Ah Park, Hye-Young Gim, An-Sik Chang, Seoung-Ho Ryu, “Analysis of Factors Contributing to Brain Drain and Work Motivation Regarding Human Resource Development in Korea: With Focus on 2015 IMD World Talent Report,” *Review of Culture and Economy* 19.1 (2015): 123-151.

³⁴ Ha-Joong Song, Who stays? Who returns?: The Choice of Korean scientists and engineers, Ph.D. diss., Harvard University, 1991; Ha-Joong Song, Coping With Brain Circulation: Current Situation and Policy Direction in Korea (The Ministry of Science, 2001).

It should be noted, however, that the brain drain is not always a problem. The circulation of the brain can play a role in bringing new ideas and knowledge in developing countries and stimulating higher education. Recently, scholars have pointed out the positive aspects of brain drain and emphasized a need to change existing perspective which framed the brain drain as a problem.³⁵ However, from the developing country's point of view, the problem is that there is a unidirectional lopsided circulation of the brain, instead of multilateral healthy circulation. In order to attain positive impact of the brain drain, he or she, who left the country, should come back at some points. Their leaving should be temporary than permanent. In order to make the brain's leaving as temporary, however, there must be a certain kind of incentives and systems that the brain would continue their research, which is not always an easy task for developing countries. Thus, in spite of recent scholarly discussions, developing countries have been still problematized the phenomenon of the brain drain, struggling to find a way to relieve the lop-sided drain of the brain. The term, brain drain, has been actively used to denote the concern of developing countries in monitoring and meditating the flow of educated people.³⁶

While many countries used the term brain drain, few however knew where the term, brain drain, originally came from. It was the U.K. which first used the term brain drain and framed it as a serious problem.³⁷ In the 1950s, after the second World War, a considerable number of scientists in the U.K. found a difficulty in pursuing their research in Europe and fled to the U.S. It raised a huge concern in U.K., even blaming "Americans' 'parasitising' of British brains."³⁸ In 1963, the Royal Society released the report on the

³⁵ Ha-Joong Song, Gi-Geun Yang, Chang-Min Kang, "A Study on the Brain Drain Circulation Model for Advanced Science and Technology Human Resources," *Korean Policy Studies Review* 13.2 (2004): 143-174; Tito Boeri, Herbert Brücker, Frédéric Docquier, and Hillel Rapoport eds., *Brain Drain and Brain Gain: The Global Competition to Attract High-Skilled Migrants* (Oxford University Press, 2012).

³⁶ See for example, Adam Tyson, *The Political Economy of Brain Drain and Talent Capture: Evidence from Malaysia and Singapore* (Routledge, 2018); Junaimah Binti Jauhar, Ahmad Bashawir Abdul Ghani, Rabiul Isam, *Brain Drain: Propensity for Malaysian Professionals to Leave for Singapore* (Springer, 2016).

³⁷ Matthew Godwin, Jane Gregory and Brian Balmer, "The Anatomy of the Brain Drain Debate, 1950-1970s: Witness Seminar," *Contemporary British History* 23.1 (2009): 35-60.

³⁸ Brain Balmer, Matthew Godwin, and Jane Gregory, "The Royal Society and the 'Brain Drain': Natural Scientists Meet Social Science," *Notes and Records of the Royal Society* 63 (2009): 339-353.

status of emigration of the U.K. scientists and its impact. The report alarmed the importance of tracing the move and flow of the brain and the serious situation of the U.K. from the 1950s. With this report, the term, brain drain, was coined and began to be widely used. It also began to spread to other countries including developing countries as an analytical term to understand the status and situation of the educated people in each country. In other words, the term brain drain itself and its following discussion had been initially shaped in U.K., a high-income country with rich tradition in research and education in science, and then spread to other countries.

Though using the same term, the brain drain in the U.K. was not the same with the brain drain in developing countries like South Korea. If comparing U.K. and South Korea, U.K. is the country where the scientific revolution was configured and took place. They had a long tradition and experience of nurturing science and scientists, and thus had a rich amount and quality of educated people in science. In contrast, in the case of South Korea, especially in the Japanese colonial period, few could experience higher education in science and it was difficult for Koreans to cultivate their own culture for nurturing scientists. In other words, South Korea and U.K. had a totally different situation in terms of experience and existing culture in nurturing scientists. Also, while in the 1960s, the number of educated people in science, who could be drained to other countries, was quite high in U.K., the number of the brain itself, who had already attained higher degree in science, in South Korea was very small. There were lots of potential brains in South Korea, not the ready-made brain. So, the kind of the brain that was drained in the name of brain drain was quite different in the U.K. and South Korea. Furthermore, while there were some well-established scientific institutes like Royal Society that governed science and technology policy, there was few organization which involved eminent scientists in shaping science and technology policy in South Korea.

So, before talking about the brain drain, it should be underlined that various countries had different experience and system in raising and handling the brain itself. They had various pre-existing conditions in terms of experience, system, and culture in making the brain flow. This is important especially in developing countries as, because of the lack of relevant experience and culture, developing countries tended to find difficulty in handling the brain itself. Before attempting to attract the brain, the crucial task for

developing countries is to know what kind of the brain they were talking about, who the brain is, what the brain wants, and how to raise and nurture the brain. Yet, developing countries lacked experience in navigating the answers to those questions; there is no well-established answer and the important thing is the way of navigating these questions. Thus, in the process of handling brain drain problem, it became developing countries' task to simultaneously cultivate a culture and system to navigate the nature of the brain. South Korea was this case. While dealing with the brain drain problem, South Korea simultaneously navigated the kind of the brain the country wanted, the kind of the culture the brain wanted, and the kind of system the country had to establish in order to nurture the brain. This is the reason why an in-depth study about brain drain in developing countries interesting and the reason why we need more empirical studies on the flow of the brain from developing countries' view instead of bringing term and framework, originated from high-income countries.

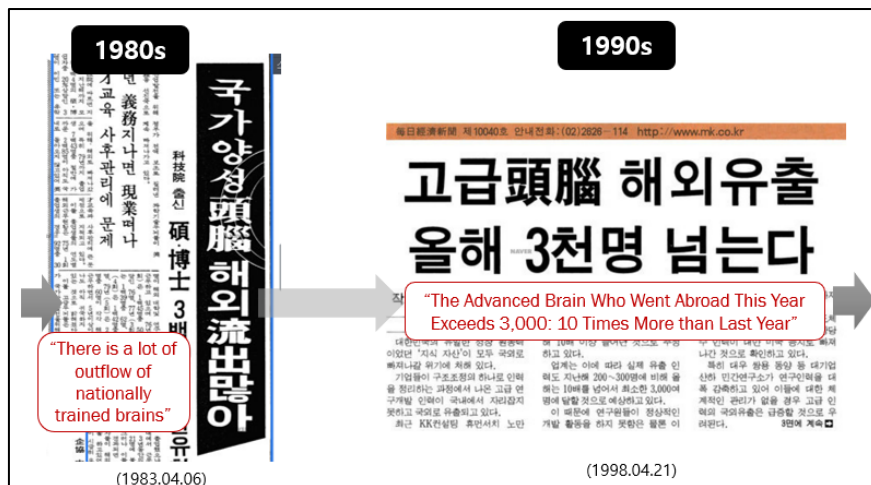
This paper examines the way South Korea has addressed the brain drain problem by focusing on the development of Korea Advanced Institute of Science (KAIS). KAIS was established in 1971 as a special educational institute for nurturing scientists and engineers in South Korea, and it was renamed into Korea Advanced Institute of Science and Technology (KAIST) in 1980.³⁹ Nowadays, KAIST is one of the top universities in science and technology in South Korea, producing more than ten thousand Ph.D. graduates since its establishment. As a science and technology oriented educational institute, KAIST became a role model for other universities like Gwangju Institute of Science and Technology (GIST), Ulsan National Institute of Science and Technology (UNIST), and Daegu Gyeonbuk Institute of Science and Technology (DGIST) in South Korea. It also became the subject of benchmarking for other developing countries. For example, in 2020, the Kenya government signed a contract with KAIST to build the Kenya Advanced Institute of Science and Technology (Kenya-KAIST). With the goal of reviving Kenya economy with talented scientists and engineers educated through Kenya-KAIST, like KAIST in South Korea, the Kenya government decided to establish it in the Konza

³⁹ KAIS was merged with the Korean Institute of Science and Technology (KIST) in the early 1980s and its name was changed into KAIST. Though KIST was separated again in the late 1980s, the institute's new name, KAIST, remained. This paper will use the term KAIS when indicating it in the 1970s and KAIST thereafter.

Technopolis near to Nairobi.⁴⁰ Likewise, KAIST has received attention as the cornerstone of South Korea's educational policy which has contributed to resolving its brain drain problem and promoting the rapid growth of Korean economy.

It should be noted however that, though the establishment of KAIS was important in relieving the brain drain problem in the 1970s, it did not totally eradicate concerns and controversies over brain drain in South Korea. In the 1980s, after the launch of KAIS, a domestic newspaper still alarmed that "there [was] a lot of outflow of nationally trained brains."⁴¹ Even in the 1990s, the concern on brain drain was resonated again as shown in [Figure 8-2].⁴² The brain drain problem did not vanish with the establishment of KAIS.

[Figure 8-2] A changing yet continuing concern over brain drain in South Korea



Source: NAVER News Library

Indeed, the brain drain problem was not something that could be "solved" in developing countries. Instead, it was what to be "cared." Focusing on the development of KAIST, this paper illuminates the meaning of caring the flow of the brain in developing countries. By showing the struggle of KAIST in dealing with changing yet continuing concern over

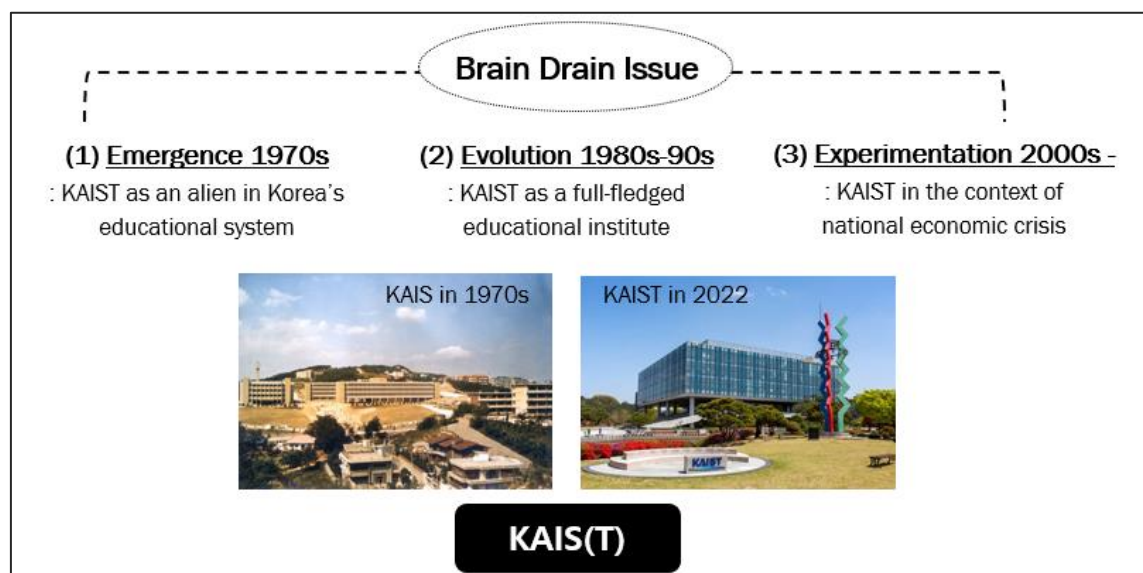
⁴⁰ The Editorial Board, "Kenya-KAIST kicks off with \$95 million funding from the Korean government," *EurekaAlert*, 13 Feb 2019, <https://www.eurekaalert.org/news-releases/835416>.

⁴¹ The Editorial Board, "There is a lot of outflow of nationally trained brains," *Maeil Economy*, 6 April 1983, 8.

⁴² The Editorial Board, "The advanced brain who went abroad this year exceeds 3,000: 10 times more than last year," *Maeil Economy*, 21 April 1998, 1.

brain drain in South Korea, the paper highlights the nature of the issue of brain drain in developing countries. This paper consists of three parts. By periodizing the development of KAIST into three – (1) emergence stage in the 1970s, (2) evolution stage in 1980s and 1990s, and (3) experimentation stage after 2000s – as shown in [Figure 8-3], the paper analyzes the way KAIST shaped its educational curriculum and strategies in line with continuing yet changing discourse over the problem of brain drain. It reveals different ways of addressing the brain drain problem in the changing economic, social, and political context of South Korea and their implications in understanding the nature of the brain drain problem in developing countries.

[Figure 8-3] Three stages of KAIST development: Emergence, Evolution and Experimentation



Source: Author

2. The Emergence Stage

Long before the establishment of KAIS(T), the brain drain has been a serious problem in South Korea. From 1953 to 1969, there were about 11,037 people who went abroad to study. Among them, only 1,160 people which was about 10% came back to the country after getting a Ph.D.⁴³ So, the Korean government tried to, on the one hand, set up a high barrier for those who want to go abroad by for example making a qualification exam mandatory. The Ministry of Education in South Korea carried out the qualification exam and only those who passed this exam could get an opportunity to go abroad to study. In 1958, the Korean government also established the inspection rule for students studying abroad.⁴⁴ While strengthening this regulation, the Korean government also, on the other hand, tried to attract the talented brain who stayed in other countries. One of the biggest problems in South Korea in attracting the brain was its poor research environment. After the devastation of the Korean War (1950-1953), it was difficult for Koreans to find a proper research laboratory with well-equipped research instrument. In the case of Seoul National University, the Minnesota Project (1955-1961) played a role in setting up basic research instruments and equipment, but it was far from enough in improving the overall research environment in South Korea.

So, in the 1950s and the 1960s, the Korean government tried to prevent the outflow of the brain by strengthening the regulation, while stimulating the inflow of the brain by improving research environment. In this context, in 1966, a new research institute, the Korea Advanced Institute of Science and Technology (KIST) was established. It aimed to bring a few eminent Korean brain in abroad by providing well-equipped laboratory and funding support. It aimed to play a role as the “reverse-brain drain center.”⁴⁵ In result,

⁴³ Manyong Moon, “Korea’s ‘Brain-Drain’ and the Role of the Korean Institute of Science and Technology,” *Korean Culture* 37 (2006): 229-261.

⁴⁴ “Regulations on Studying Abroad,” National Archives of Korea, accessed September 9, 2022, https://www.archives.go.kr/next/search/listSubjectDescription.do?id=008652&pageFlag=&sit_ePage=1-2-1; Hwang-Hee Cho, Eun-Kyoung Lee, Chun-Keun Lee, Sun-Woo Kim, *Science and Technology Human Resource Policy in South Korea* (Seoul: Science and Technology Policy Institute, 2002).

⁴⁵ For more about the KIST, see Manyong Moon, *The Formation of Modern Research System in South Korea: The Establishment and the Development of KIST, 1966-1980* (Seonin, 2010).

about 25 Korean scientists and engineers who had been mostly in the U.S. came back to the country. However, KIST was a research institute. It meant that, though KIST could recruit several prominent scientists and engineers, it was difficult for KIST to continuously supply and raise research assistants, i.e. graduate students or postdoc, for KIST researchers. It should be noted that a researcher did not do research alone. A research needs not only well-equipped laboratory and instruments but also a group of great researchers and assistants. In the case of developed countries like U.K., there was a great deal of well-established educational institutes which could supply well-educated students and assistants for research institutes. Yet, there was no educational back-up system in South Korea in the 1960s.

Thus, in 1968, KIST conducted “a study on the promotion of S&T universities and graduate schools.”⁴⁶ Based on this, in 1969, KIST proposed “a plan to establish a KIST-affiliated science and technology graduate school.” It was a plan to build a new graduate school within KIST. By doing so, KIST attempted to nurture human resource by itself. Regarding KIST’s proposal, the domestic media noted that:

“the reason why [KIST] could not but having an interest in graduate school is because that what can determine the success and failure of our research institute depends on the securing of the talented people. Currently, domestic graduate schools are terribly under-developed, and as a result, excellent students turn their eye to foreign countries. In fact, the issue of nurturing graduate school has been a topic that has been debated in academia for long time.”⁴⁷

By building its own graduate school, KIST tried to resolve this situation. Ultimately, KIST aimed to perform both as a research institute and educational institute in South Korea. It was the result of realizing the importance of supporting human resource to the brain

⁴⁶ The Editorial Board, “Strengthening Graduate Schools: To Prevent the Brain Drain,” *Kyunghyang Shinmun*, 13 December 1969, 5.

⁴⁷ Ibid.

coming back to the country in addition to material resources like instruments and equipment.

However, KIST's plan faced harsh opposition from other universities and the Ministry of Education. From other universities' point of view, KIST's proposal to launch a new graduate school could distract the government's attention in improving existing universities' condition. The government's funding was always limited. In this situation, if KIST established a new graduate school, there was a plausible possibility that the material and social support of the government to traditional universities would go to KIST to some extent. In addition, KIST was not under the Ministry of Education. As a research institute, KIST was operated under the jurisdiction of the Ministry of Science and Technology. In that sense, building a new graduate school in KIST meant, from the perspective of the Ministry of Education, that the Ministry of Science and Technology would expand its role toward the educational system. The Ministry of Education could not easily accept this idea. So, the Ministry of Education and traditional universities urged the government to invest more money in improving the existing educational environment instead of attempting to build a new graduate school in KIST. In the end, KIST's proposal was dismissed.

Yet, almost exactly at this moment, another new impetus to build a science & technology graduate school in South Korea emerged not within South Korea but in the U.S. by a Korean scientists, Kun-Mo Chung. Chung was a physicist, who earned his Ph.D. degree in Michigan State University in 1963. Right after his graduation, Chung became the assistant professor in the department of physics at the University of South Florida. Yet, in 1966, Chung dropped his professorship to pursue more research in nuclear engineering at MIT as a research professor. In 1967, he became the associate professor in the department of electrophysics at Polytechnic Institute of New York University. In short, Chung was one of the Korean scientists who went abroad for study and achieved a stable academic career in science in the U.S.⁴⁸ As the brain drained out of South Korea, Chung himself actually had a huge interest in understanding the phenomenon of brain

⁴⁸ For more about Kun-Mo Chung, see Youjung Shin, "1970, To live as a scientist or as a science policy maker," *Horizon* (forthcoming); for Chung's autobiography, see Kun-Mo Chung, *A Scientist in the County which Made Miracle* (Korea.com, 2020).

drain – its causes and effects – and the way to relieve the brain drain phenomenon for South Korea. So, during the period when he conducted research in nuclear engineering at MIT, he also made time for studying science and technology policy at Harvard Kennedy School. In the late 1960s, at Harvard Kennedy School, the two eminent scholars, Don K. Price and Harvey Brooks, run the seminar series on science, technology, and public policy, and Chung participated in it.

As a term paper for the seminar course at Harvard Kennedy School, Chung wrote a paper on “a policy measure to prevent brain drain in a developing country.” It explored the reasons of brain drain in developing countries, which Chung would reflect his own experience, and a policy measure to relieve the brain drain phenomenon. Based on what he had experienced as a scientist in the U.S. and what he had studied as a student at Harvard Kennedy School, he speculated what would be needed for developing countries to reverse the flow of the brain and attract them. One of his ideas was to build a new graduate school specialized in science and technology, equipped with unprecedented material as well as social support in South Korea. However, he did not expect his paper to be actually realized. It was more like an idealistic vision rather than practical guideline. He had no concrete idea on the way of drawing funding or support for a new graduate school that he proposed. So, Chung did brainstorm the idea of preventing the brain drain but he left it behind, and moved to Polytechnic Institute of New York University to continue his career as a scientist.

However, in 1969, the time came to Chung. When he was in Polytechnic Institute of New York University, he happened to see the news that John A. Hannah, who was the president of Michigan State University, became the head of the United States Agency for International Development (USAID).⁴⁹ Chung had a close relationship with John A. Hannah when Chung was a graduate student in Michigan State University; Hannah even participated in Chung’s wedding. Chung saw Hannah’s interview addressing that USAID would change its way of aiding developing countries. While before the 1960s, the USAID and foreign aid agencies tended to aid developing countries by directly delivering goods

⁴⁹ For more about the beginning of KAIS, see Dong-Won Kim and Stuart W. Leslie, “Winning Markets or Winning Nobel Prizes?: KAIST and the Challenges of Late Industrialization,” *Osiris* 13 (1998): 154-185.

or investing money, some concerns were raised in terms of its effectiveness in the late 1960s. Also, especially, in the context of continuing war in Vietnam, the U.S. witnessed the increased amount of money investing for winning the war and unsettled domestic economy. In this context, John A. Hannah as the new head of the USAID in the late 1960s said that “instead of giving fish, [USAID] will teach them how to fish.” It was an attempt to change the way USAID provided foreign aid in developing countries other than directly supplying products. Chung thought that his paper which stressed the need of establishing graduate school would be in line with USAID’s new policy direction, and met Hannah in 1969.⁵⁰

Chung’s personal connection with Hannah and the changing context over the U.S. foreign aid policy came to merge in the late 1960s. After meeting Chung, Hannah proposed him to draft policy proposal based on the paper that he wrote in Harvard Kennedy School. So, on October 1969, Chung wrote “The Establishment of a New Graduate School of Applied Science and Technology in Korea” and submitted to USAID. Then, the work progressed quickly. USAID delivered this policy proposal to the Korean government, indicating that once the Korean government decided to proceed this policy, USAID would offer financial support. To explain the policy proposal, Chung was invited to the Korean government in 1970. Chung stressed that: “the brain goes where the brain is and the brain goes where it is challenged. The problem of the brain drain from developing countries to advanced countries can be solved **by creating the center of knowledge in developing countries**” (emphasis is added).⁵¹ A new graduate school specialized in science and technology would be (and should be) built as “the center of knowledge” in South Korea.

However, again in South Korea, there was an opposition as in the case of KIST’s proposal. Since what Chung proposed was to build a new graduate school, it was quite similar with KIST’s proposal though it would be not KIST-affiliated. Similarly, traditional universities and the Ministry of Education opposed to this idea as it could distract existing government’s support on higher education. The Ministry of Education emphasized that,

⁵⁰ Ibid.

⁵¹ Kun-Mo Chung, “The Establishment of a New Graduate School of Applied Science and Technology in Korea” (KAIST, 1969).

instead of creating a new graduate school, it would be more effective and efficient if the government increased funding and support on existing universities. Yet, this time, the Ministry of Education could not win over. As USAID indicated that it would provide financial aid, around 6 million dollars, once the Korean government pushed forward this agenda, it was difficult to simply deny it. It was a different situation compared to when the KIST proposed a new graduate school. Also, for the advocates of a new graduate school, the opposition from the Ministry of Education was predictable and its logic had been debated before with the KIST's proposal. So, in the end, despite conflicts, the then president Chung-Hee Park decided to establish a new graduate school in science and technology under the Ministry of Science and Technology: Korea Advanced Institute of Science (KAIS).

In short, KAIS was the result of struggling to build a new graduate school in science and technology in South Korea. It was, in a nutshell, 'a new wine in a new bottle' strategy. Instead of investing more money for existing universities, the Korean government decided to launch a new graduate school in science and technology. However, it was not achieved at the first attempt. The KIST's proposal to build a new graduate school could not be realized due to harsh opposition from existing universities and then the Ministry of Education. However, a new initiative to build a specialized graduate school in science and technology in South Korea came from the changing context over foreign aid policy in the U.S. in the 1960s. Chung's idea and his personal connection with Hannah along with the contingent U.S. political and social environment resulted in the policy proposal to build KAIS. It should be noted that a policy needs appropriate timing. USAID's suggestion was important because of not just its financial support but also its symbolic meaning. In repelling oppositions from traditional universities and the Ministry of Education in building a new graduate school, which was predictable as in the case of KIST, USAID played a symbolic role.⁵² In the end, KAIS was built in 1971 as the "center of knowledge" in South Korea.

⁵² For more about the role of USAID in building KAIS, Buhm Soon Park, "The Role of USAID in the Establishment of KAIST," presentation slide, February 2021, STEPI (unpublished).

By building KAIS under the Ministry of Science and Technology, KAIS could bypass traditional regulations that the Ministry of Education imposed on traditional universities in terms of professors' salary, promotion system, budget, and etc.⁵³ This was the advantage of 'a new wine in a new bottle' strategy. In result, KAIS provided unprecedented economic incentive to draw the brain. KAIS's professor received almost four times higher salary than the average Korean in the 1970s. It was also even two times higher than, for example, the Seoul National University professors' salary. KAIS also provided housing subsidies; like apartment which was not common in South Korea.⁵⁴ No wonder traditional universities' professors were annoyed yet somewhat envious of KAIS's professor. However, it was difficult for professors in traditional universities to move to KAIS, since KAIS mostly recruited the brain in overseas, especially in the U.S. KAIS's utmost goal was to solve the brain drain problem. So, KAIS did not consider domestic professors as the candidate of KAIS's professors. Instead, it searched for the brain who were outside of the country.

Again, it was 'a new wine in a new bottle' strategy: a new graduate school under a non-traditional ministry's jurisdiction with a new people coming from abroad. Again, no wonder a tension arose between traditional universities and KAIS. Chung played a role in searching the candidate for KAIS in the U.S. as he himself was a professor at Polytechnic Institute of New York University. Though KAIS offered unprecedented economic incentive, it should be noted that it did not naturally draw the brain. A lot of persuasion processes were needed. Chung and other KAIS builders selectively and carefully approached to a few candidates and interviewed them one by one. Thereby, they explained KAIS's advantages, like what KAIS would provide to them, as well as KAIS's characteristic. KAIS put much emphasis on meeting "the needs of Korean industry and Korean industrial establishment" instead of "add[ing] the world's store of basic knowledge."⁵⁵ For researchers in the U.S., which enjoyed research freedom and academic curiosity, KAIS seemed like imposing intellectual constraints while providing

⁵³ This was the reason why KAIS, though it was an educational institute, did not use the term, university.

⁵⁴ KAIST's Quarter of a Century: Endless Effort for the Future (KAIST, 1996).

⁵⁵ Donald L. Benedict, Kunmo Chung, Franklin A. Long, Thomas L. Martin, Frederick E. Terman., "Survey Report on the Establishment of the Korea Advanced Institute of Science," December 1970, <http://large.stanford.edu/history/kaist/docs/terman/summary/>.

economic incentive. To attract them, thus, a series of explanation and persuasion process were needed. It should be remembered that economic incentive did not just naturally attract the brain but required a central figure who would contact the brain and persuaded them with various options and incentives. In the end, a few who were persuaded decided to come to South Korea as the professor of KAIS. The below is the recall of two early professors who came back from the U.S. in the 1970s.

*“KAIST has been **a groundbreaking university** in many ways since its establishment. There has never been a university **that gives me money and a house, and let me just study as much as I want**. The treatment of the professors was also exceptional” (Prof. Ho-Gi Kim, emphasis is added)”*

*“When I got my Ph.D., I **was planning to settle down in the US...** But the more I heard Dr. Chung's suggestion, the more attractive it became...I thought that, if I accepted the professorship at KAIST, there was **a great advantage that I can set up a good laboratory**” (Prof. Jung-Woong Ra, emphasis is added)*

3. The Evolution Stage⁵⁶

On March 1973, the first freshman which was around 100 students came to KAIS. There were six departments: mechanical engineering, material engineering, electrical engineering, industrial engineering, chemistry and chemical engineering, mathematics and physics; in 1974, biological engineering department was added. Twenty-nine tenure-track professors worked at KAIS in 1973; 44 professors in 1975. In line with KAIS's emphasis on meeting the needs of Korean industry – instead of adding “the world's store of basic knowledge” – new courses and curriculums were designed in a way to emphasize the learning of practical skill and problem-solving ability. For example, in mechanical engineering department, it stressed the need to “nurture eminent engineers who have a top problem-solving skill” and encourage students’ “curiosity which can apply principle to real practice.”⁵⁷ The entrance competition rate of KAIS was quite high – 5:1 in 1973 – due to KAIS's exceptional support on students for example tuition exemption, dormitory, military exemption, and etc. On September 1973, the first class was opened at KAIS in a newly built building like [Figure 8-4] shows the KAIS's first building constructed in a very modern way in the 1970s.

[Figure 8-4] The KAIS's First Building



⁵⁶ This section was written based on the author's published article; Youjung Shin, “Making the Brain for Daeduk Science Town: The Transformation of KAIST into an University in Daeduk,” *Journal of Science and Technology Studies* 21, no. 3 (2021): 4-37.

⁵⁷ The Introduction of Department: Mechanical Engineering, KAIS, 1973, KAIST Archive.

Source: KAIST Archive

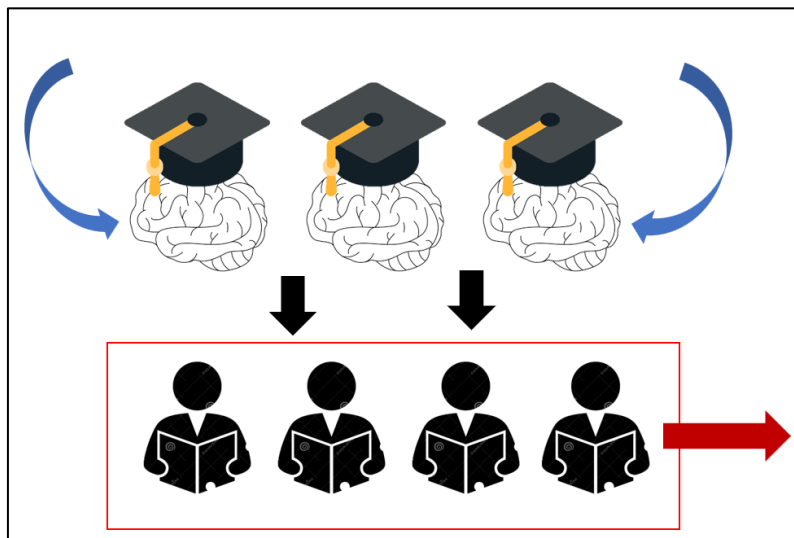
However, two problems awaited KAIS. First, it should be remembered that KAIS's 'new wine in a new bottle' strategy costed a lot. By detouring existing regulations of the Ministry of Education, KAIS offered unprecedented support to its professors and students; two-times higher salary to KAIS's professor compared to Seoul National University professors. Yet, the number of the KAIS's professor would increase and so would the number of students. Then, the problem was how KAIS would be able to sustain the funding for exceptional support to its professors and students. In that sense, using economic incentive for attracting the brain could be effective, but it had a sustainability issue. Without sustainability, economic incentive can be just a temporal measure. One of the reasons that KAIS could provide unusual support to its professors and students was because of the USAID's financial aid, i.e. 6 million dollars, though there was the matching fund from the Korean government. Yet, the USAID planned to support KAIS for its initial five years. After that, it was solely the KAIS and the Korean government's responsibility to run its new institute, which was not always easy.

Economic incentive for attracting the brain in a developing country had a sustainability issue not just because of the increased number of professor and students at KAIS, but also because of its effect on stimulating the needs of better support from traditional universities. Professors at traditional universities, who had witnessed the increased salary and support to KAIS professors, came to argue for the similar support, in terms of salary, sabbatical leave, and etc, for the Ministry of Education. In the end, it resulted in like a gradual increase of salary for professors at traditional universities. In that sense, one can evaluate that KAIS played a role as a stimulant in improving the overall environment of the Korean higher education system. Yet, one can also say that KAIS raised a contentious tension between universities making dispute, discrepancies, and barriers. It should be remembered the advantages and the pitfalls of 'a new wine in a new bottle' strategy. Using economic incentives in 'a new wine in a new bottle' strategy can be effective, as it did not bother existing old bottles, yet it could raise social tension resulting in the sustainability issue.

The second problem KAIS faced after its establishment was a generation problem. So, KAIS was launched by recruiting a series of eminent professor who had stayed outside

the country. By doing so, it contributed to relieving the brain drain problem in South Korea to a certain extent. However, if we think about the next generation that these brains would raise at KAIS, the brain drain problem was not the dead issue. On April 1983, a domestic newspaper alarmed that “there is a lot of outflow of nationally trained brains.”⁵⁸ Here, the “nationally trained brains” referred to the graduates of KAIS as they received government support with tuition, dormitory, and scholarship. The media reported that “20% of KAIS graduates which was about 330 people went abroad for immigration or study, and especially, by 1979, almost half of the graduates – 285 students among 743 – had not yet returned to the country.”⁵⁹ It alarmed the continuing phenomenon of the brain drain in South Korea. Though KAIS had reversed the flow of the brain to certain extent by recruiting a few eminent professors, the new brain that was raised out of KAIS began to flow toward outside of South Korea. This was the generation problem in brain drain phenomenon.

[Figure 8-5] The Generation Problem in Brain Drain



Source: Author

However, the outflow of the students at KAIS was not accidental but more of the result of KAIS's plan and strategy. It was KAIS's decision to encourage the outflow of the

⁵⁸ The Editorial Board, “There is a lot of outflow of nationally trained brains,” *Maeil Economy*, 6 April 1983, 8.

⁵⁹ Ibid.

students. It is important to understand that the brain drain in the 1980s in South Korea was more like intentional than natural with the hope of controlling its flow. In the process of answering questions over, for example, what kind of the brain KAIS should raise and what kind of education KAIS would offer, the outflow of KAIS's students was shaped.

In the context of changing political regime – the long-ruling dictator Chung-Hee Park was assassinated in 1979 and Doo-Hwan Chun seized the power through an internal coup in 1980 – KAIS began to evolve. In the process of drafting and implementing the KAIS's 2nd 5-year plan in the late 1970s, two visions were collided in terms of setting its educational direction. One was to focus on raising the so-called “technologist” who would be basically technicians yet with the ability of higher-level thinking. It emphasized the importance of tinkering skills with problem-solving and manipulative abilities. As KAIS focused on meeting the industrial need for Korean economy, its utmost concern was to educate students accordingly, expecting them to play an important role in industry. To this end, in 1977, KAIS announced its plan to launch a new “specialized technical program,” which was consisted of “specialized master program” and “service technology program.” This program aimed to offer advanced skills and knowledge to working engineers. Its curriculum focused on the “project-based” class which would be “directly related to the fields.”⁶⁰ The launch of this program showed the KAIS's efforts in educating practice-oriented technologists.⁶¹

While strengthening its curriculum emphasizing tinkering skills, a bit conflicting vision arouse at KAIS in terms of its educational direction in the early 1980s. In 1983, a group of KAIS professors wrote the report on “the project plan for launching science gifted

⁶⁰ KAIS, *The 2nd Five-Year Working Plan 1977~1981*, May 1977, 11, KAIST Records Management System, Non-Public Online Archive, KAIST, 21.

⁶¹ KAIS, “The Release of the 1978 Admission Guide (Including the Specialized Technical Program),” June 22, 1977, 67, KAIST Records Management System, Non-Public Online Archive, KAIST; this plan was also related to the plan of building KAIS's branch campus in Daeduk. For more about controversies over the move of KAIS to Daeduk, see Youjung Shin, “Making the Brain for Daeduk Science Town: The Transformation of KAIST into an University in Daeduk” (2021).

education course.”⁶² It was the plan for promoting early and accelerated education for KAIS’s students in the name of gifted education. The report addressed that an early and accelerated education would be important in maximizing the thinking ability of the brain. By analyzing the Nobel laureates in science, the report concluded that many Nobel laureates received their Ph.D. in 20s – for example, E. Fermi, W. Pauli, M. Planck, who got their Ph.D. at 21, I. M. Frank, W. Heisenberg, J. J. Thomson at 22, F. Bloch, W. Bothe, R. Hofstadter at 23, A. H. Compton, R. P. Feynman, D. Glaser, E. O. Lawrence at 24 – and their most brilliant research was carried out in their twenties. With this particular understanding of the history of science, the report concluded that most of “the creative works...were accomplished in their 20 years, between 22 and 44...particularly a lot of works around the age of 29.”⁶³ Thus, in order to maximize the thinking ability of the brain and its creativity, according to the report, it was important to educate and produce Ph.D. in their 20s.

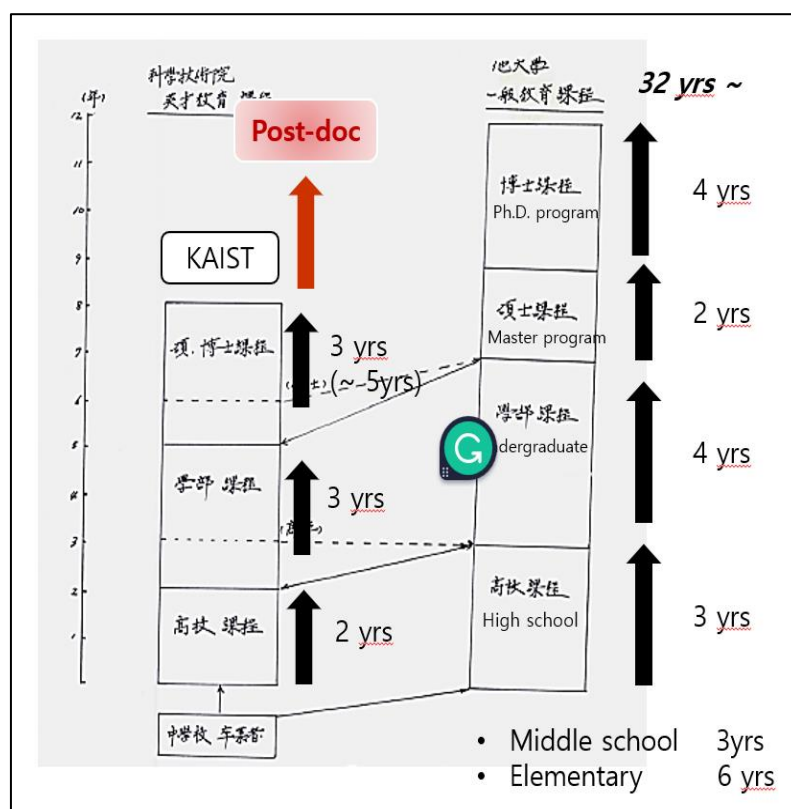
However, with the Korean traditional education system, it was nearly impossible for one to complete a Ph.D. degree within their 20s. [Figure 8-6] (the right column) shows the traditional higher education system in South Korea. After graduating from middle school, one usually attended high school (3 years), then college (4 years), then a master’s program (2 years), and a Ph.D. program (3 years). This amounted to at least 12 years. In case one attended to KAIS after college, this process can be reduced a bit, but it was still difficult to get a Ph.D. within their 20s. It was a serious problem according to the report. What the report suggested was instead to enable accelerated education by launching a new undergraduate program at KAIS which could be completed within three years. Based upon the assumption that those who graduate from the undergraduate at KAIS “would continue their study as a Ph.D. at KAIS,” the report hoped to let students finish

⁶² KAIST, *The Educational System and Programs for Gifted Students in Science and Engineering* (Seoul: Korea Science and Engineering Foundation, 1982).

⁶³ *Ibid*, 28.

their undergraduate courses in three years and get a Ph.D. in three years as shown in [Figure 8-6] (left column); the maximum would be five years.⁶⁴

[Figure 8-6] KAIS's Accelerated Education



Source: KAIST Archive

By doing so, it encouraged students to receive their Ph.D. within their 20s so that they could maximize their thinking ability. It was a plan emphasizing KAIS students' ability to think with creativity so that they could achieve remarkable discoveries, like Nobel laureates.

However, if one followed this accelerated education, there was a possibility that they did not have enough time to study and acquire research capability. Regarding this

⁶⁴ For more about high school gifted education system, see Chang-Wook Lee, "The Establishment of Gyeonggi Science High School and Formation of Korean Science Gifted Education, 1969-1986," Master thesis., (Seoul National University, 2017).

problem, what the KAIS report thought was to encourage post-doctoral training abroad. After an accelerated education at KAIS in South Korea, one was expected to receive further training as a post-doc overseas. Actually, in 1981, the Korea Science and Engineering Foundation established a new Post-Doctoral Overseas Training Fellowship. It supported Korean Ph.D. to go abroad to study science and technology.⁶⁵ Here, it was notable that, at that time, the head of the Korea Science and Engineering Foundation was Chung-Oh Lee, the KAIS's professor in the mechanical engineering department. Lee also served as the minister of science and technology. A considerable number of KAIS's graduates received the Korea Science and Engineering Foundation's Post-Doctoral Overseas Training Fellowship. Along with this fellowship, 54 Korean Ph.D. went abroad in 1982; 135 Ph.D. in 1986, and 200 in 1988. Most of them, 68%, went to the U.S. and 12% chose Japan; 7% to the UK, 5% to Canada, and 3% to Germany.⁶⁶

In short, in the 1980s, there were two conflicting visions over educating KAIS's students. While the specialized technical program emphasized the importance of nurturing tinkering ability to KAIS students with project-based courses, the accelerated education program put more focus on maximizing students' thinking capability with early and accelerated courses. While the former oriented its course toward real industrial fields, concerning performativity at practice, the latter oriented its education expecting students to deepen their study with a post-doctoral training, concerning the creativity of the brain. This conflict resulted from the lack of experience of setting up the educational direction and system for the brain in South Korea. It was more about education than research per se. As mentioned in the introduction, unlike the U.K., which had a long tradition and culture in nurturing the brain in science and technology, South Korea lacked the experience and knowledge about the way of educating and raising the brain in science and technology. The particular understanding on the Nobel laureates in KAIS's report showed how insufficient knowledge the KAIS professors had in terms of the history of science, and the way scientists carried out science in society. In this situation, conflicting visions and directions were developed which shaped the evolution

⁶⁵ The Editorial Board, "Overseas Training for 200 Premium Brains Each Year," *Maeil Economy*, September 3, 1984, 9.

⁶⁶ Hwang-Hee Cho, Eun-Kyoung Lee, Chun-Keun Lee, Sun-Woo Kim, *Science and Technology Human Resource Policy in South Korea* (Seoul: Science and Technology Policy Institute, 2002), 263.

of KAIS in the 1980s. It showed the way the developing country muddle through to grasp the nature of the brain while handling the brain drain problem.

In the end, the vision of early accelerated education won over the specialized technical program. KAIS launched its undergraduate program in 1984. A considerable number of KAIS undergraduates directly entered its graduate program, completing a Ph.D. at an early age. In 1989, the domestic media reported the KAIS “as the birthplace of the Ph.D. in 20s.”⁶⁷ Compared to this, the specialized technical program was not expanded as planned. Though the original plan was to launch seven specialized technical programs, only three programs were launched in production engineering, chemical engineering, and industrial engineering. Other than that, no further programs were initiated. In 1983, even those three programs were closed. In the late 1980s, KAIS became the symbol of accelerated education in science and technology in South Korea. A considerable number of KAIS graduates went abroad for post-doctoral studies. With the Korea Science and Engineering Foundation’s new Post-Doctoral Overseas Training Fellowship, the Korean government generally relieved the regulation of studying abroad. In 1981, the Regulations on Studying Abroad was signed by presidential executive order. It abolished the qualification test and certification, thereby encouraging the outflow of the brain. With the hope as well as the confidence of controlling the flow of the brain, the KAIS intentionally encouraged the outflow of the brain with its accelerated education while the Korean government lowered the barrier of its flow in the 1980s.

⁶⁷ The Editorial Board, “The ‘Birthplace’ of the Ph.D. in 20s, KAIST’s Graduation Ceremony,” *Maeil Economy*, 18 February, 1989, 15.

4. The Experimentation Stage

In the 1980s, the number of Korean students in the U.S. rapidly increased as shown in [Figure 8-7] almost tripled from 6,150 in 1980 to 18,660 in 1985. The annual Korean science and engineering Ph.D. recipients from U.S. institutions also increased from 131 students in 1980 to 323 students in 1985. The domestic media reported as the reemergence of the brain drain problem yet it should be noted that it was the result of delicate decision and choice. It was partly affected by the Korean government's deregulation on studying abroad from the 1980s. In this context, between conflicting visions over the kind of ability that KAIS would nurture, KAIS's educational direction was shaped in a way to emphasize accelerated studies supplementing further studies as a post-doc. While, in the 1970s, at the KAIS's emergence stage, the Korean government focused on capturing the brain abroad, and locking up them within South Korea, from the 1980s, it gradually unlatched the gate and let the brain flow though controlling its flow. It partly stemmed from the confidence that South Korea in the 1980s would be able to control the brain flow properly as it had been achieving remarkable economic growth: 7.25% GDP growth in 1981, 13.38% in 1983; remind the [Figure 8-1] which showed higher brain drain rate of lower-middle income countries than low-income countries.

[Figure 8-7] Korean Students and Ph.D. Recipients in America and Korea, 1960-93

Korean Students and PhD Recipients in America and Korea, 1960-93				
Year	Korean Students in America*		Annual Korean Science and Engineering PhD Recipients	
	Number in the USA (All Fields)	Rank as Foreign Students	From US Institutions*	From Korean Institutions
1960	2,310	6	28	NA
1965	2,666	7	66	6
1970	3,857	10	107	76
1975	3,260	11	105	259
1980	6,150	14	131	168
1985	18,660	3	323	562
1991	25,720	5	1,107	1,355
1992	28,520	5	1,119	1,402
1993	31,080	5	1,118	1,832

* For years before 1980, the figures shown are the average for the five years following the years in question (1960-64 for 1960).
Sources: Institute of International Education (1995).
National Science Foundation, Ministry of Education, Seoul, Korea.

Source: Song (1997)

In the late 1980s, with the June Democratic Struggle, South Korea laid the foundation of democratic society. Along with the continuing economic growth – 11.99% GDP growth in 1988, 7.97% in 1989, 9.88% in 1990, and 10.78% in 1991⁶⁸ – the political stability came to South Korea. In addition, with the fall of the Berlin Wall in 1989, the geopolitical tension between North and South Korea was also relieved. On September 1990, the South-North ministerial-level talk was held for the first time. Against its backdrop, in the early 1990s, there was a few reports concerning the brain drain problem in South Korea. Though the number of the brain who went abroad kept growing, so did the number of the brain who returned to the country. Not just the economic development but also the rise of democratic society as well as the relieved geopolitical tension in South Korea played a role in bringing the brain back to the country. In 1993, the international science exposition “Daejeon Expo” was also held. In 1994, Samsung released the world’s first 256M-DRAM chip. In the 1990s, both economically and culturally, the brain found South Korea attractive than in 1970s. In 1997, South Korea joined the Organization for Economic Cooperation and Development (OECD) as its 29th member.

However, the harsh economic crisis hit South Korea in the late 1990s. In the middle of the Asian financial crisis started from Thailand, Indonesia, Philippines with the rapid outflow of foreign capital, the foreign exchange reserves of South Korea got to be drained. In 1997, the Korean government received the International Monetary Fund (IMF)’s bail-out package. In this context, a lot of companies including big companies like Daewoo Group, SsangYong Group, and Haetae Group went bankrupt. Among top 30 companies in South Korea in the 1990s, 17 companies went bankrupt or dissolved. The remaining companies also suffered greatly. It brought a huge confusion and shock in Korean society.⁶⁹ As South Korea had just entered to the OECD which symbolized the rapid development of the nation, the receive of IMF’s bail-out package was not just an economic but also cultural and social shock to South Korea. In this context, the brain drain and the concern over it were raised again. In 1998, the domestic media reported

⁶⁸ “South Korea GDP Growth Rate, 1961-2022,” Macrotrends, accessed 19 September 2022, <https://www.macrotrends.net/countries/KOR/south-korea/gdp-growth-rate>.

⁶⁹ Sang-Hwan Jang, “South Korea’s Capitalism, Why Faced the Era of IMF,” *Critical Review of History* (1998): 97-118; Jae-Min Lee, “The Nature and Consequences of the Korean Foreign Exchange Crisis,” *The Korean Economic Forum* 9.2 (2016): 79-135.

that “the advanced brain who went abroad this year exceeds 3000” which was “10 times more than last year.”⁷⁰ Another media also reported that “the brain which was attracted to South Korea made a U-turn” and went back to their country.⁷¹

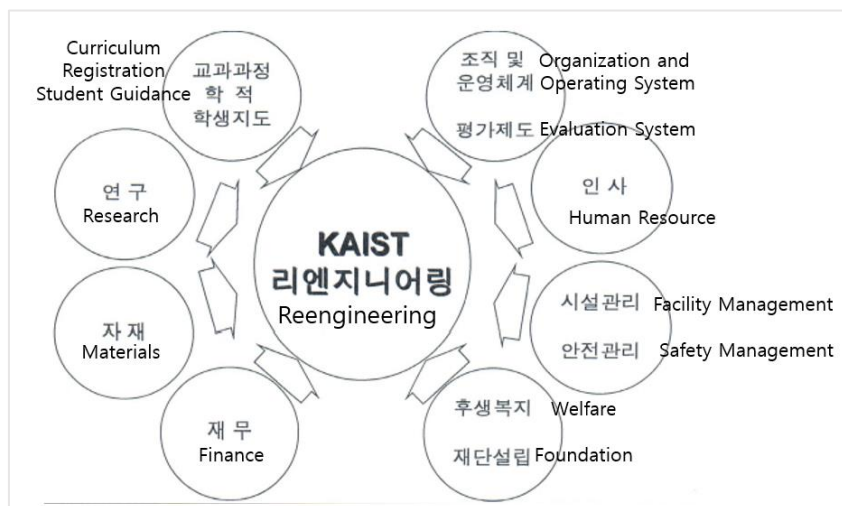
In this changing situation, KAIST adjusted itself in terms of its educational direction and institutional strategy. In 1997, KAIST released a new report called “KAIST Re-Engineering Report.” By addressing the term “re-engineering,” KAIST showed its willingness to change its component, mechanism, and system. The “KAIST Re-Engineering Report” stressed that “this is the time that the change and the repair are inevitable in order to leap toward world’s top-10 university by accommodating rapidly changing domestic and global environments.”⁷² It emphasized that, now, KAIST “could not settle for its 25 years achievements.” From the late 1990s, KAIST sought for different strategies and mechanisms in raising and circulating the brain in South Korea. In other words, from the late 1990s, KAIST entered to the stage of experimentation. The scope of KAIST’s re-engineering project was wide. From education and research to welfare as shown in [Figure 8-8], KAIST conducted an experiment.

⁷⁰ The Editorial Board, “The advanced brain who went abroad this year exceeds 3,000: 10 times more than last year,” *Maeil Economy*, 21 April 1998, 1.

⁷¹ The Editorial Board, “The brain which was attracted to South Korea made a U-turn,” *Chosun Il-bo*, 26 September, 1998, 3.)

⁷² KAIST Re-Engineering Task Force, *KAIST’s Re-Engineering Project* (KAIST, 1997).

[Figure 8-8] KAIST's Re-Engineering Project



Source: KAIST Archive

In relation to the brain drain, the interesting impetus that KAIST pushed forward was the consideration of the kind of the brain that KAIST would attract. So far, when KAIST as well as the Korean government concerned on reversing the brain from outside the country, the major target was the Korean brain, i.e. well-educated Koreans who leave outside the country. There was less concern on the foreign brain per se, i.e. non-Korean yet who is very talented. In 2001, however, KAIST decided to open its entrance to foreigners.⁷³ More specifically, before 2001, it was possible for foreigners who graduated from foreign universities could apply to KAIST. So, KAIST allowed foreigners to apply for its graduate programs, but not for its undergraduate program. However, in 2001, KAIST revised its rule so that for those who graduated from domestic and foreign high schools yet foreigners could also apply for KAIST. It meant the opening of KAIST's undergraduate entrance to foreigners. As a result, the first foreign undergraduate student, Yan Long, who was a Chinese-Korean, entered to KAIST in 2003.⁷⁴ The number of foreign students in KAIST's graduate program also got rapidly increased in 2000s.

⁷³ Sang-Hee Kwon, "KAIST Undergraduate System Allows the Entrance of Foreigner," *Electronic Times*, 14 May, 2001, <https://www.etnews.com/200105110038>.

⁷⁴ Chun-Ho Oh, "A Foreigner Entered to KAIST as an Undergrad for the First Time," *Hankyung*, 2 April, 2006, <https://www.hankyung.com/society/article/2002102221531>.

While 77 foreigners entered to KAIST's graduate program in 2003, there were 142 foreigners in 2004.⁷⁵

In this context, in 2004, KAIST decided to elect a foreigner as its president. Robert Laughlin, who was the professor of physics and applied physics at Stanford University and won the Nobel Prize in physics in 1998 with Horst L. Störmer and Daniel C. Tsui, became the KAIST's president in 2004. It was the first moment in which non-Korean international scholar led KAIST as its president. By doing so, KAIST attempted to make it as the center of knowledge not just within South Korea but also in a global world. With this change, KAIST aspired to attract not only Korean brain but also diverse international brain.

In the context of the government's emphasis on the internationalization of the higher education system in overall in 2000s, KAIST was in the forefront of its reformation. Laughlin tried to make KAIST global by, for example, increasing the number of international faculty, encouraging class in English, the diversification of funding of KAIST. He also conceived to enlarge the number of students enrolled annually from 700 to 2000. By doing so, he thought to establish not only science and engineering department but also the department of law, art, pre-med, humanities et ac. His ultimate goal was to transform KAIST from science oriented specialized school to full-fledged university. He thought that, to attract diverse international brain, KAIST should offer diverse intellectual stimulus by enlarging the kind of department and courses while expanding the kind of students enrolled to KAIST. To this end, he sought to diversify the KAIST's finance in addition to the Korean government's support. Laughlin took the California State University as a model for KAIST.⁷⁶

However, Laughlin's plan faced harsh opposition from KAIST's professors. First of all, his idea of diversifying the course of KAIST – including pre-law or pre-med courses – had raised concern. For KAIST's professors, the identity of KAIST was science and technology

⁷⁵ Yo-Han Choi, "Foreign Students, Coming to Daeduk Research Park," *The Science Times*, 28 April, 2005, <https://url.kr/vibpnj>.

⁷⁶ Ho-Jae Jung, "Honest Comments from Robert Laughlin, the President of KAIST," *Dong-Ah Il-bo*, 25 October, 2005, <https://shindonga.donga.com/3/all/13/104912/1>.

oriented specialized university. Once KAIST expanded its area toward art, law, humanities, medicine, etc, KAIST's identity would get to be blurred; difficult to differentiate from other universities. In addition, it would be difficult for KAIST to focus on nurturing scientists and engineers as some of students would be distracted toward other majors. However, Laughlin thought that the government or the institute should not intervene the students' education and career per se. Instead, the university should just offer a wide range of excellent courses, let students explore, and find a few students who want to major in science and technology. In Laughlin's view, that's the way the U.S. has done for nurturing scientists and engineers in universities. Instead of concentration, the diversification of course, students, finance, and professors were what Laughlin want at KAIST.

In addition to the conflict over the vision of KAIST, Laughlin were in conflict with some of KAIST's professors due to his communication and management style. In the process of drafting KAIST's vision, including the expansion of enrolled students, he seldom made a place for discussing his ideas with KAIST's members. He made his vision and plan in his office and just released it publicly. Thus, KAIST's members criticized his communication style as dogmatic and undemocratic. Yet, regarding this criticism, Laughlin often just stated that it was inevitable as the reformation would be always followed by criticism. It should be remembered that, as a foreigner, Laughlin was unfamiliar with Korean culture and system. Then, it was needed for Laughlin to put effort into learning local culture and environment. Without this effort, his vision – regardless of how much good it was – was difficult to be realized. Some of KAIST professors who even sympathized with Laughlin's vision on KAIST criticized his communication style and leadership.⁷⁷

In the end, after a lot of controversies and conflicts, Laughlin failed to serve a second term and his reformation was stranded. Laughlin's rise and fall showed, on the one hand, the attempt of KAIST to take the experimentation and reformation in the changing social context, but, on the other hand, their confusion over how to make it with what by whom. Laughlin's vision emphasized the importance of diversity in terms of education and

⁷⁷ Yo-Sep Kim, "Laughlin, What on Earth Did He Wrong," HelloDD, 6 April 2006, <https://www.hellodd.com/news/articleView.html?idxno=14966>.

students in order to make KAIST as a global university. While KAIST has been emerged and evolved as science and technology-oriented school, in the experimentation stage, it got challenged to be a full-fledged university; diversification instead of concentration. What would it take for KAIST to attract not only Korean brain but also diverse international brain? This was the question that KAIST dealt with in the stage of experimentation. In addition, who would be able to lead this KAIST's experimentation? By foreigners? In the early 2000s, KAIST thought so and invited Laughlin, a Nobel-Prize winner, as its president, but it turned out to be very contentious.

Laughlin left the KAIST in 2006 but the KAIST's experimentation did not stop. In 2006, the new president, Nam-Pyo Suh, came to KAIST. Nam-Pyo Suh was born in Korea in 1936 but he attained the citizenship in the U.S. in 1963. So, Suh was a Korean-American. KAIST elected the foreigner again as its president, but, this time, searched for someone who would be familiar with Korean culture and system. Before leaving the country in 1954, he went to middle and high-schools in South Korea. Then, he entered to the Massachusetts Institute of Technology receiving his bachelor degree in mechanical engineering; he did not experience Korean university system. After receiving Ph.D. in Carnegie-Mellon University in 1964, he spent four years as a professor at the University of South Carolina, then, in 1970, moved to Massachusetts Institute of Technology's department of mechanical engineering. There, Suh established his reputation as a talented engineer serving, for example, the head of the MIT's department of mechanical engineering for ten years. As the foreigner born in Korea having achieved well-established scientific career, Suh was chosen as the right person to carry out the experimentation of KAIST.

After becoming the KAIST's professor, Suh released his five-year development plan of KAIST. Like Laughlin, Suh also emphasized to make KAIST as a global university. To this end, Suh, on the one hand, strengthen the evaluation system of students and professors and, on the other hand, broadened the qualification requirement to enter to the KAIST; diversify KAIST's students and professors while evaluating them with tough criteria. For professors, Suh strengthened their promotion evaluation system including tenure examination. In 2007, among 38 professors who underwent tenure examination, 15 professors could not pass the examination which was a big shock to the Korean

academic society.⁷⁸ For students who had received the government's scholarship – so, tuition exemption with the support of the cost of living – Suh obligated them to receive the certain GPA in minimum. Those who did not receive the minimum GPA had to pay a certain amount of tuition. It contributed to spurring competition among students at KAIST.⁷⁹

In addition, he pushed forward making courses at KAIST in English. First, Suh obligated professors to offer KAIST's major courses in English, then made all the courses including the liberal art classes in English. Again, it was an attempt to attract not only the Korean brain but also the international brain. Yet, it raised a huge tension among KAIST members. As English is not the first language for Koreans, both professors and students found difficult to interact in class in English. Some of KAIST's professors opposed Suh's initiative by saying that "100% English teaching in KAIST is a crazy thing."⁸⁰ They stressed that "there are students who could not follow the class because of English [not because of the class's contents per se]." Students also found confusing. Some students sent a letter or posted a hand-written poster urging for Korean class at KAIST.⁸¹ However, Suh adhered to the opinion that 100% English class is necessary to make KAIST global, and continued his initiative.

However, Suh's experiment could not last long. In 2011, a series of students (as well as a professor) killed themselves in a row. Though they might have different reasons, it invoked a huge concern on the KAIST's ongoing experiments on students. To make KAIST global, Suh carried out several experiments like strengthening evaluation systems, diversifying KAIST's entrance exam, and encouraging 100% English class. Yet, it was criticized as, first, spurring competitive environment instead of collaborative

⁷⁸ The Editorial Board, "There Is No Anymore "Cheolbaptong" for KAIST professors and students," NoCut News, 26 March, 2007, <https://www.nocutnews.co.kr/news/269606>.

⁷⁹ Doo-Ho Kim, "A challenge for World 1st University, Nam-Pyo Suh, the President of KAIST, as the Leader of University Reformation," Interview 365, 7 November 2008, <https://www.interview365.com/news/articleView.html?idxno=1869>.

⁸⁰ Hwan-Joo Heo, "KAIST 100% English Class is a Crazy Thing," Pressian, 11 April 2011, <https://www.pressian.com/pages/articles/104017>.

⁸¹ The Editorial Board, "[Letter] KAIST Student's Appeal, 'Please Offer Korean Class Too,'" ScienceON, 1 February, 2012, <http://scienceon.hani.co.kr/32497>.

environment at KAIST, second, unifying and standardizing KAIST's classes despite the diversification of entrance exam, and, third, enforcing impractical 100% English class despite the opposition from KAIST's members. Suh also struggled with communication problem. For example, right after a series of tragic incidents in 2011, there was an emergency meeting that students requested for Suh yet Suh not only came late but also hardly communicated with students. It represented the way Suh communicated with KAIST stakeholders.

In the end, in 2013, Suh left KAIST before his official term ended. It was the second time that the foreigner president could not serve its term at KAIST. It indicated how difficult it was for foreigner presidents to make a good rapport with KAIST members. To realize some vision – regardless of how good it was – it was necessary for those who carried out experimentation to have a close communication and interaction with its members. Without it, experimentation would be difficult to be achieved. To make KAIST global – in other words, to attract not only Korean brain but also international brain as the center of knowledge in the world (not just in South Korea) – KAIST members in the 2000s used to think that bringing the leader outside the country would be important. Yet, after undergoing Laughlin and Suh, they realized that KAIST's experimentation should be derived from inside KAIST; not from outside of the country. After Laughlin and Suh, no foreigners were further elected as the president of KAIST. Nevertheless, KAIST's experimentations to make it as the global center of knowledge have been still kept going on.

5. Discussion & Conclusion

So far, this paper traced the way KAIST dealt with the brain drain problem in South Korea from the 1970s to 2010s. To date many people remembered that KAIST ‘solved’ the brain drain problem in the 1970s by offering unprecedented support and attracting the brain from overseas. Yet, as this paper shows, the brain drain problem was not ‘solved,’ though it could be temporarily relieved in the 1970s. Instead, the brain drain problem in South Korea kept rising over and over in the 1980s, in the 1990s, and even in the 2000s. KAIST, on the one hand, contributed to outflowing the brain while also putting effort in stimulating the inflow of the brain. In the changing sociopolitical contexts over the flows of the brain in South Korea, KAIST has kept concerning and shaping the way the brain flows in/out of South Korea. This paper regarded KAIST as a medium to examine the evolving ways of addressing the brain drain problem in South Korea. It divided KAIST’s development into three stages: (1) emergence, (2) evolution, and (3) experimentation stage.

At the emergence stage, KAIST focused on attracting the brain in abroad by offering material incentive with the support of the USAID. In the context of changing domestic as well as foreign context over the foreign aids, the USAID provided \$6 million to establish KAIST though the Korean government also provided the matching fund. It made KAIST possible to offer unprecedented material support for those who would come to KAIST. It should be noted, however, material incentive did not naturally draw the brain in overseas. Continuous persuasion and negotiation processes were needed which for example Chung played a role as a Korean yet a professor in the U.S. KAIST emerged as the result of ‘new wine in new bottle strategy:’ a new institute in a non-traditional ministry. By establishing KAIST under the supervision of Ministry of Science and Technology - not the Ministry of Education which supervised most of the higher education system in South Korea - the government enabled KAIST to use unprecedented material incentives including salary, housing subsidies, and research support in attracting the brain from overseas.

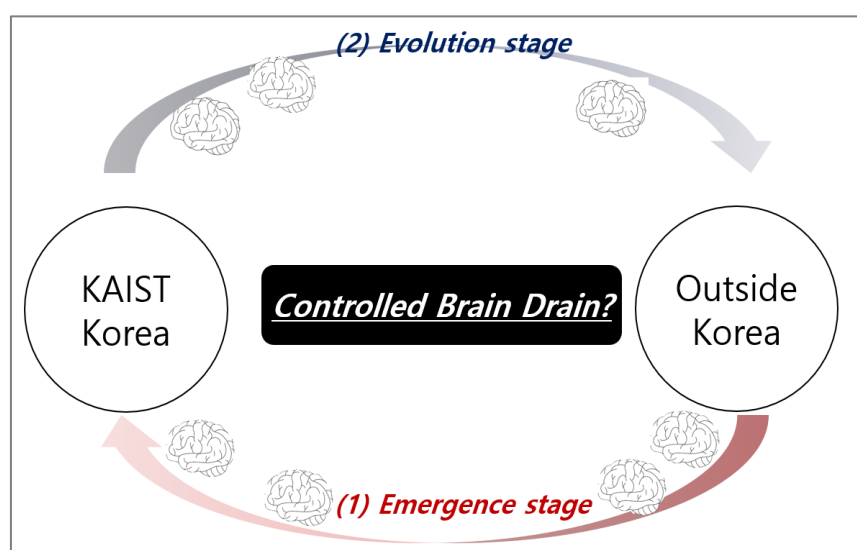
Yet, the strategy of ‘a new wine in a new bottle’ had limitations. First, offering unusual material incentive had a sustainability issue. The USAID’s support was planned to be ended within five years. The number of students and professors kept growing. How would KAIST sustain its material incentive in this situation? In addition, partly due to KAIST, there were increasing demands for increasing material support for traditional universities as well. It should be remembered that, if using material incentive for some institute, it could raise a social tension among stakeholders as this paper showed, and it would raise sustainability issue. From the late 1970s while addressing this problem in the context of the changing political regime, different visions were shaped and collided in deciding the way KAIST would evolve in the 1980s. Though, in South Korea, in the context of rapidly growing economy in the 1970s and the 1980s, KAIST could sustain its material support to both students/professors to certain degree, it would not always happen to other countries.

The second problem KAIST faced in the evolution stage was the generation problem. Though KAIST attracted talented brain from overseas in the emergence stage, the next question remained: how they were gonna raise and nurture the next generation? Unlike advanced countries, like U.K., where the term, brain drain, originally came from, South Korea lacked the experience and culture in nurturing science and scientists. Thus, South Korea could not but learning the brain while handling the brain drain problem. What kind of the brain South Korea needed? What those brain needed in order to do brain activities? What would be the role of KAIST and the Korean government? These were the example questions that KAIST should have solved in its evolution stage. As this paper revealed, amid handling these questions in the 1980s, the new flow of the brain was shaped around KAIST.

As shown in [Figure 8-9], while at the emergence stage, KAIST focused on attracting the brain from outside, at the evolution stage, KAIST intentionally made some outflow of the brain by promoting postdoc education in abroad. With the launch of the gifted education in the conflicting vision over it – technologist vs Ph.D. in 20s – KAIST began to offer an early and accelerated education based on the belief that the brain’s creativity would be maximized in their 20s; which was not quite true if seeing the history of science more closely. In order to complement the accelerated education, KAIST encouraged

students to conduct postdoc study in abroad along with the government's de-regulation on studying abroad in the 1980s. Thanks to the formation of the Korean Science and Engineering Foundation's Post-Doctoral Overseas Training Fellowship, which Lee who was the professor of KAIST as well as the minister of science and technology played a role, the controlled circulation of the brain was shaped at KAIST in the 1980s. Concerns on the brain drain were raised in the 1980s yet, this time, it was more of the result of intention and deliberation with the confidence of controlling it.

[Figure 8-9] The Emergence and Evolution Stage of KAIST



Source: Author

There was a downside of this strategy, though. As KAIST outsourced the postdoc study in abroad, the domestic system for postdoc education lost the opportunity to grow. In the 2000s – even currently – criticisms have pointed toward the poor postdoc system in South Korea; see for example the report of “the size and characteristics of post-doc researchers in South Korea” published in 2020.⁸²

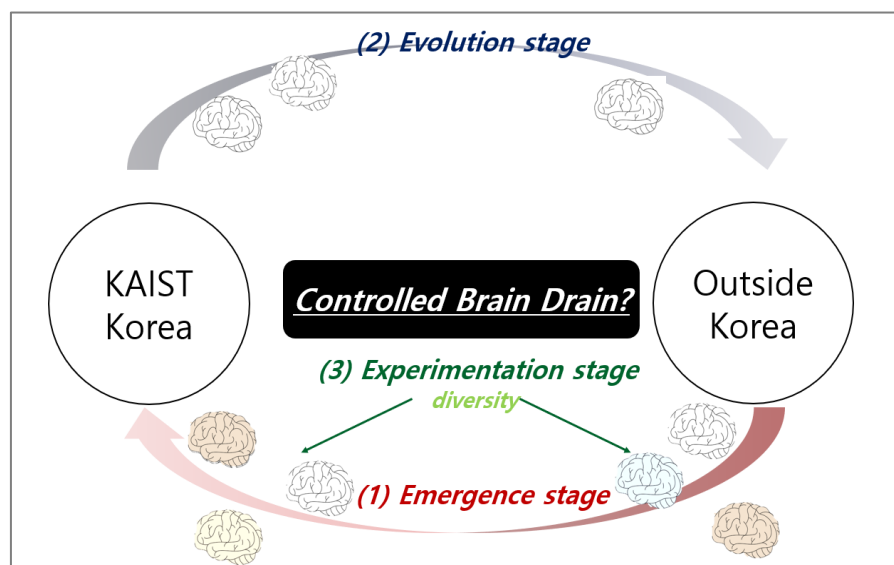
Though the aspiration of controlling the brain flow could work to certain degree in the 1990s at which South Korea could attain not only economic but also political and cultural

⁸² Kibeom Park and Hyunjoon Park, The Size and Characteristic of Domestic Postdoctoral Researchers, STEPI Insight (2020), <https://www.stepi.re.kr/site/stepiko/report/View.do?reidx=255&cateCont=A0501>.

development – i.e., democratization movement, the host of Daejeon Expo, the invention of 256M-DRAM chip by Samsung – it began to dismantle in the wake of IMF crisis from the late 1990s. Again, the concern on the brain drain was highly reported in domestic media. It alarmed the rapidly outflowing of the brain from South Korea to abroad. In this situation, KAIST also readjusted itself. While at the emergence and evolution stage, KAIST mostly concerned about the Korean brain, from the 2000s, not only the Korean but also various international brain began to be actively attracted. For example, by revising the rule of its entrance, KAIST broadened the scope of students and the first foreign undergraduate had entered to KAIST in 2003. Also, by inviting the foreigner as the president of KAIST, it carried out several experiments to make KAIST as the global center of knowledge.

As shown in [Figure 8-10], KAIST's experimentation stage was the stage at which not only the quantity but also the diversity of the brain in the flow of the brain in South Korea. While at the emergence and evolution stage, most of concerns over the brain drain focused upon the number of the brain that went in or outside South Korea, at the experimentation stage in the 2000s, KAIST took experiments to diversify the kind of the brain including international brain in the campus. However, KAIST struggled with figuring out, in what way, and, by whom, they should carry out experiments. It was also related to the lack of experience and culture in nurturing the brain and its flow within South Korea as the newly industrialized country. KAIST attempted to make it global by bringing some global figures like Robert Laughlin, a Nobel prize winner in physics, and Nam-Pyo Suh, a mechanical engineer from MIT. Yet, a series of struggles and failures of those two presidents in completing their term showed the importance of driving experiments from inside KAIST; instead from outside of the country. KAIST's experimentation is still going on nowadays.

[Figure 8-10] The Experimentation Stage of KAIST



Source: Author

As shown in this paper, the brain drain problem was not what could be easily 'solved' in the developing country. Despite the launch of the KAIST which offered privileged support to attract the brain, the brain drain problem was not eliminated; though temporarily resolved. In South Korea, the brain drain problem came over and over in different forms. In this sense, it should be noted that, instead of trying to 'solve' the brain drain problem, the better approach to this problem is to put effort in 'caring' the brain flow. To this end, it is needed to (1) have a prepared mindset to keep watching the brain flow since this problem would take long time (2) keep an eye on examining changing local and global contexts that affected the circulation of the brain, and (3) keep monitoring and caring the quantity as well as diversity of the brain flow. This would be the first step in approaching the brain drain in developing country.

In this process, as emphasized in this paper, it is important to build the developing country's own experience and viewpoint in nurturing the brain and its flow. Unlike U.K., where the term, brain drain, came from, developing countries mostly lacked experience, system, and culture in nurturing the brain per se. So, before talking about the problem of brain drain, developing countries should take time in realizing more about the brain; for example, who is the brain, what the brain wants, what kind of the brain they are

talking about, how to nurture the brain, what is the role of the society, and how they are gonna answer those questions in what ways. These are the questions that developing countries should grasp while dealing with the brain drain problem, simultaneously. In addition, it should be underlined that these are not the questions whose answers could be found in textbook or scholarly articles. Rather, answers to those questions should be continuously found and shaped amid the specific local and global context. So, the issue is whether the developing country is ready for 'caring' the flow of the brain instead of 'solving' it.



CHAPTER 9

Summary of the National Master Plan Draft for STI R&D and Application in Lao PDR

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 9. Summary of the National Master Plan Draft for Science, Technology and Innovation R&D and Application in Lao PDR

Soumana CHOULAMANY, Anousone INTHAVONG, Vilavsone BOUPHALATH, Khamnhong SICHANTHAVONG, and Thanva Sisoulath

Ministry of Technology and Communication (MTC)

1. Background

Research and Development (R&D) is a key factor to improve the national economy, particularly R&D on science and technology (S&T). Policymakers have taken R&D and S&T for innovation to strengthen commercial productivity. It aims to transform economic structure to industrialization and modernization in achieving the sustainable development goals (SDGs) by 2030.

In order to carry out the National Socio-Economic Development Plan for the 9th year (2021-2025), in the past, the government has planned and expanded the science, technology and innovation infrastructure of the country from the center down to the local area so that people can access and use science, technology and innovation in life, business and socio-economic development. Therefore, there is a need to establishing and improve legal system and policies on science, technology and innovation to suit the growth of the socio-economy particularly formulate National Master Plan for STI and application in Lao PDR as one of the most important drivers of the 9th NSEDP achievements.

In recent years, science and technology development has advanced rapidly. Therefore, Lao PDR should seize this opportunity for socio-economic development, especially the application of science and technology in production, services, and human resource

development to achieve higher quality growth and to be able to compete at the regional and international levels.

In Lao PDR, it is essential to develop the National Master Plan for Science, Technology and Innovation Development and Application as well as the National Policy on Science and Technology Research. The plans should focus on survey and study development indicator of relevant sectors, human resource, science and technology research innovation networking and infrastructure development. It is specially to increase competitiveness of trade, capability of research, and industrial development to be more innovative. Recently, there are several areas that are important for developing of science and technology such as technology commercialization, small and medium-sized enterprises (SME) innovation, international science, technology and innovation (STI) Collaboration, etc.

1.1 International environment

They are constrained for STI activities due to a lack of financing and a shortage of skilled labor

With respect to resources, the government invests in human resource management by establishing higher education institutes, vocational institutes, and skill development institutes over the past decades. They have been increased in number, but limited qualification

With respect to institutions, R&D implementation are at the early stage of STI development, which focus on technology utilization rather than invention or creation. There are numbers of research and education institutes, but applied research to improve products and services is poor. Research based course is insufficient to build research base and human capital. Research fund is limited

Despite these, in terms of R&D, policies lack of a term development, particularly in science and technology research, and instrument are not clear and complex. Implementation periods of R&D and S&T projects are not appropriate to support a continuation of researches, which obstruct to accumulate knowledge and human capital.

Therefore, it is essential to modify research term. For example, short-term, mid-term, and long-term projects could be determined as 1 year, 2-3 years, and 4-5 years, respectively.

1.2 Domestic environment

Despite many achievements in STI, it is still seen that the research, management and use of research results in development is not yet as wide as it should be due to research and development sectors are not deepening their responsibility for R&D.

The Lao government established the Ministry of Science and Technology (MOST) in 2011, in regard with the decree number 309/PM, officially issued on September 28, 2011. It indicated the government's interest in fostering R&D. However, Lao PDR sought to reduce the overlap of R&D policies. In February 2021, the government settled the MOST and transferred its roles and responsibilities to Ministry of Education and Sports (MOES), Ministry of Post and Telecommunications (MPT), and other relevant line ministries. In October 2021, the government established the Ministry of Technology and Communications (MTC) under the Decree on Organization and Activities of Ministry of Technology and Communications, No.625/PM, officially issued on October 22, 2021. It is responsible mostly for MOST's responsibilities and roles in terms of governance and management of technology, innovation, telecommunication, internet, post, and cyber security.

Lao PDR's 7th Socio-Economic Development Plan (2011-2015) included the development technology and promotion of technology transfer was one of the significant elements in such development plan. The technology and technology development for SMEs was supported under the Development Strategy of Science and Technology (2013-2020) of the Ministry of Science and Technology (MOST).

A legal and policy framework supporting development of technology in universities, R&D labs and support centers on IP rights exist has also been established by the Government (ERIA and OECD, 2014).

There is limitation on the financial support schemes funded by the government, donors, and other organizations. The pilot public funds supporting R&D activities are also provided with a commercial orientation and limited allocation.

1.2.1 Future Vision 2030

- Transitioning to the emerging middle-income country status from a least-developed country in 2020 (5 year NSEDP, period VIII, (2016-2022))
- Transforming Lao PDR to a Sustainable Lower Income Economy by 2030 to promote R&D and application of STI into all sectors.

1.2.2 Role of STI to achieve the vision 2030

1) **Strategy 1:** Technology Commercialization – Promote the linkage of the supply and demand for technologies.

- **Objective 1-1:** Development of human capacity and facilities related to technology commercialization and technology transfer.
 - **Action 1-1:** Skill development for the staffs of both public and private sectors related to TC and TT;
 - **Action 1-2:** Allocate the stable budget for development of human capacity and facilities related to TC & TT (e.g., funding for GoL, private sectors, ODA, etc.)
 - **Action 1-3:** Improve the management system for TC & TT (i.e., organization system and information system including statistic data related to TC & TT).
- **Objective 1-2:** Promotion the awareness on technology commercialization, technology transfer, and IPRs knowledge for both public and private sectors.
 - **Action1-4:** Improve public interest in technology and innovation including TC, TT, and IPRs.
 - **Action 1-5:** Support SMEs and industries to identify, adopt, and utilize existing technologies from foreign or domestic sources (i.e., agriculture, handicraft, eco-tourism, foods, technology services, etc.).
 - **Action 1-6:** Sharing information or data related to TT, TC, and IPRs by making available to public

- **Action 1-7:** Establish the policies to support the promotion and activities related to technology & technology transfer/commercialization in Lao PDR.
 - **Objective 1-3:** Establishment of the cooperation mechanisms between universities or research institutions with industries.
 - **Action 1-8:** Promote and strengthen the public-private-partnership (PPP) related technology commercialization
 - **Action 1-9:** Develop the technology/industrial parks (physically) that facilitate interactions among enterprises, entrepreneurs, and researchers (i.e., this can attract investment from private sectors and foreigner countries.
 - **Action 1-10:** Establish agency and/or organization that plays a role of linker between technology owners (universities, research institutes, etc.) and technology users (SMEs, industries, etc.).
 - **Objective 1-4:** Strengthening cooperation, integration and interconnection related to technology regionally and internationally.
 - **Action 1-11:** Improvement of the joined project related to technology & innovation in order to attract funds from foreign countries.
 - **Action 1-12:** Improving the effectiveness of policies aimed at attracting and facilitating entry of Foreign Direct Investment (FDI) related to technology and innovation.
- 2) **Strategy 2: MSME Innovation** – Promote the development MSMEs to generate demands for technologies.
- **Objective 2-1:** Improved human resources in Science and Technology R&D
 - **Action 2-1-1:** Promotion of scientific R&D with a concentration of basic research
 - **Action 2-1-2:** Qualified human resources and skilled labor to enhance entrepreneurship and improvement of MSMEs' competitiveness
 - **Action 2-1-3:** Strong and modern cooperation and integration with the region and internationally
 - **Objective 2-2:** Effective policy ecosystem to facilitate MSMEs
 - **Actions 2-2-1:** Effective technology and innovation utilization in accordance with green and environmentally friendly

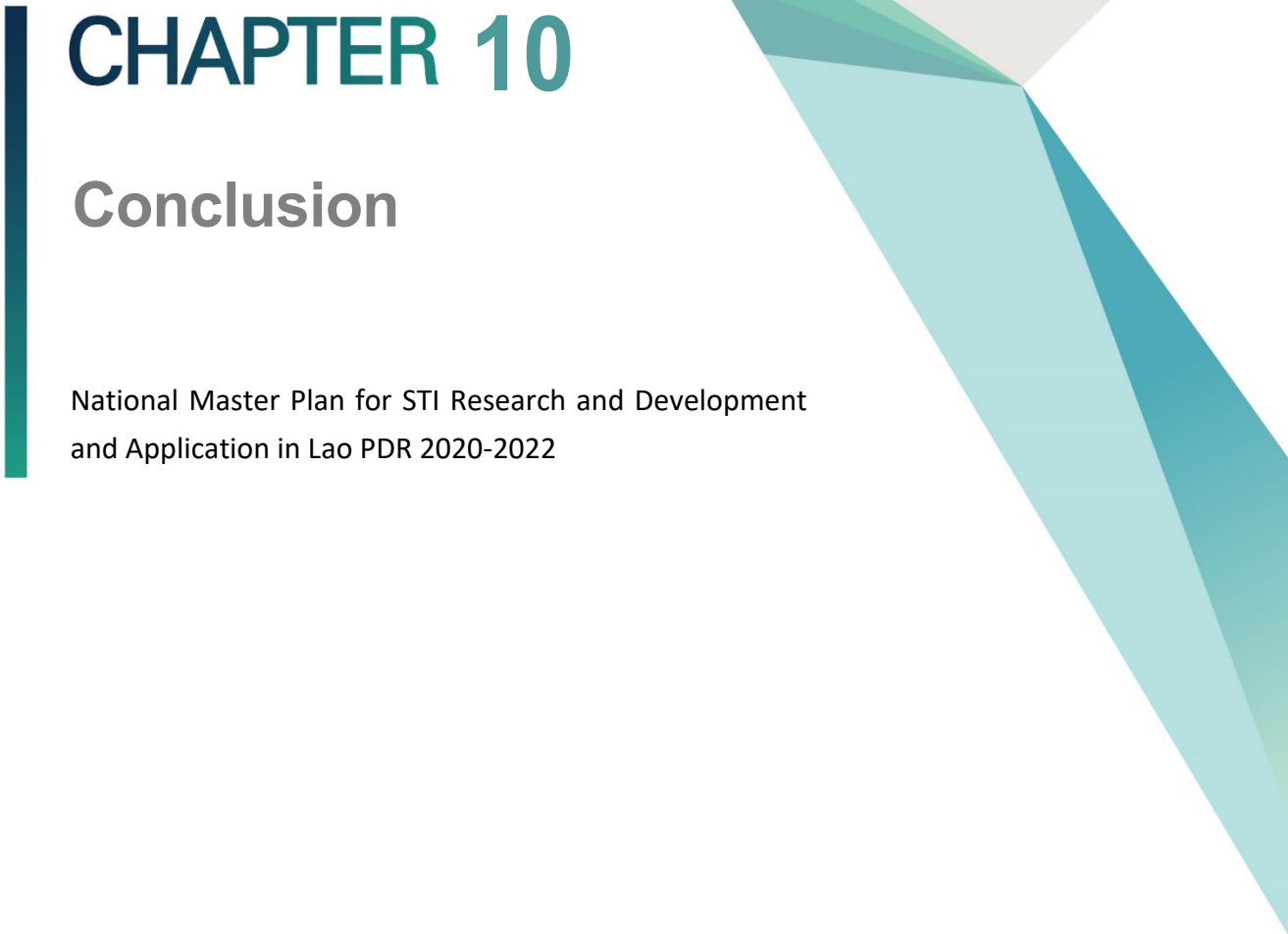
- **Actions 2-2-2:** Efficient public administration in accordance with the direction of public administration by law
- **Action 2-2-3:** Strong and efficient cooperation between the public and the private sectors

3) Strategy 3: International STI Cooperation – Take advantages for the strengthening of S&T capacity.

- **Objective3-1:** Capacity building and human resource development
 - **Action3-1:** “Development of human resources in science and technology by enhancing the development of skills and knowledge for employees of the relevant ministries and universities, institutions, including those of the industrial zone”
- **Objective3-2:** Introduction of effective technology and utilization of innovation
 - **Action3-2 :** Promotion of Technology Transfer
 - **Action3-3:** Promotion of joint R&D
- **Objectiv3-3: Infrastructure in Science and Technology**
 - **Action3-4:** Organization structure and management of Science and Technology
 - **Action3-5:** Strengthening infrastructure on STI

1.2.3 Master Plan Implementation.

- Use this national master plan as a guide to creating a 5-year development Plan of DTI (2021-2025)
- It is a research base for creating National Master Plan for Technology and Innovation in (2026-2030)
- In three themes indicated: technology commercialization, SMEs innovation and International STI collaboration are the important strategy in national master plan to achieve the vision 2030
- Pursuing the dissemination of this NMP for STI R&D to relevant ministries, public and private institution, university in order to build cooperation mechanism for development of STI in Laos through joint project, program together.

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CHAPTER 10

Conclusion

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022



Chapter 10. Conclusion

Inkyoung SUN, STEPI

The K-Innovation Partnership Program with Lao PDR (2020-2022) was launched with the title of the “National Master Plan for STI R&D and Application in Lao PDR” in the spring of 2020. A couple of unexpected events hindered the project teams-STEPI and Ministry of Science and Technology (MOST) in Lao PDR-of both countries from lively communicating and operating the project, such as the COVID-19 pandemic and Laotian government restructuring followed by the dissolution of the project counterpart MOST in early 2021. Nevertheless, the project was successfully resumed with the Ministry of Technology Communications (MTC) in the midst of the second year of the project 2021 thanks to the great support both from the Laotian Embassy in Korea and the Korean Embassy in Lao PDR. Following the hybrid High-Level Meeting in September 2021, STEPI conducted the STI Survey targeting civil servants studying in Korea and STI experts of Lao PDR in order to identify top priorities and key STI challenges facing the Laotian government that can be referenced for bilateral collaboration with Korea. According to 46 respondents, the Laotian government officials and STI experts stressed that the lack of human resources and financial support and the limited international network are critical challenges in STI development.

During a series of workshops and meetings conducted throughout the year 2022, our bilateral research team explores several key questions on STI master planning to develop the framework of the master plan; why is the STI master plan necessary in Lao PDR? How will the master plan be different from existing policies and plans? What would be the urgent and long-term issues to be solved by MTC to achieve the 9th National Socio-economic Development Plan? How does MTC mobilize the financial and human

resources to solve current challenges and implement the master plan? How will MTC monitor and evaluate the implementation of the master plan?

In the course of examination of the national innovation system of Lao PDR and current challenges facing MTC officials in developing STI policy, the initial suggestions were made to align the development of the national STI master plan with DTI 5-year development plan and MTC policies and development plan (2021-2025) under the umbrella of the 9th National Socio-Economics Development Plan (2021-2025). The suggestions particularly apply to the vision and objectives sections of the master plan. They also emphasize a clear direction and objectives of the plan and identification of key players and available resources for actionable and effective implementation of the plan in the future.

The suggestions from the Korean experts have been echoed in Chapter 1. Starting with the concept and critical components of a national STI plan and its importance in the context of national development, Chapter 1 provides the background on why and how the Korean government switched the national development plan to the Science and Technology Framework Plan. It covers the whole dynamics of the national STI masterplan of Korea in rich from the development of rationales of the master planning, organization of planning processes, structure of contents, and to implementation and ex-post evaluation. The chapter may work as a key reference for the policy community to re-visit and gain insights into STI policy planning and implementation.

Three strategic areas of the Master Plan were selected based on the analysis of the national innovation system of Lao PDR, which was delivered in the 2021 project report, with the consideration of the main tasks and roles of MTC as well as the result of the survey: technology commercialization, SME innovation, and international STI collaboration. From Chapter 2 to Chapter 7, each chapter explores the international and domestic environments of the designated strategic area and identifies challenges and opportunities in the area by examining statistical data, governance, stakeholders and required resources, and current policy issues of both countries.

The proposed Master Plan presents its vision for 2030, “transforming Lao PDR to a sustainable lower-income economy by 2030 through promoting R&D and application of STI into all sectors.” Under the vision, three strategies are highlighted; technology commercialization to promote the linkage of the supply and demand for technologies, MSME innovation to develop MSMEs to generate demands for technologies, and international STI cooperation to take advantage of the strengthening of S&T capacity. Each strategy also delineates several objectives and actions of each strategic area in the proposed Plan to achieve the vision. In terms of ‘learning by doing,’ this project is especially meaningful for all participants to dedicate themselves to drafting the Master Plan. It has been a great journey for all experts to learn mutually and deliberate on STI policy challenges and solutions in the unique circumstance of Lao PDR.

While there are needs to be continuously updated and revised until the implementation of the Plan, the Department of Technology Innovation (DTI) in MTC plans to use this National Master Plan as a guide to creating a 5-year development plan of DTI and to vision a more active role of STI in the next 10th National Socio-Economic Development Plan (2026-2030). In addition, the DTI is going to pursue the dissemination of the Plan to relevant STI ministries, public and private institutions, and universities in order to build a cooperation mechanism for STI development in Lao PDR. Despite the termination of the K-Innovation program with Lao PDR this time, STEPI will continue supporting the dissemination and implementation of the National STI Master Plan of Lao PDR and further developing this valuable bilateral partnership between Korea and Lao PDR